สยามชำนาญกิจ
SIAM CHAMNAN KIT

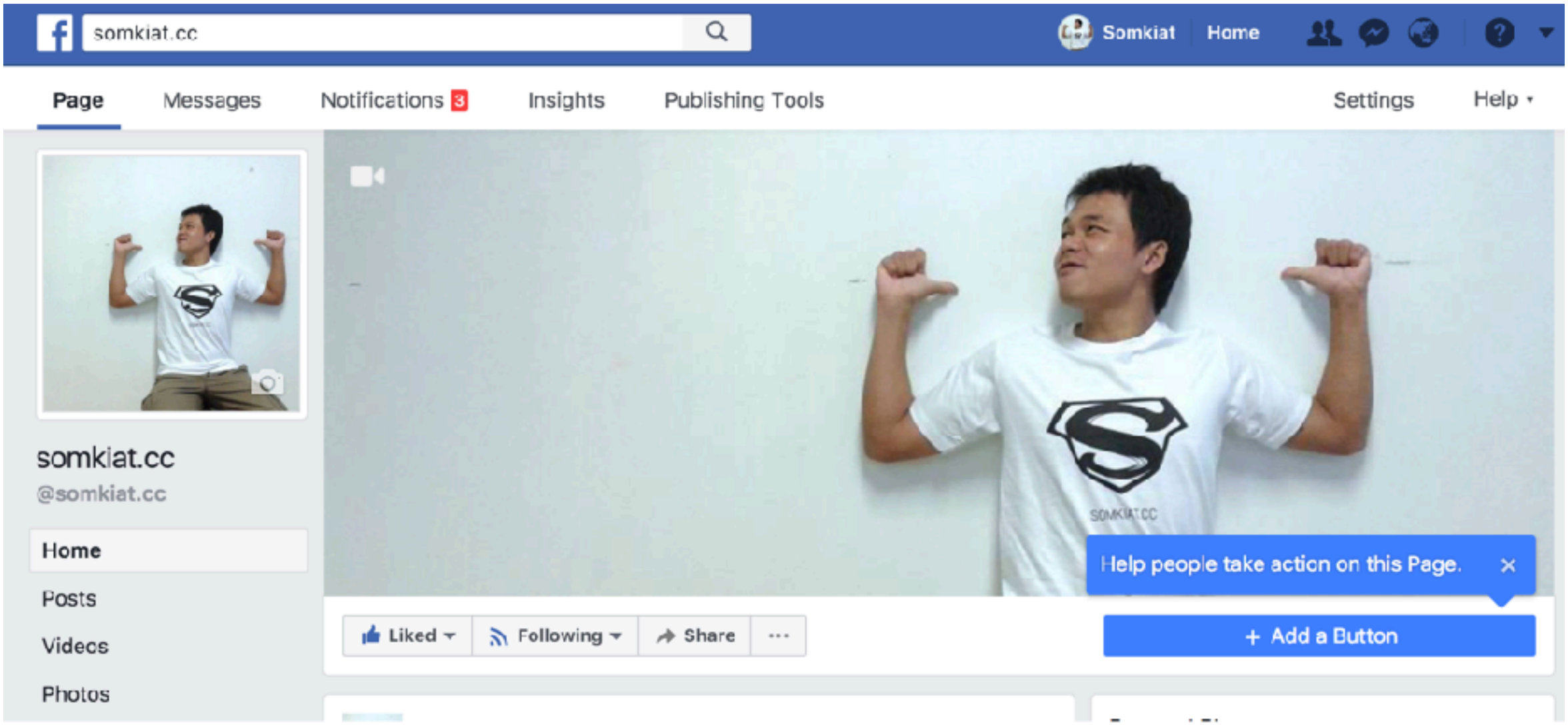
Generative AI for Software Development 2026

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https://github.com/up1/workshop-ai-with-technical-team

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Goals

Integrate Generative AI in Software Development
Optimize code quality
Team up with AI on coding tasks
Develop innovative solutions



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Learning Path

AI/ML

AI Model

Prompt Engineer

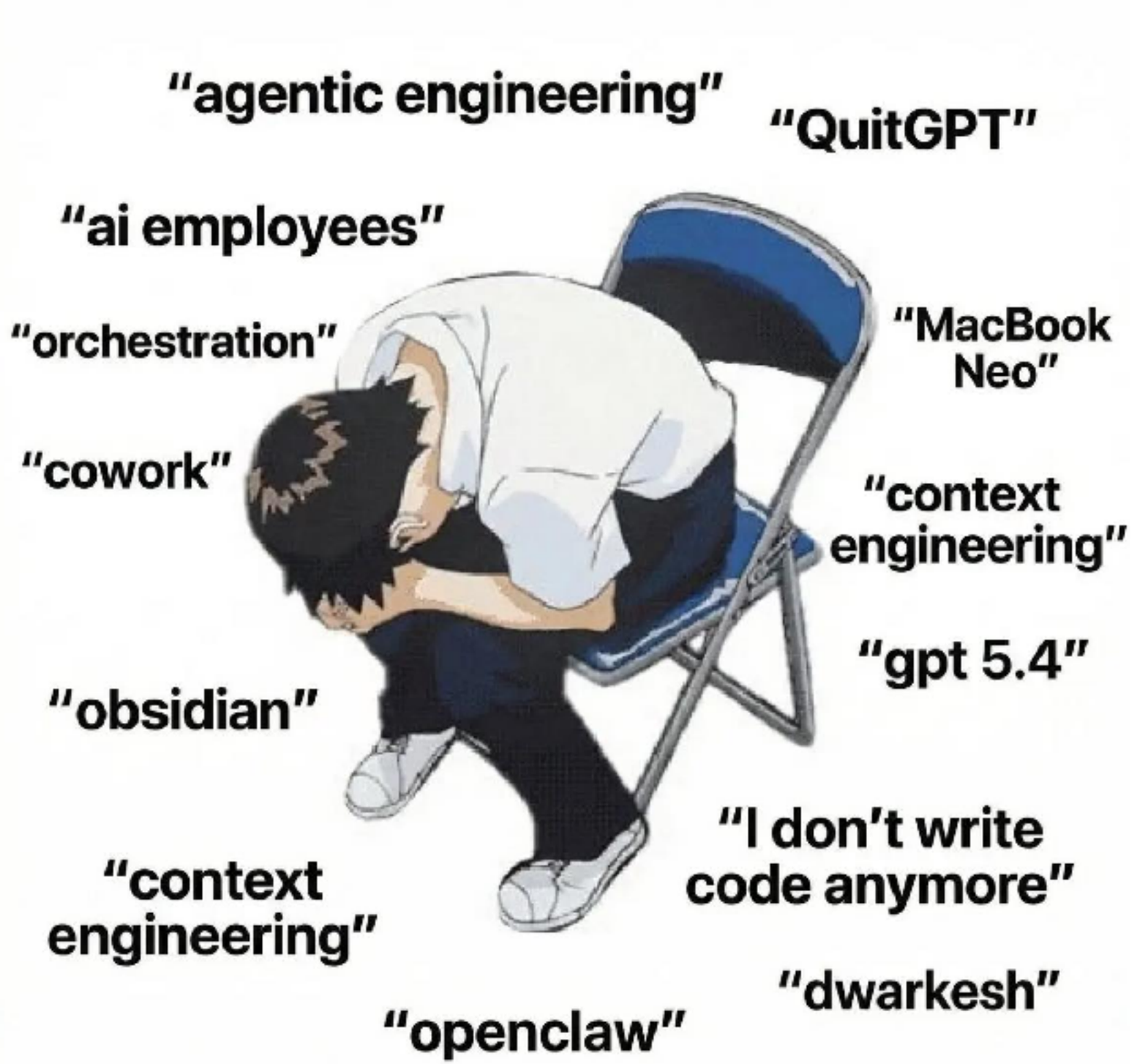
AI in Software development

Develop AI/LLM app

RAG

Use cases and workshops

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How to use AI ?

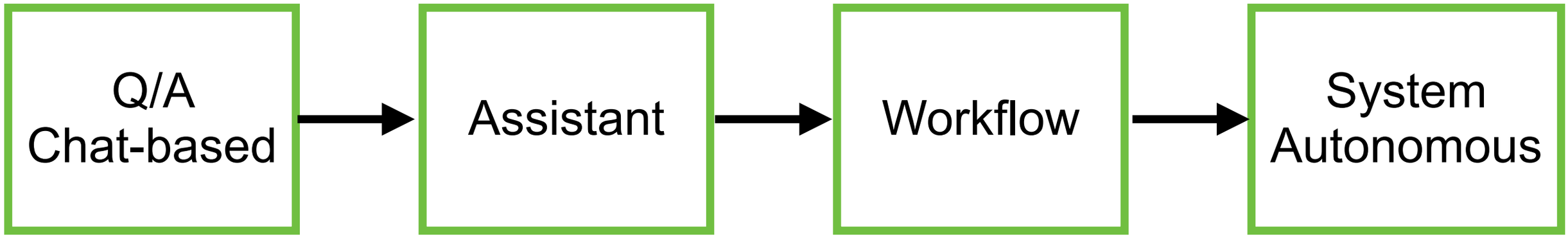
AI as
a **tool**

AI as
an **assistant**

AI as
an **agent**

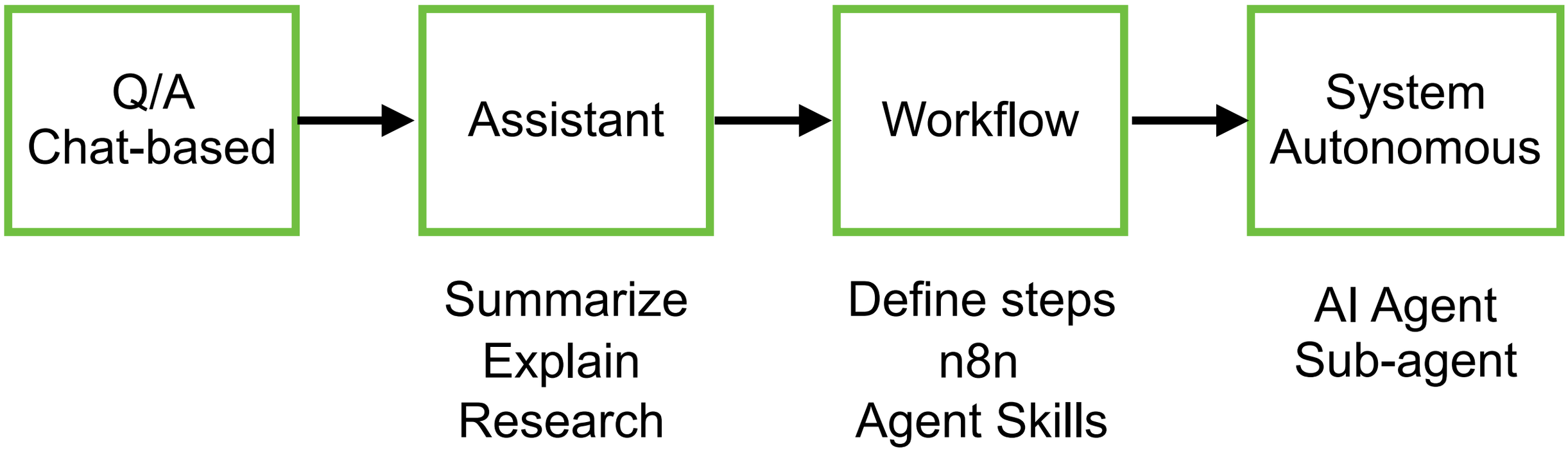
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Steps to use AI ?



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Steps to use AI ?



Prompt engineering

Manage context and knowledge for AI (Context engineering)
Human in the Loop (HITL)

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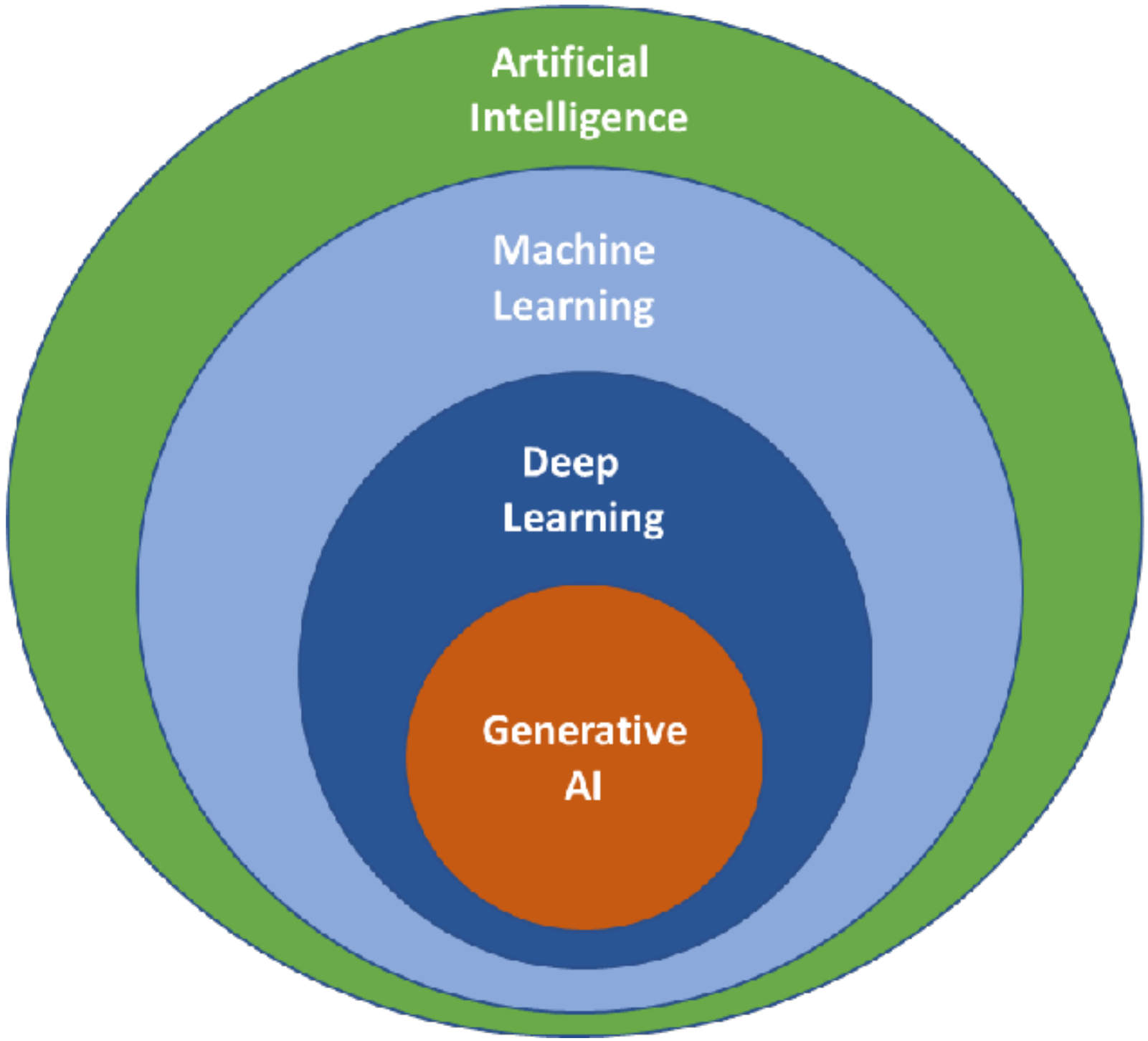
Software Development

Requirement | Design | Develop | Testing | Deploy

Generative AI

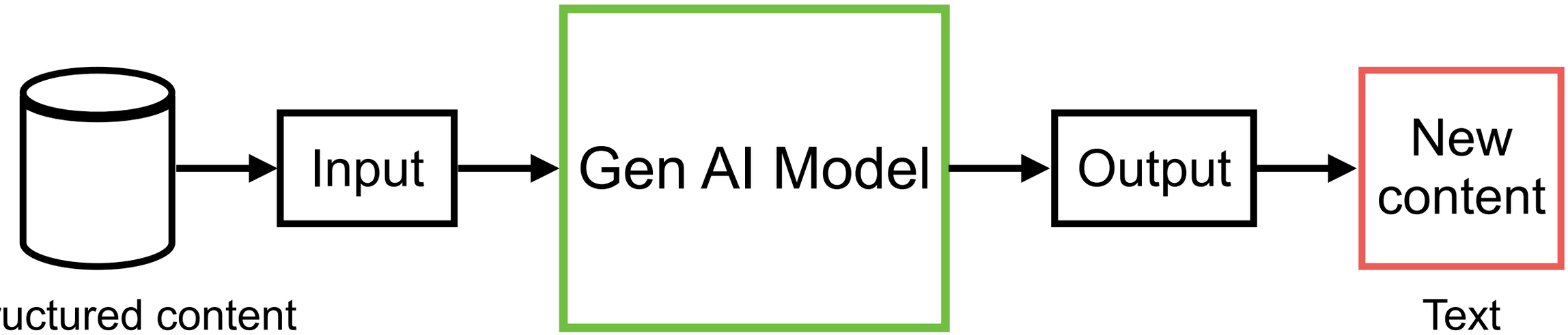
Improve Productivity … (Replace human !!)

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Generative AI



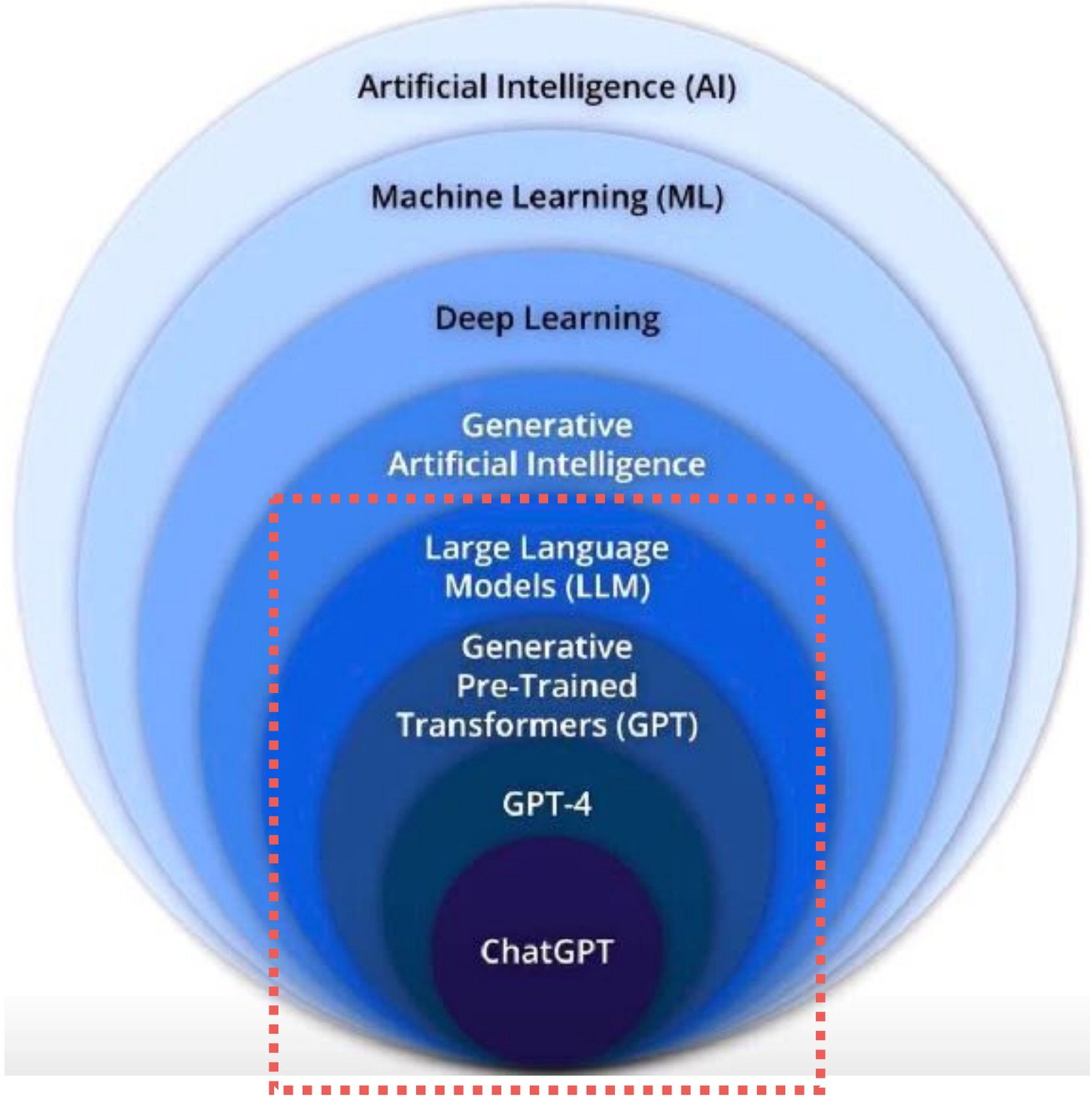
Unstructured content

Learn patterns from Unstructured content

Text
Image
Video
3D
Task

https://grow.google/ai-essentials/

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Large Language Model (LLM)

Type of AI model
Process, understand
Generate human readable data

LLM

Training data !!

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LLMs in industry

Model name	Company
Bidirectional Encoder Representation from Transformers (BERT)	Google AI
Generative Pre-trained transformer-5 (GPT-5)	OpenAI
Pathways Language Model-E (PaLM-E)	Google AI
BLOOM	NVIDIA AI
Llama 4	Facebook
Claude’s Sonnet/Opus 5.5	Anthropic

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LLM Development timeline

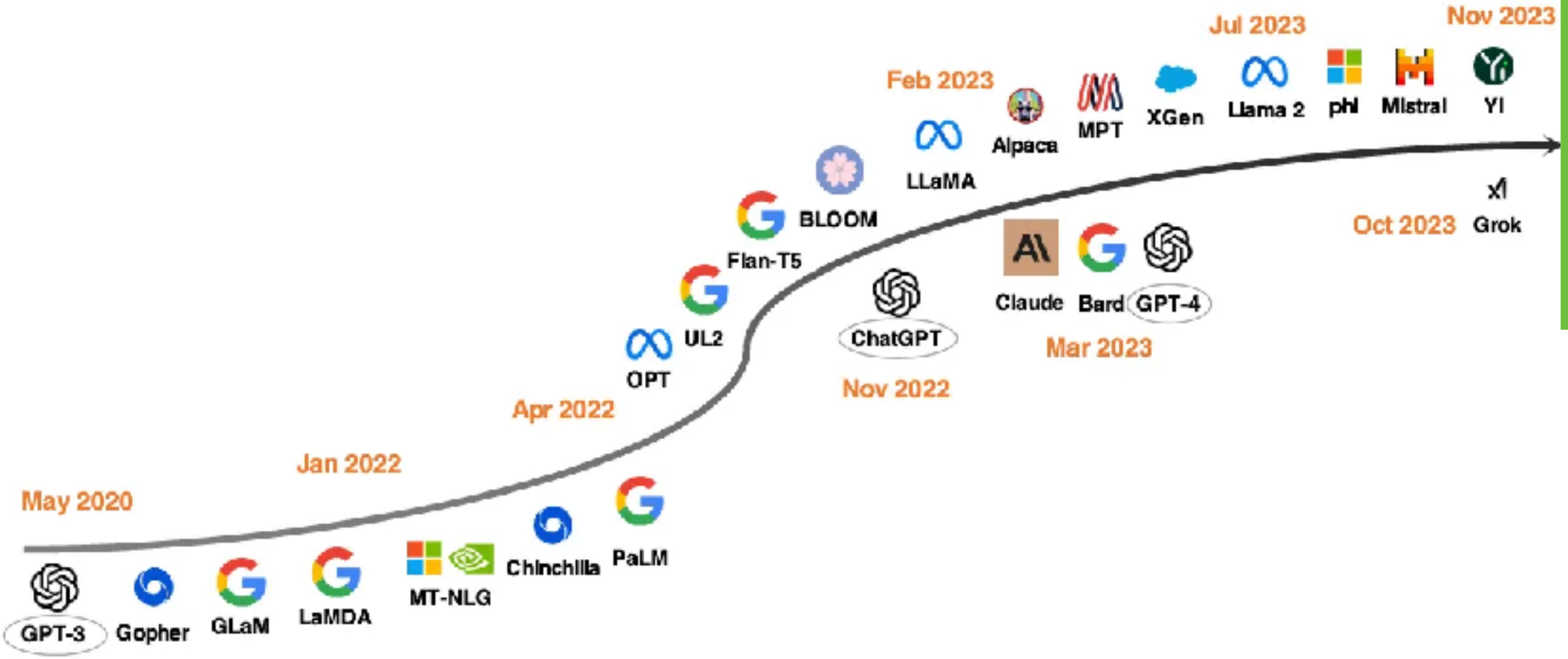


Figure 3: LLM development timeline. The models below the arrow are closed-source while those above the arrow are open-source.

https://arxiv.org/abs/2311.16989

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Let’s go !!

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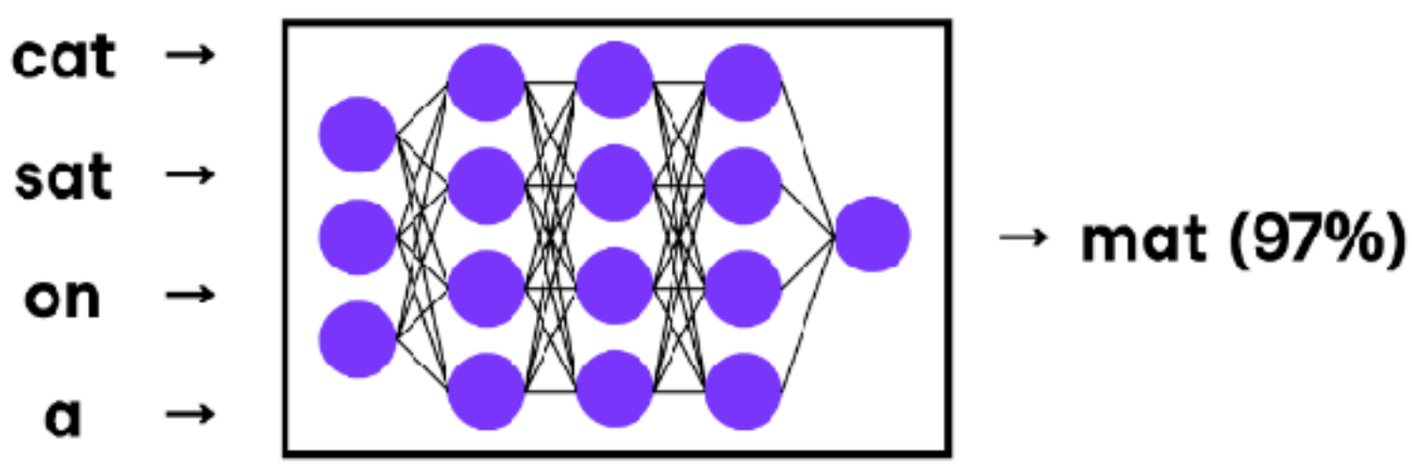
สีเหลือง

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Large Language Model (LLM)

Neural network

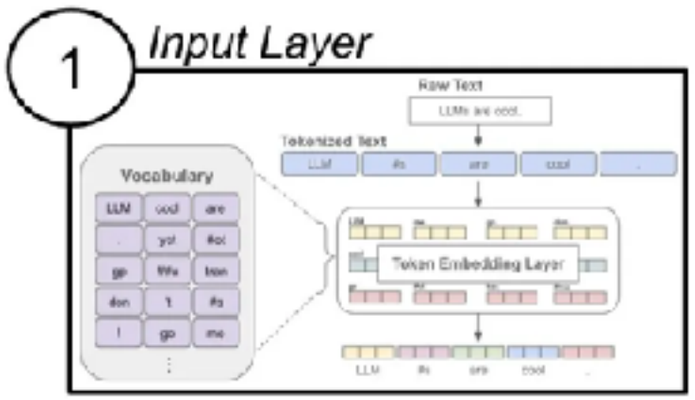
Predicts the next word in a sequence

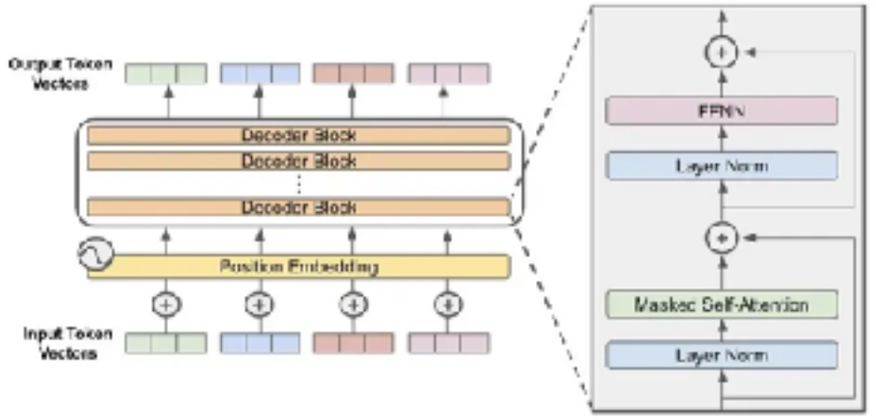


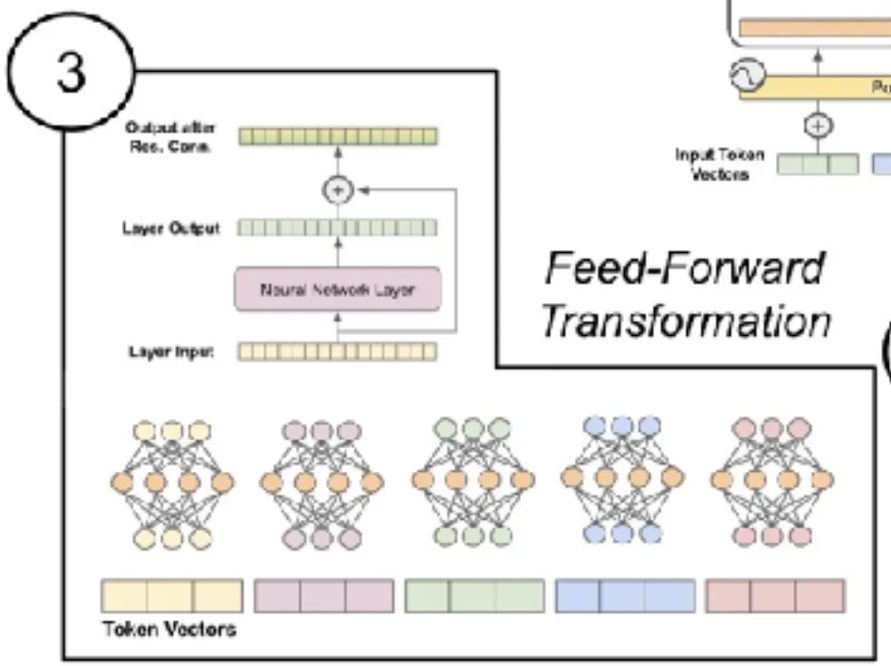
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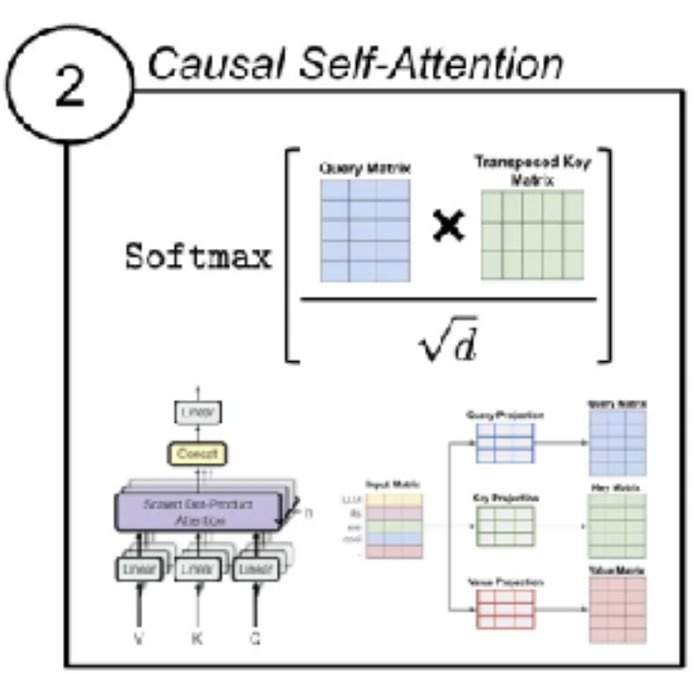
LLM components !!

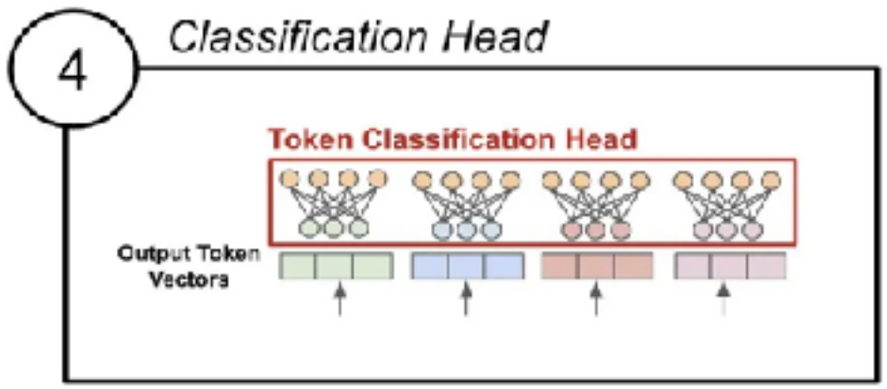
Components of the Decoder-only Transformer









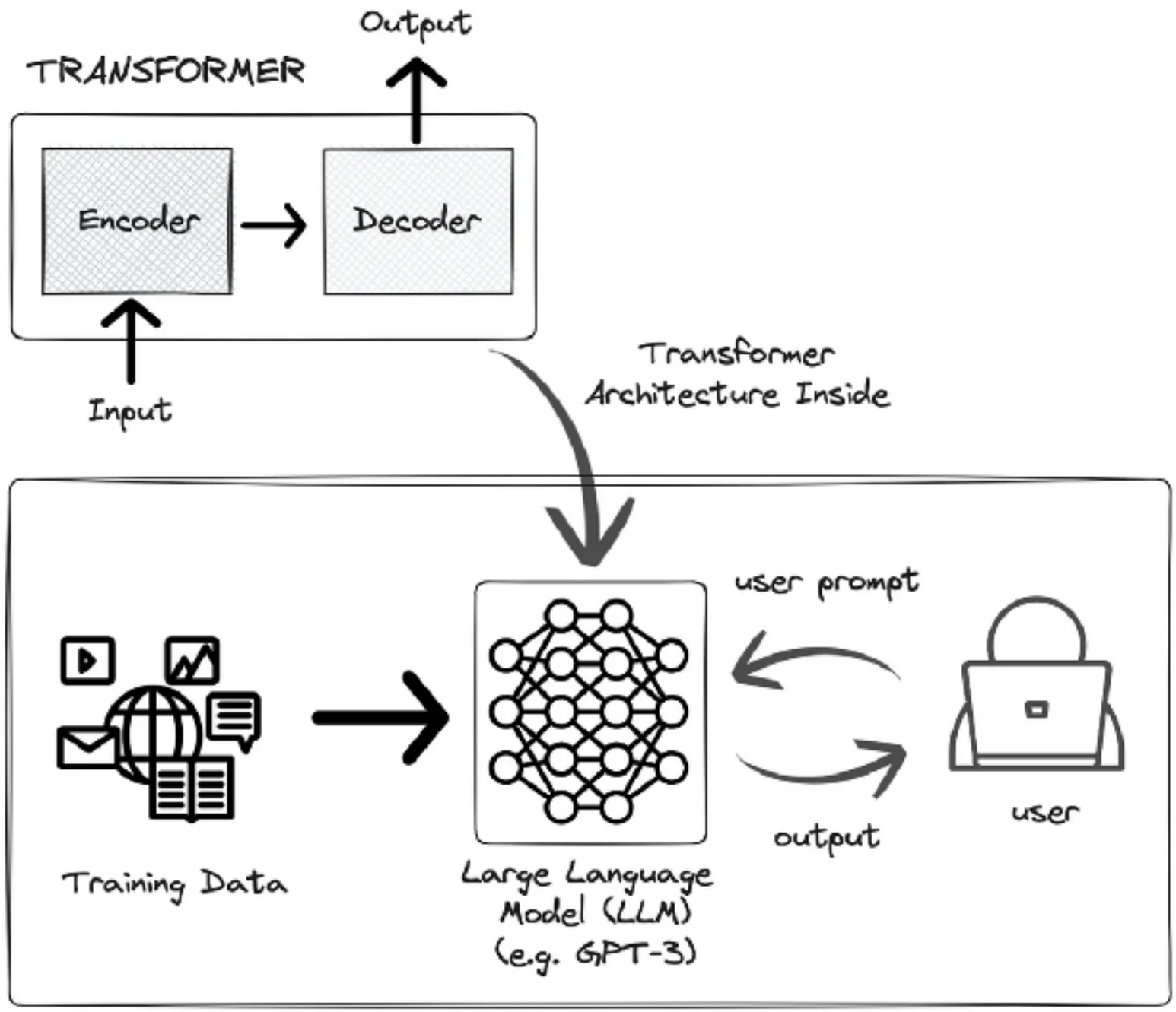




https://stackoverflow.blog/2024/08/22/llms-evolve-quickly-their-underlying-architecture-not-so-much

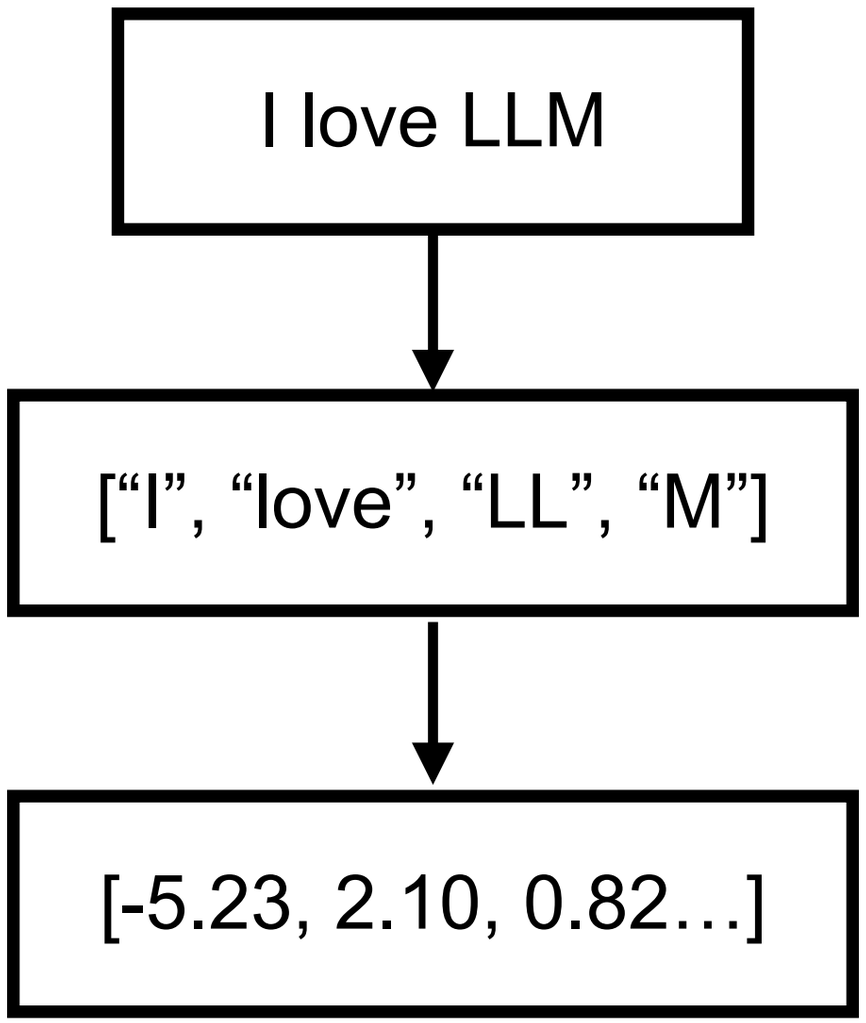
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Transformer inside



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Transformer process



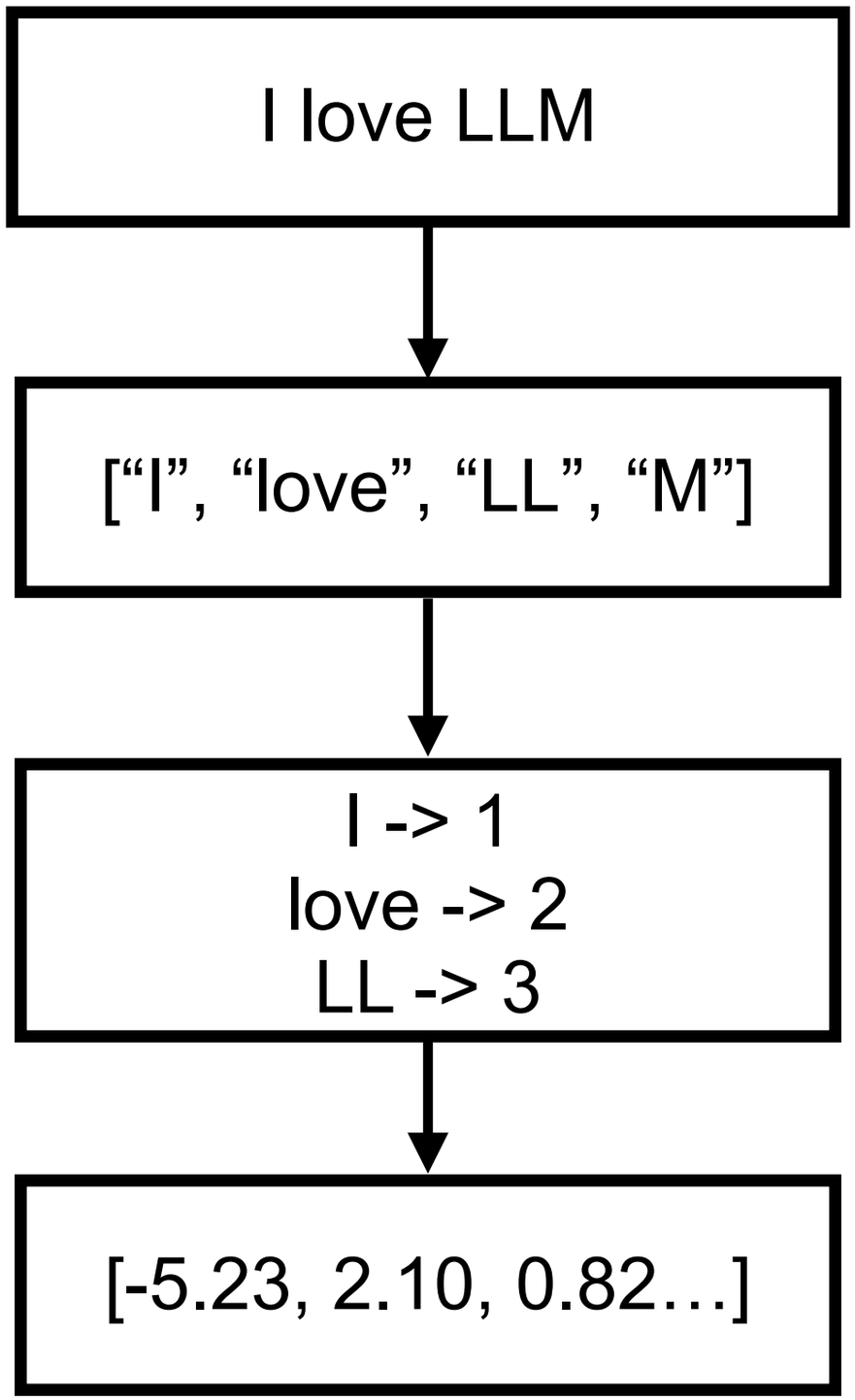
Sentence

Tokens

Embedding

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Transformer process



Sentence

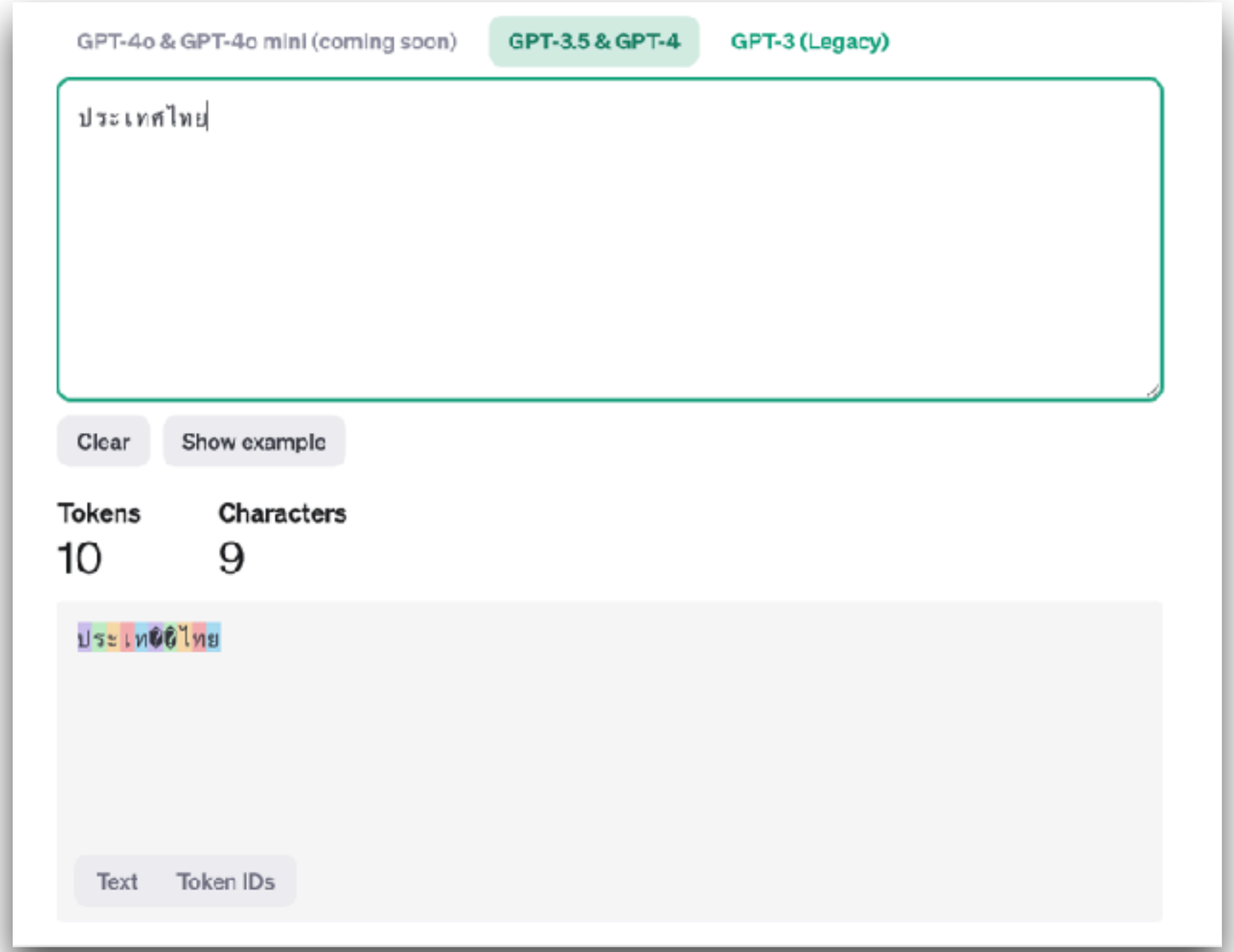
Tokens

Token index

Embedding

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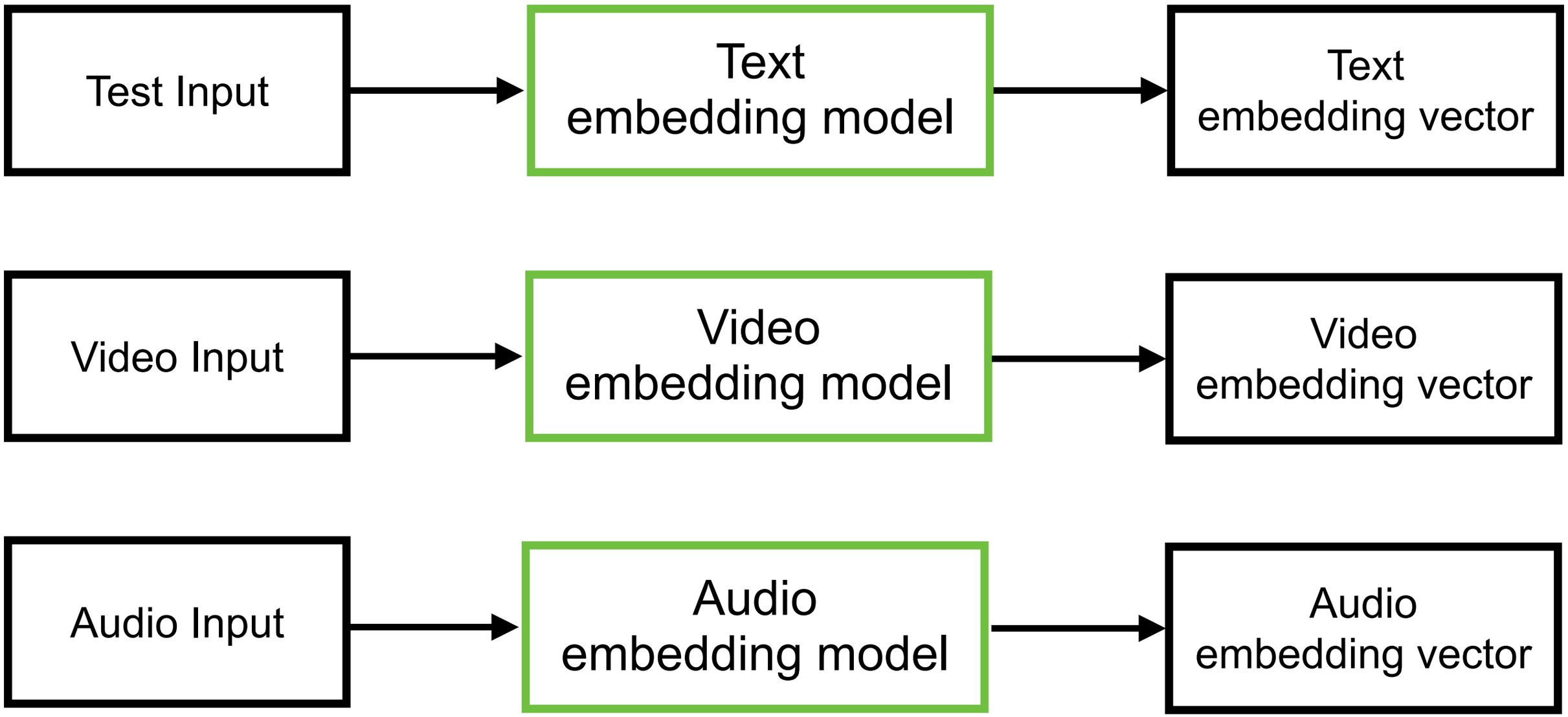
OpenAI Tokenizer



https://platform.openai.com/tokenizer

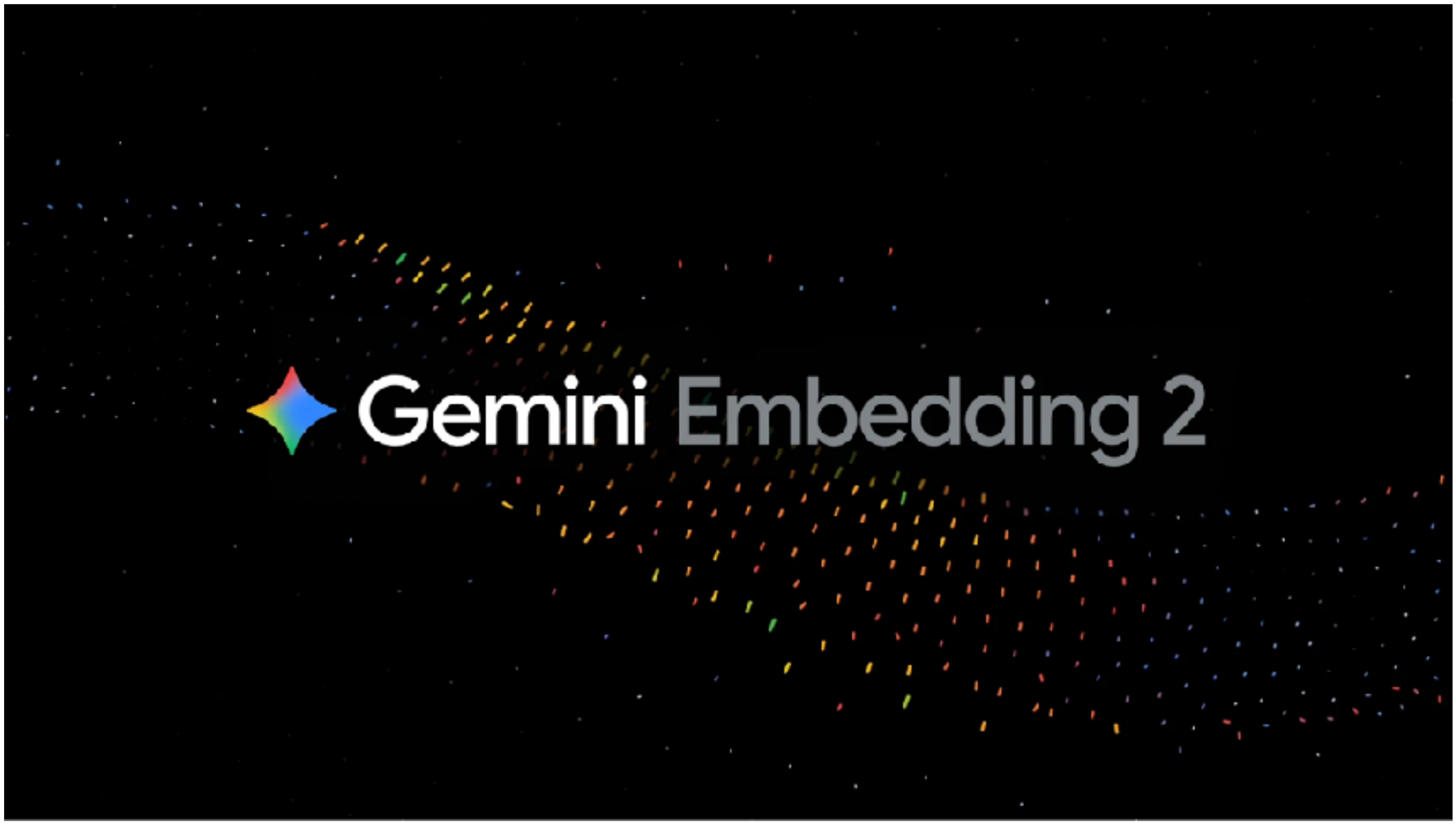
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Embedding



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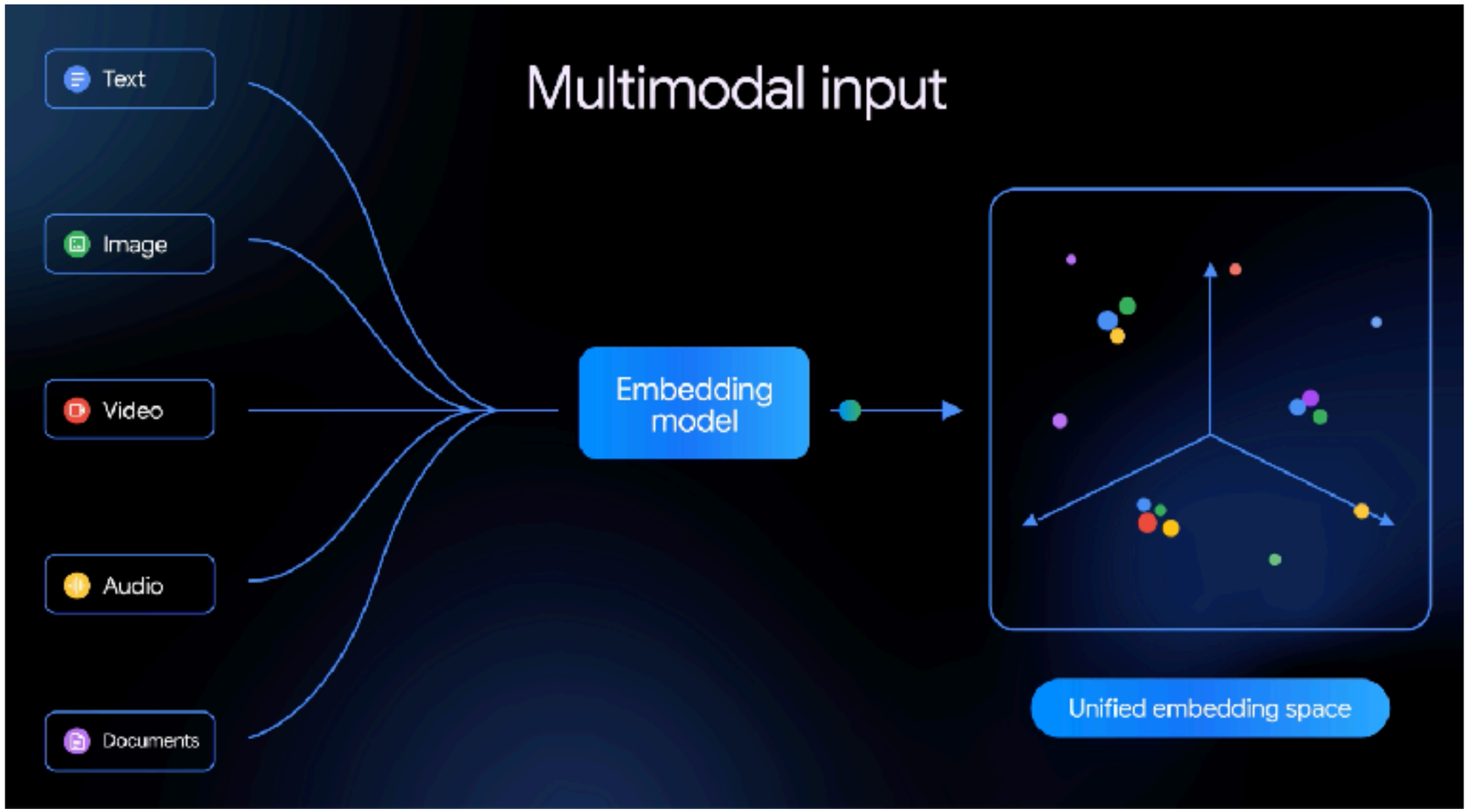
Multimodal Embedding Model



https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-embedding-2/

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Multimodal Embedding Model

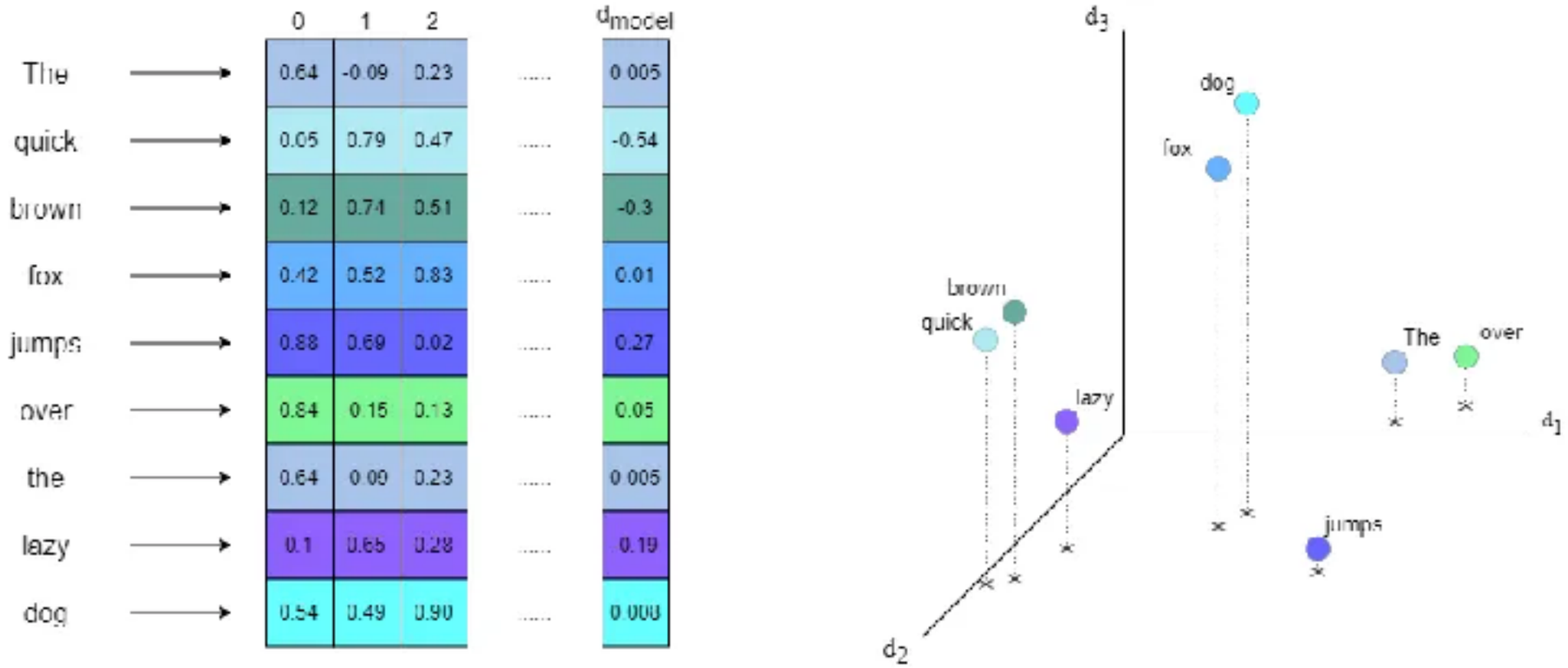


https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-embedding-2/

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Embedding ?

Map items of **unstructured data** to high-dimensional real vectors



https://towardsdatascience.com/transformers-in-depth-part-1-introduction-to-transformer-models-in-5-minutes-ad25da6d3cca

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Embedding Leaderboard

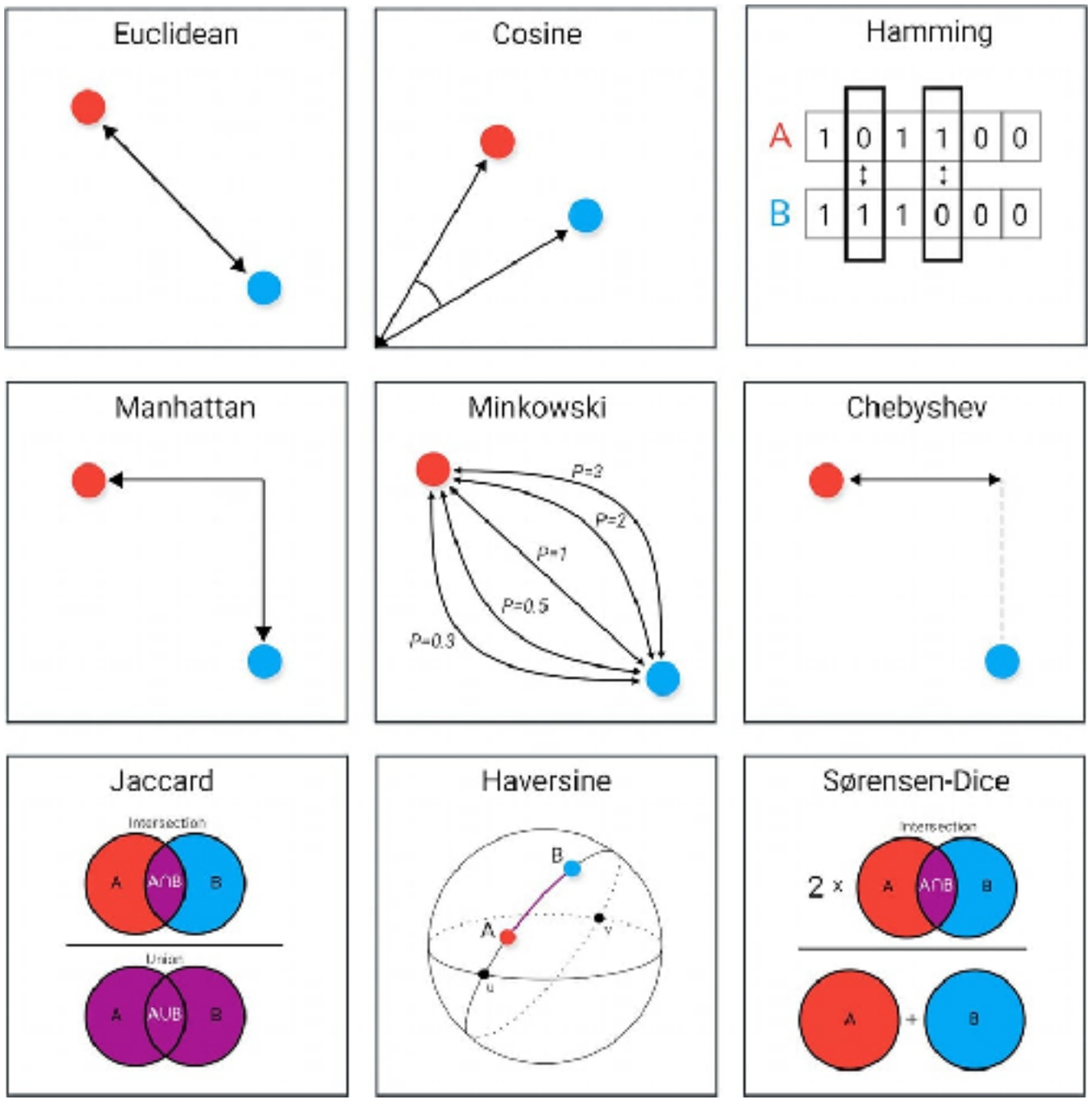
Rank (Bor…	Model	Embedding D…	Max Tokens	Mean (T…	Mean (TaskT…	Bitext …	Classification	Clustering	Instructi
1	KaLM-Embedding-Gemma3-12B-2511	3840	32768	**72.32**	**62.51**	**83.76**	**77.88**	55.77	5.49
2	llama-embed-nemotron-8b	4096	32768	69.46	61.09	81.72	73.21	54.35	10.82
3	Qwen3-Embedding-8B	4096	32768	70.58	61.69	80.89	74.00	57.65	10.06
4	gemini-embedding-001	3072	2048	68.37	59.59	79.28	71.82	54.59	5.18
5	Qwen3-Embedding-4B	2560	32768	69.45	60.86	79.36	72.33	57.15	11.56
6	Octen-Embedding-8B	4096	32768	67.84	60.28	80.35	66.68	55.68	8.98
7	F2LLM-v2-14B	5120	40960	68.74	59.45	77.15	73.60	**60.91**	0.62
8	F2LLM-v2-8B	4096	40960	68.09	58.99	75.96	71.93	60.62	0.85
9	Seed1.6-embedding-1215	2048	32768	70.26	61.34	78.68	76.75	56.78	-0.82
10	F2LLM-v2-4B	2560	40960	67.06	58.25	74.49	70.73	59.53	2.25
11	jina-embeddings-v5-text-small	1024	32768	67.00	58.90	69.71	71.32	53.41	1.35

https://huggingface.co/spaces/mteb/leaderboard



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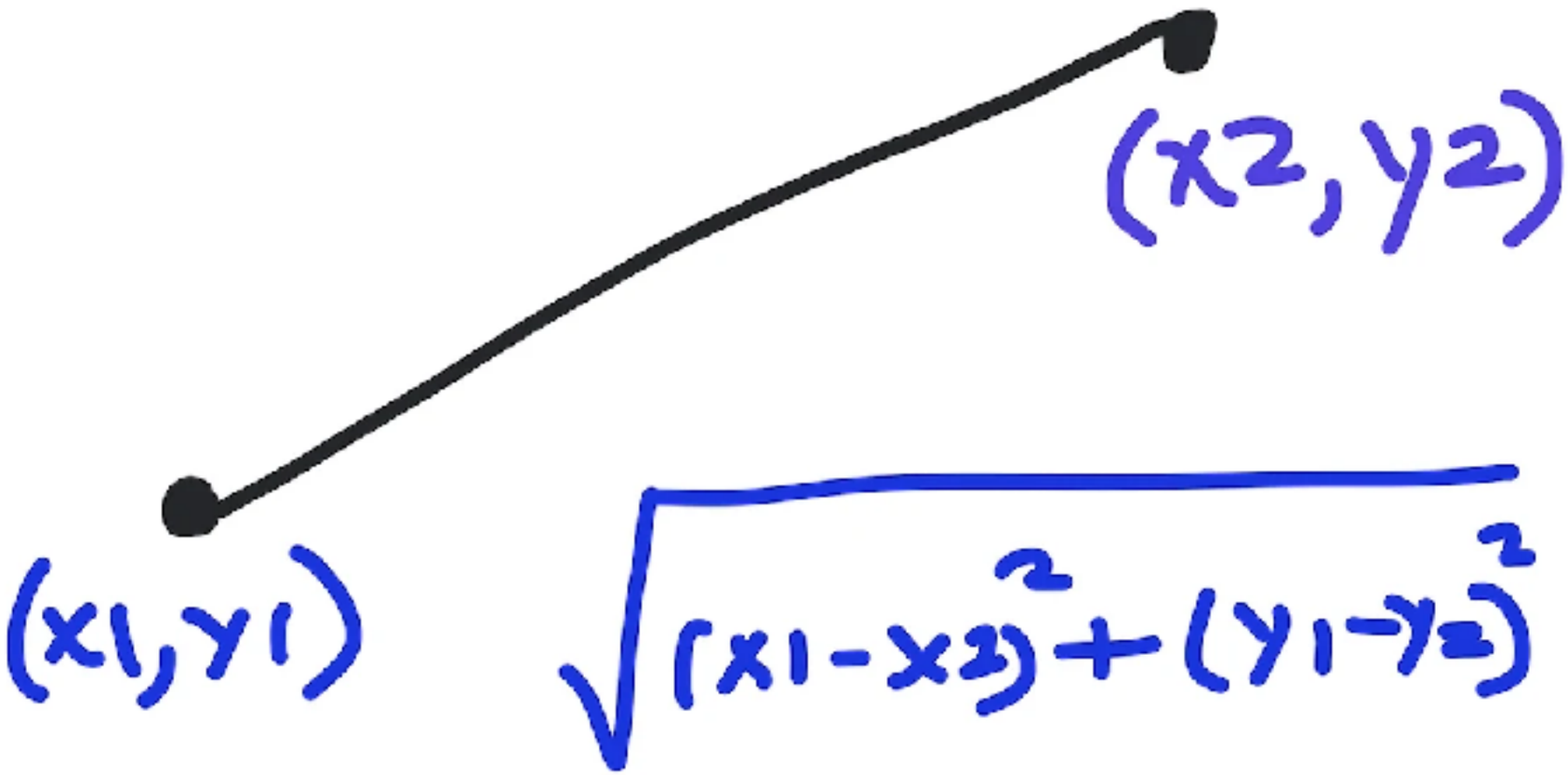
Distance measure in Data Science



https://www.maartengrootendorst.com/blog/distances/

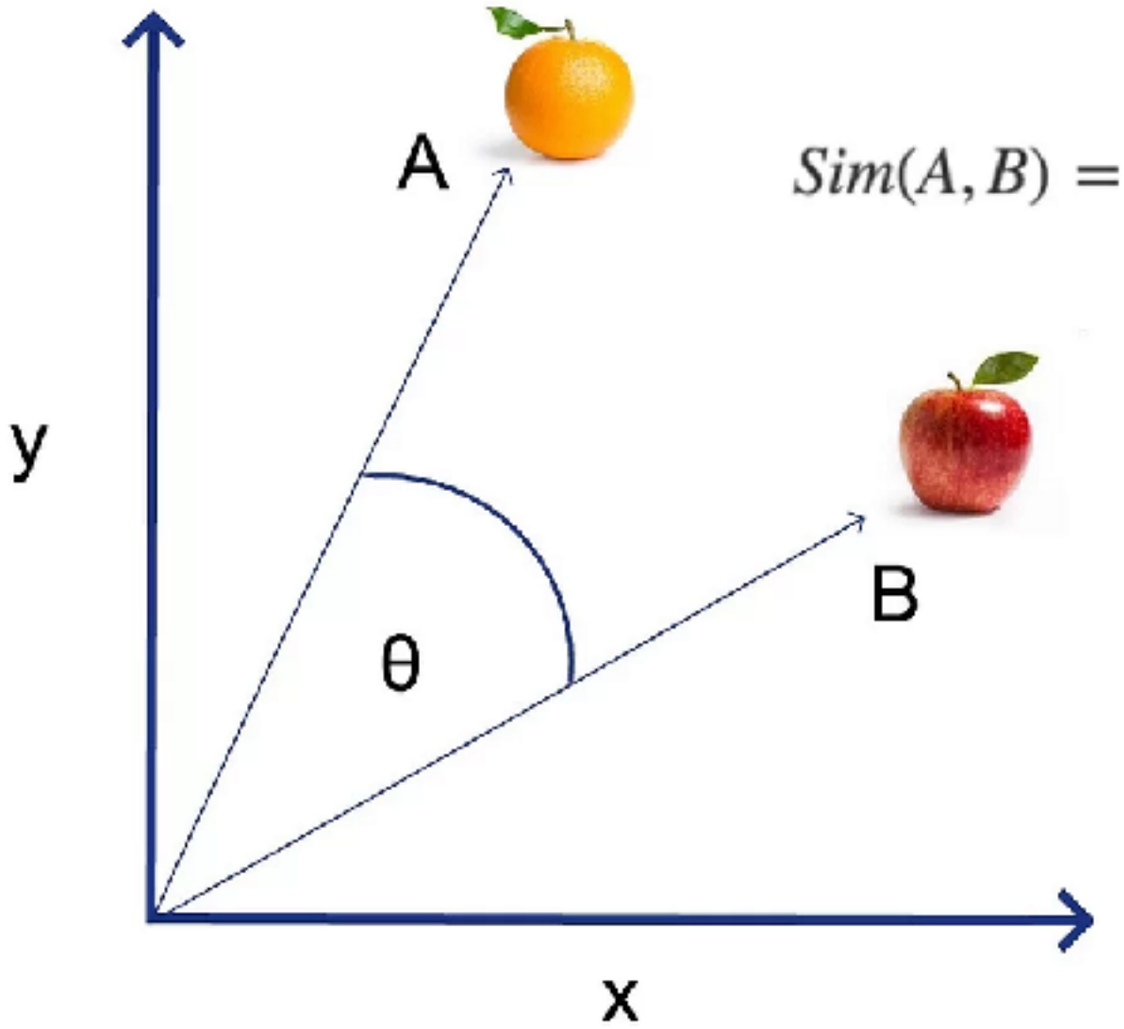
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Euclidean Distance



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Cosine Similarity

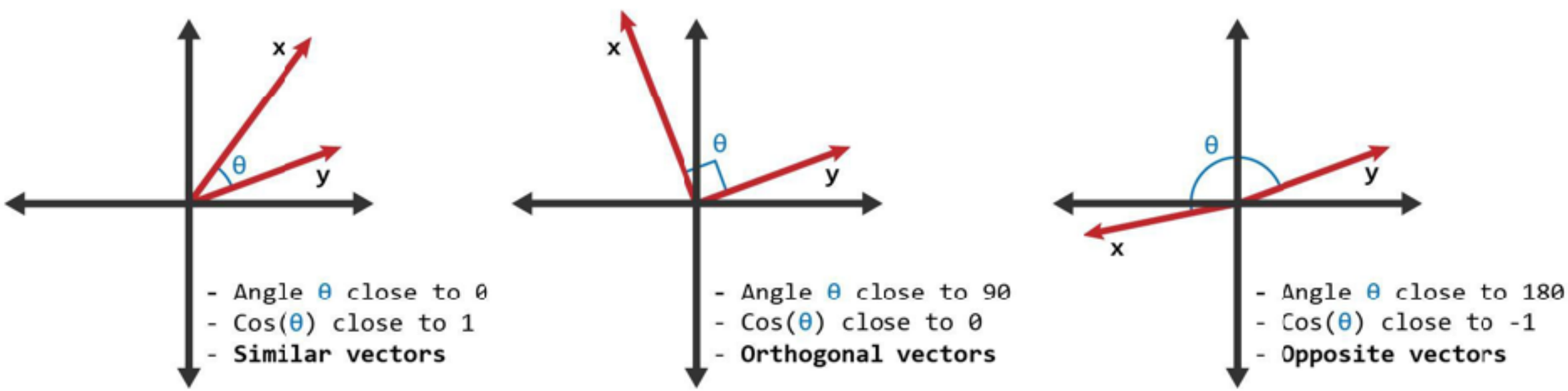


$$Sim(A, B) = \cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|}$$

https://python.plainenglish.io/mastering-cosine-metrics-a-data-scientists-essential-toolkit-3ce95eda91f9

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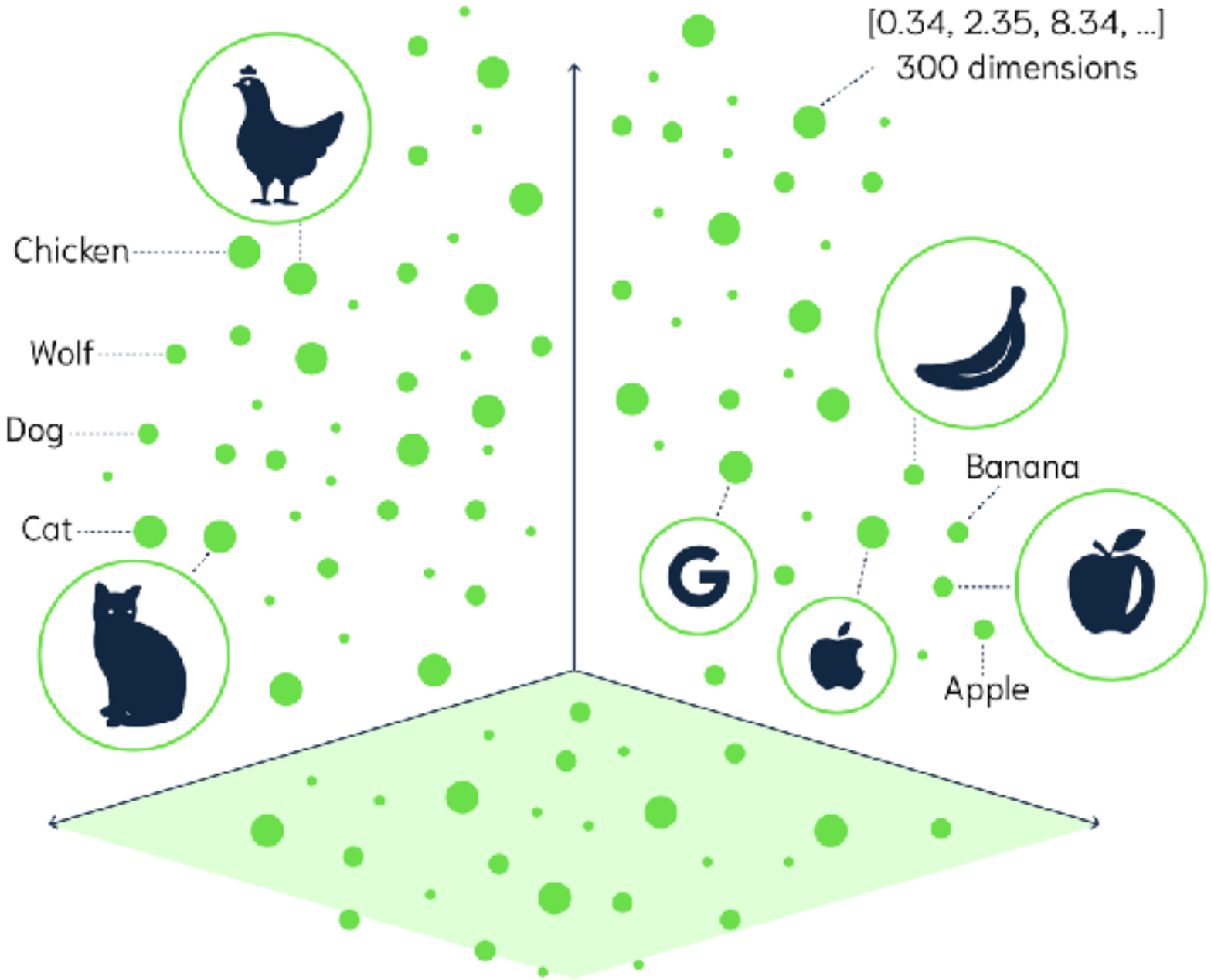
Cosign Similarity



https://www.learndatasci.com/glossary/cosine-similarity/

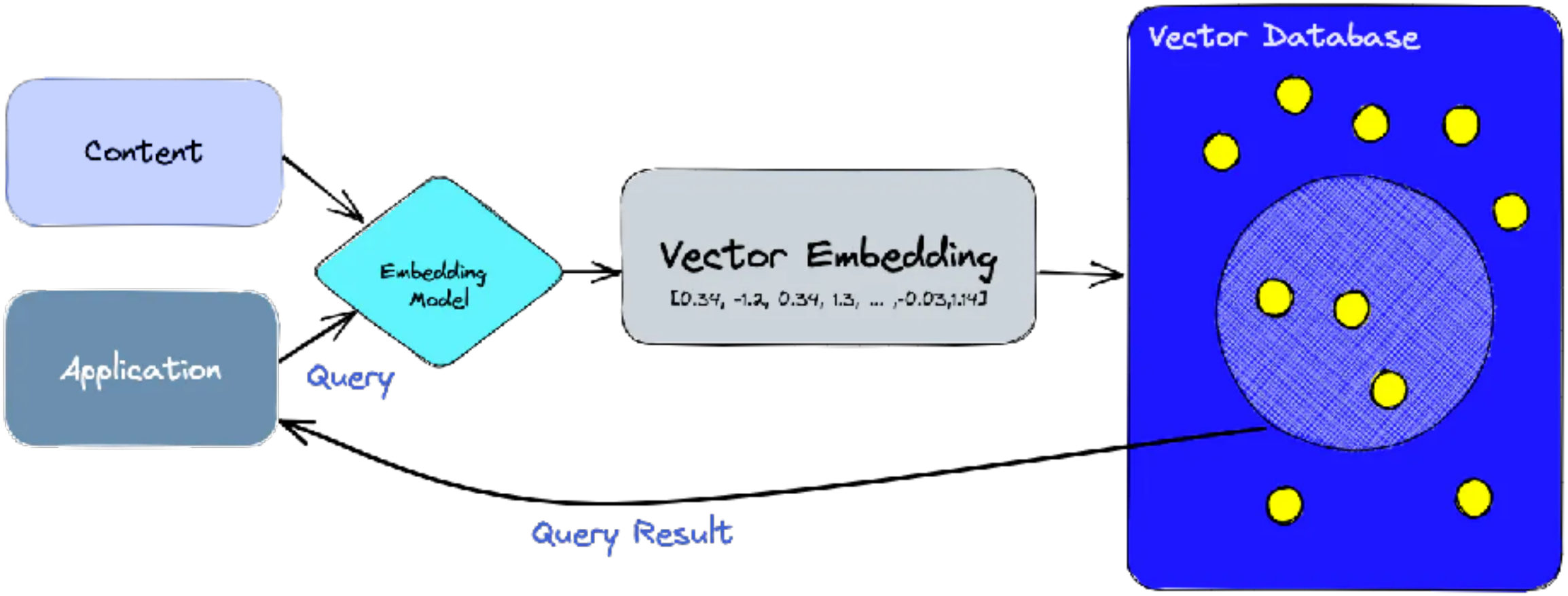
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Visual of Vector space



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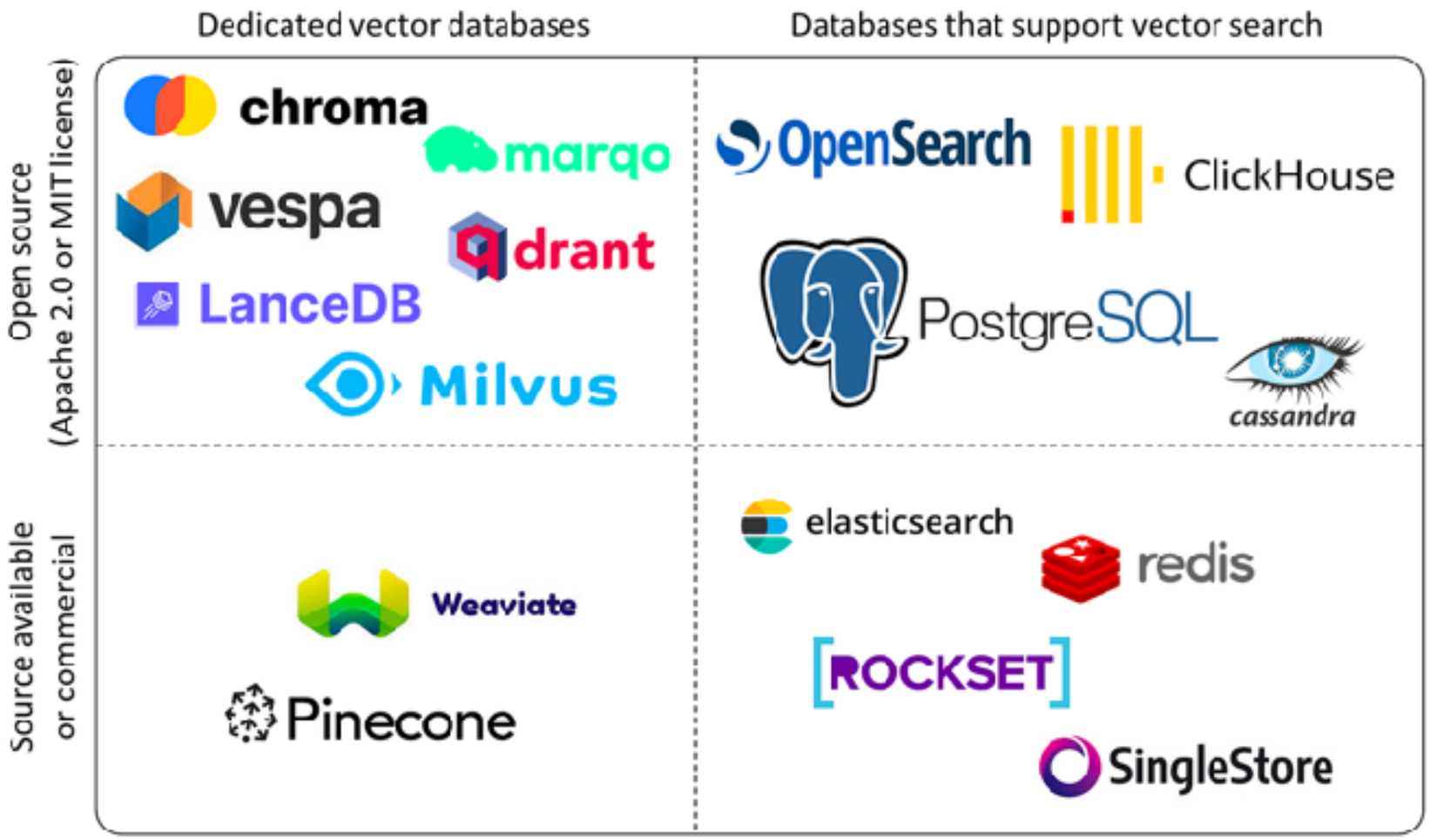
Store data in Vector Database



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Vector Database ?

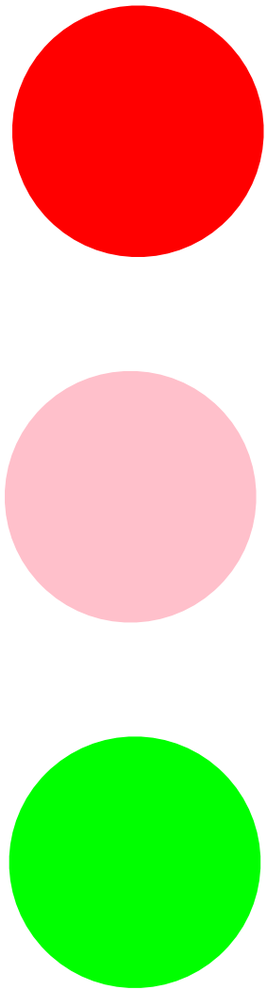
Map items of unstructured data to high-dimensional real **vectors**



https://towardsdatascience.com/transformers-in-depth-part-1-introduction-to-transformer-models-in-5-minutes-ad25da6d3cca

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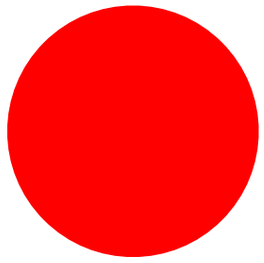
RGB ?



https://github.com/up1/workshop-ai-with-technical-team/tree/main/workshop/demo-rgb

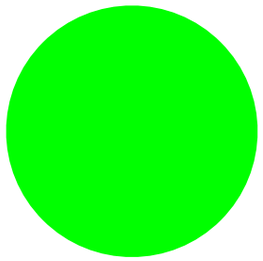
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RGB ?



[255, 0, 0]

[255, 192, 203]

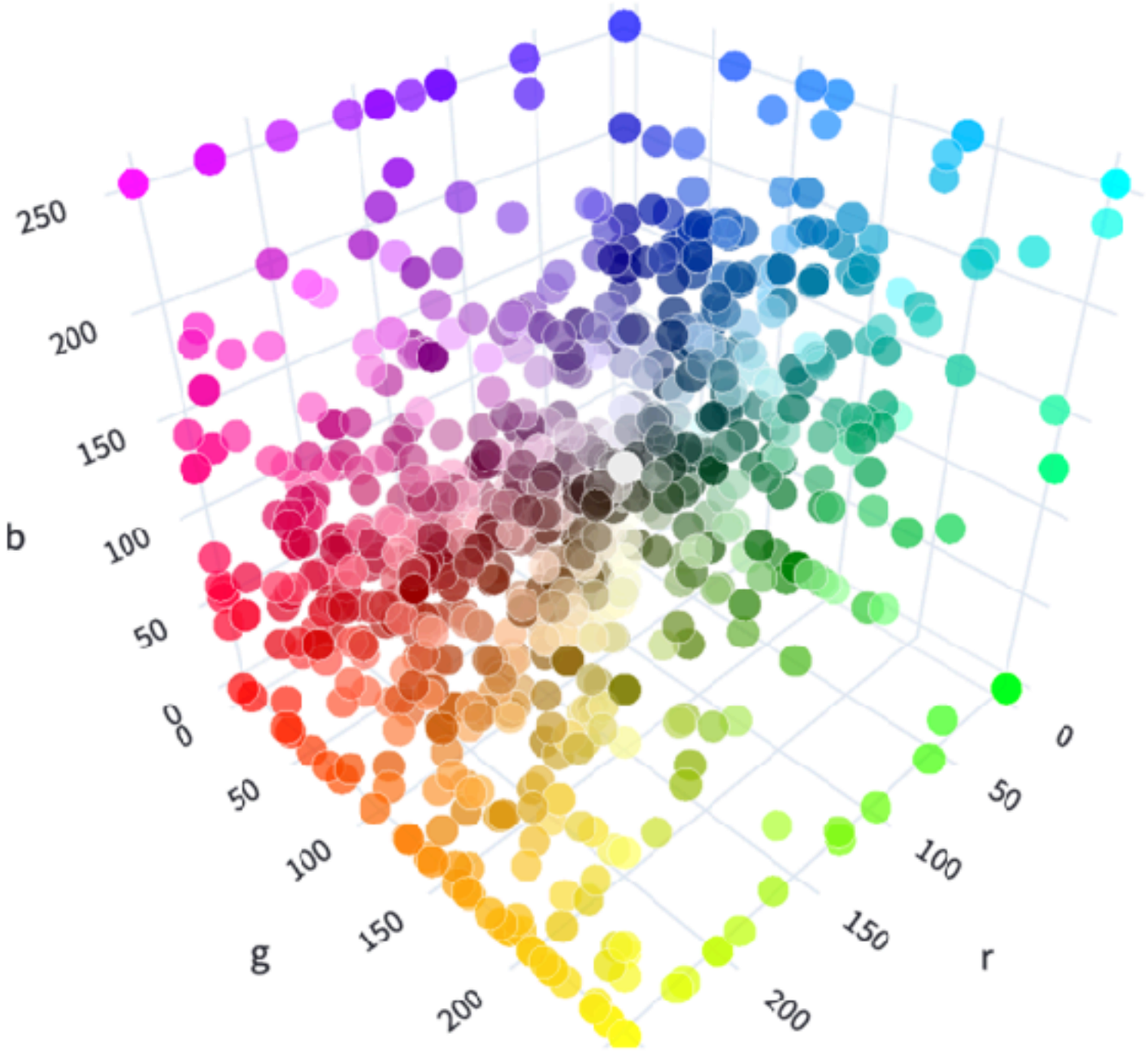


[0, 255, 0]

https://github.com/up1/workshop-ai-with-technical-team/tree/main/workshop/demo-rgb

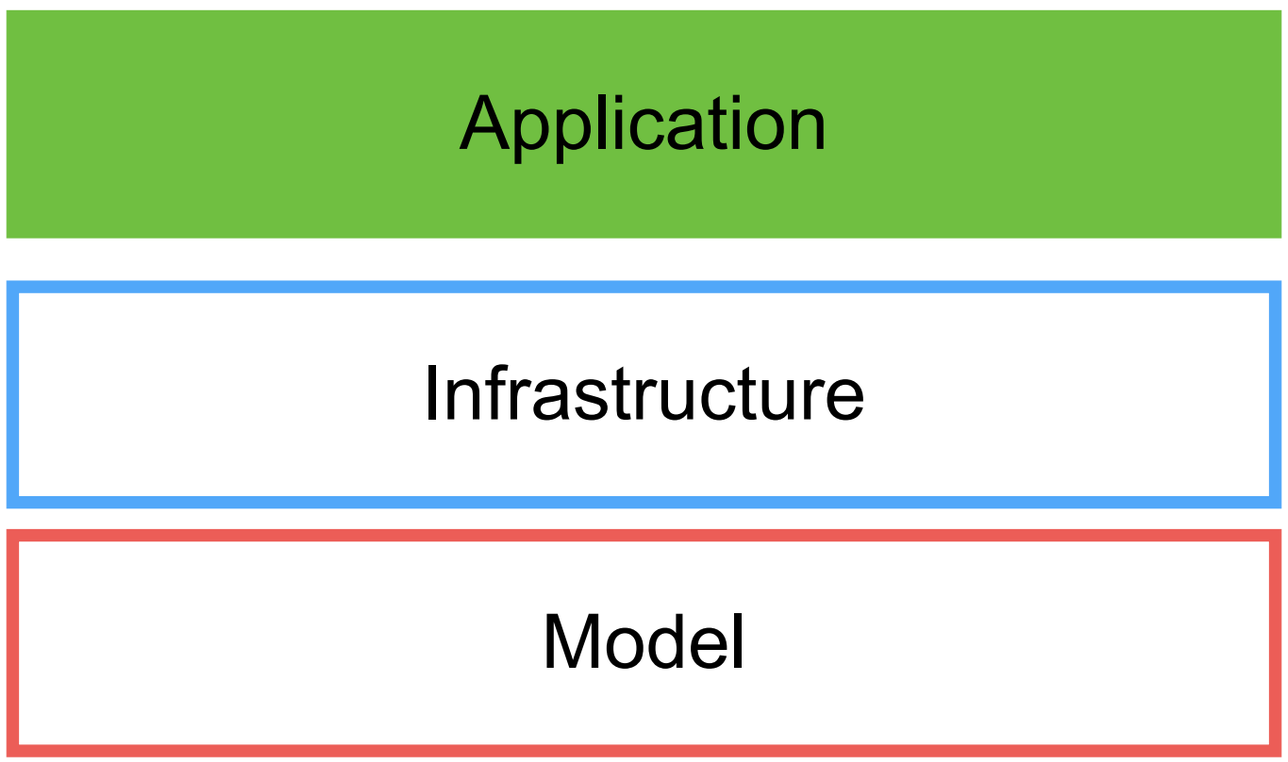
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Visualize RGB

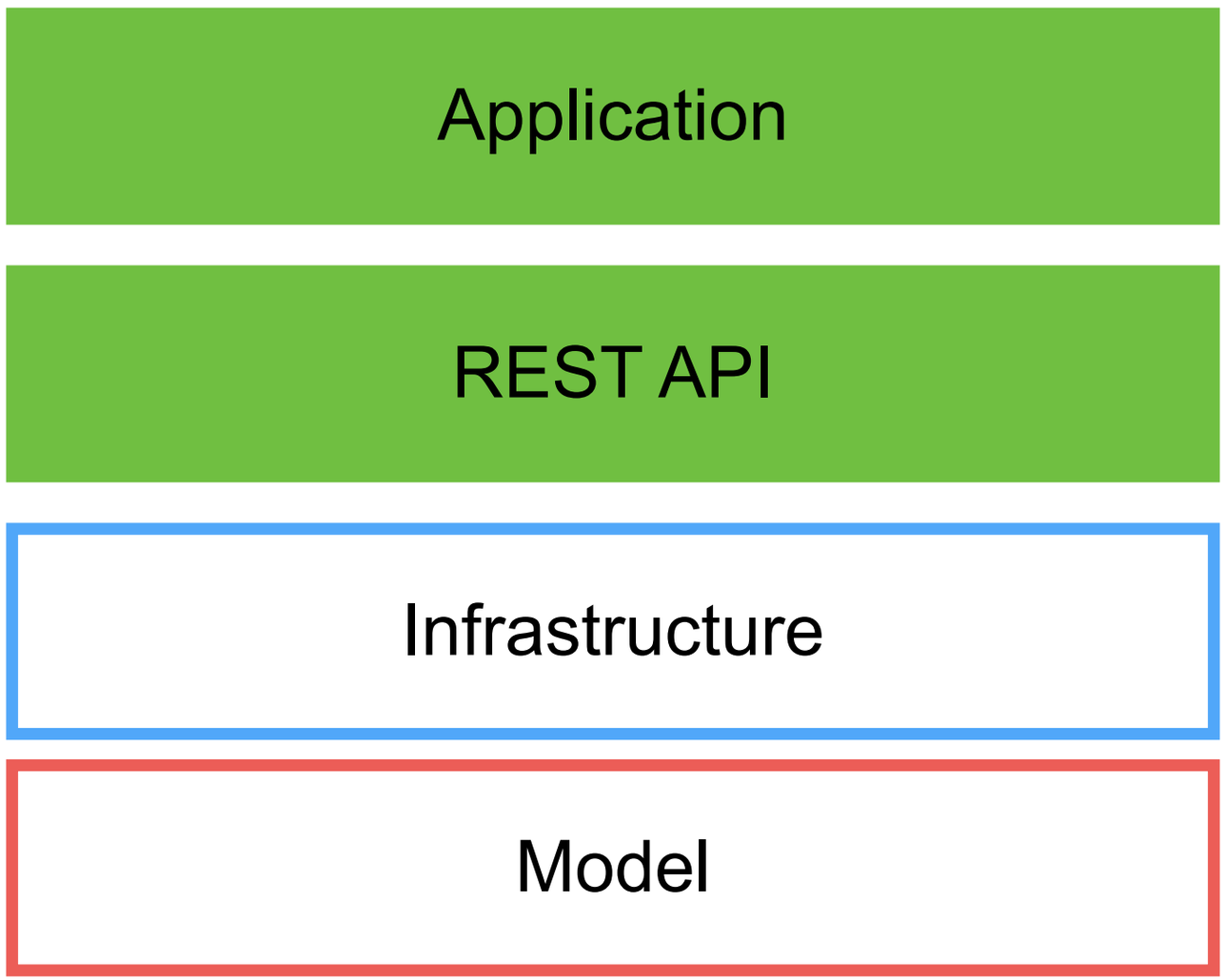


https://huggingface.co/spaces/jphwang/colorful_vectors

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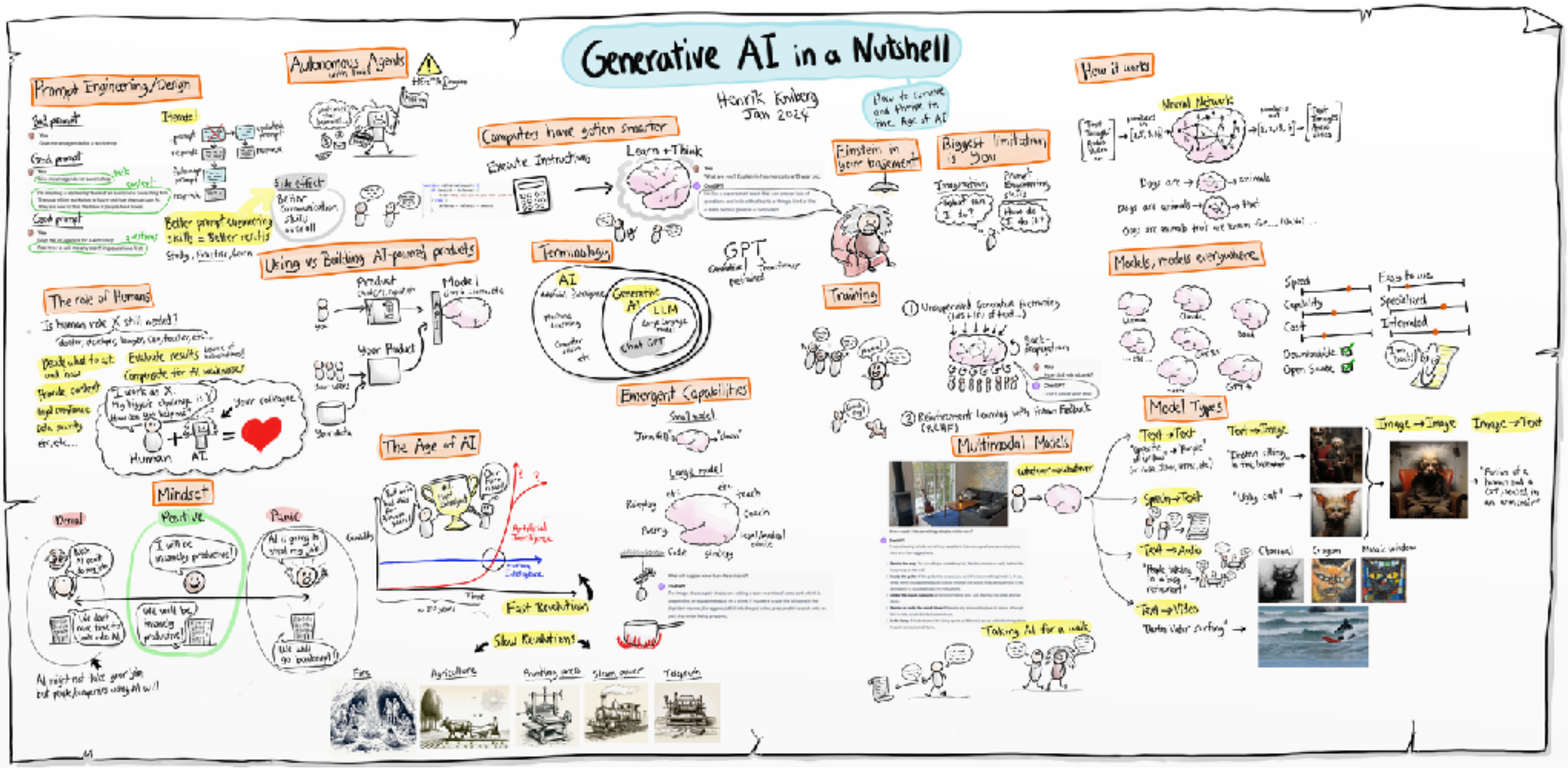


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Generative AI in Nutshell

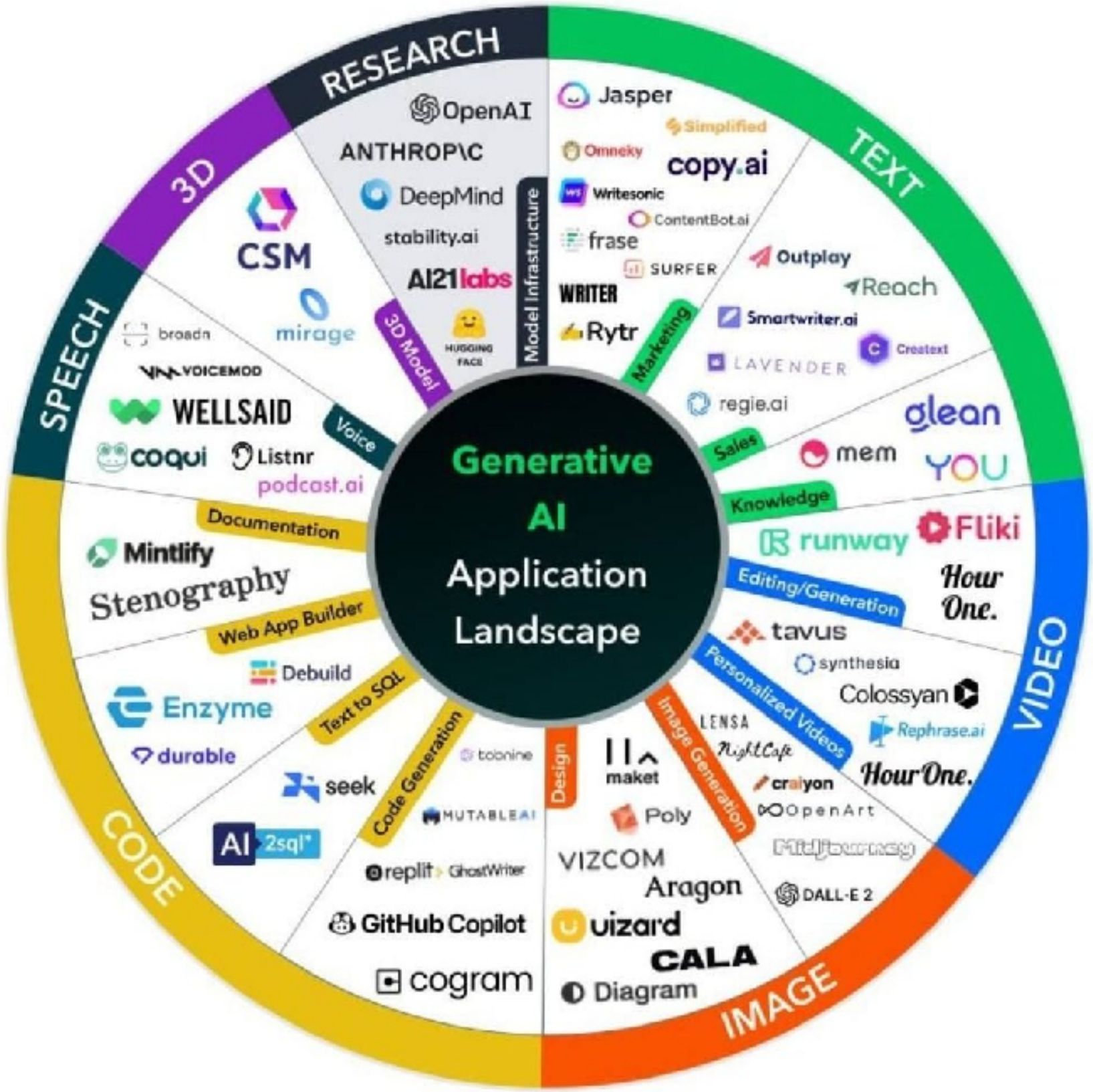


https://www.youtube.com/watch?v=2IK3DFHRFfw

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Application
Tool chains

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Tool chains category

Assist tasks

Interaction modes

Prompt composition

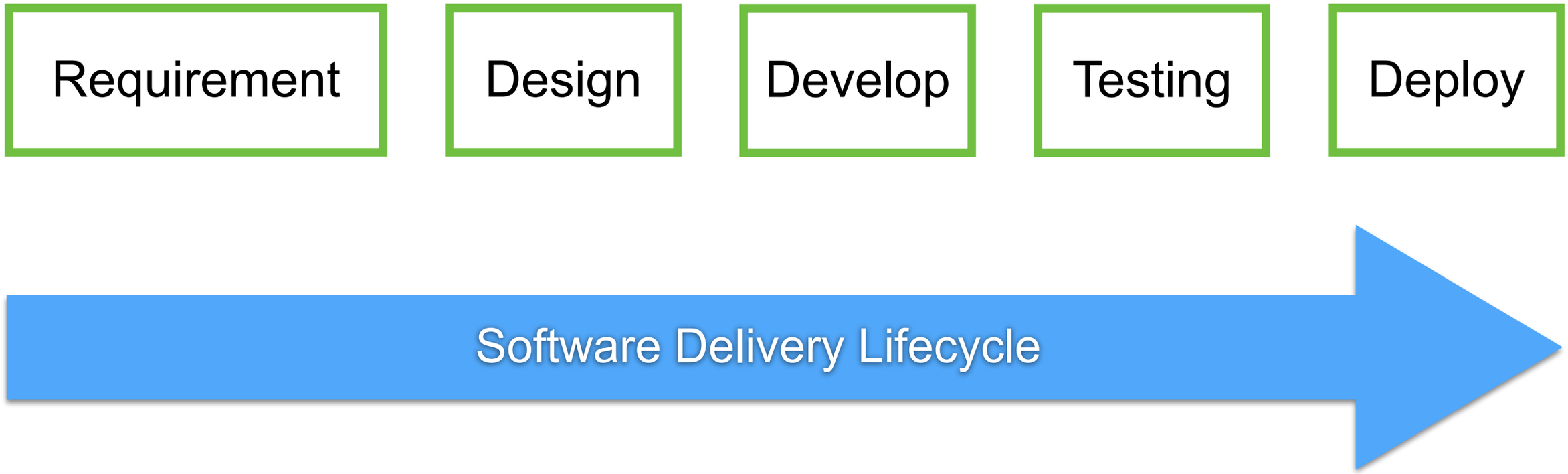
Properties of model

LLM models

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Assist tasks

Finding information faster in context
Generating code
Reasoning about code
Transforming code into something ..



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Interaction modes

Chat interfaces

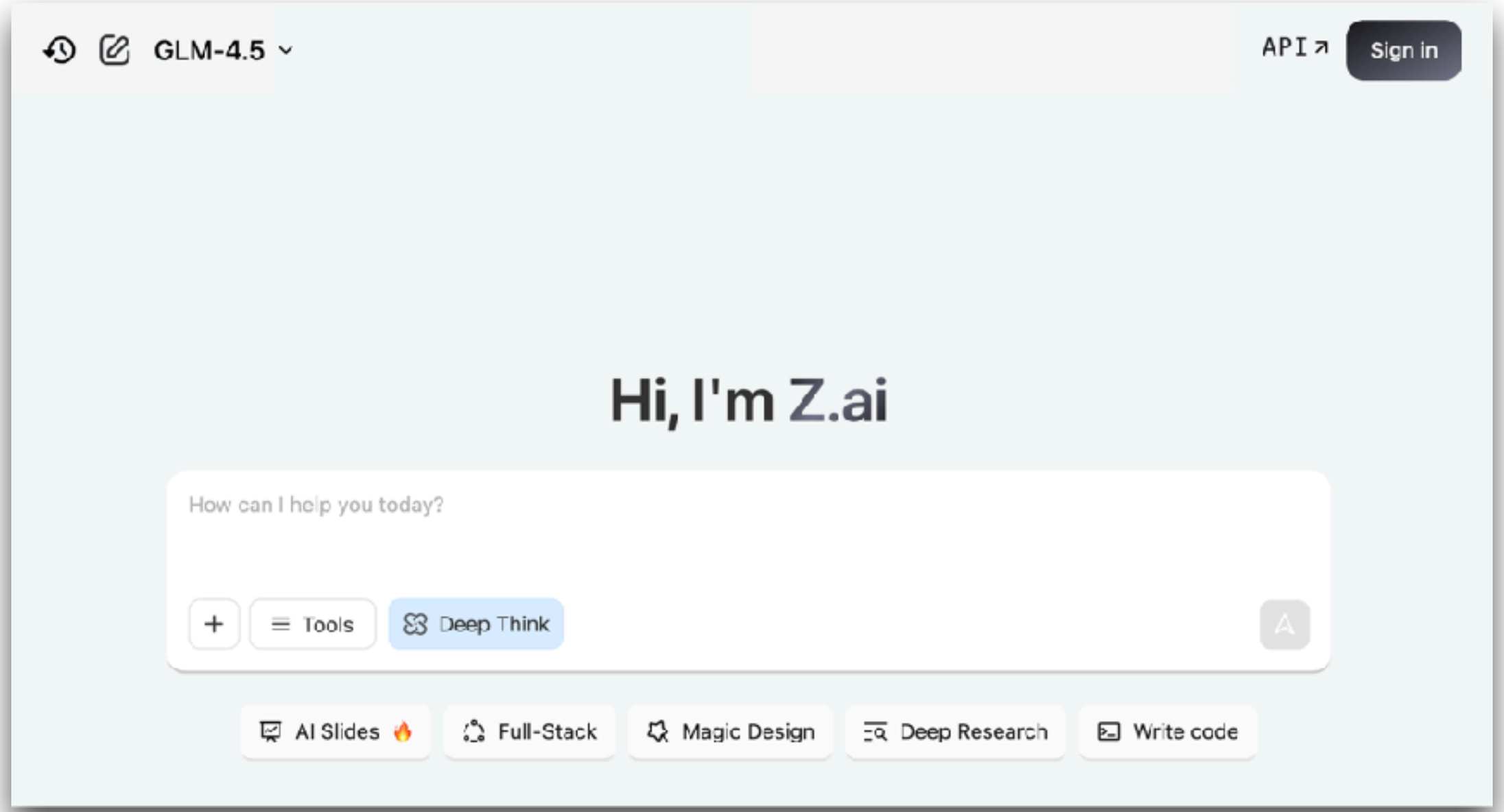
In-line assistance (typing in code editor)

CLI (command-line interface)



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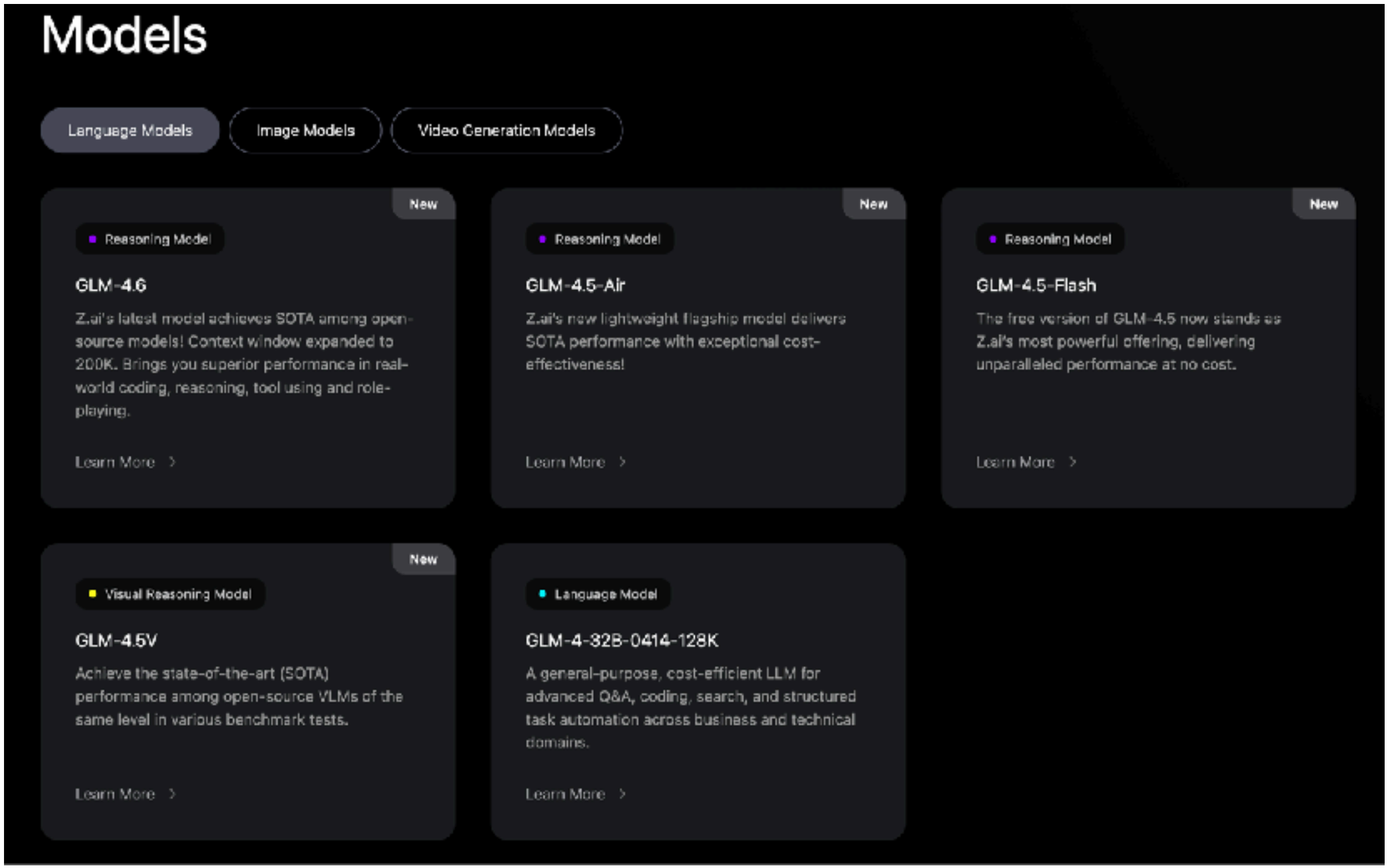
Z.ai



https://z.ai/

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Models

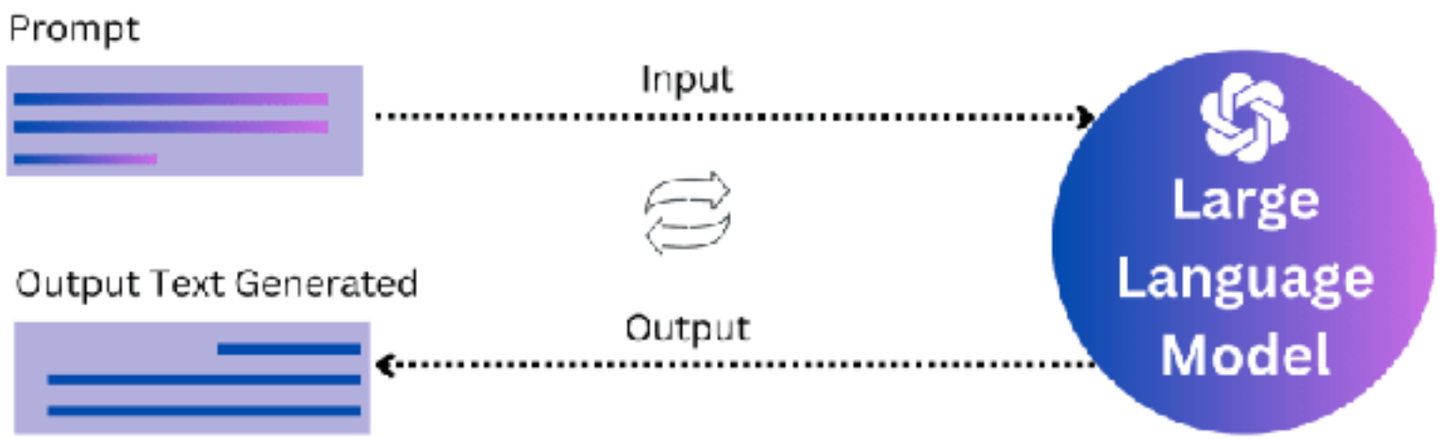


https://z.ai/

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Prompt composition

Prompt engineering
Compose prompts from user inputs and context



https://platform.openai.com/docs/guides/prompt-engineering

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Better Prompt

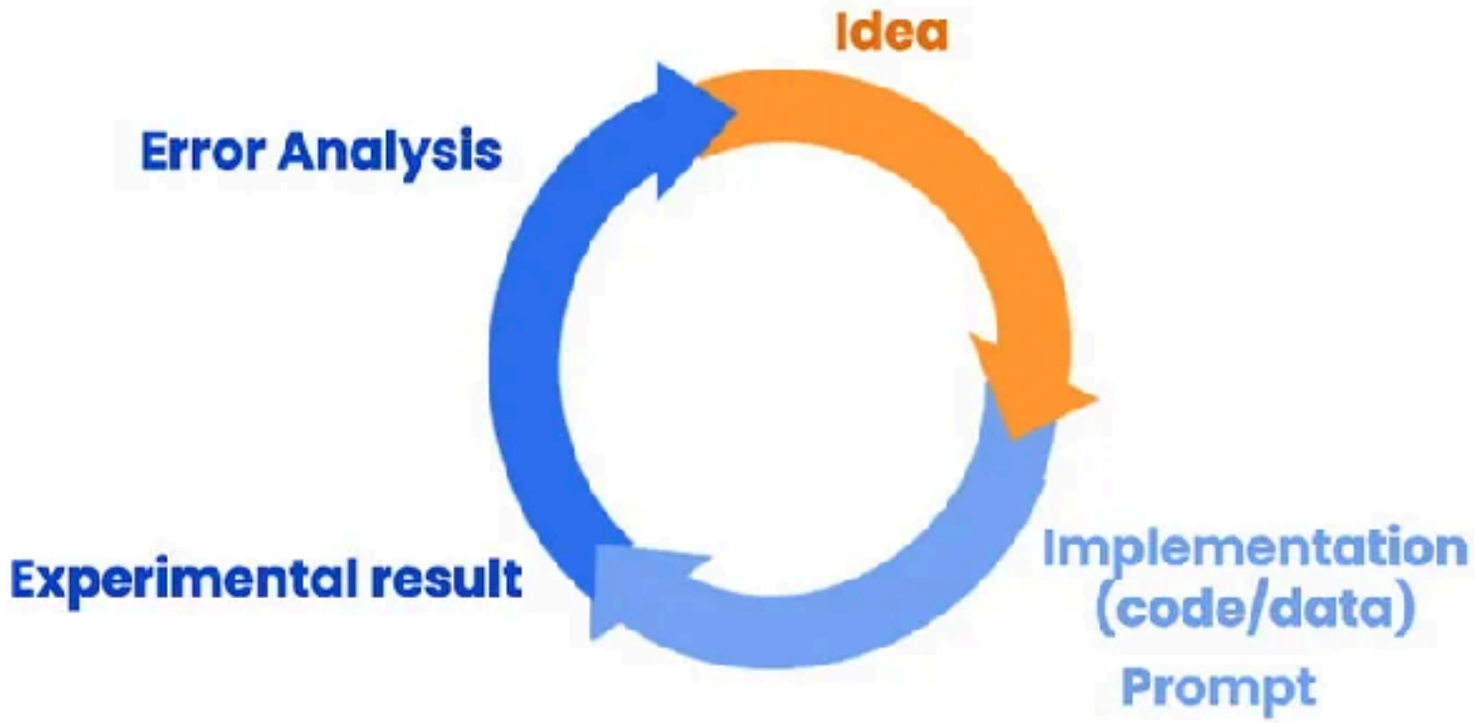
Write clear instructions
Provide reference text
Split complex tasks into simpler subtasks
Give the model time to think
Testing and improve …

https://platform.openai.com/docs/guides/prompt-engineering



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Iterative Prompt Development



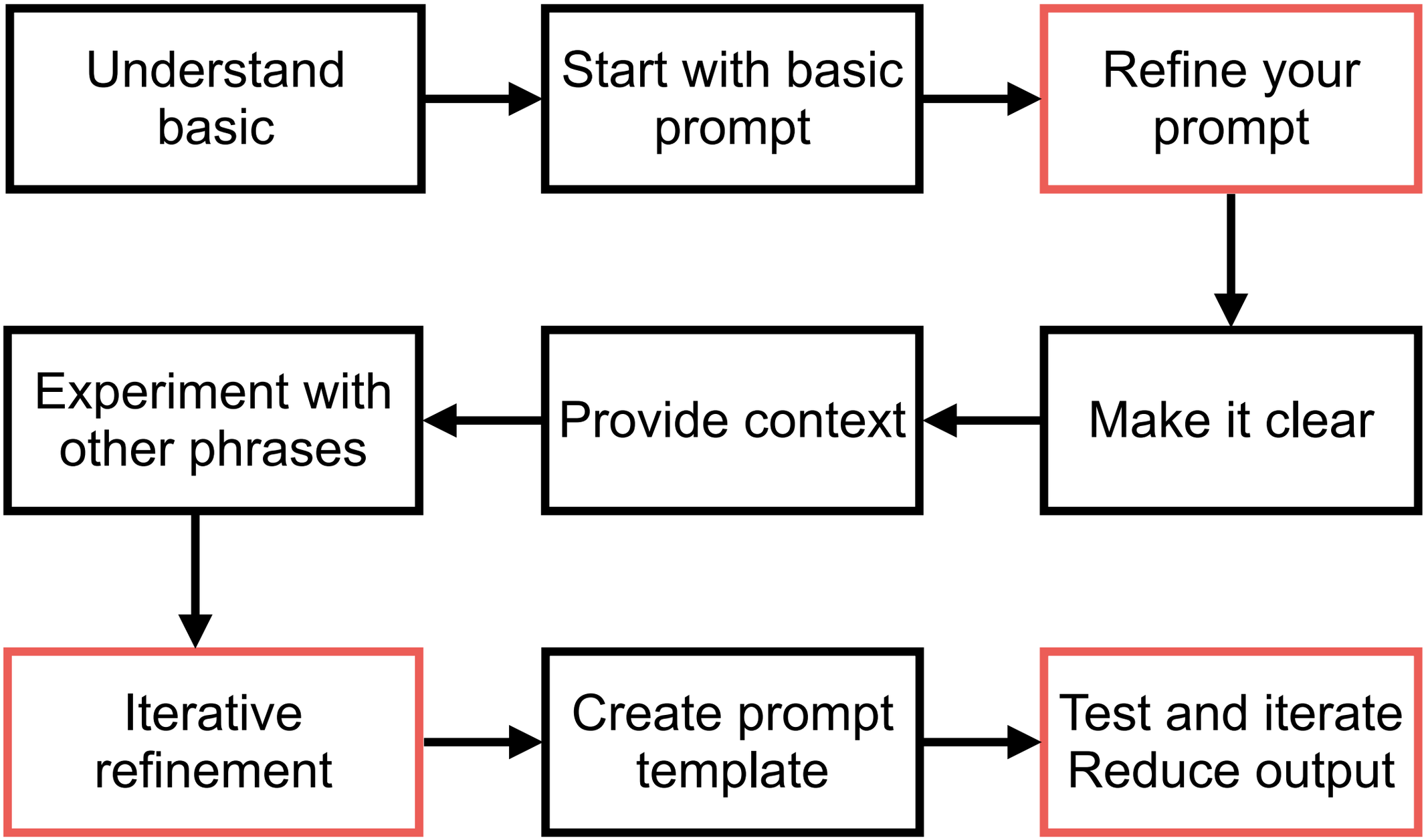
Iterative Process

- Try something
- Analyze where the result does not give what you want
- Clarify instructions, give more time to think
- Refine prompts with a batch of examples

https://www.deeplearning.ai/short-courses/chatgpt-prompt-engineering-for-developers/

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Basic of Prompt Engineer



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Better Prompt

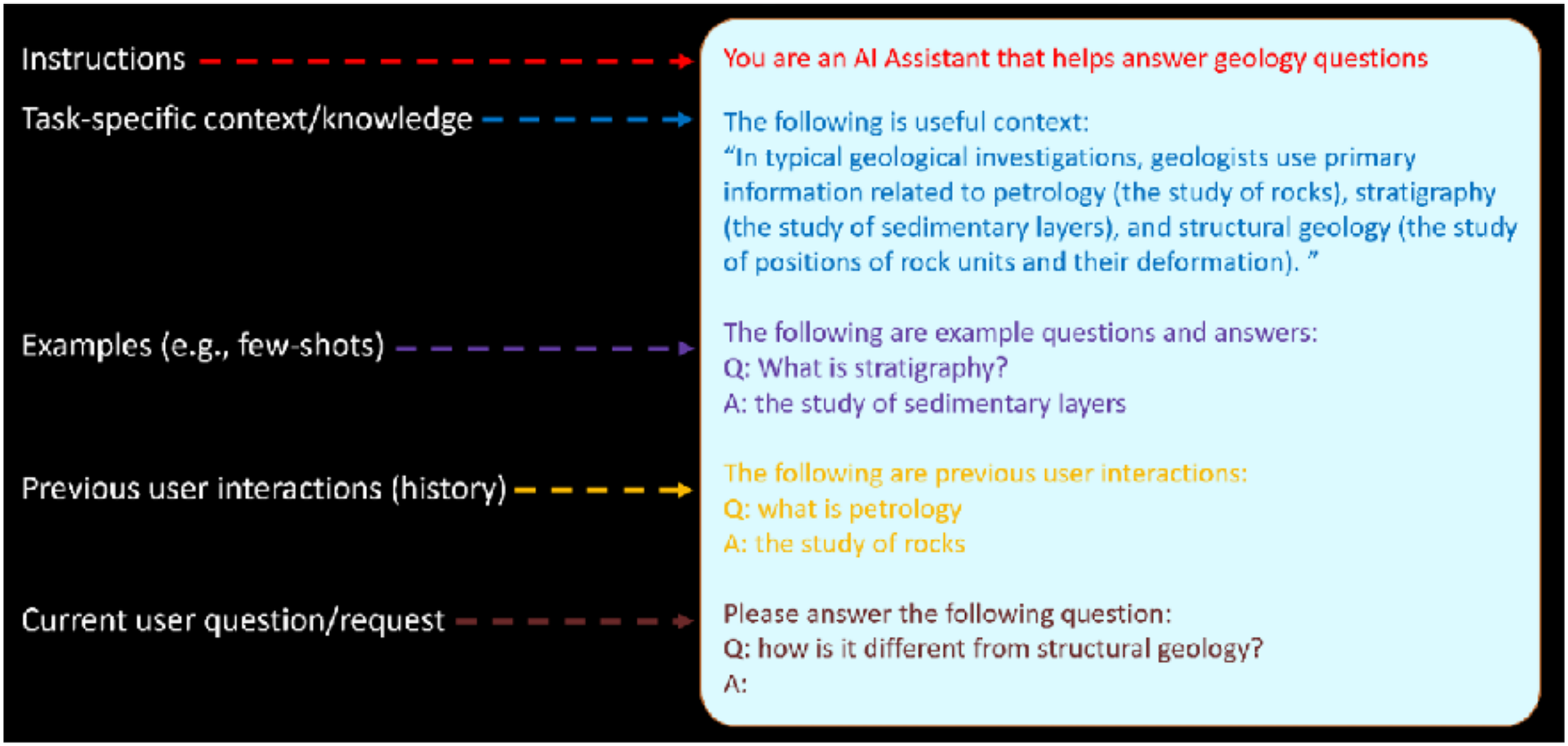
Write clear instructions
Provide reference text
Split complex tasks into simpler subtasks
Give the model time to think
Testing and improve …

https://platform.openai.com/docs/guides/prompt-engineering/six-strategies-for-getting-better-results



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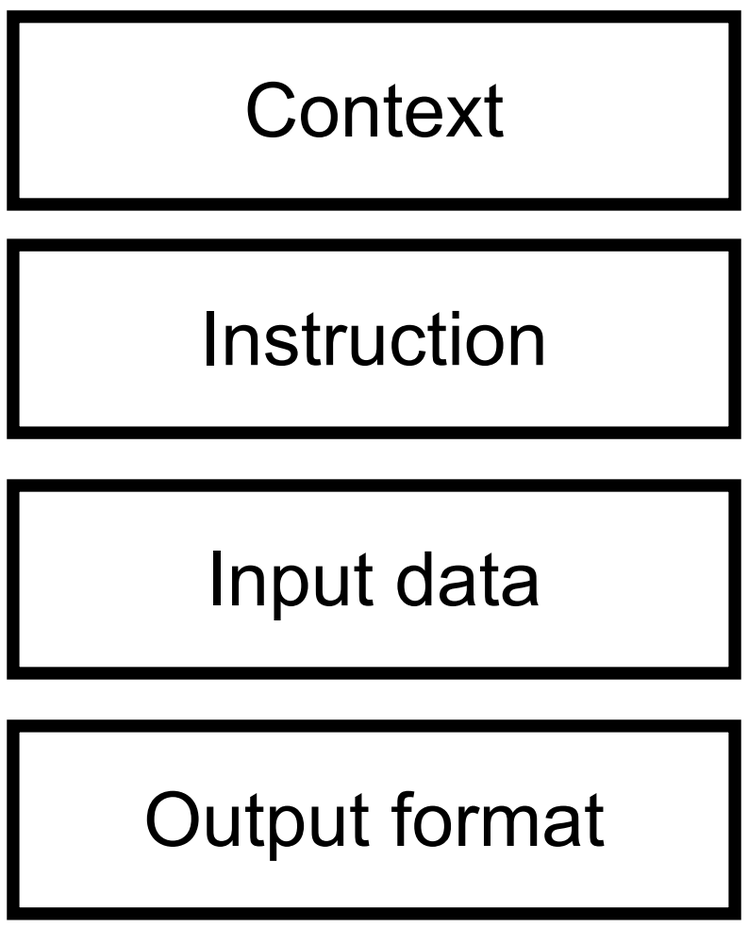
Prompt Structure



https://learn.microsoft.com/en-us/ai/playbook/technology-guidance/generative-ai/working-with-llms/prompt-engineering

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Structure of Prompt



https://platform.openai.com/docs/guides/prompt-engineering

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Prompting Guide

Prompt Engineering

Prompt Engineering Guide

Prompt engineering is a relatively new discipline for developing and optimizing prompts to efficiently use language models (LMs) for a wide variety of applications and research topics. Prompt engineering skills help to better understand the capabilities and limitations of large language models (LLMs).

Researchers use prompt engineering to improve the capacity of LLMs on a wide range of common and complex tasks such as question answering and arithmetic reasoning. Developers use prompt engineering to design robust and effective prompting techniques that interface with LLMs and other tools.

Prompt engineering is not just about designing and developing prompts. It encompasses a wide range of skills and techniques that are useful for interacting and developing with LLMs. It's an important skill to interface, build with, and understand capabilities of LLMs. You can use prompt engineering to improve safety of LLMs and build new capabilities like augmenting LLMs with domain knowledge and external tools.

https://www.promptingguide.ai/



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Structure of Prompt !!

APE
Action, Purpose, Expectation

RACE
Role, Action, Context, Expectation

TAG
Task, Action, Goal

COAST
Context, Objective, Action, Scenario, Task

RISE
Role, Input, Step, Expectation

https://twitter.com/pradeepeth/status/1673271866696544257

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Prompt Techniques

Zero-shot

Chain-of Thought (CoT)

Few-shot

Meta or structure

https://www.promptingguide.ai/techniques

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Chain of Thought Prompting (CoT)

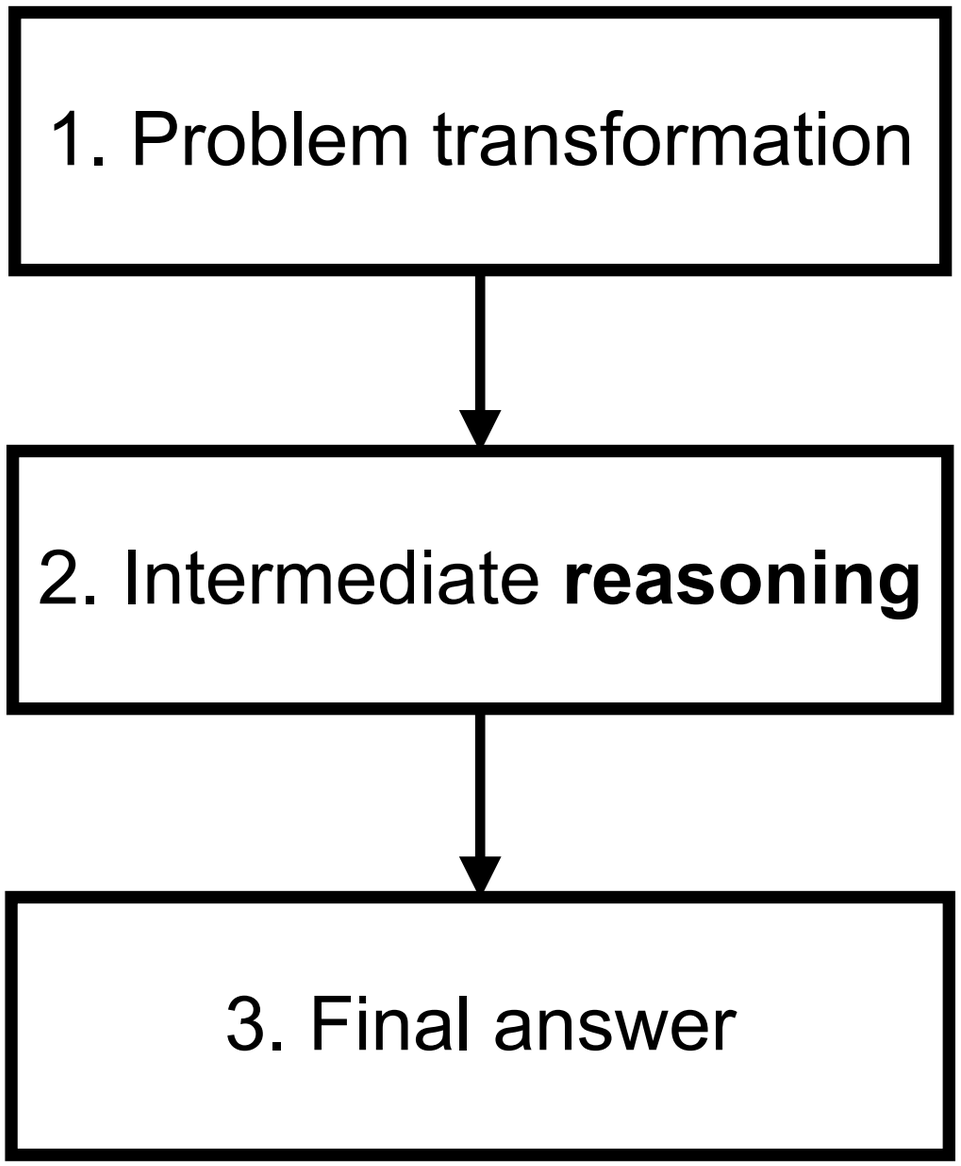
Technique used to improve the reasoning ability of LLM

Try to break down a complex problem into smaller, More manageable steps, lead to final answer

Reasoning model !!

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Chain of Thought Prompting (CoT)



eg. step-by-step

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Chain of Thought Prompting (CoT)

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A:

(Output) The answer is 8. ✗

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are 16 / 2 = 8 golf balls. Half of the golf balls are blue. So there are 8 / 2 = 4 blue golf balls. The answer is 4. ✓

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: The answer (arabic numerals) is

(Output) 8 ✗

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: ***Let's think step by step.***

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

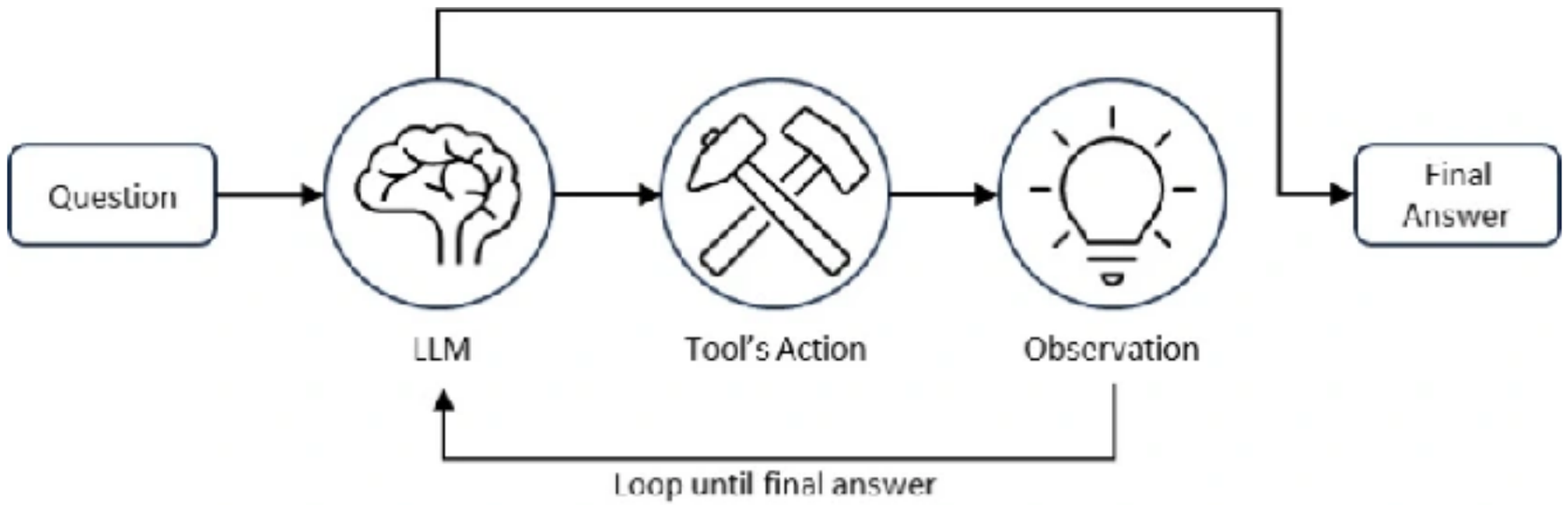
https://www.promptingguide.ai/techniques/cot



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ReAct Prompt

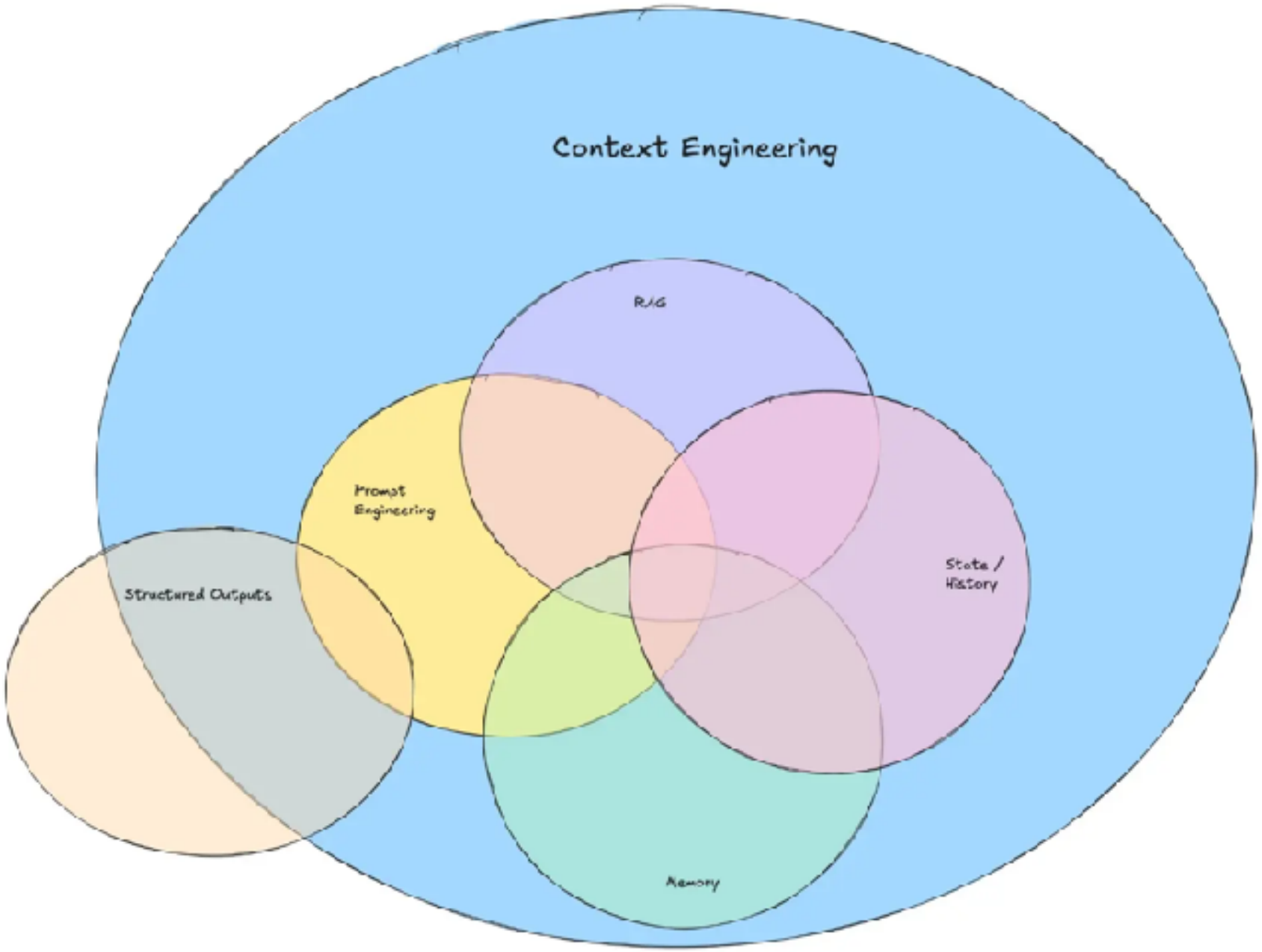
LLM reasoning and additional tools (expert)
Improve better answer



https://www.promptingguide.ai/techniques/react

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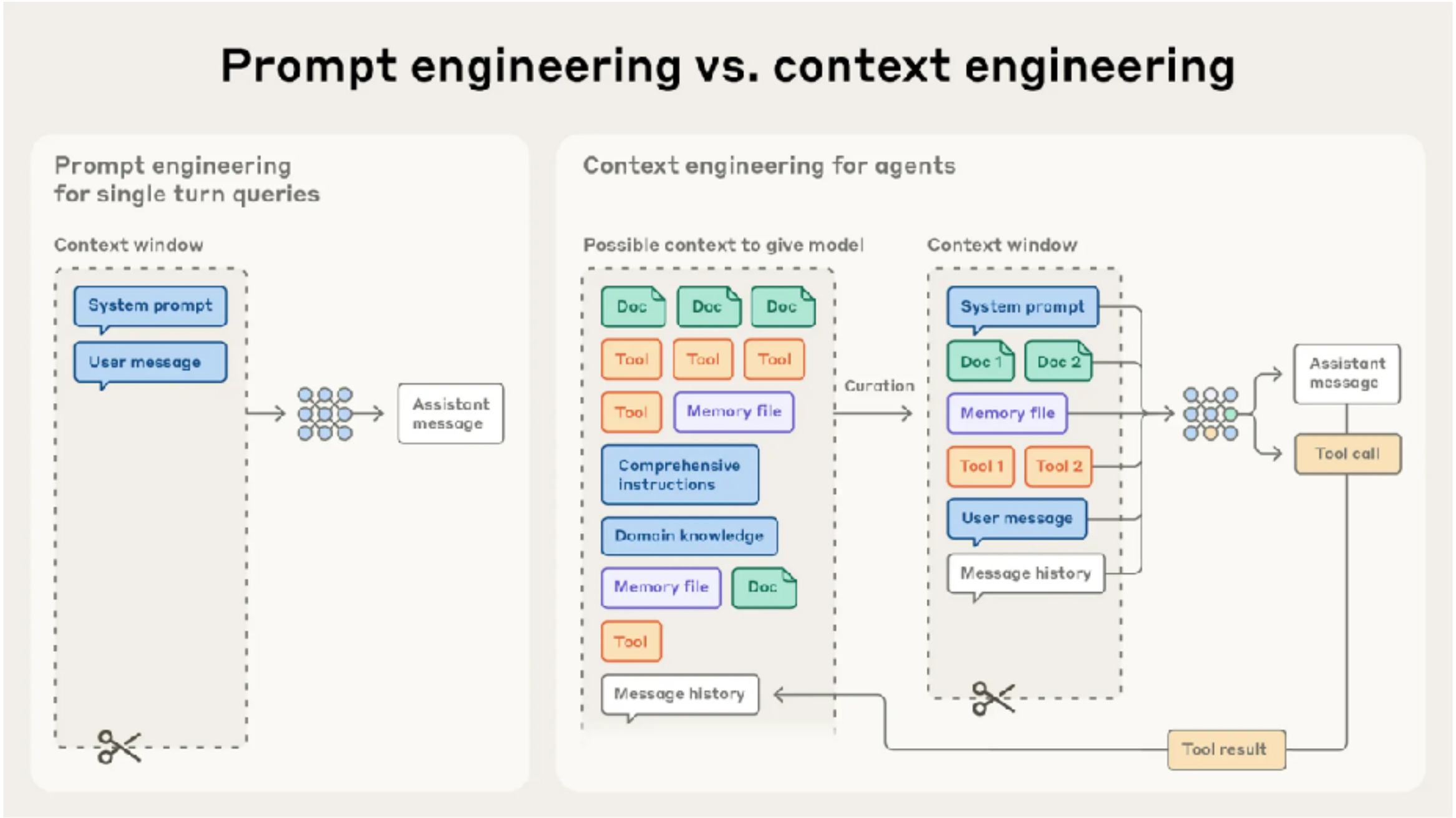
Context Engineering



https://www.promptingguide.ai/guides/context-engineering-guide

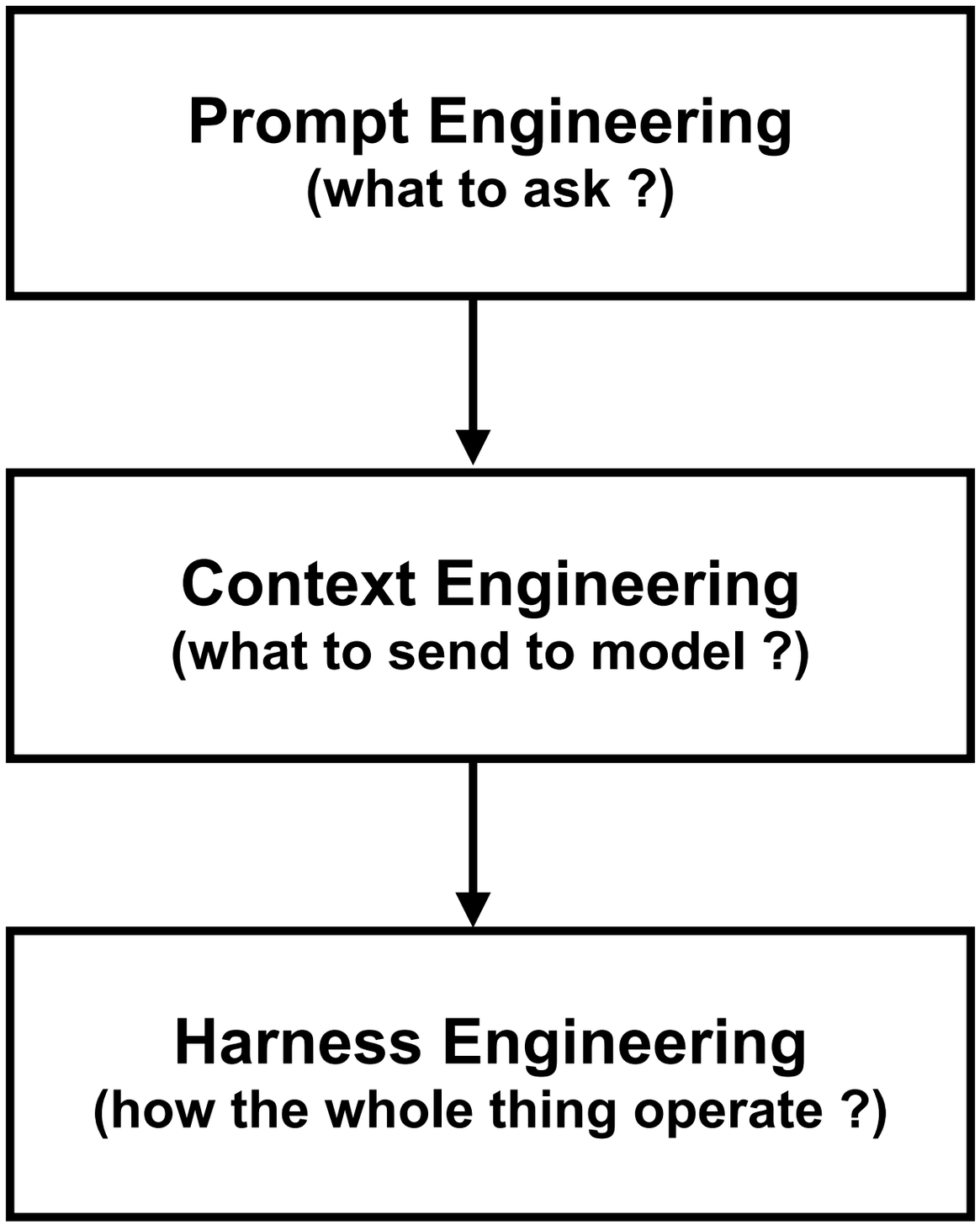
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Effective Context Engineering

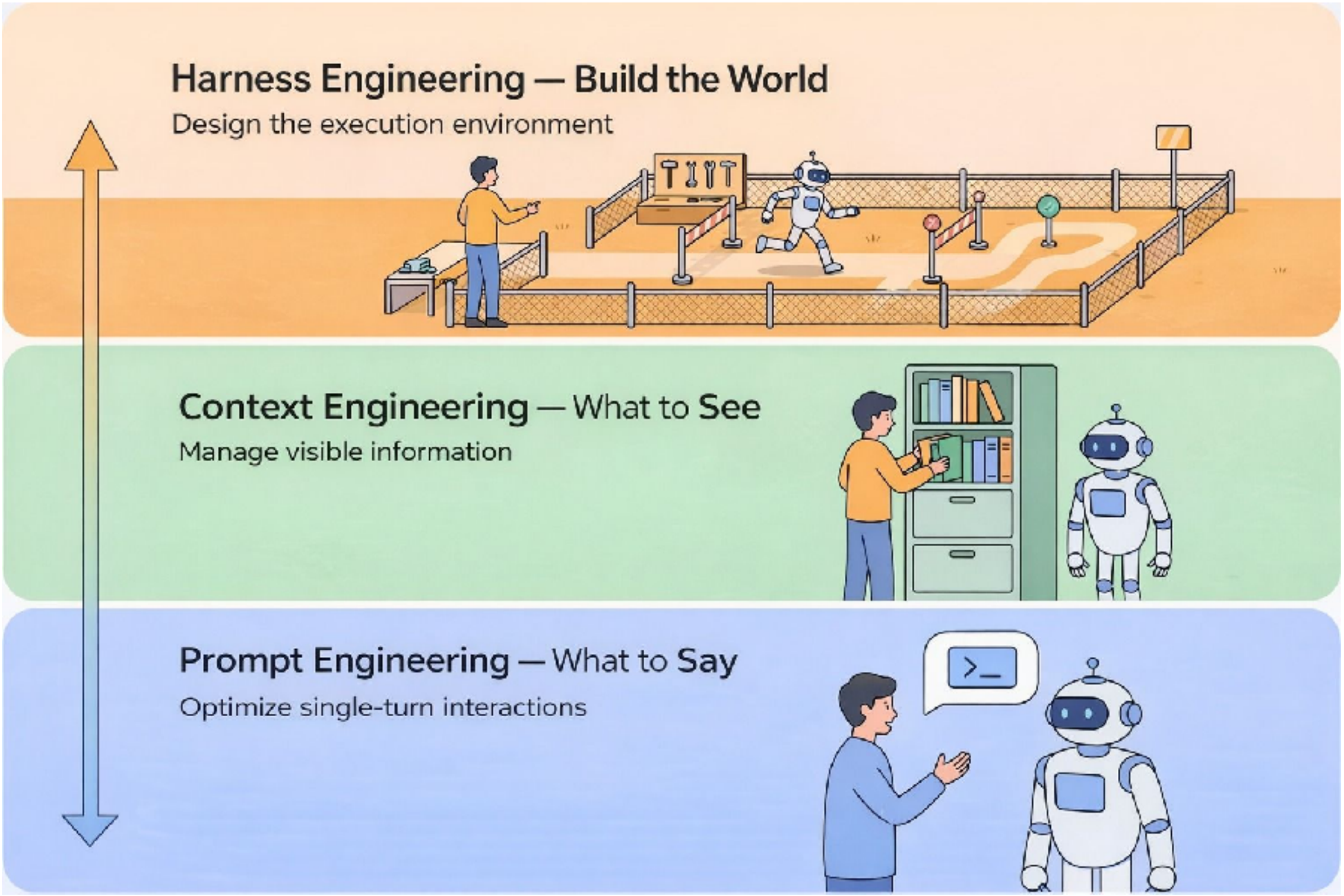


https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents

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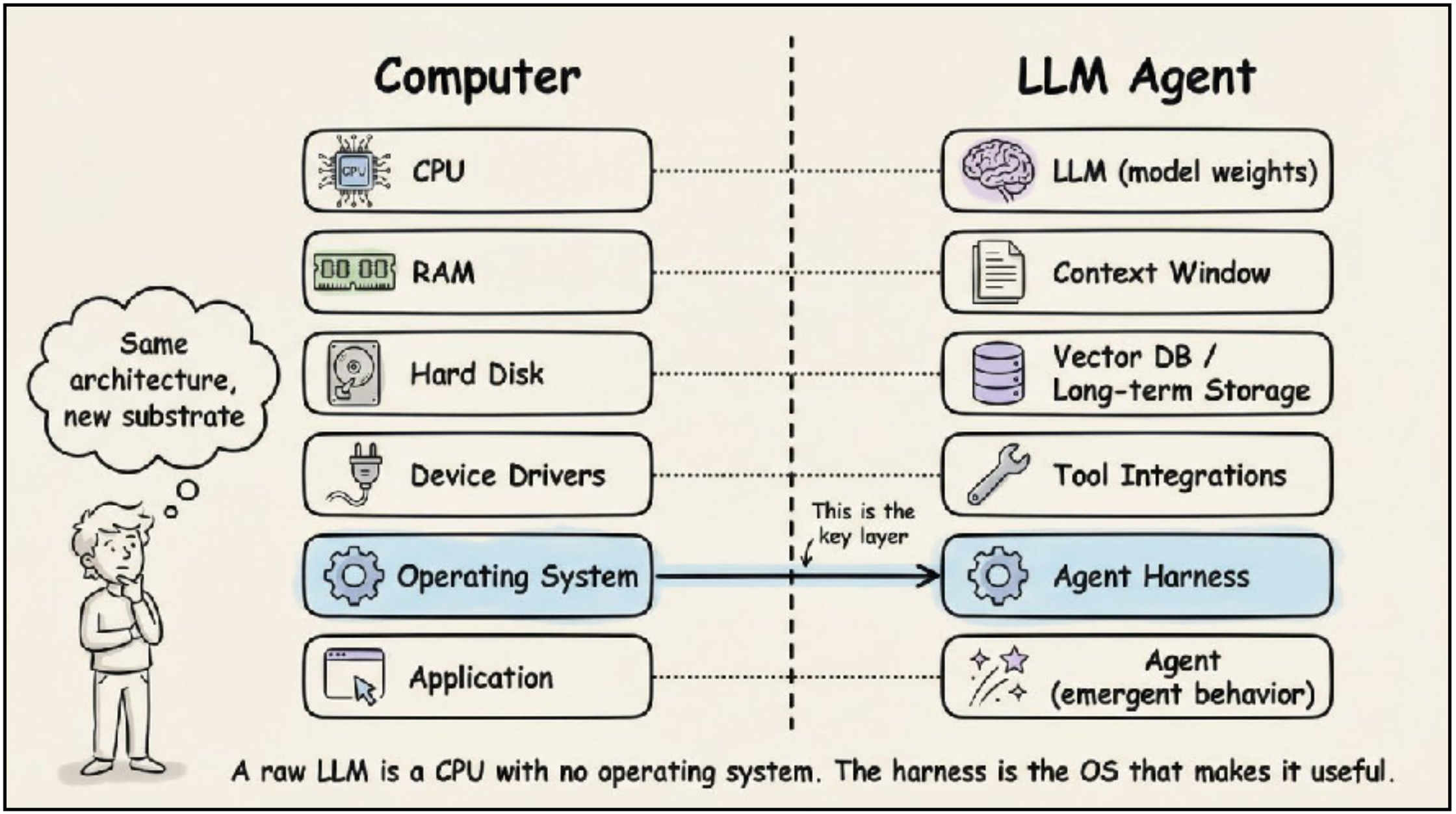
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https://milvus.io/blog/harness-engineering-ai-agents.md

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Harness Engineering



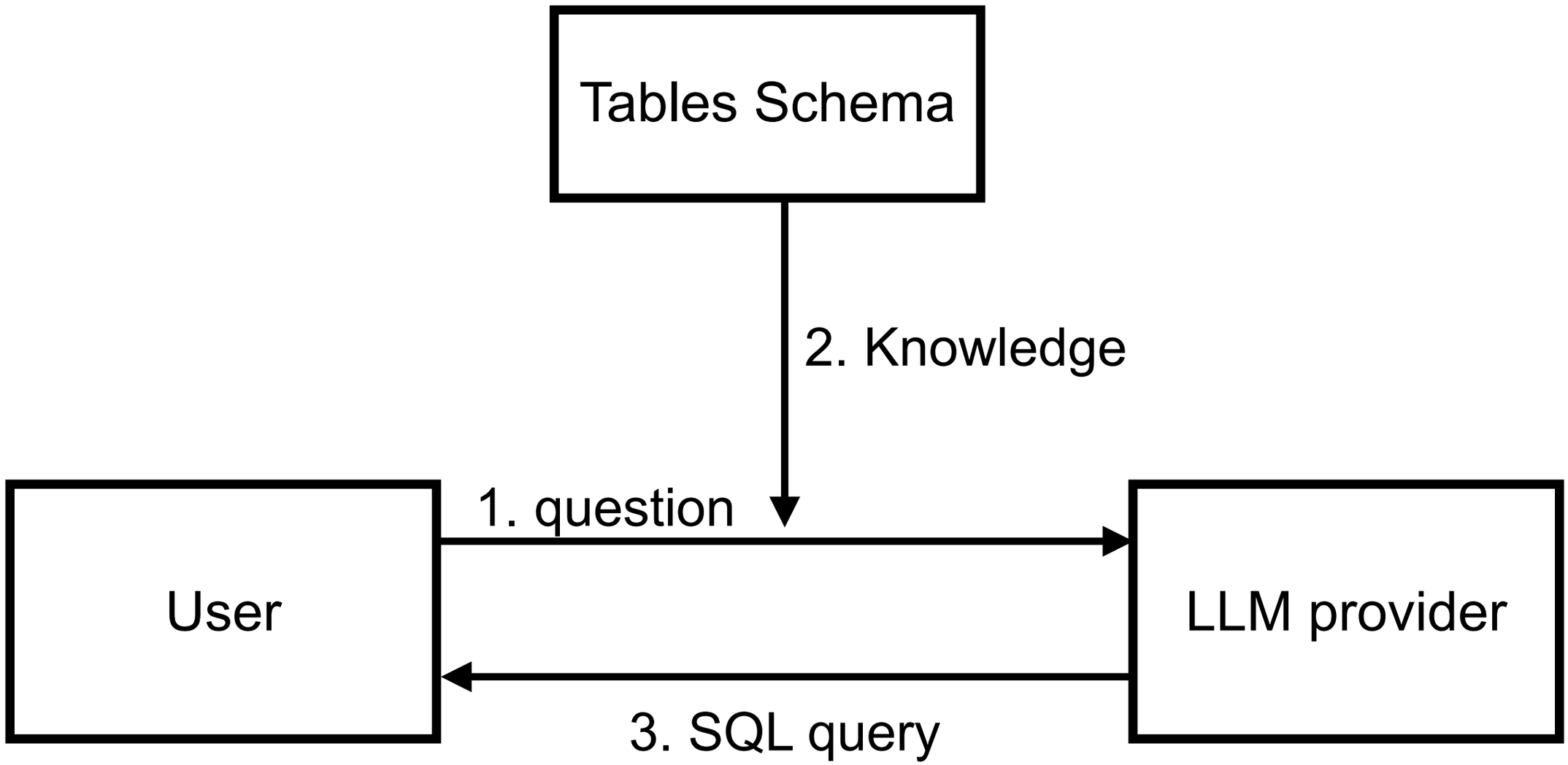
https://x.com/akshay_pachaar/status/2041146899319971922

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Workshop

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Text-to-SQL



https://github.com/up1/workshop-ai-with-technical-team/tree/main/workshop/demo-sql

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Software Development Life Cycle

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SDLC



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Planning

- Written planning process
- Arch diagrams

Testing

- Automated testing
- TDD
- Testing environments
- Testing in prod
- Performance testing
- Load testing
- Generate test data

Development

- Automated dev env
- CI/CD
- Prototyping
- Code review
- Code generation
- Templates
- Cross-platform dev
- Preview env
- Post-commit code review
- Linting
- Static code analysis
- Project mgmt

Shipping

- FF & experimentation
- Logging
- Monitoring & alerting
- Staged rollouts

Maintenance

- Debug production
- Documentation
- Runbook examples
- Mitigation runbook

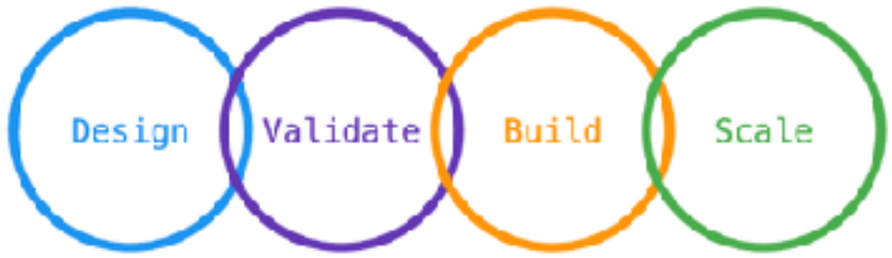
pragmaticengineer.com



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AI in Software Development

BEFORE

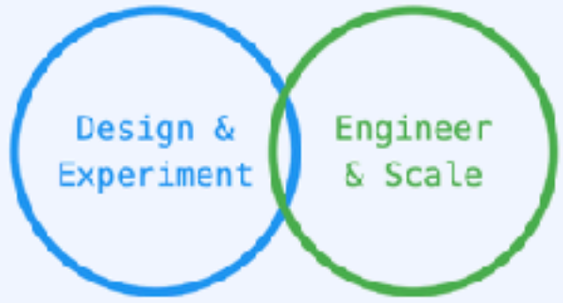


→ Fragmented SDLC Ownership;

→ Scaling Challenges;

→ Siloed Teams Increasing friction.

NOW



→ Democratized development;

→ Bridge between business and IT;

→ Composability for business agility.

AI-native SDLC drives faster, customer-centric, and higher-quality innovation.

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Impacts with productivity ?

Automated simple tasks

Improve quality and reliability

Improve communication

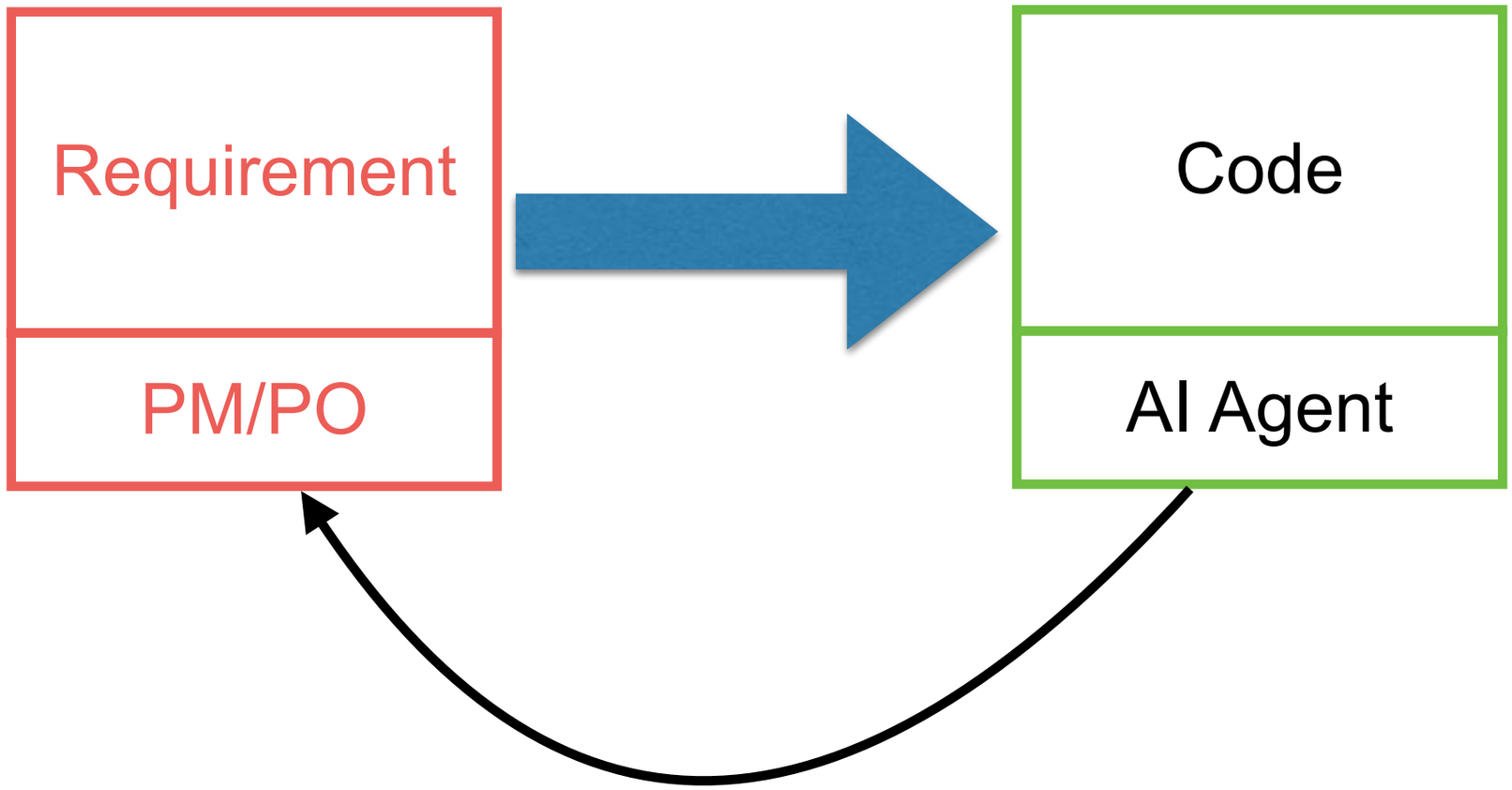
Faster prototype



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New Bottleneck ?

Decide what to **build**, not code



Get feedback faster

https://www.theaiopportunities.com/p/is-ai-slowing-down-the-new-bottleneck

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Generative AI isn't just a tool

it's your **team member**

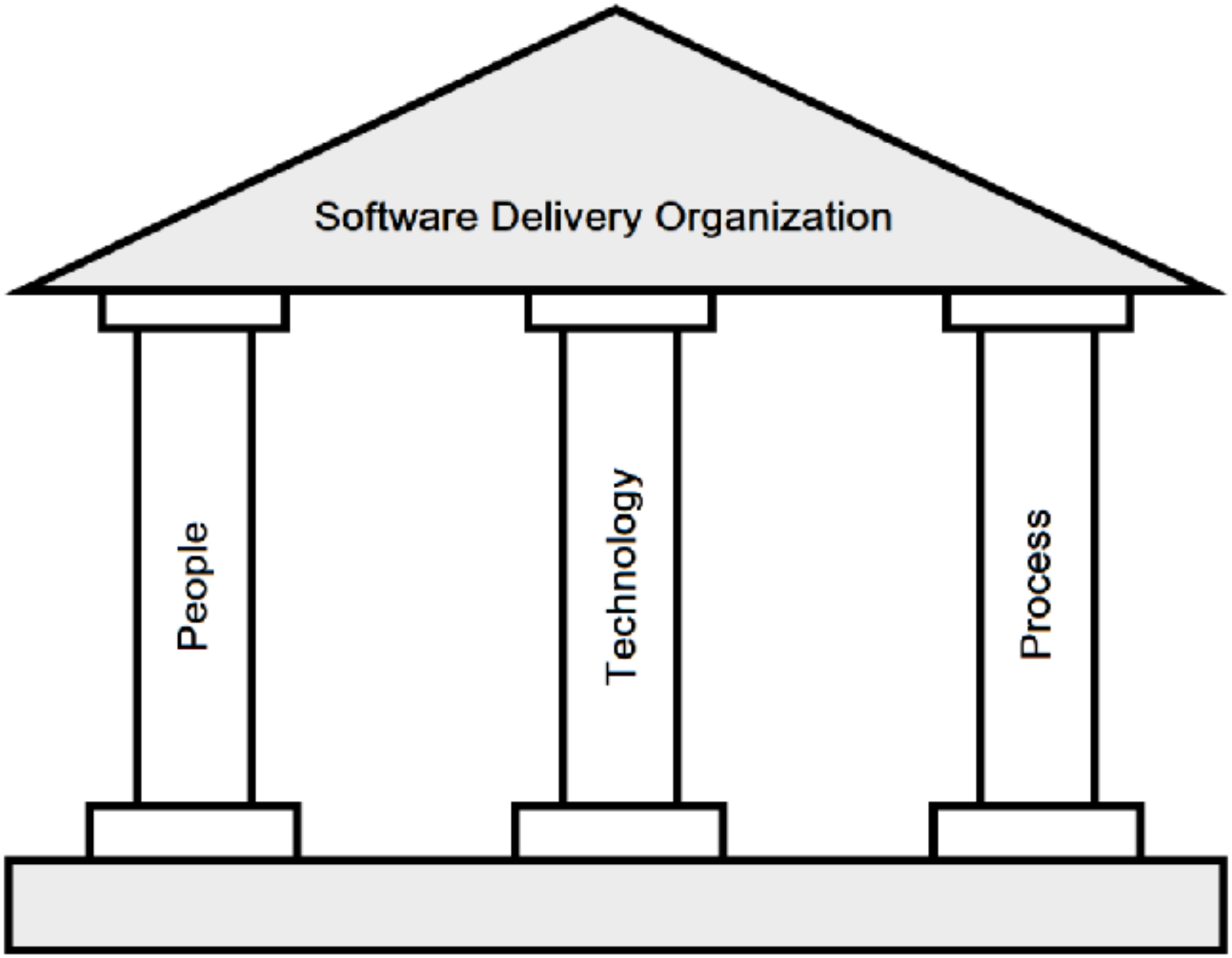
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Trust, but verify output !!



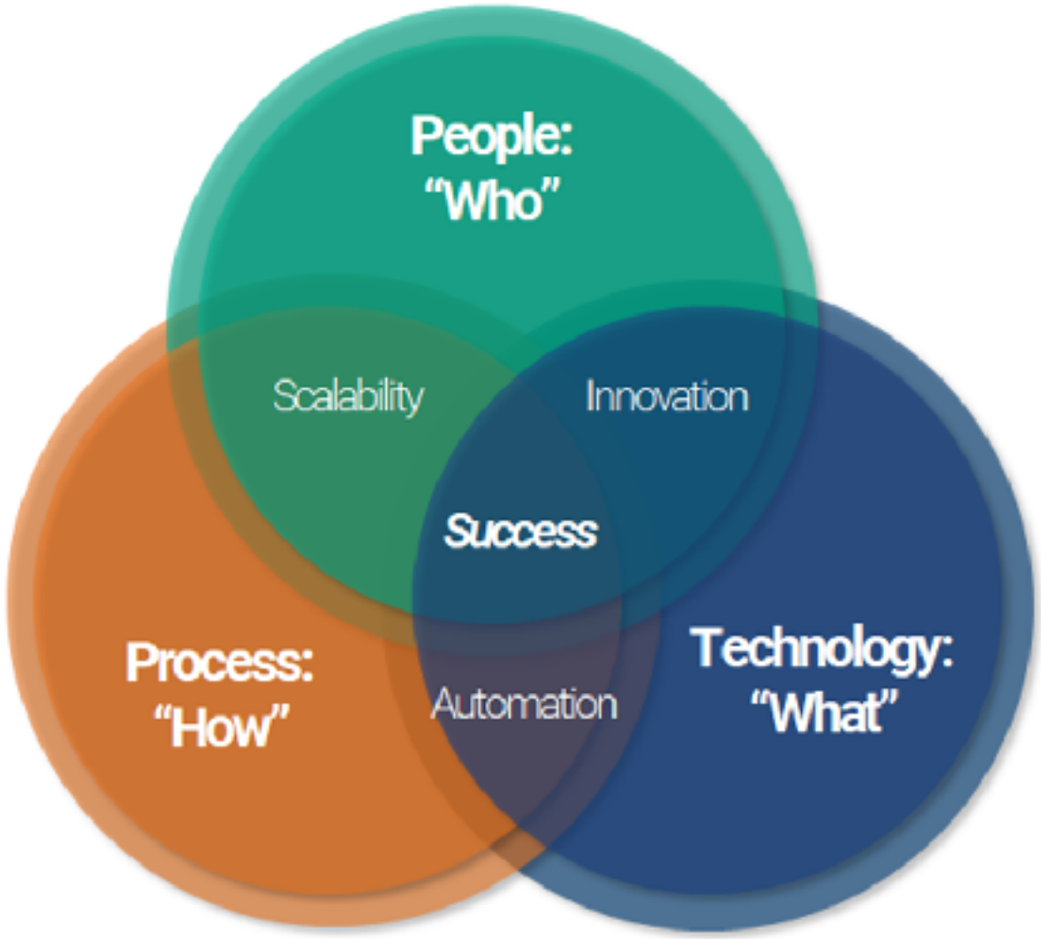
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3 Pillar of Software Development



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3 Pillar of Software Development



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Requirement and Analysis



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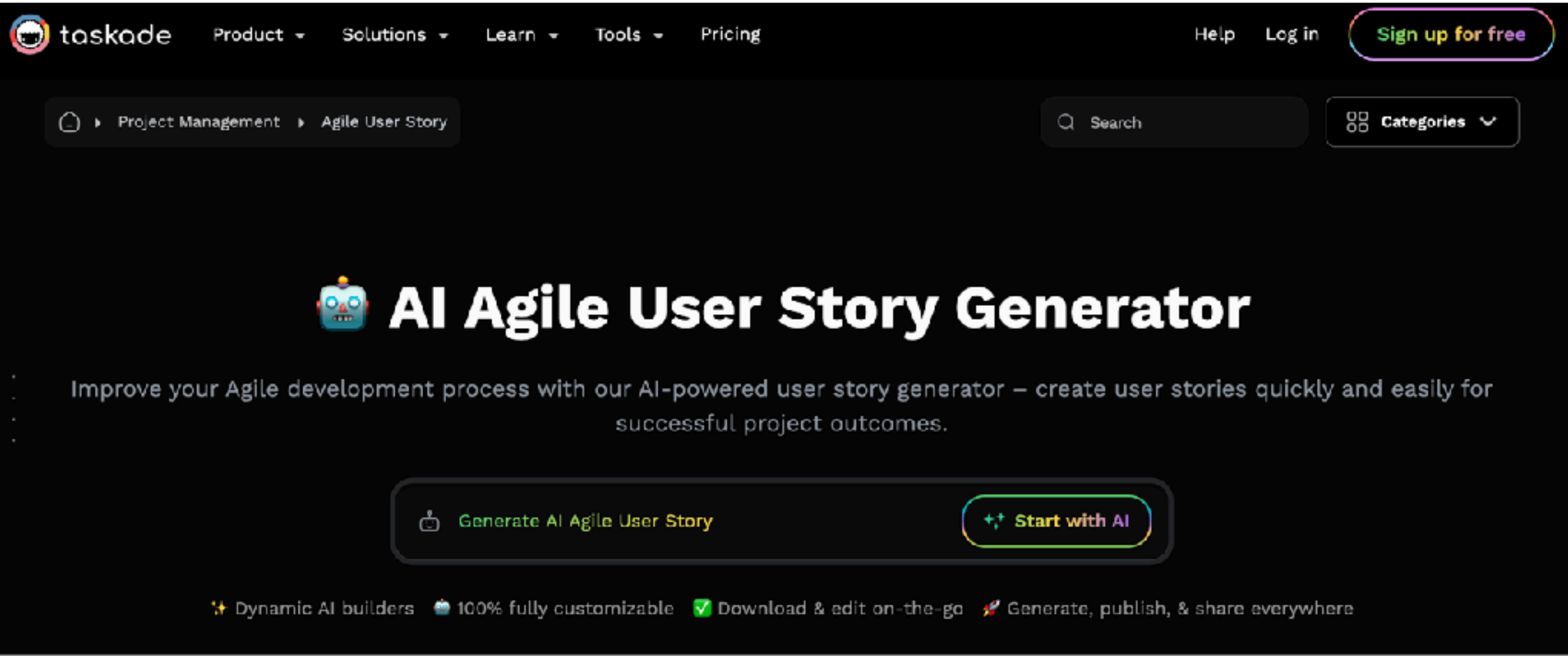
Requirement and Analysis



Requirements writing and analysis
User story generation

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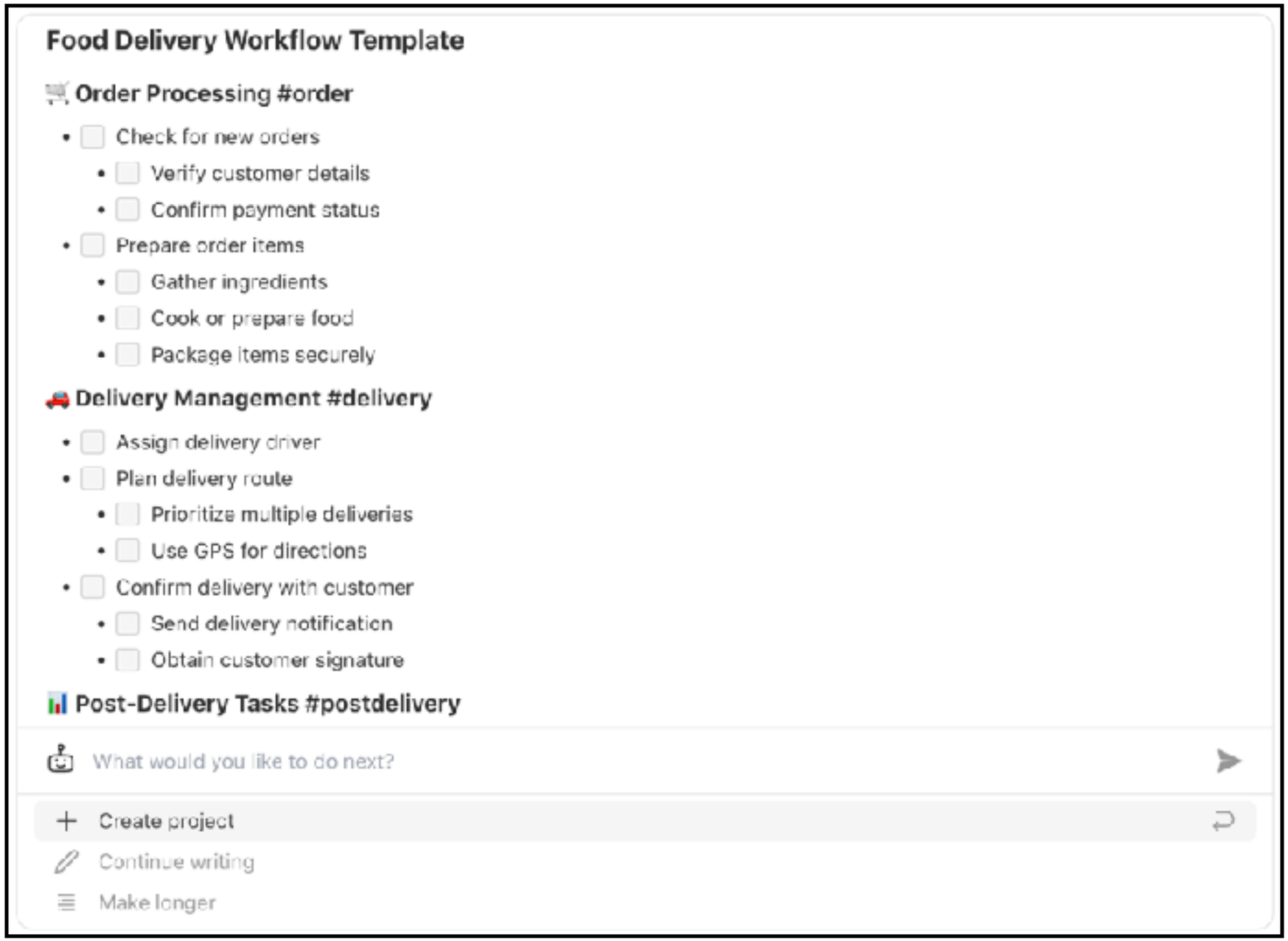
Agents AI for Automated tasks



https://www.taskade.com/generate/project-management/agile-user-story

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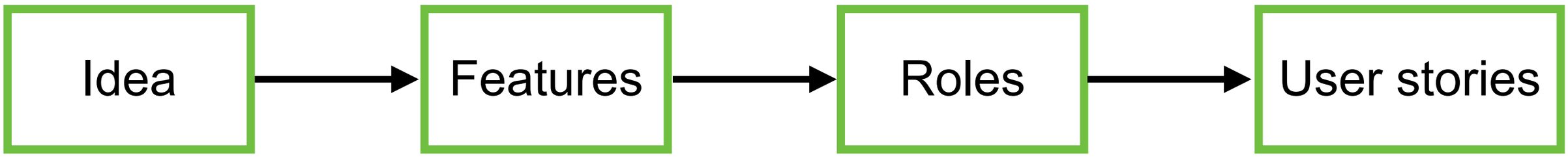
Example with food delivery



https://www.taskade.com/generate/project-management/agile-user-story

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User Story Generator

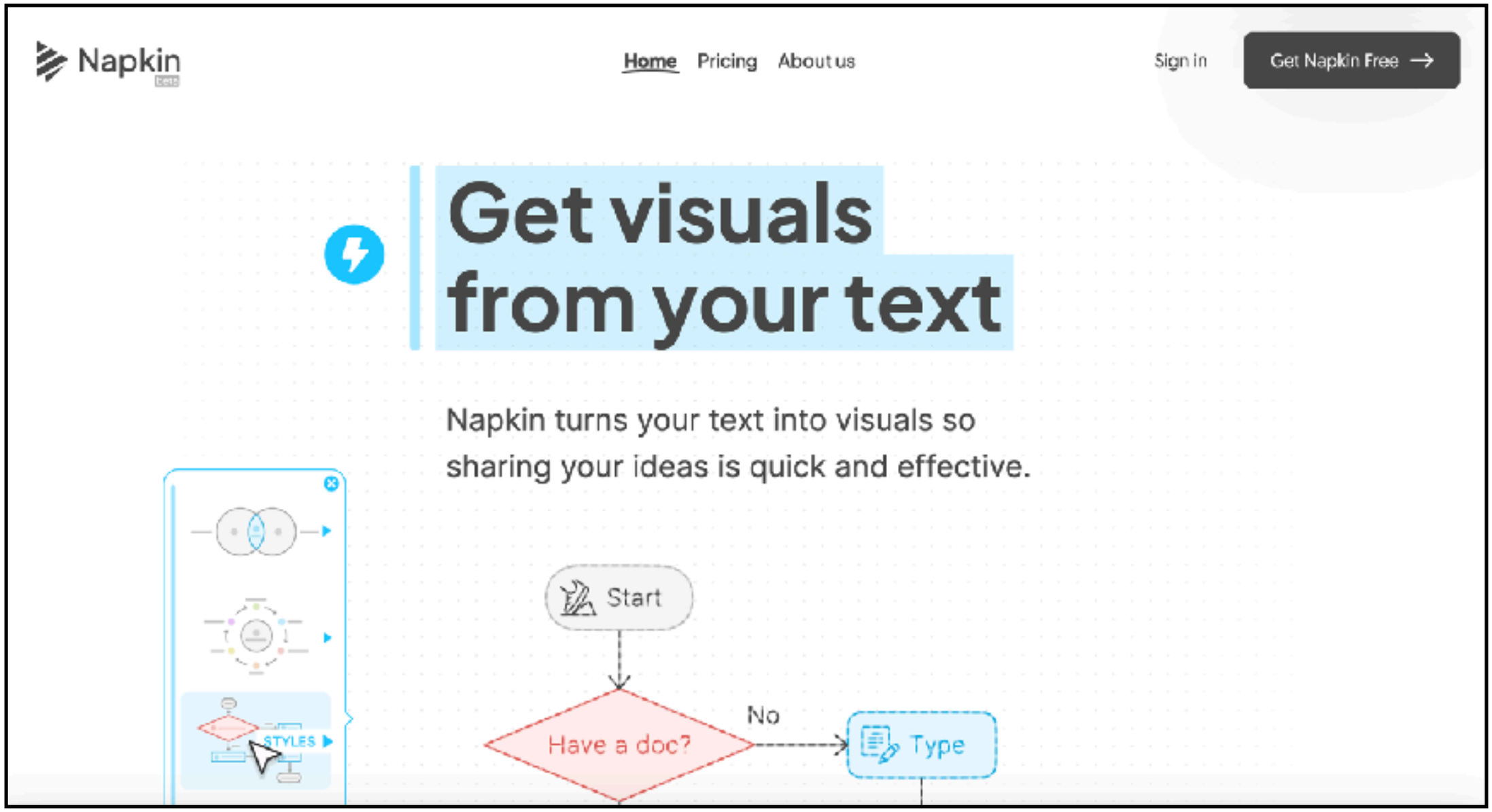




https://userstorygenerator.ai/

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Napkin



https://www.napkin.ai/

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Requirement analysis

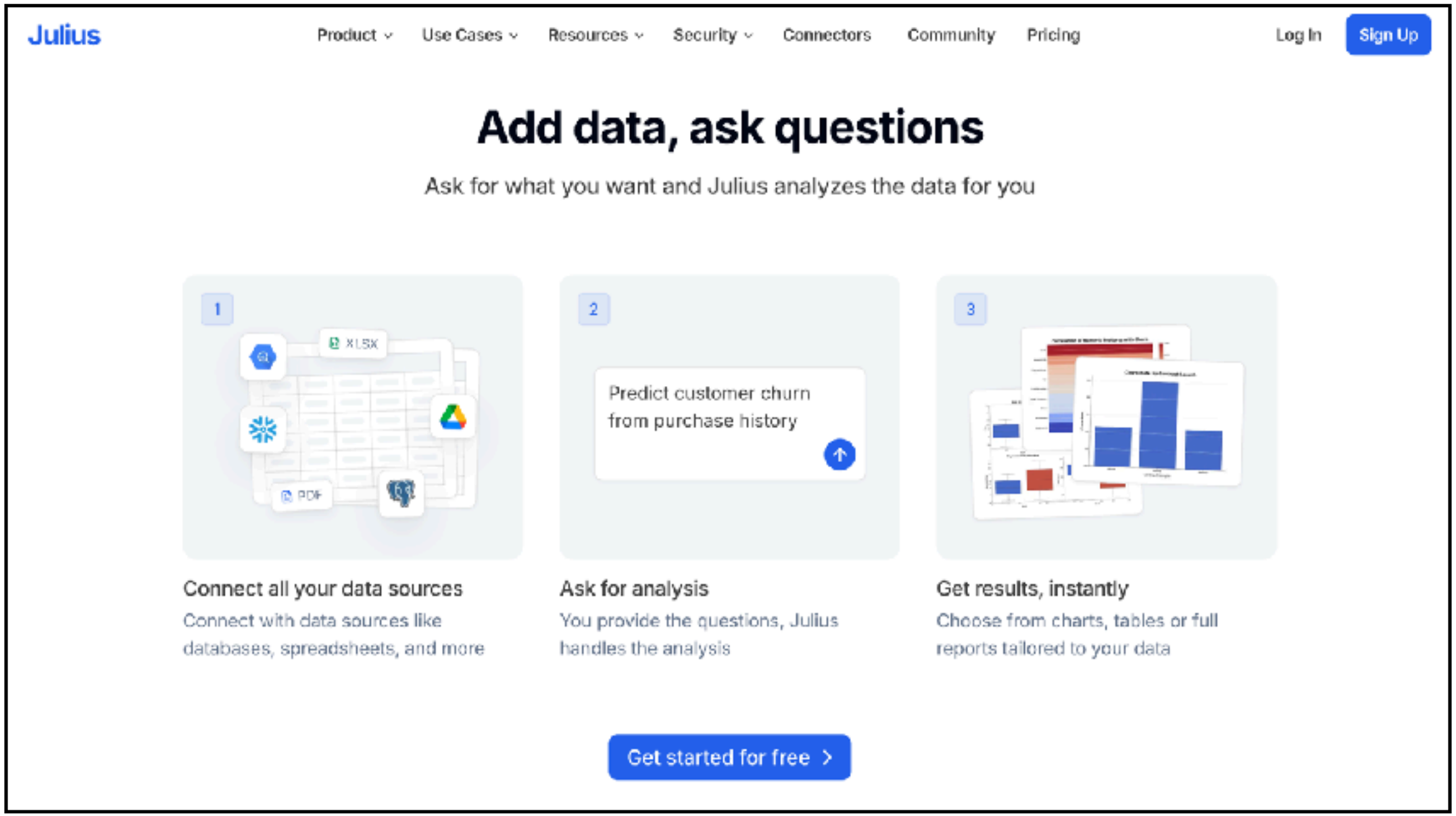
Clarify of User requirement ?

https://github.com/up1/workshop-ai-with-technical-team/wiki/Requirement-analysis



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Data Analysis



https://julius.ai/

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Design Process

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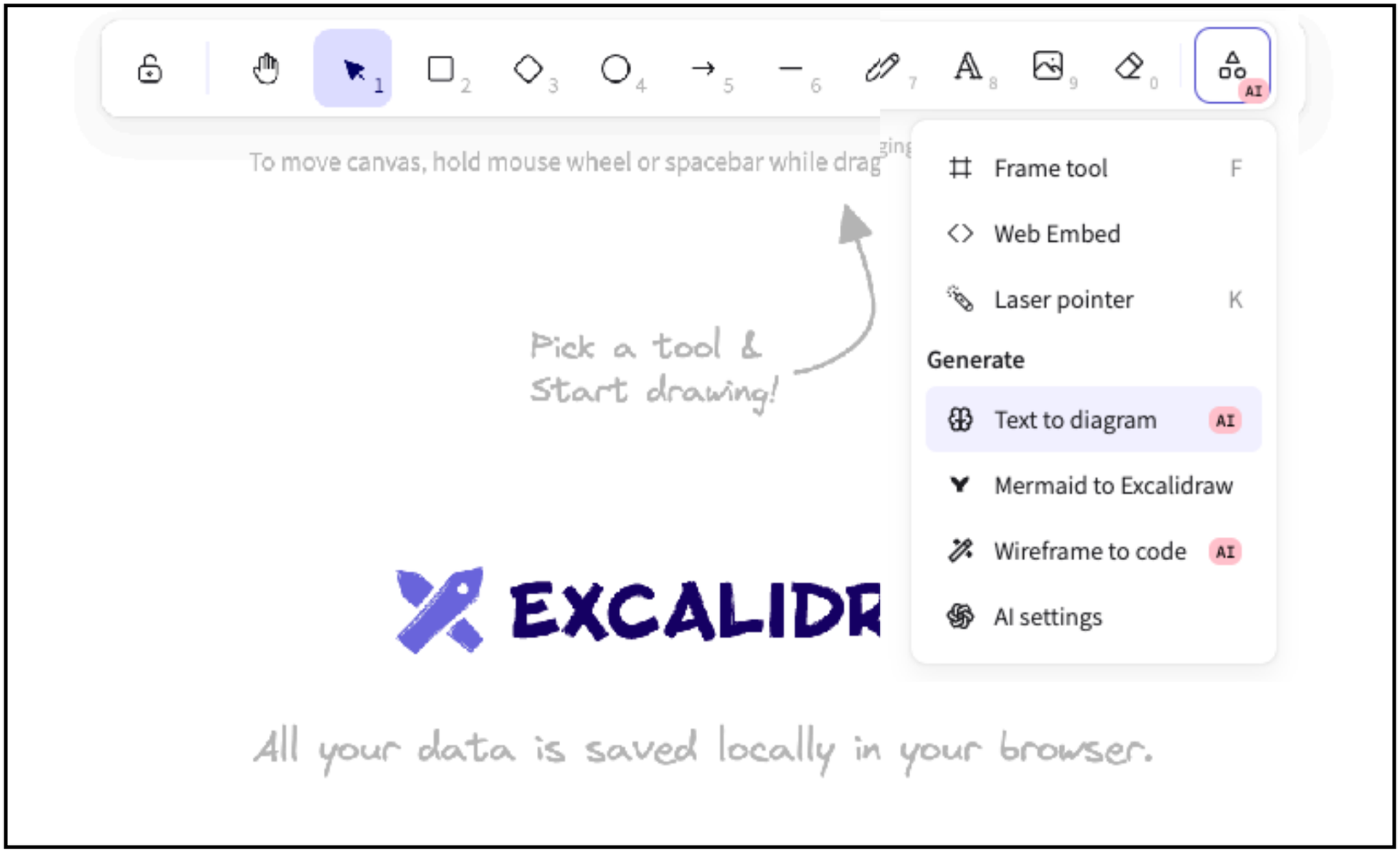
Design



Architecture writing assistance
Sequence flow diagram generation
Data modeling
UX/UI design assistance

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Excalidraw with AI



https://excalidraw.com/

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Demo

Text to diagram AI Beta | Mermaid

Currently we use Mermaid as a middle step, so you'll get best results if you describe a diagram, workflow, flow chart, and similar.

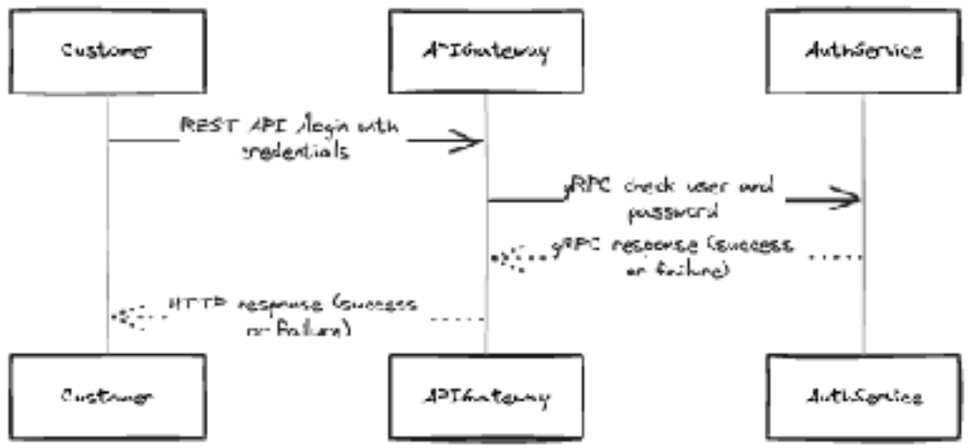
Prompt 9 requests left today

```
Try to generate authentication service from below
1. Customer call api gateway with REST API /login
2. Api gateway check user and password from auth service via
gRPC
3. API gateway send response to client
```

Generate → Cmd Enter

View as Mermaid →

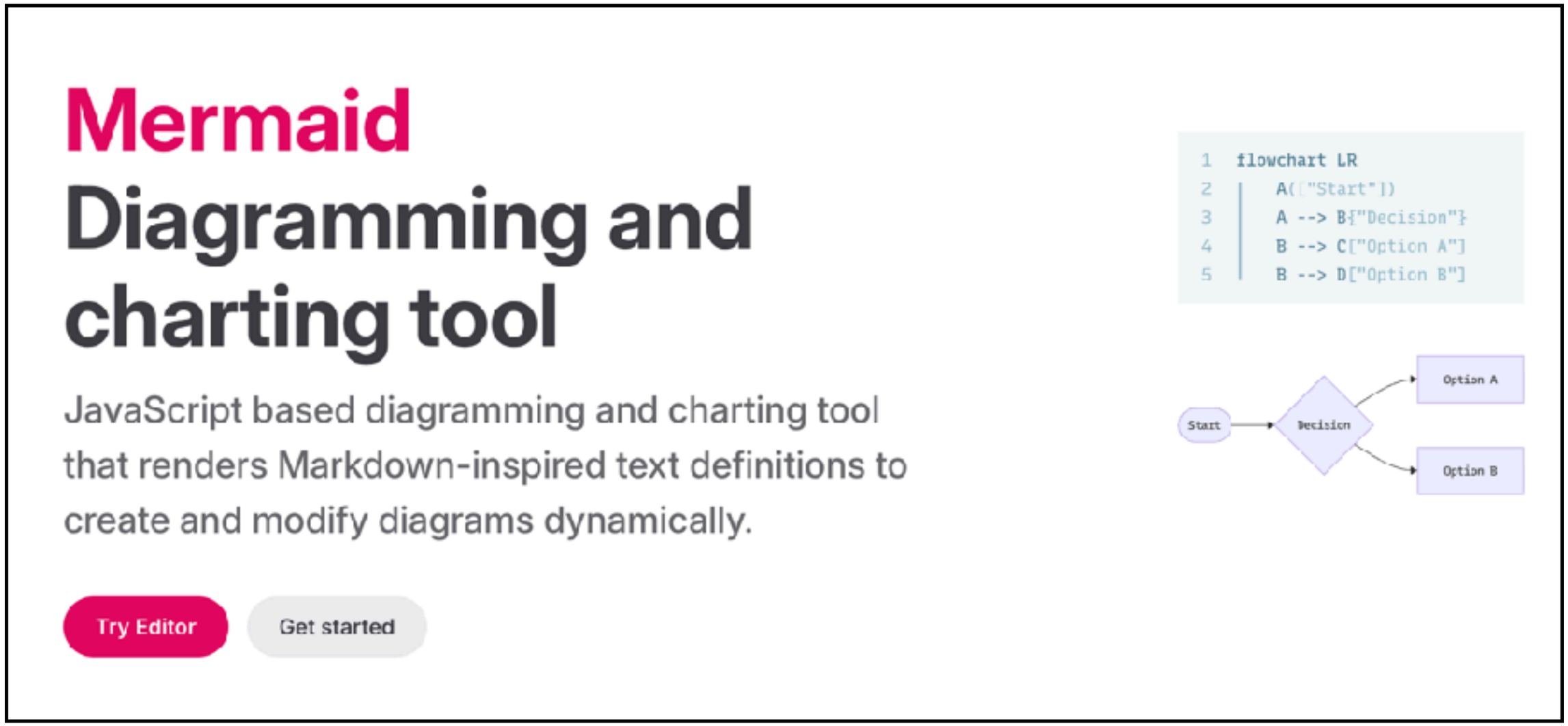
Preview



Insert →

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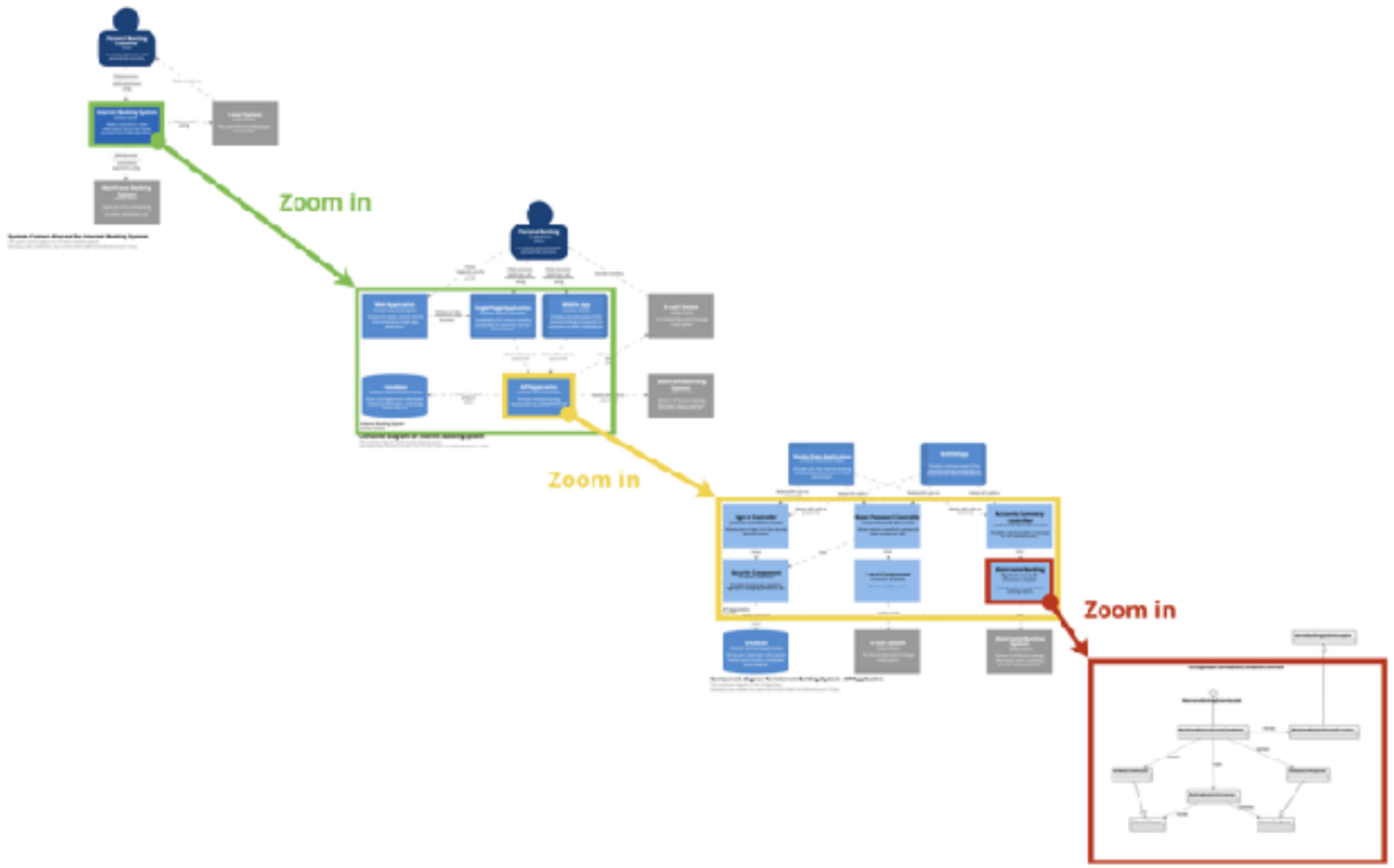
Mermaid Diagram



https://mermaid.js.org/

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Software Architecture



Level 1
Context

Level 2
Containers

Level 3
Components

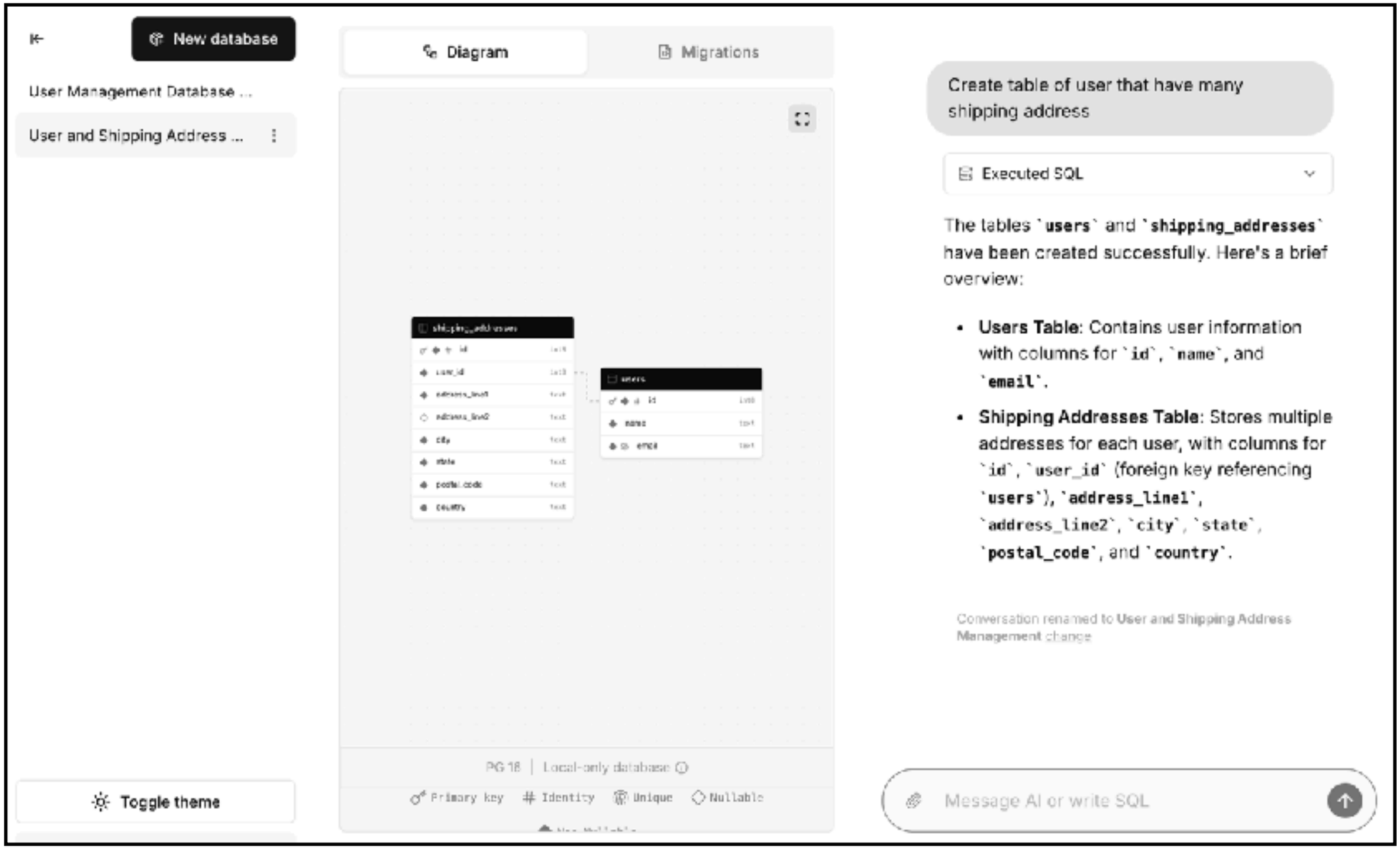
Level 4
Code

https://c4model.com/



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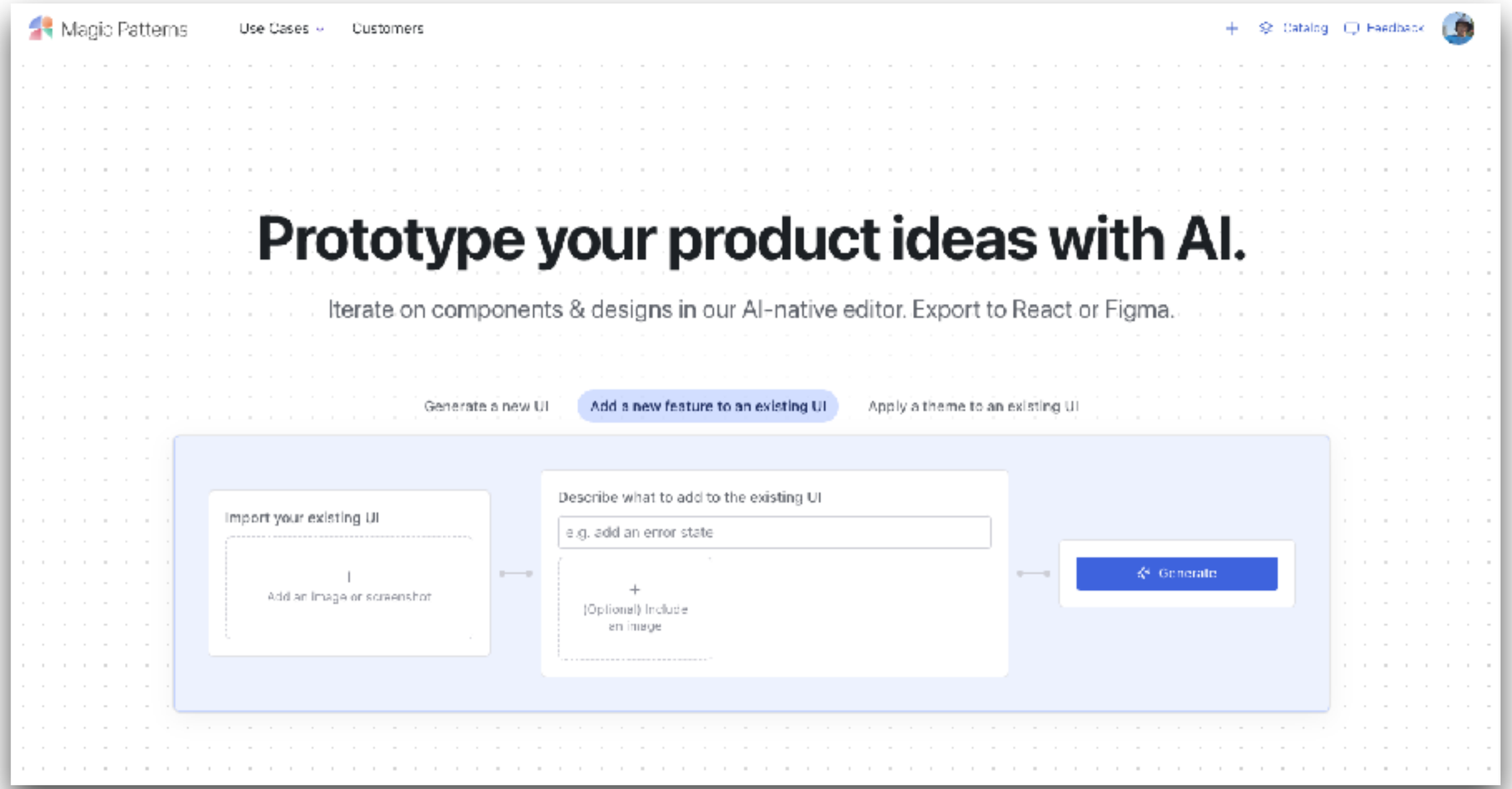
Database



https://database.build/

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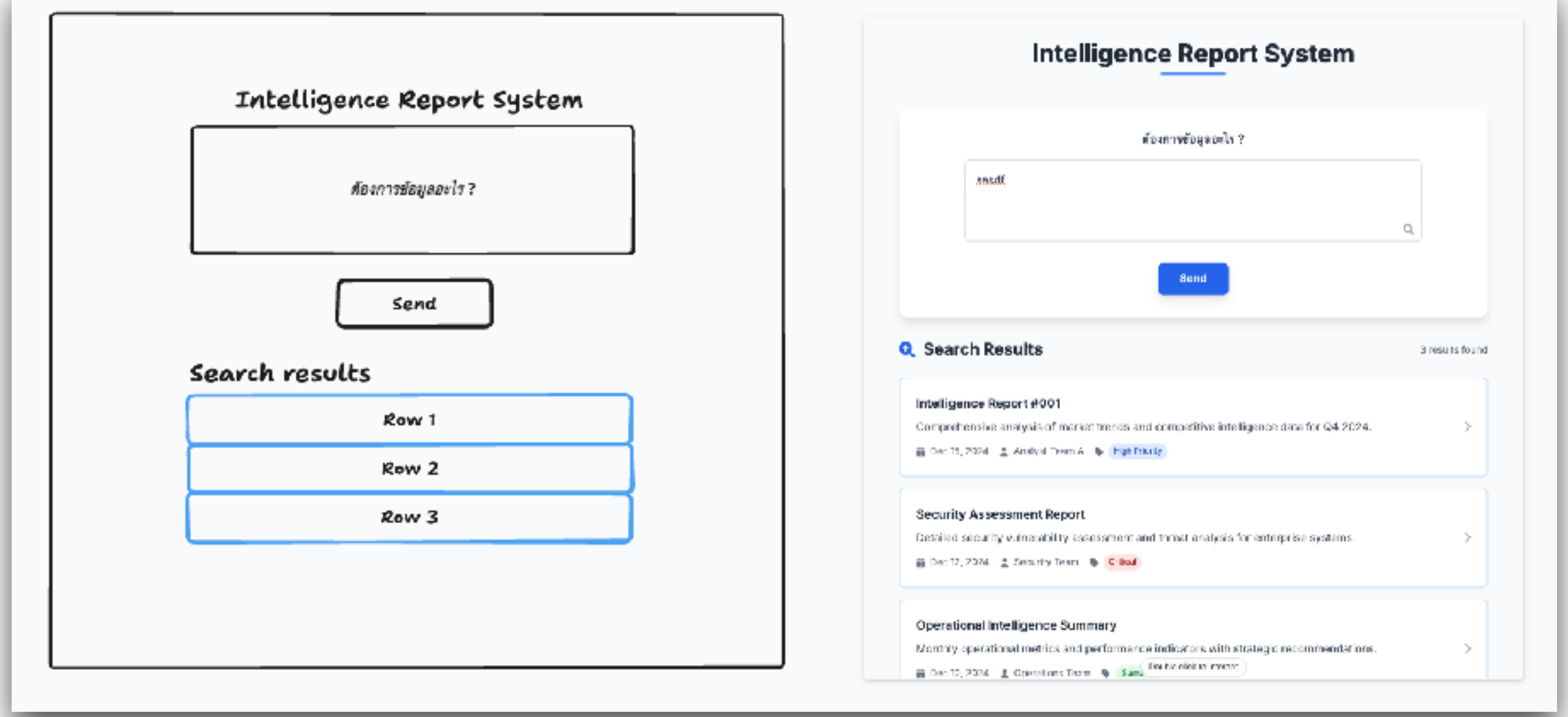
Magic Pattern



https://www.magicpatterns.com/

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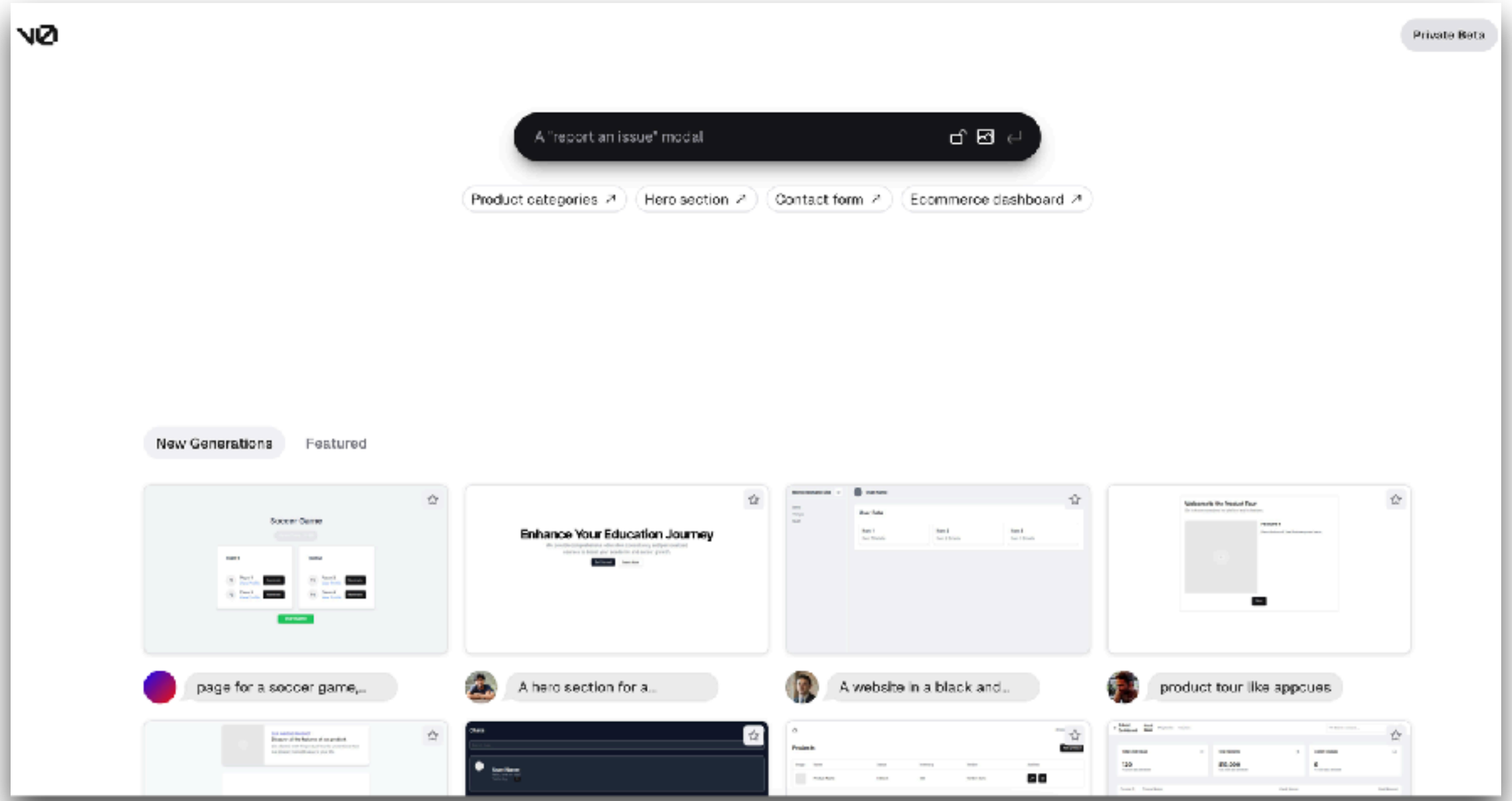
Make Real



https://github.com/tldraw/make-real

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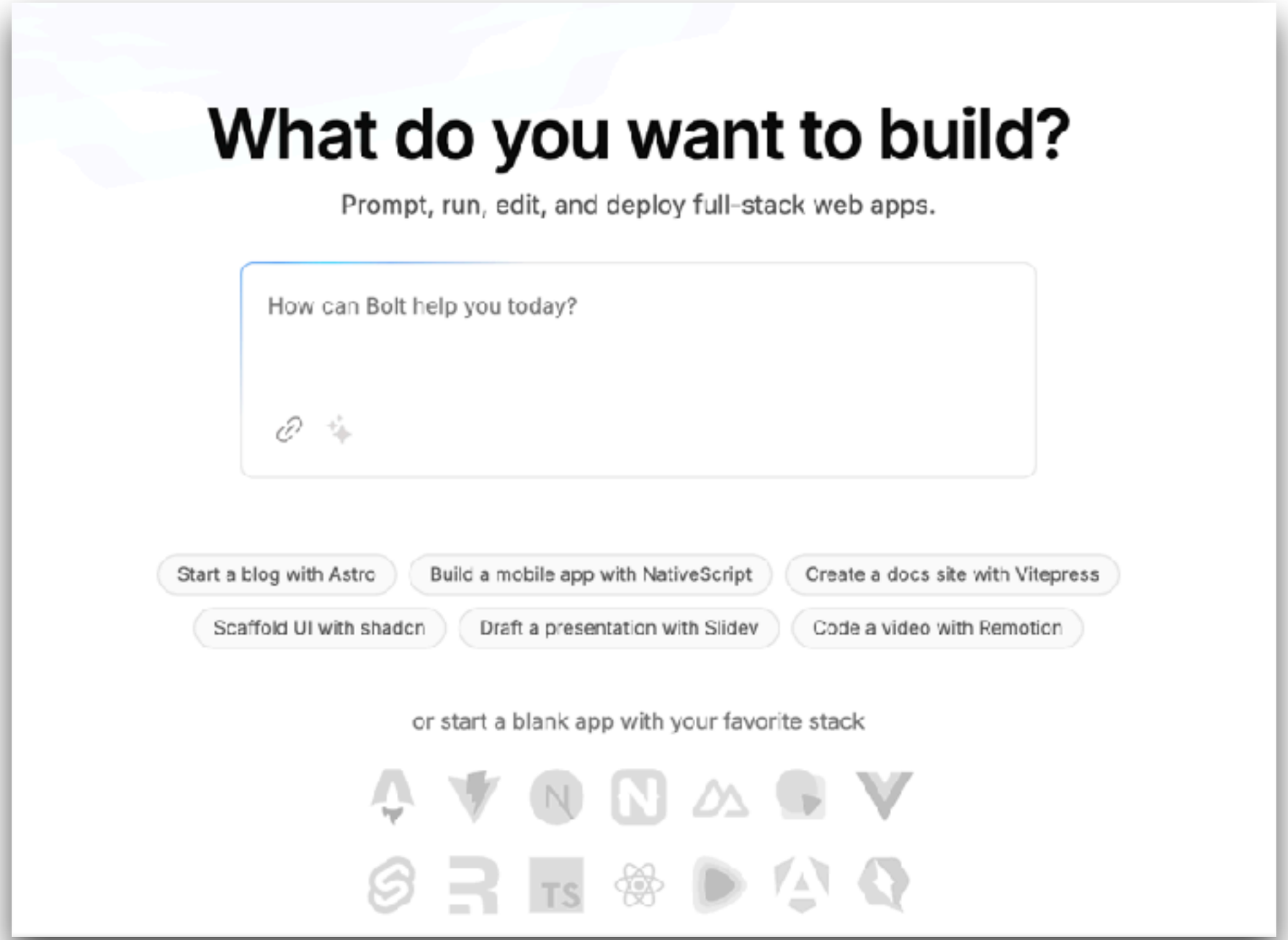
V0.app



https://v0.dev/

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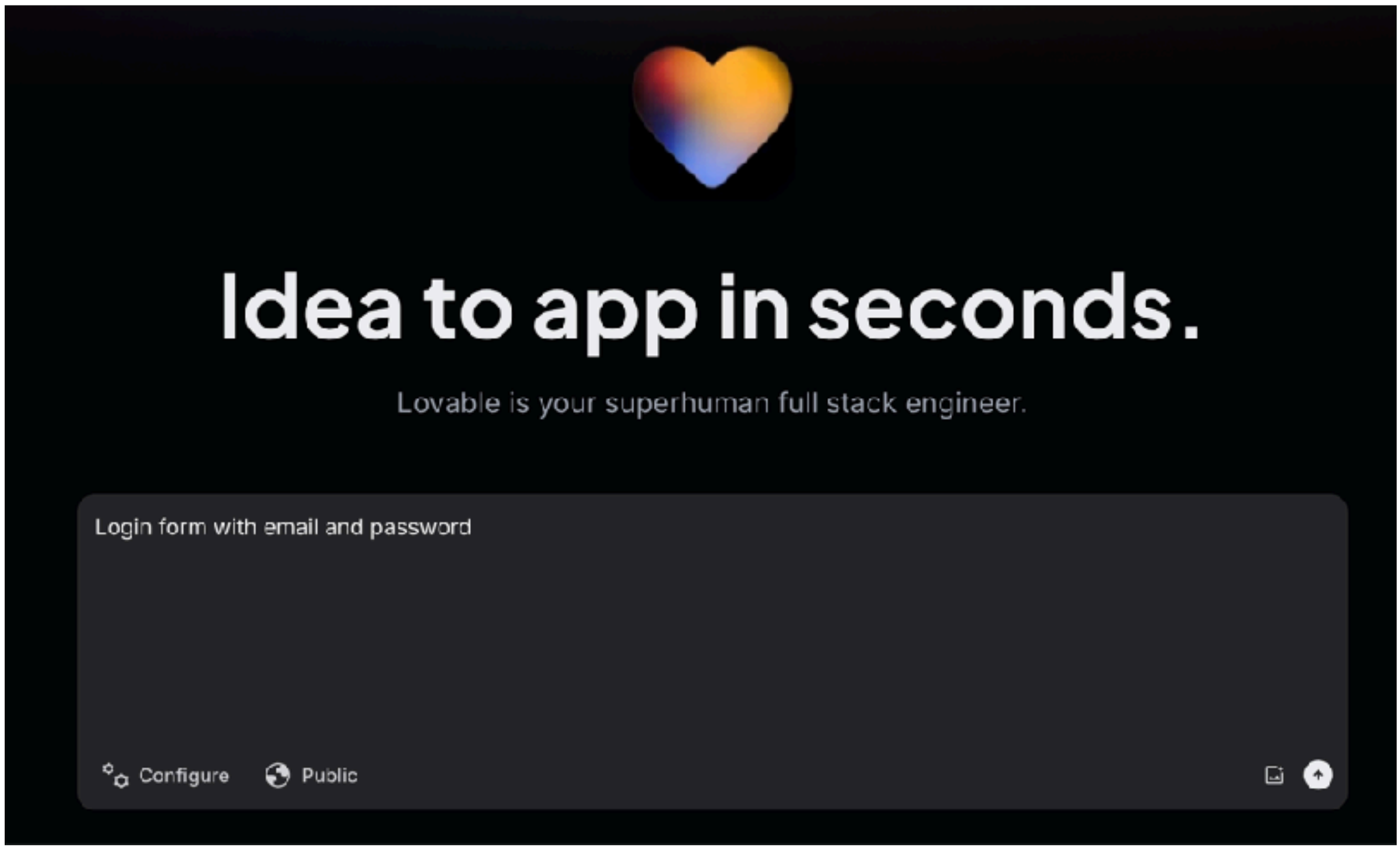
Bolt.new



https://bolt.new/

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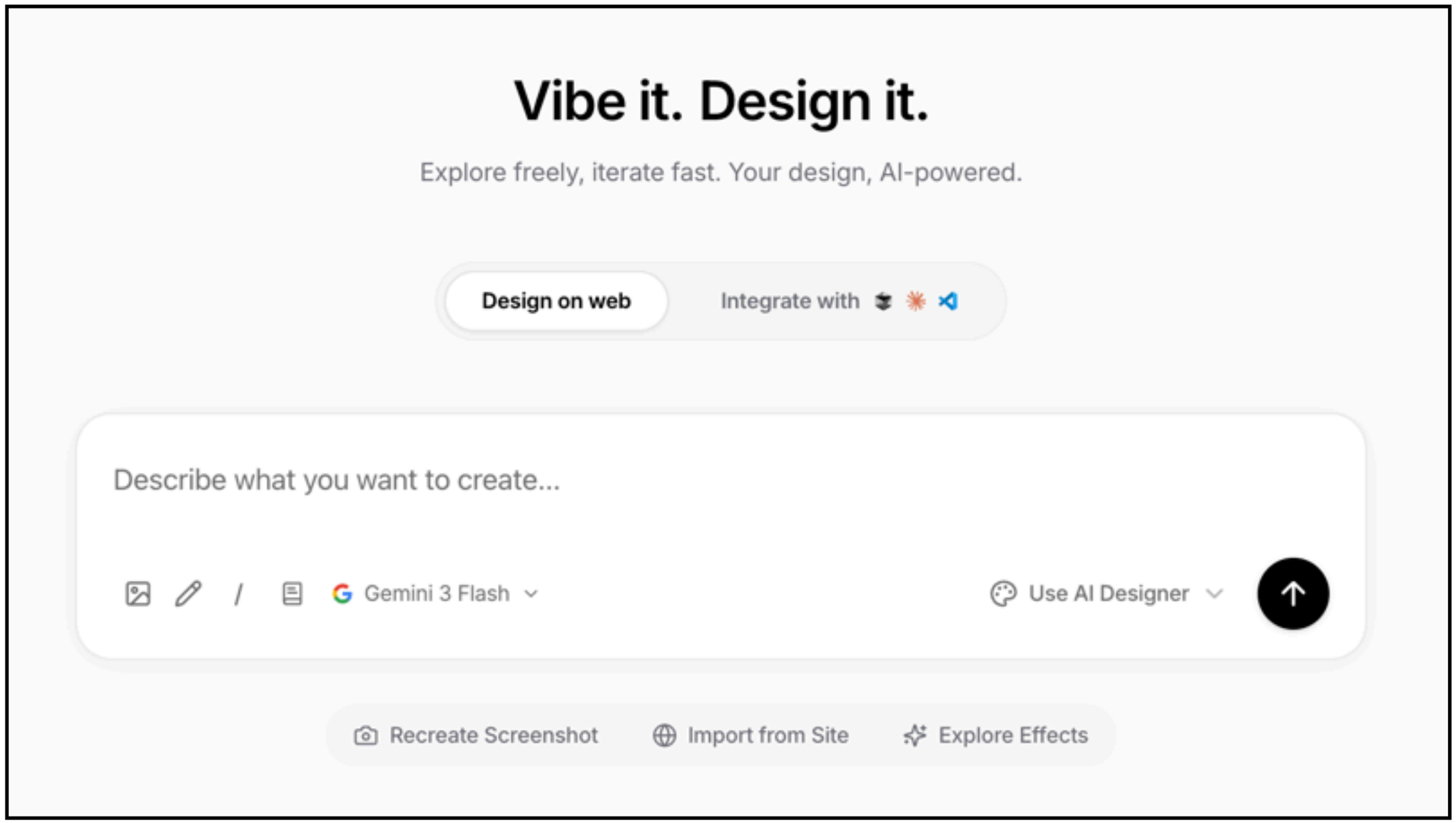
Lovable



https://lovable.dev/

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SuperDesign



https://app.superdesign.dev/

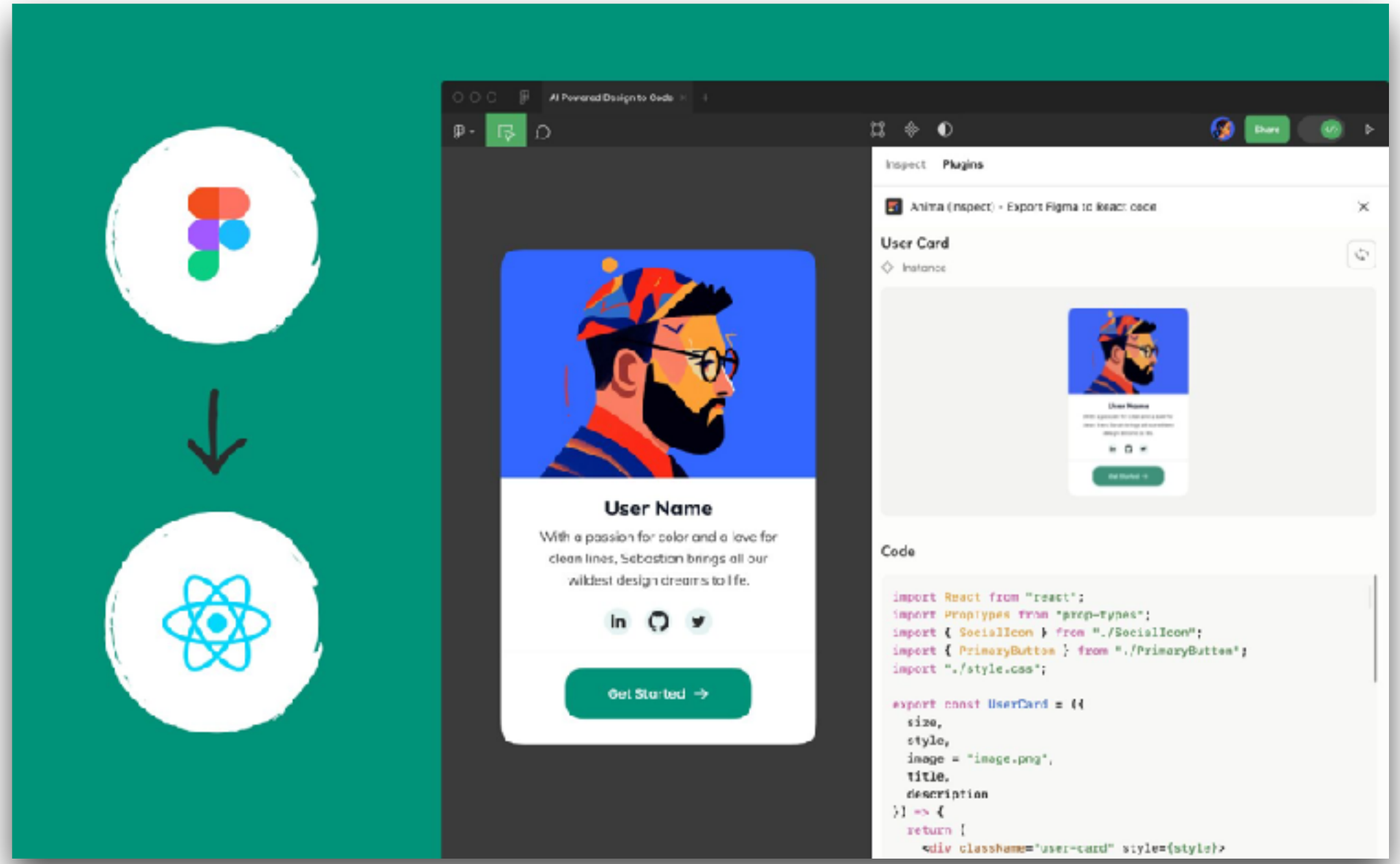
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Design to Code



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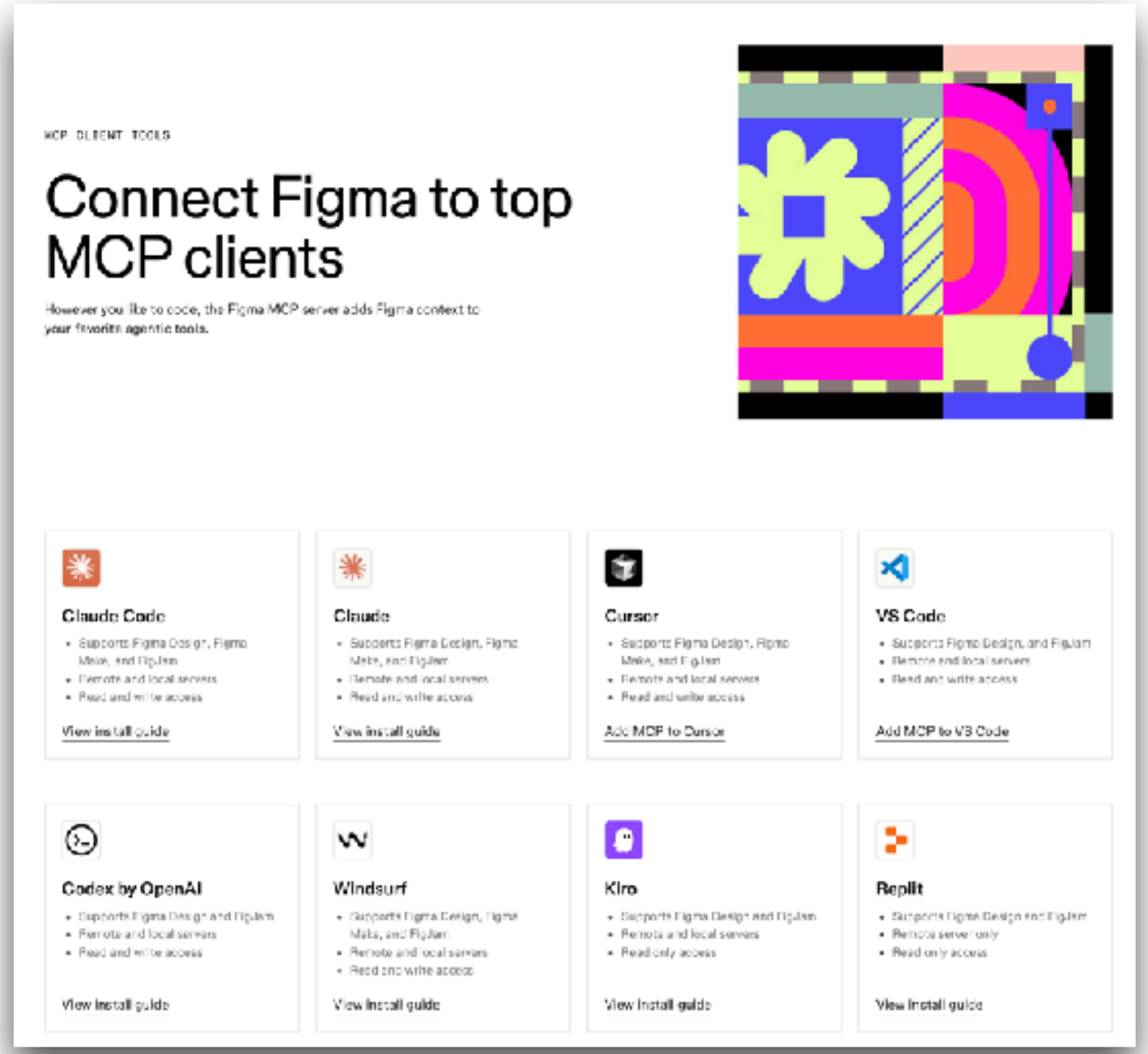
Figma



https://www.figma.com/mcp-catalog/

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Figma MCP

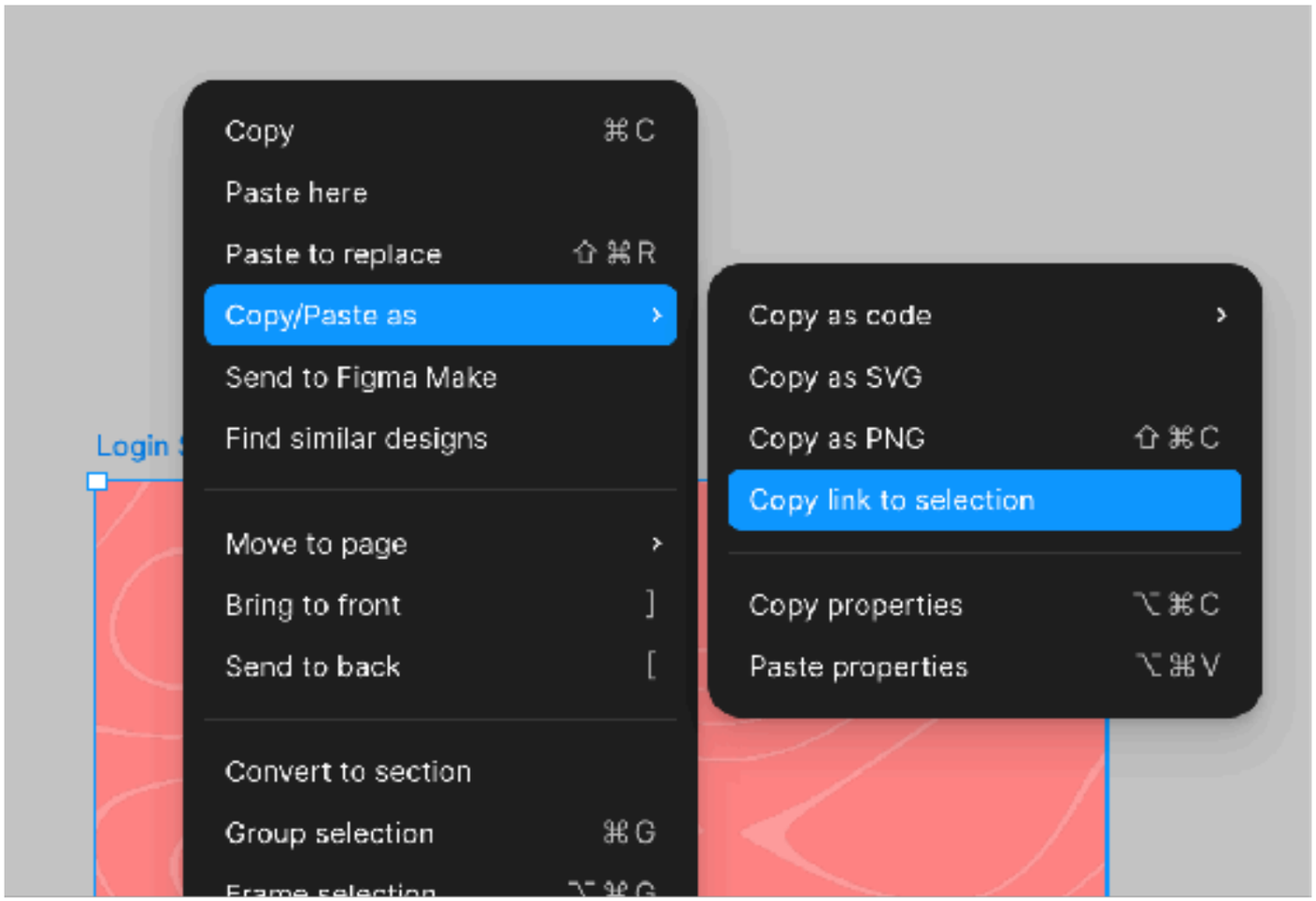


https://www.figma.com/mcp-catalog/

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Design to Code in Figma

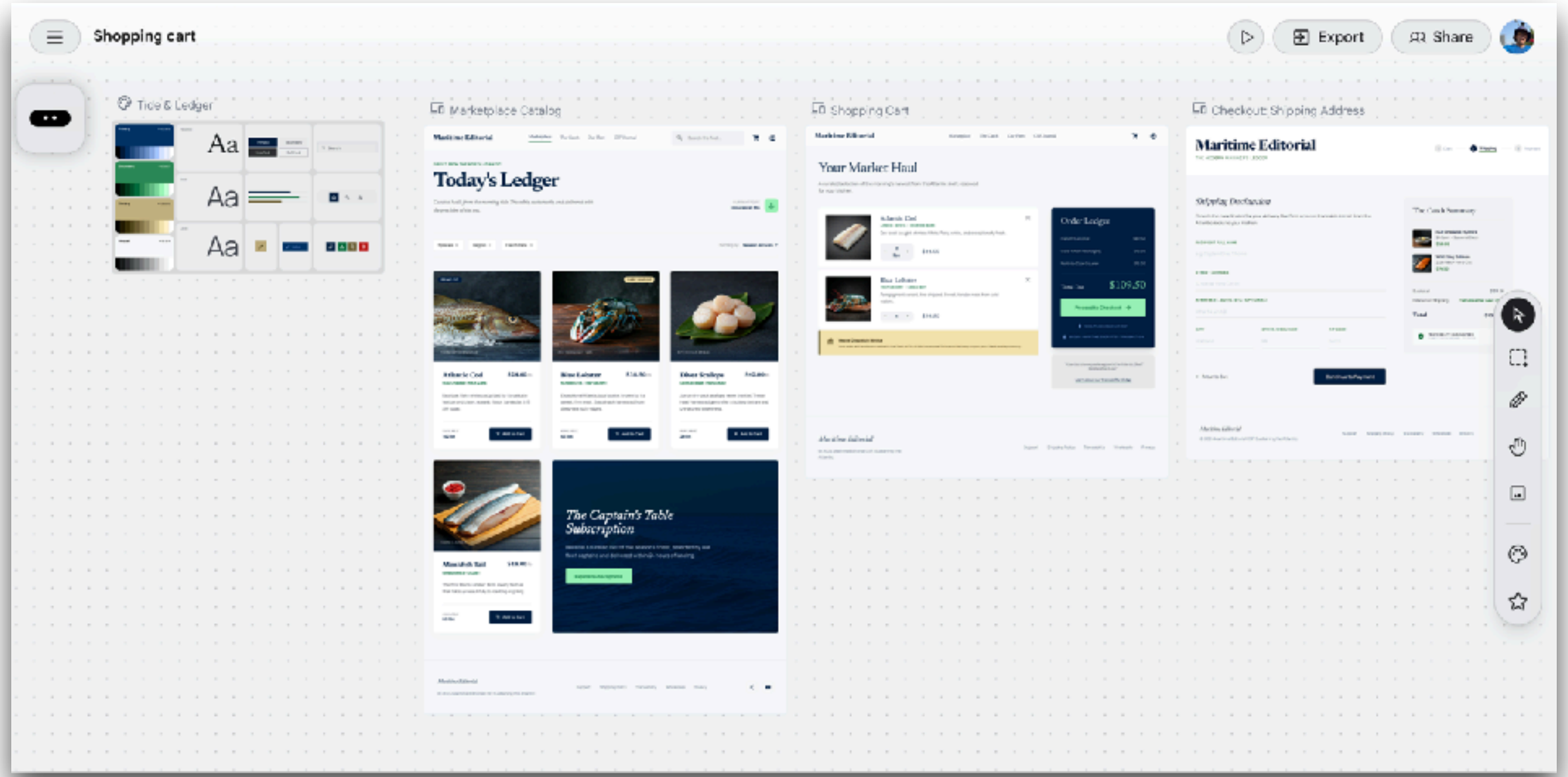
Select frame -> Copy/Paste as -> link to selection



https://www.figma.com/mcp-catalog/

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Google Stitch

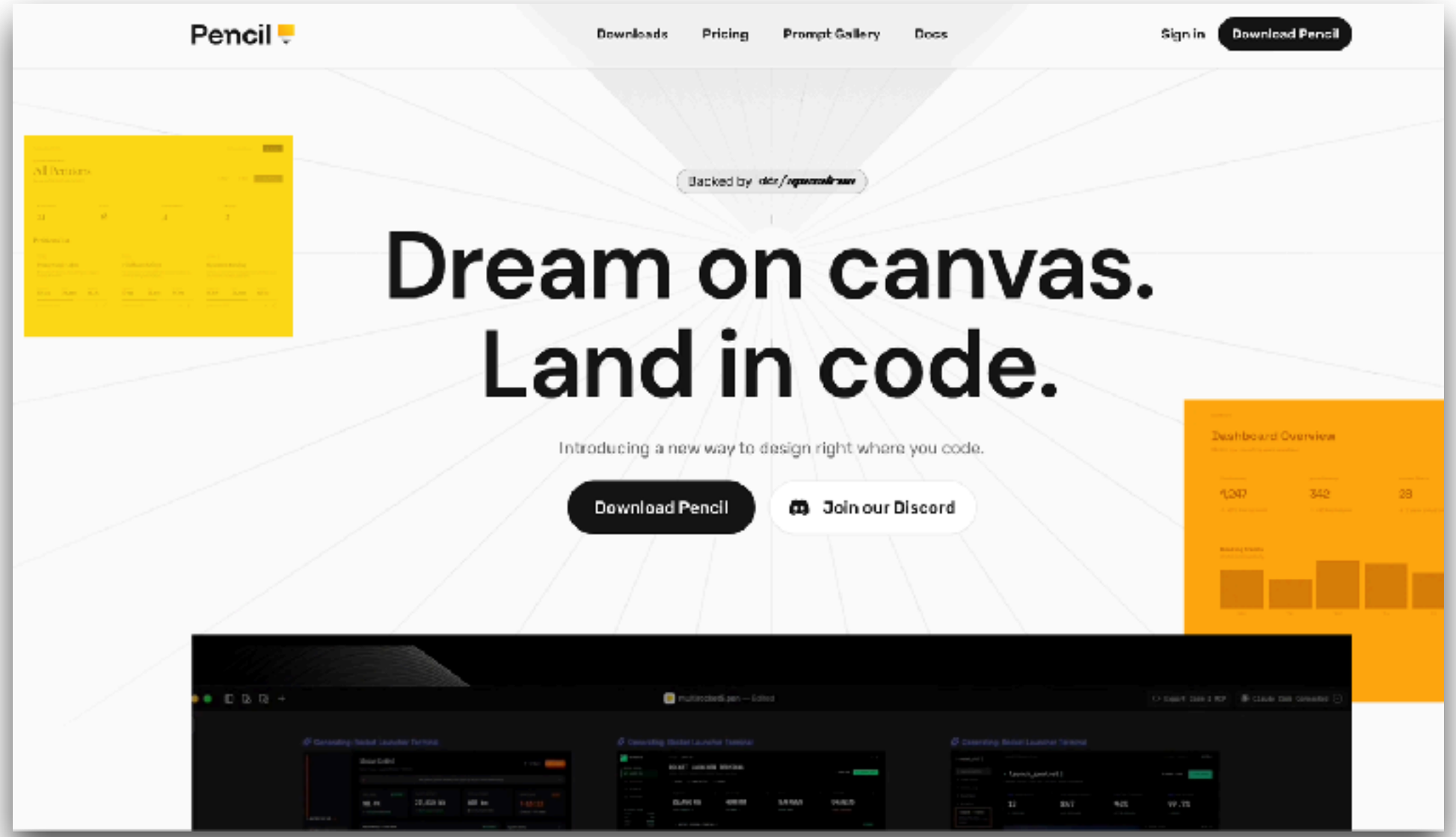


https://stitch.withgoogle.com/

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Pencil

Bridge the gap between design and development



https://docs.pencil.dev/getting-started/ai-integration

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Development Process



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Develop

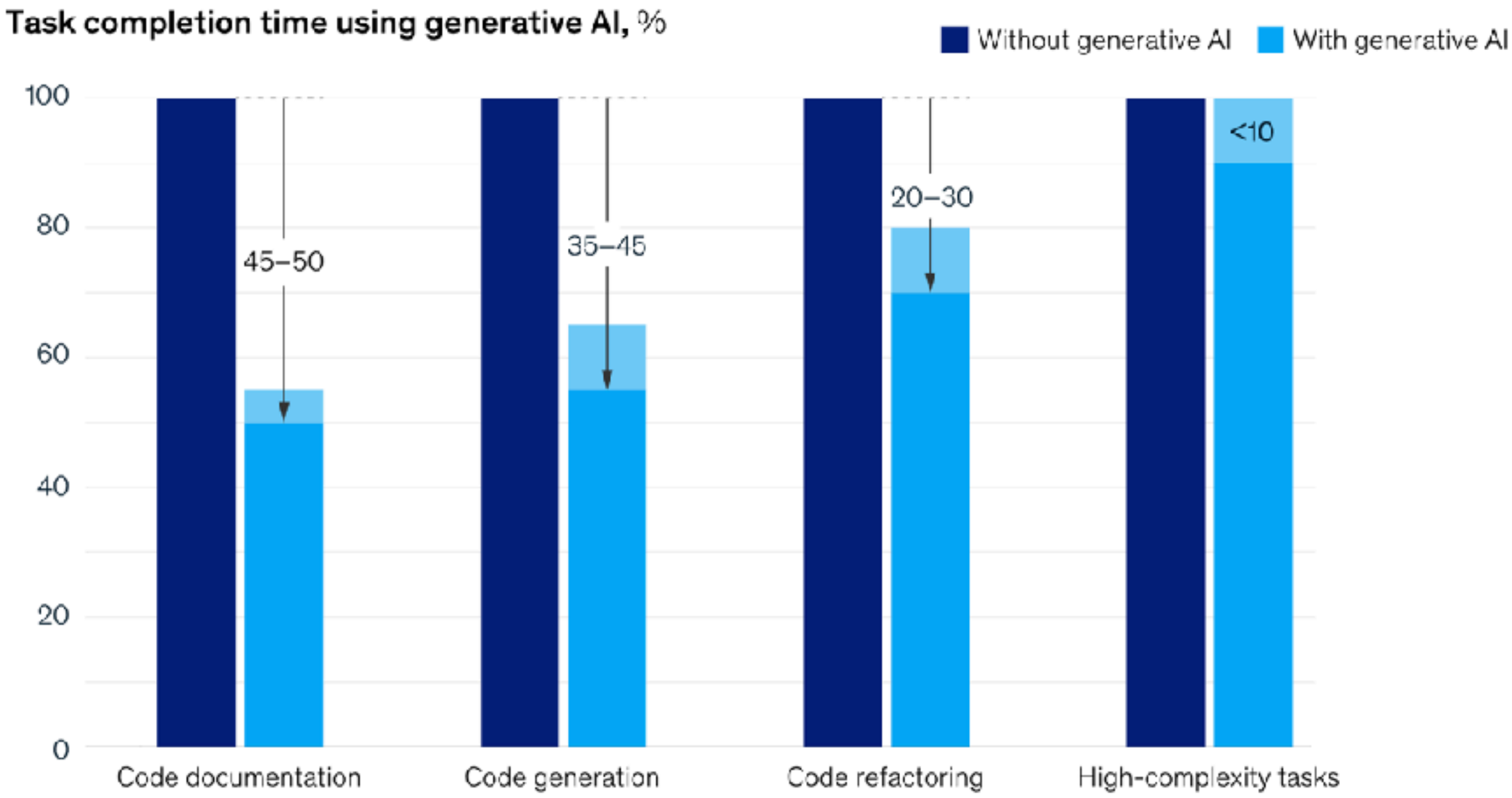
Requirement	Design	Develop	Testing	Deploy

Code generation
Review and explain code
Debugging code
Improve consistency
Code translation

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Development

Generative AI can increase developer speed, but less so for complex tasks.



https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/unleashing-developer-productivity-with-generative-ai

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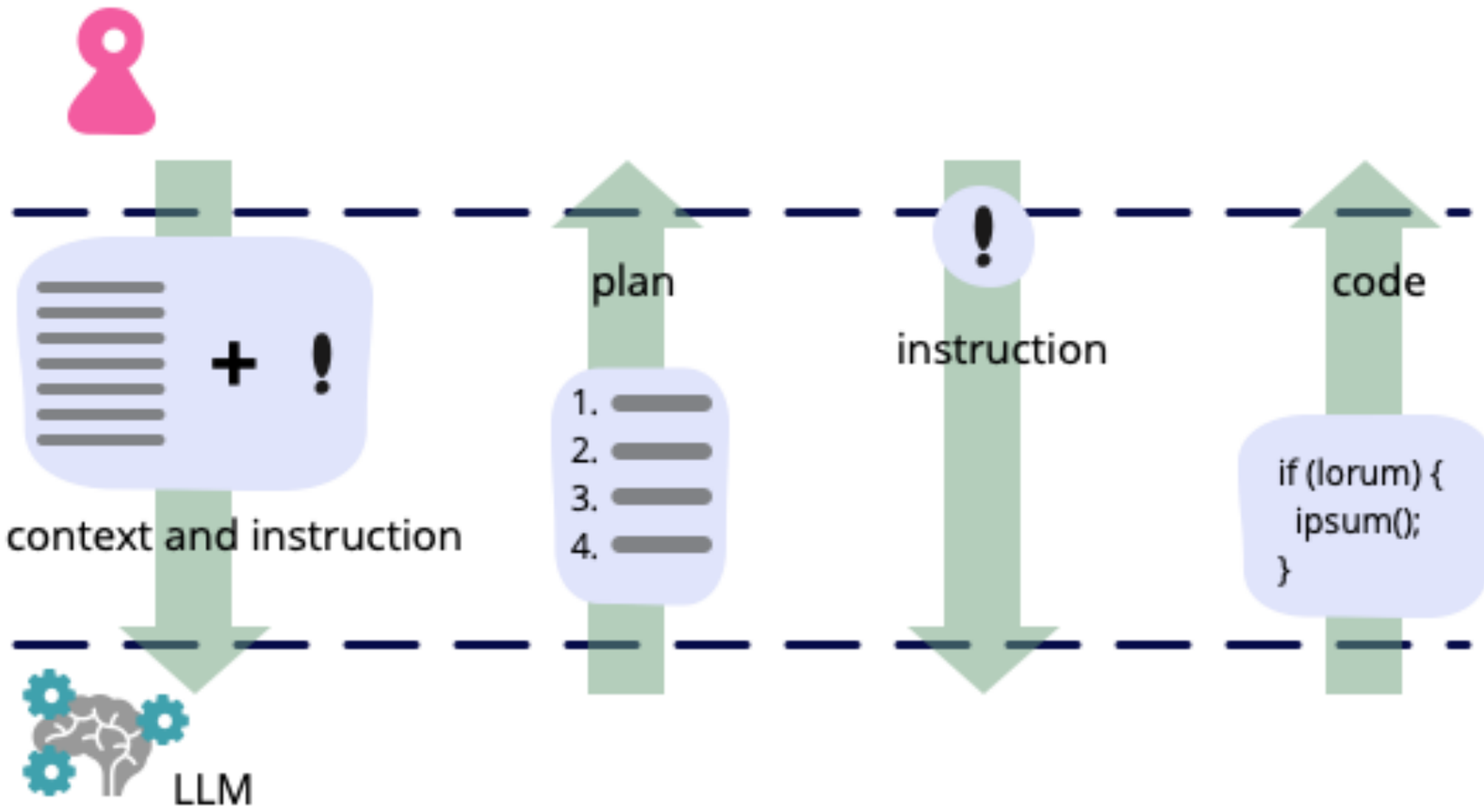
Category of Tools

Chat AI	Code AI	AI Agent Public and Local
ChatGPT	Copilot in VSCode	Aider
Gemini	Cursor IDE	Claude code
Claude.ai	Windsurf	Gemini CLI
DeepSeek	Zed	Codex CLI
		Qwen3-coder



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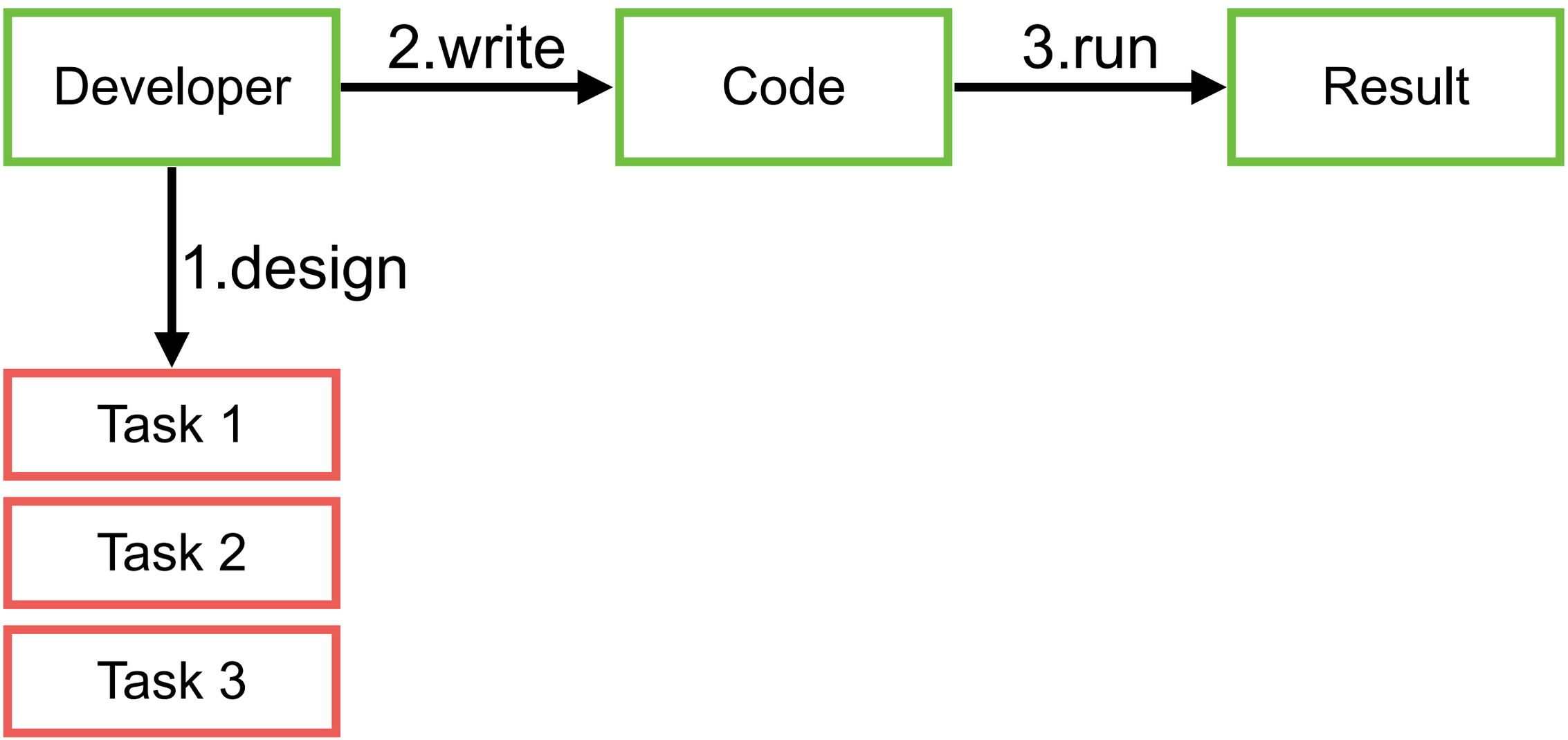
Development



https://martinfowler.com/articles/2023-chatgpt-xu-hao.html

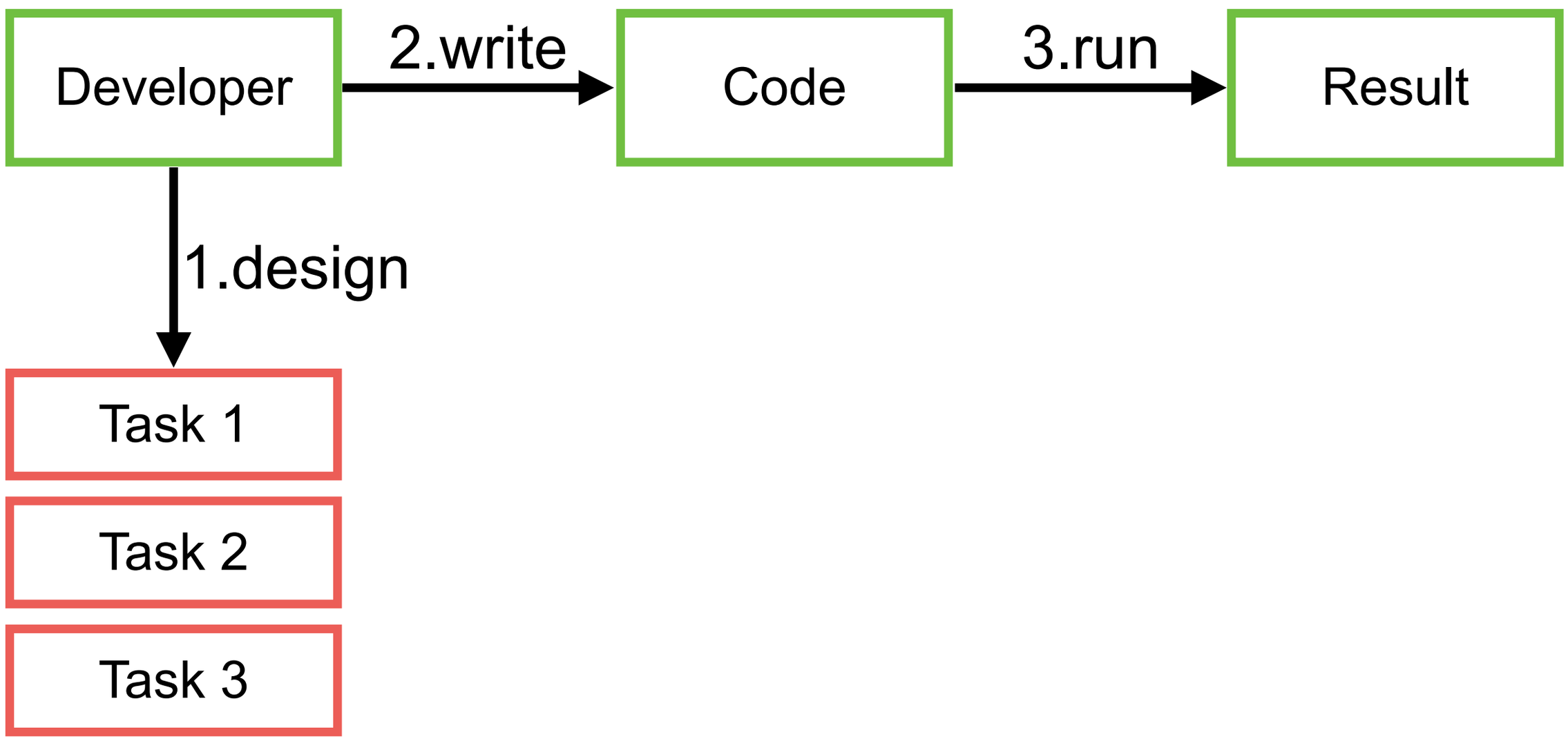
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Development



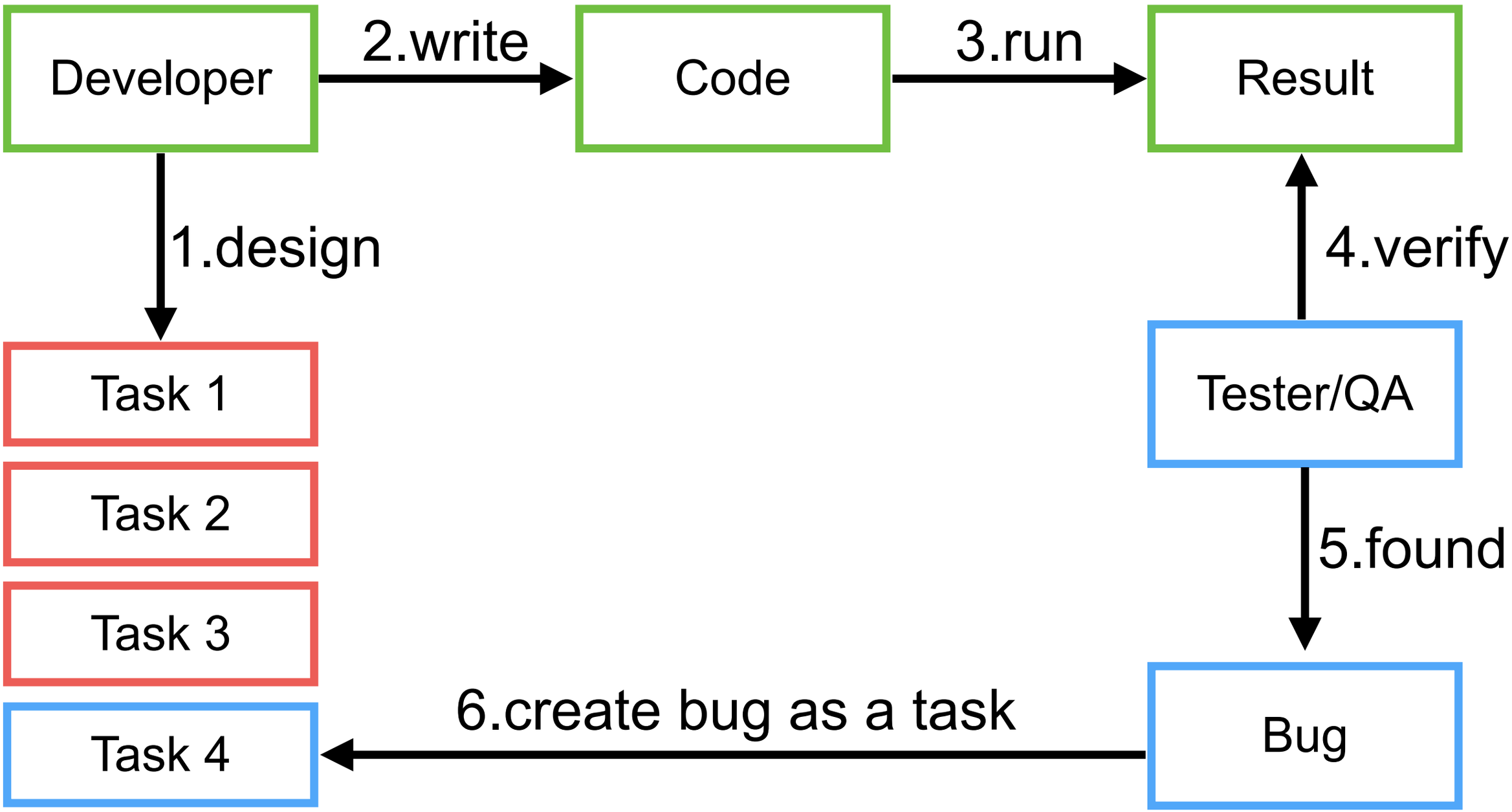
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Development



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Development + Testing



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Main Features

Ask question

Generate code

Refactor code

Document code

Find problems in your code

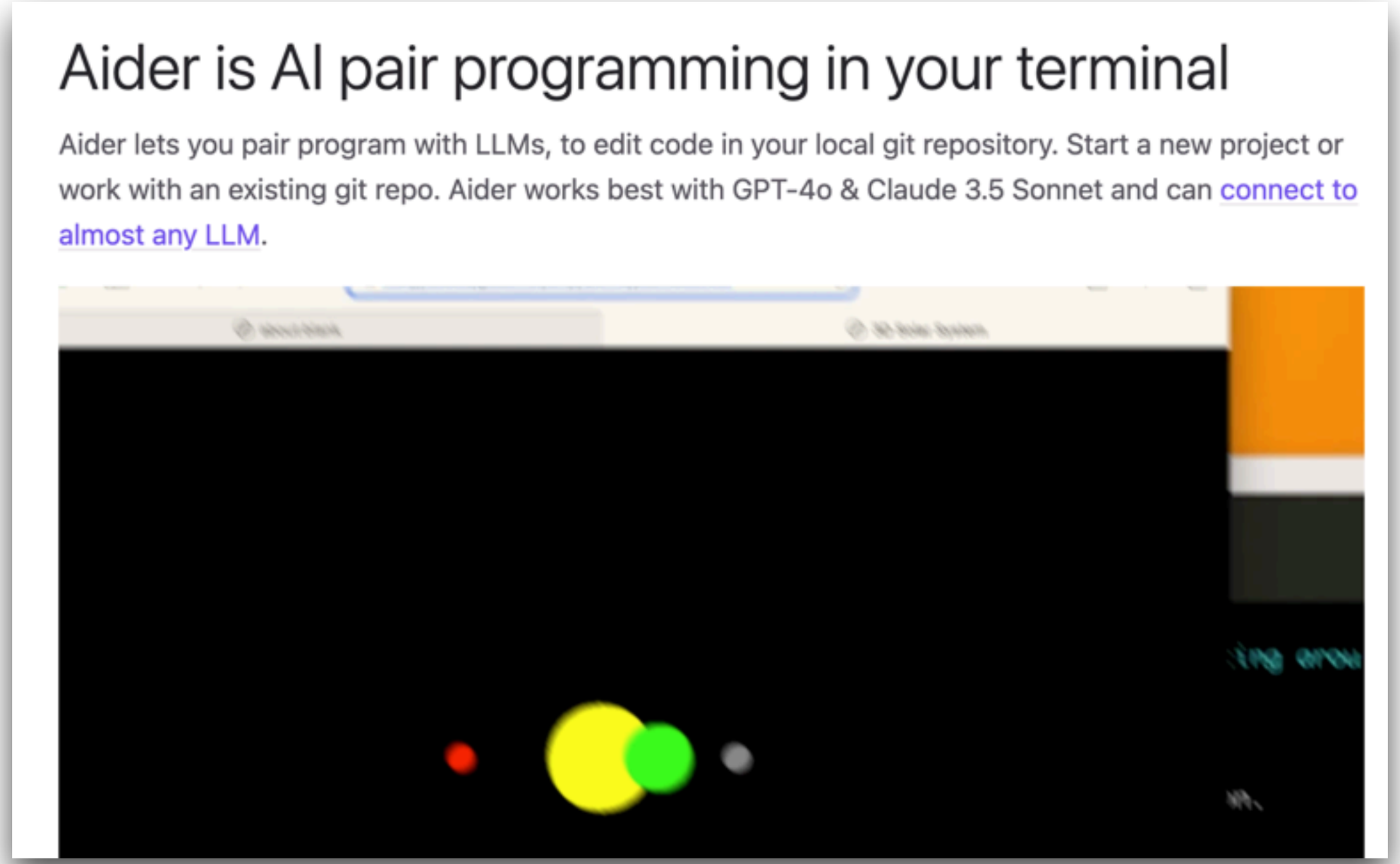
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Pair programming with AI



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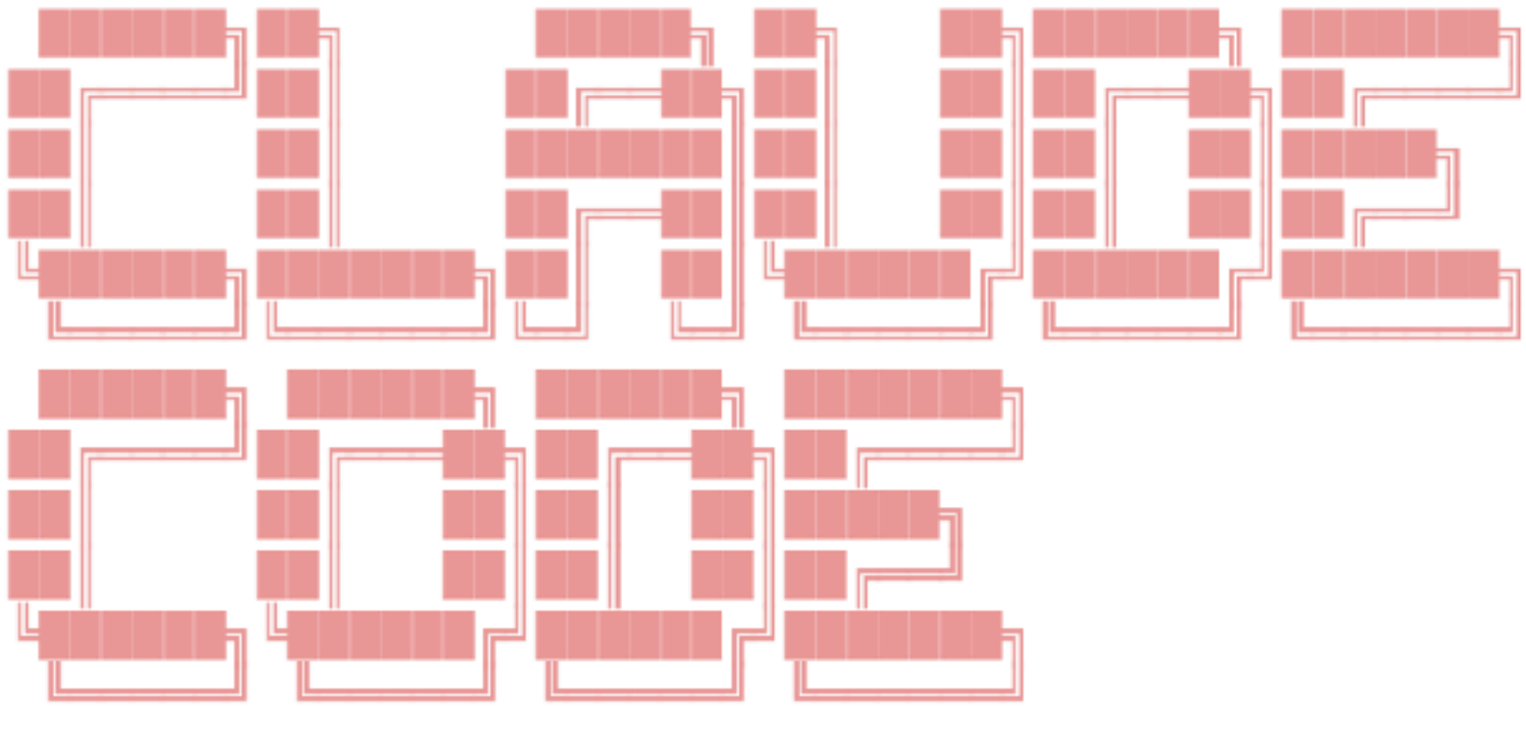
AI Pair Programming



https://github.com/paul-gauthier/aider

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```
* Welcome to Claude Code research preview!
```



```
Claude Code is billed based on API usage through your Anthropic Console account.

Pricing may evolve as we move towards general availability.

Press Enter to login to your Anthropic Console account…
```

2025/02

https://www.anthropic.com/news/claude-3-7-sonnet



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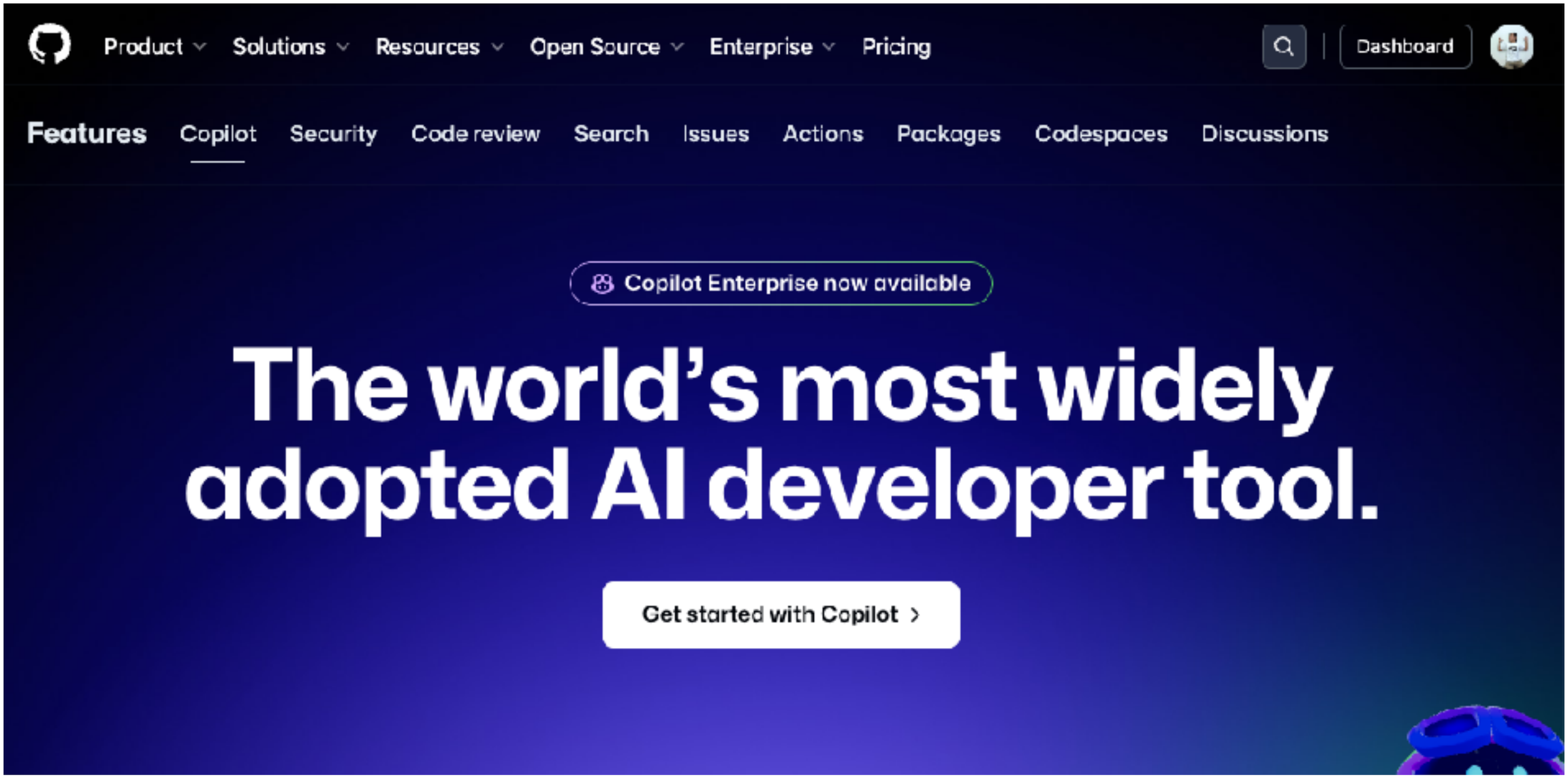
2025/06

https://github.com/google-gemini/gemini-cli



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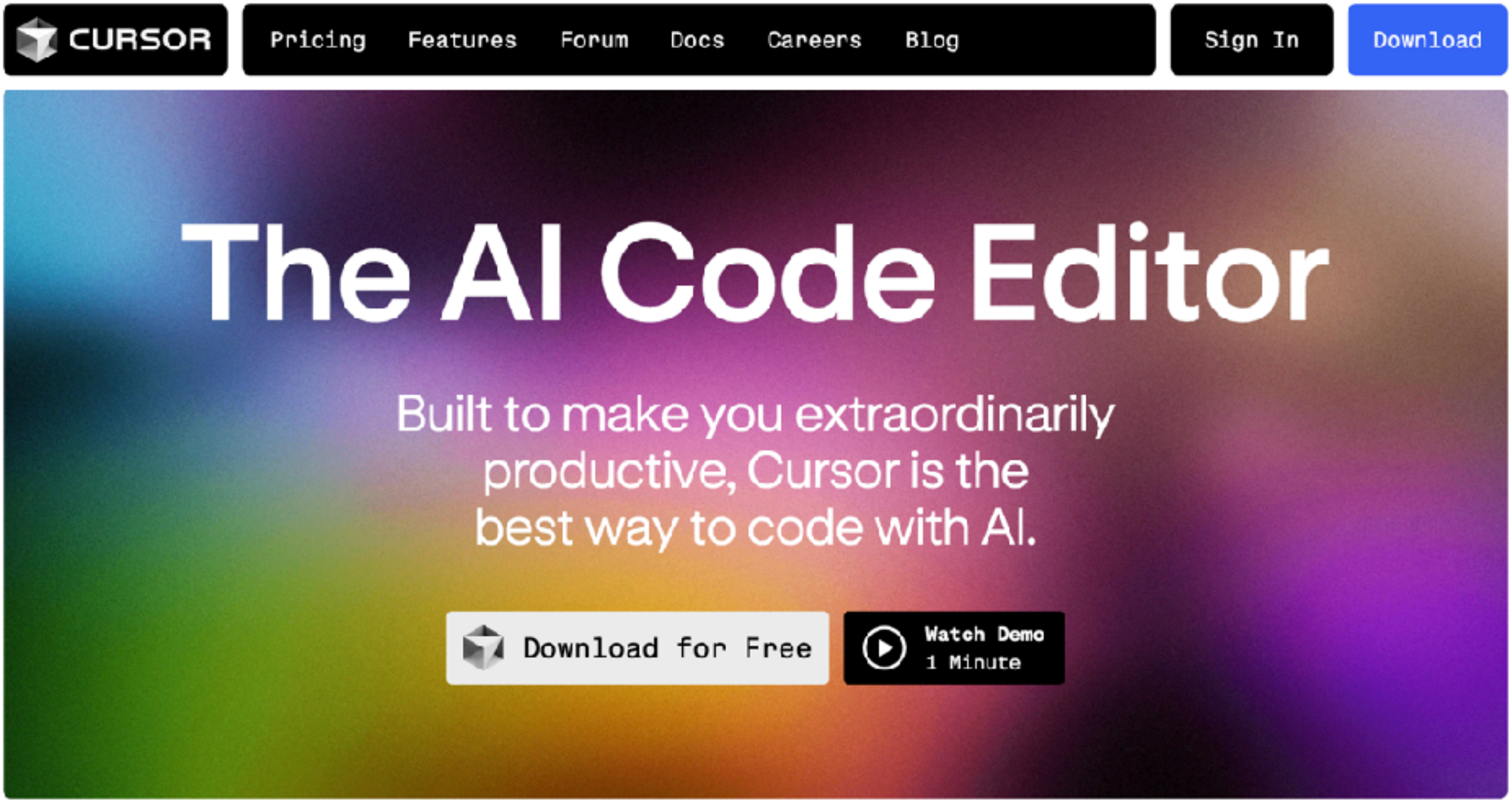
GitHub Copilot



https://github.com/features/copilot

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Cursor.sh



https://www.cursor.com/

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Tips and Techniques with AI

Scope of prompt

Error in result

Setup and config project

Latest information

Use technical keywords

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Specification-Driven Development (SDD)

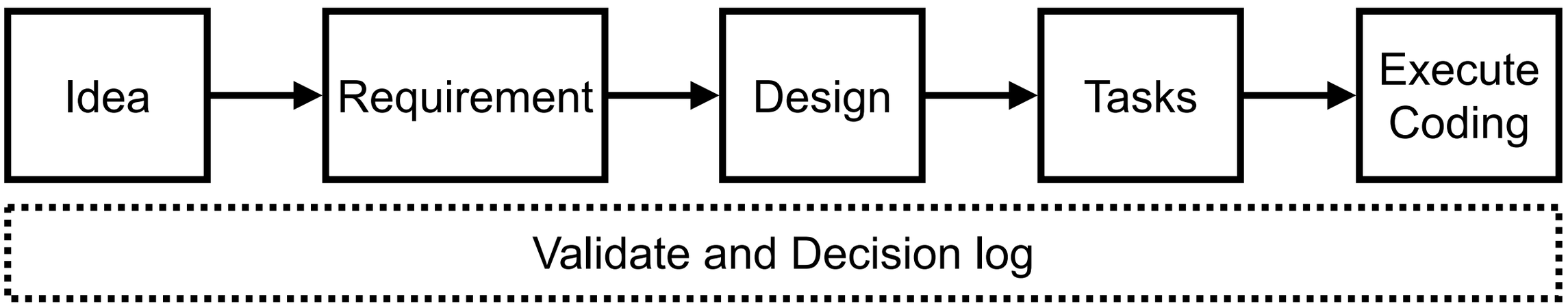


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SDD ?

Specification or Documentation-First

Modern software development approach where detailed, structure specifications are created before writing code



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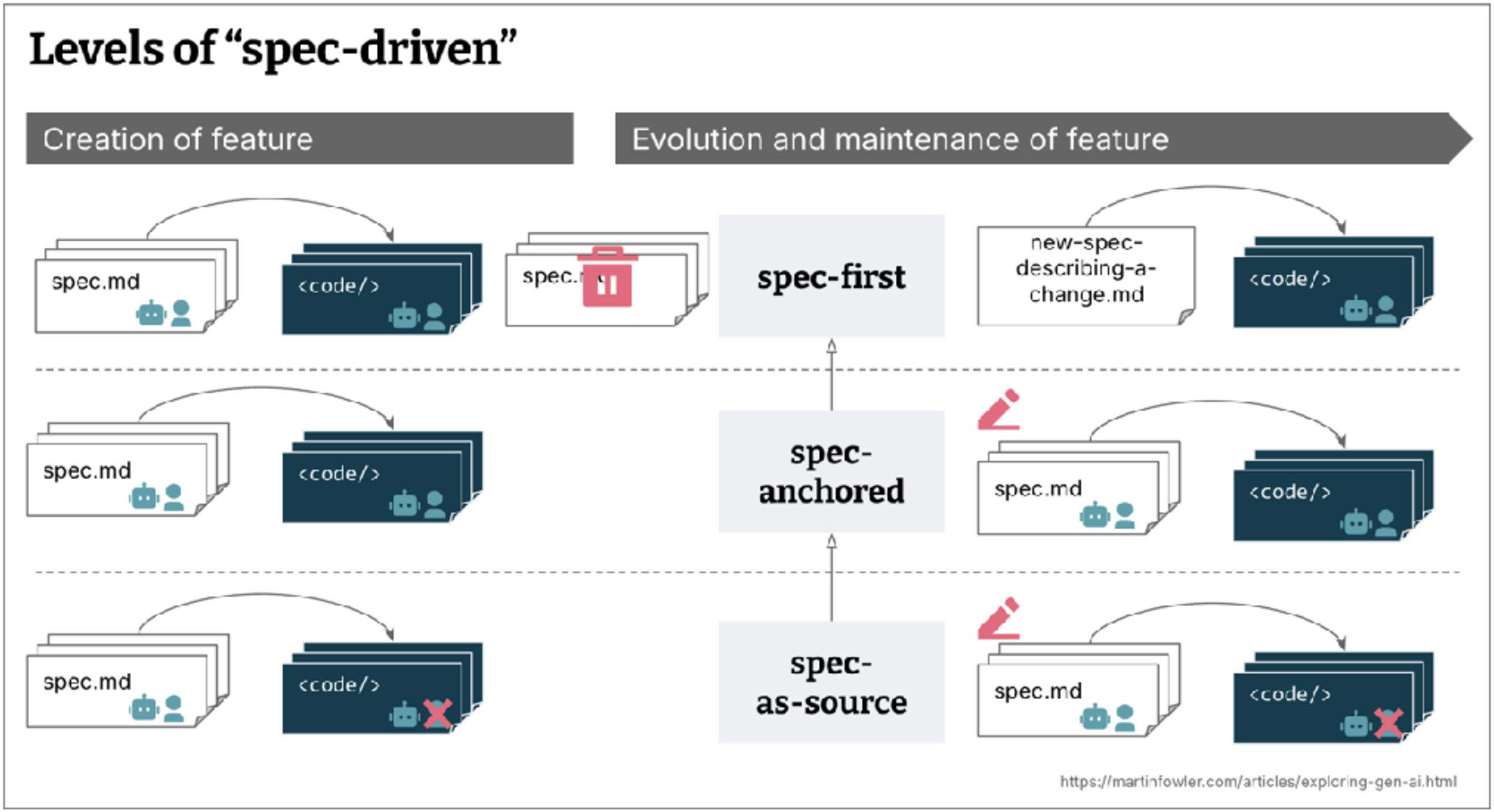
Types of Specification ?

User requirements
Design documents
Interface contracts (UI, API)
Data schemas
Validation rules
Data/business flows
Security policy and constraints
Resources and performance constraints



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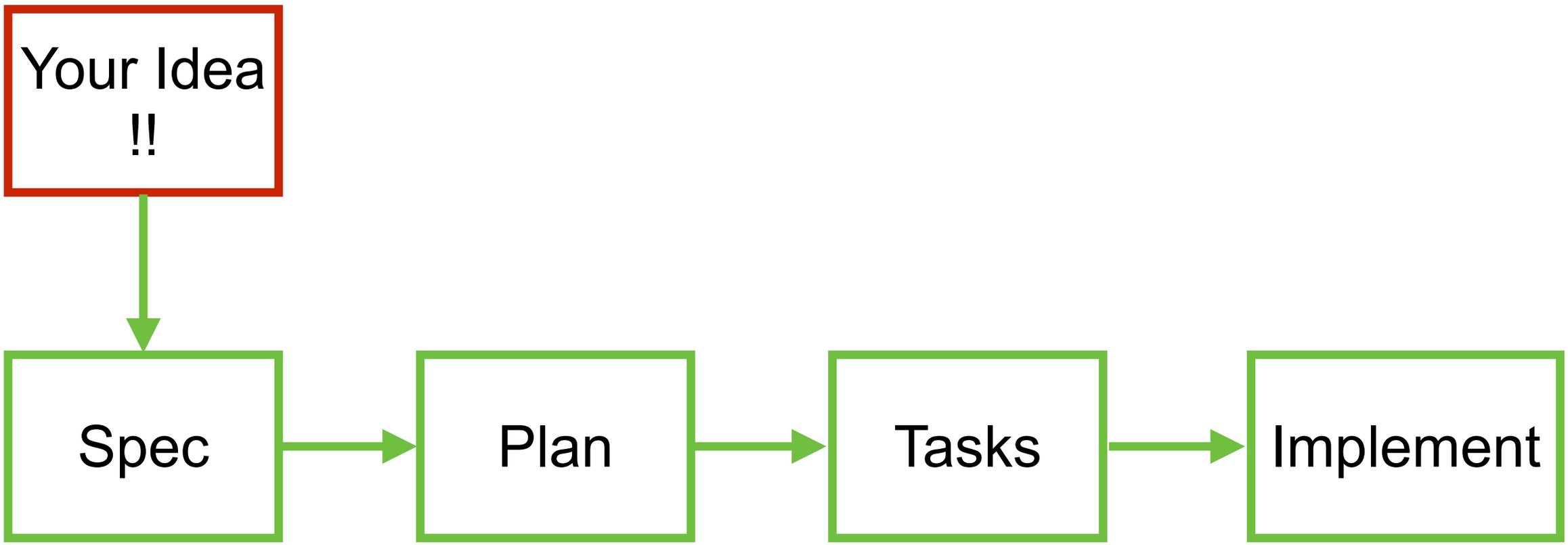
Workflow SDD



https://martinfowler.com/articles/exploring-gen-ai/sdd-3-tools.html

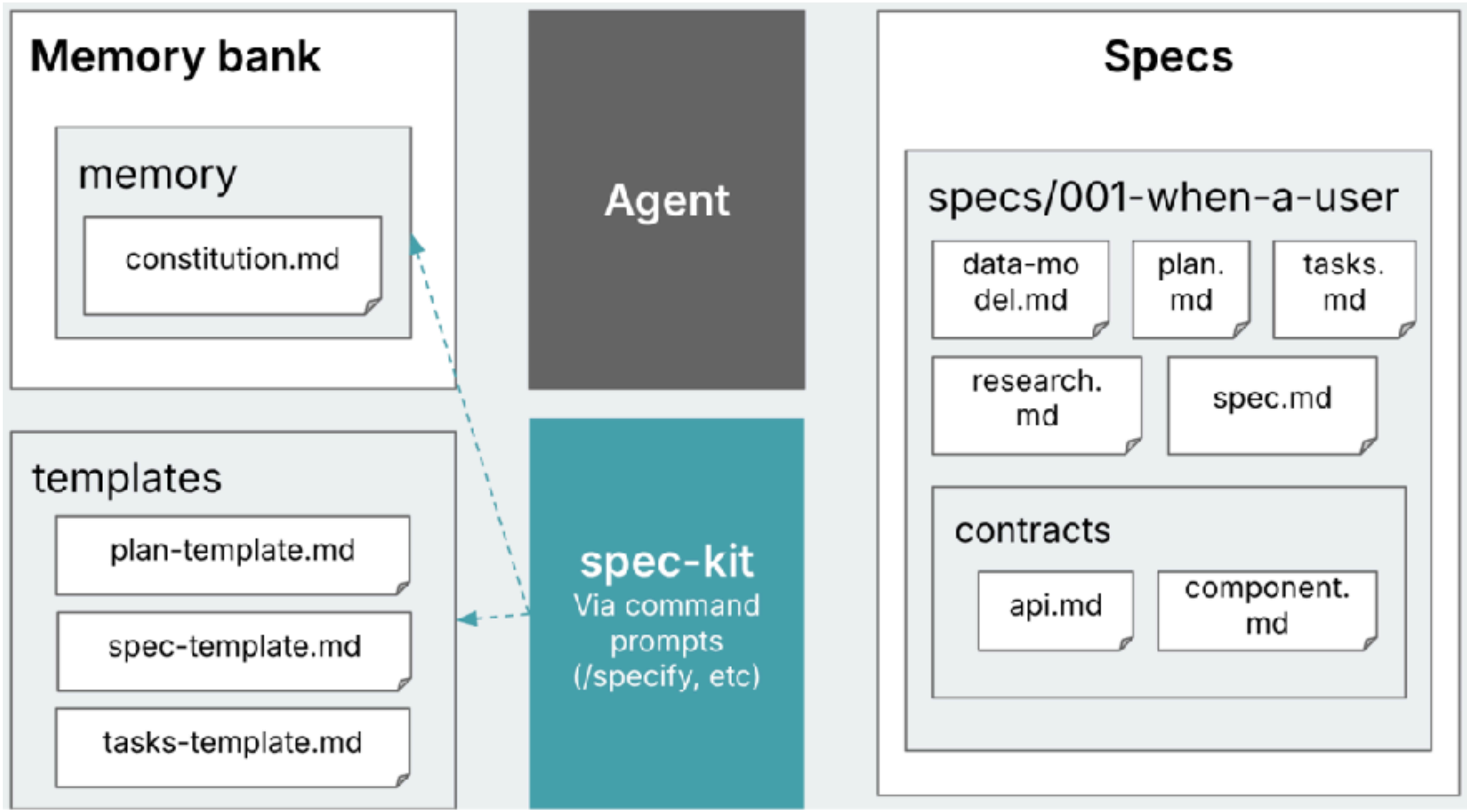
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Workflow SDD



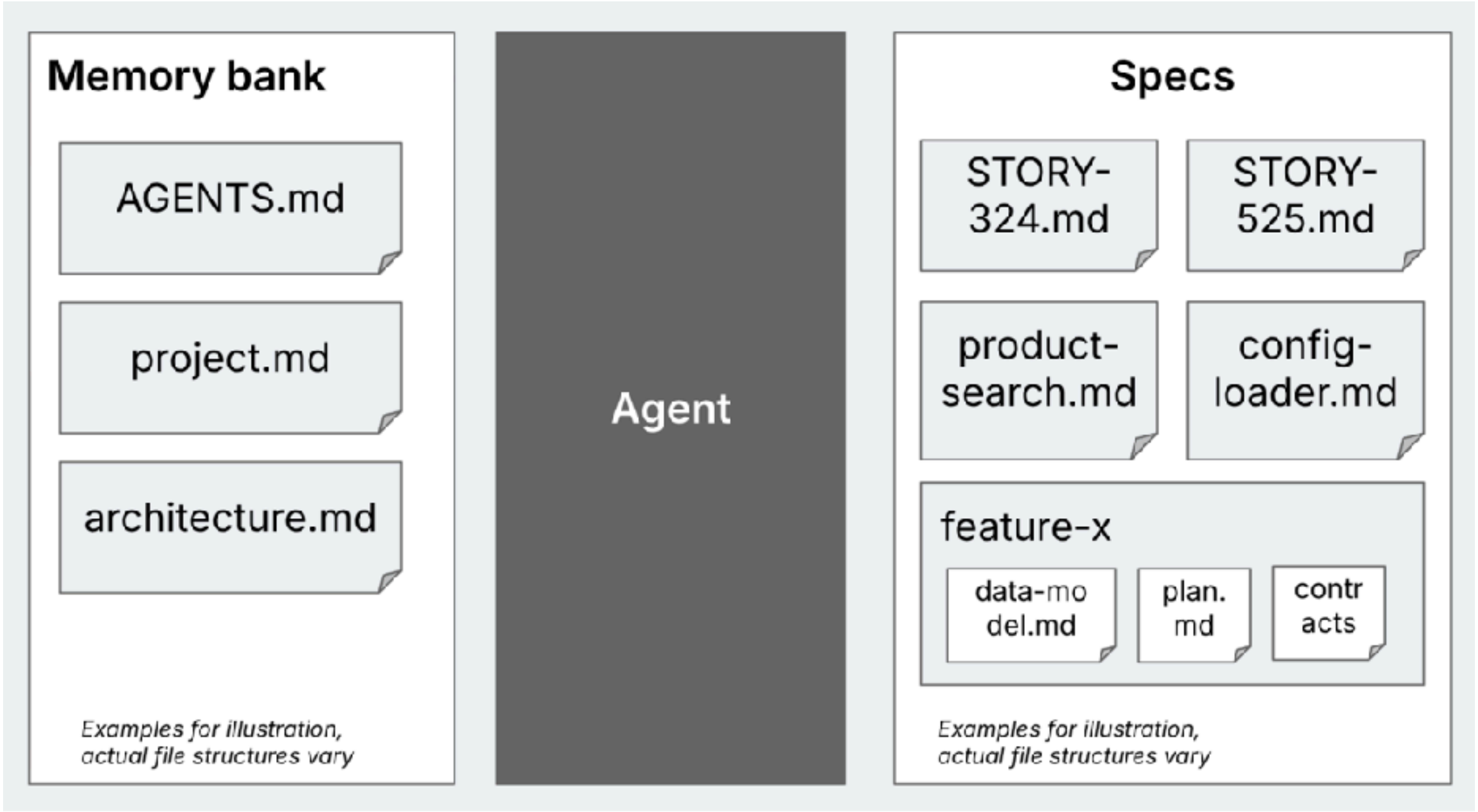
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Spec-Driven Development



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Memory Bank



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Workshop with Coding

Chat

Text Editor with AI

Pair programming with AI

SDD and Rules

BMAD-METHOD

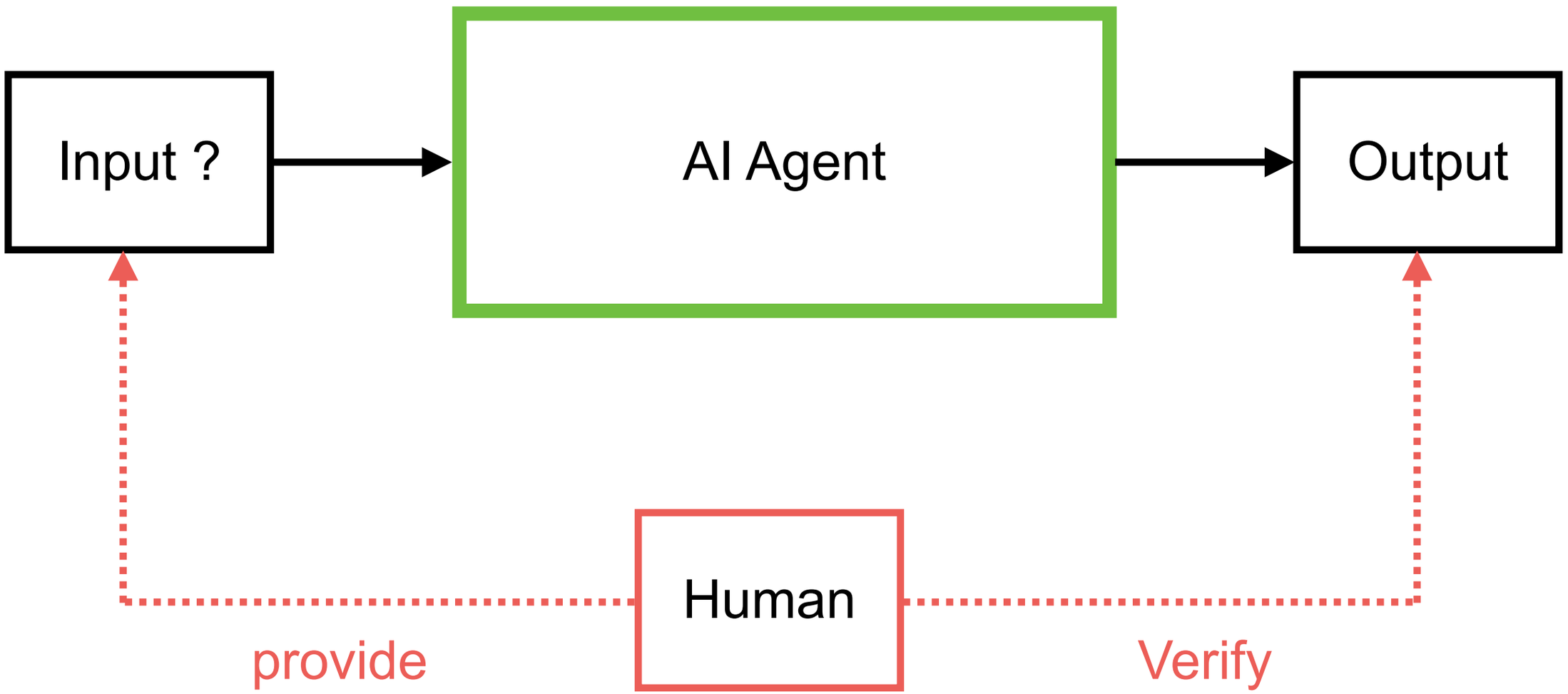
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AI Agent for Coding

Claude Code
Github Copilot Chat
Cline
Qwen code
Gemini CLI
Pi coding agent

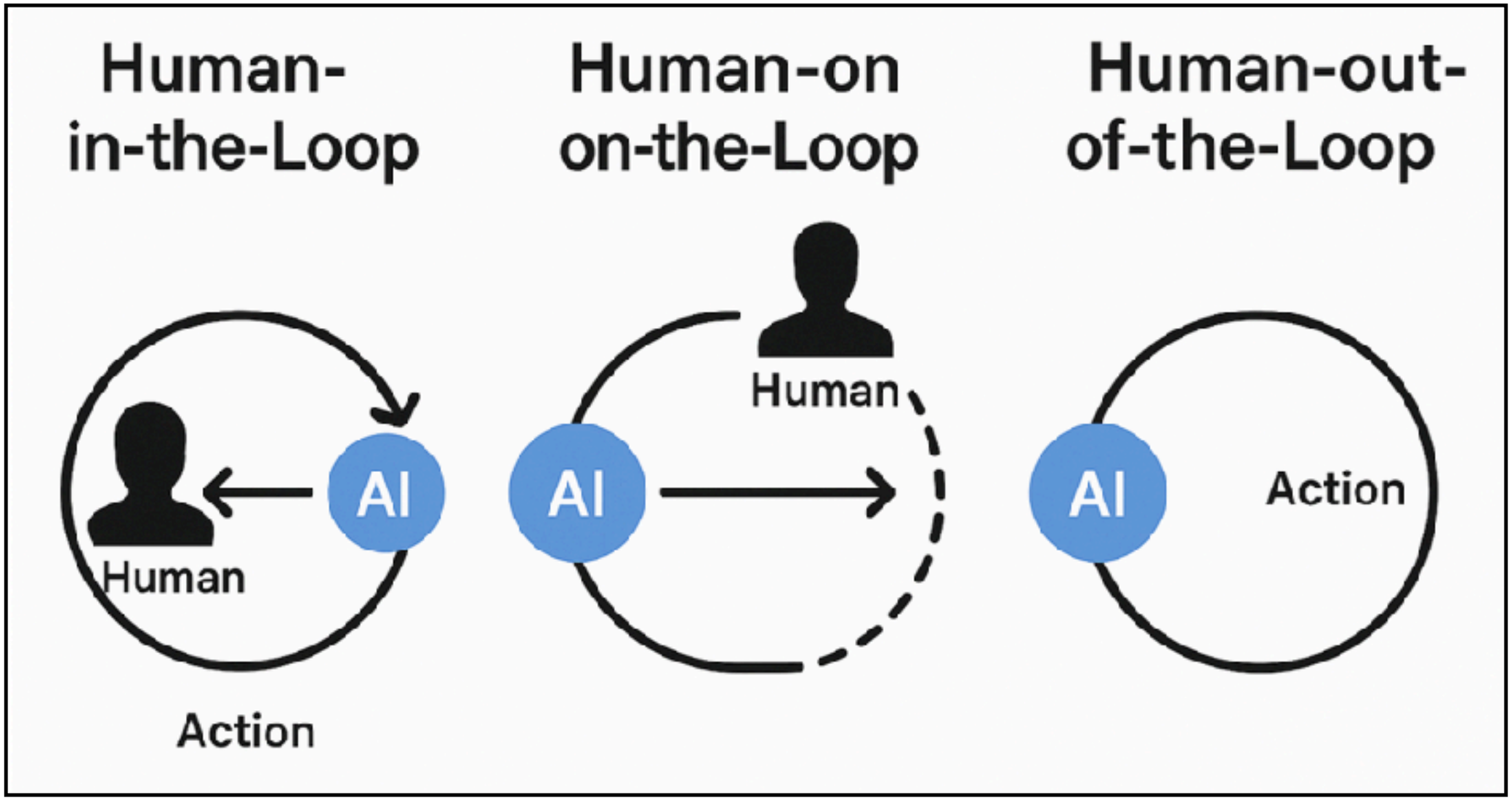
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Coding with AI Agent



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AI Agent with/without Human



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AI Agent for Coding

AI Agent

Controller loop (reasoning loop)

LLM

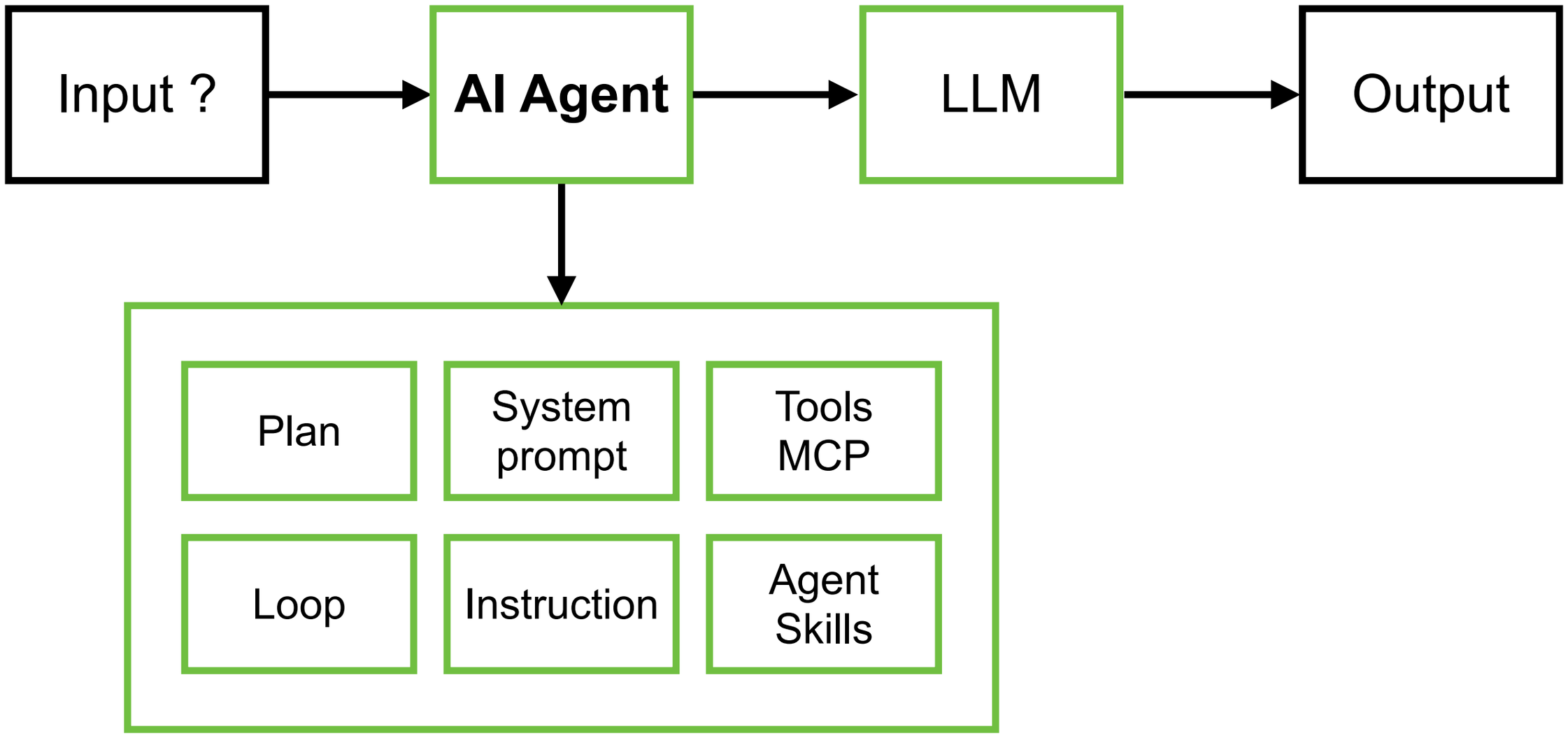
Code writing tools

Memory and Context

Other tools

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Working with AI Agent



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Testing Process
with High quality process

Functional

Non-Functional

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Testing



Test cases writing
Test code generation
Bug detection
Test planning
Data test generation

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6 Keys Software Quality

Defect density

Code duplication

Hardcode token/key

Security vulnerabilities

Outdated package

Non-permissive opensource libraries



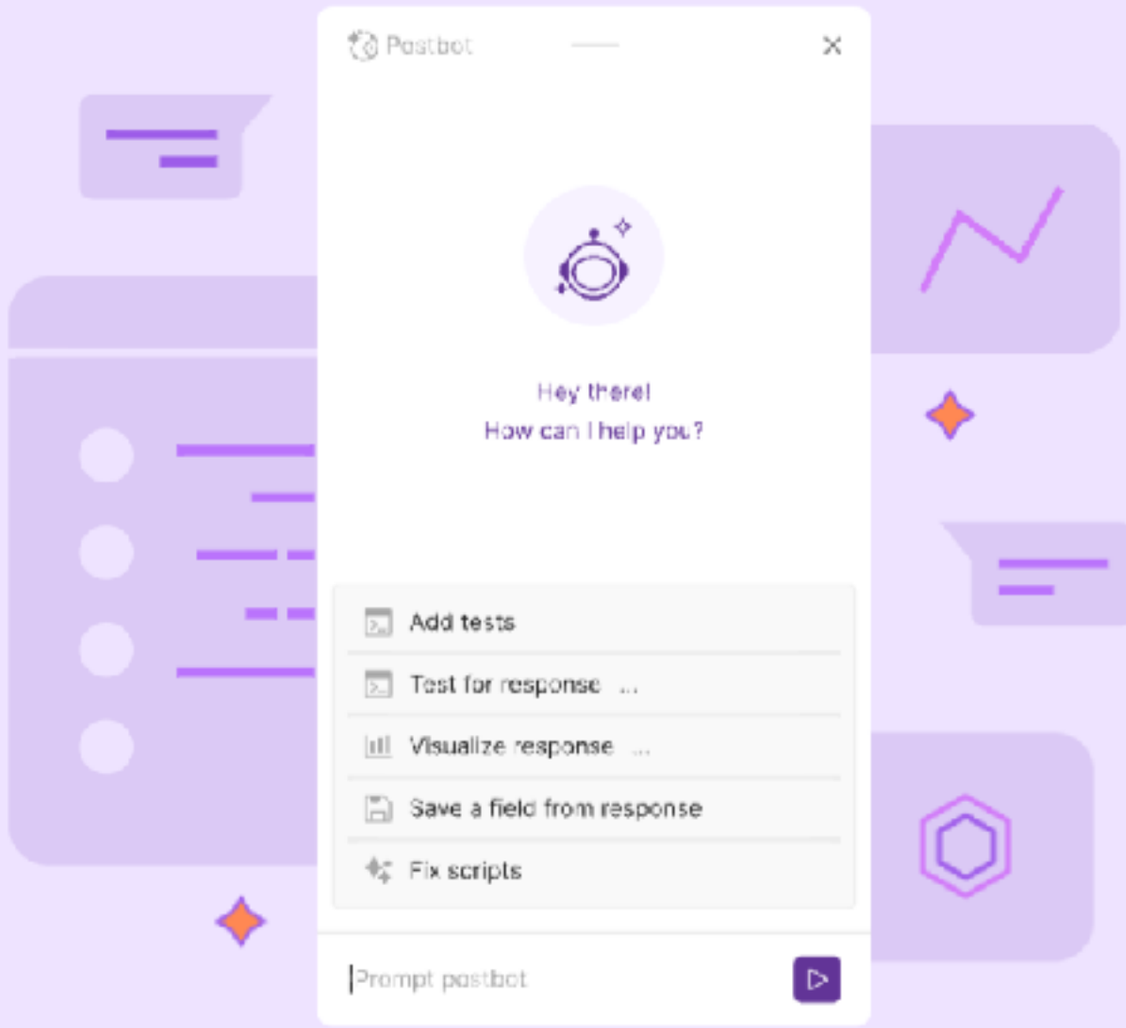
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Testing with Postman

Postbot, our AI-powered assistant, will supercharge your API development.

Speed up your most common API development workflows with natural-language input, conversational interactions, and contextual suggestions.

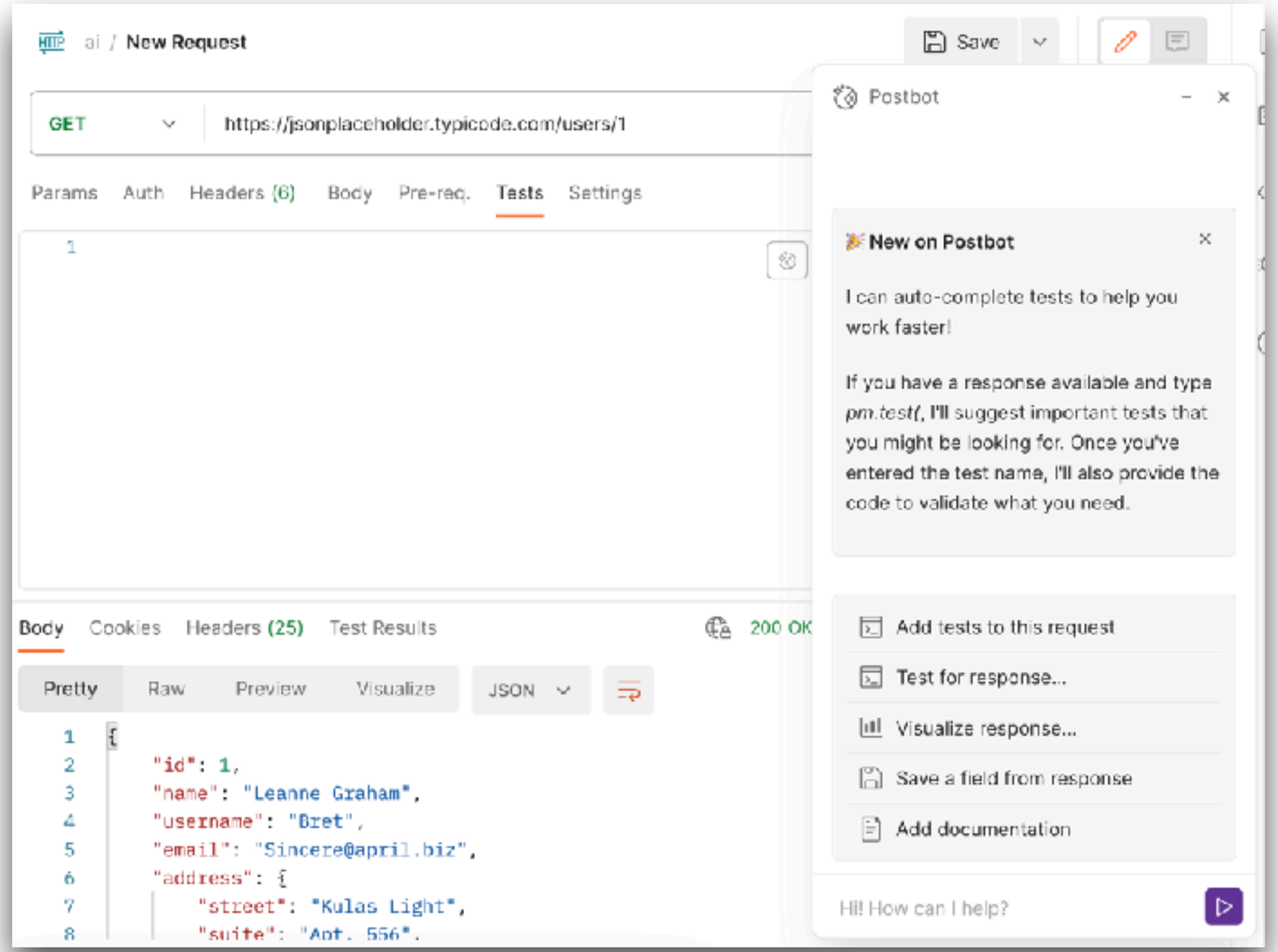
Get Started



https://www.postman.com/product/postbot/

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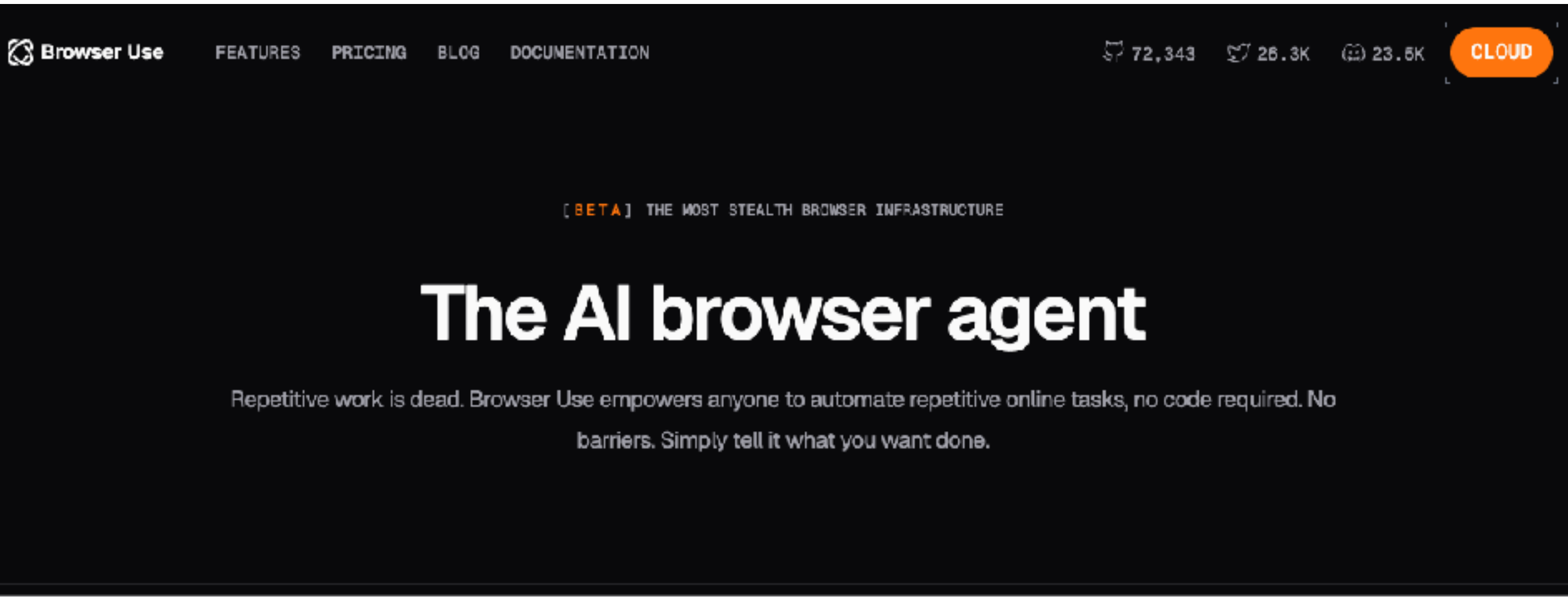
PostBot



https://www.postman.com/product/postbot/

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Browser Use



https://browser-use.com/

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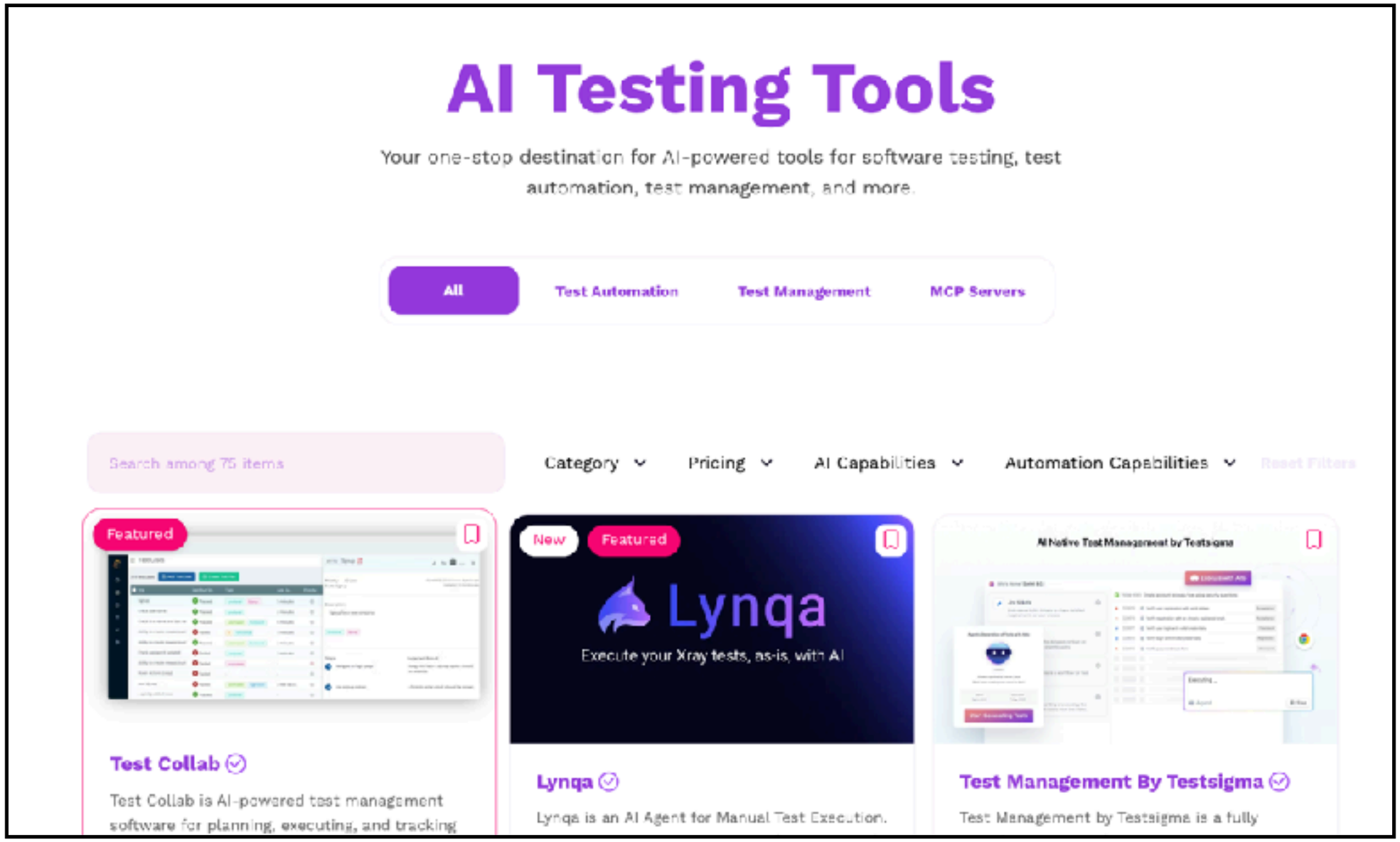
Playwright Test Agent



https://playwright.dev/docs/test-agents

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AI Testing Tools



https://www.testingtools.ai/

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Deployment Process



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Deploy and manage



CI/CD pipeline
Infrastructure as a code
Automated script
Performance and monitoring suggestion
Document generation
AI-assist support
ChatOps, AIOps

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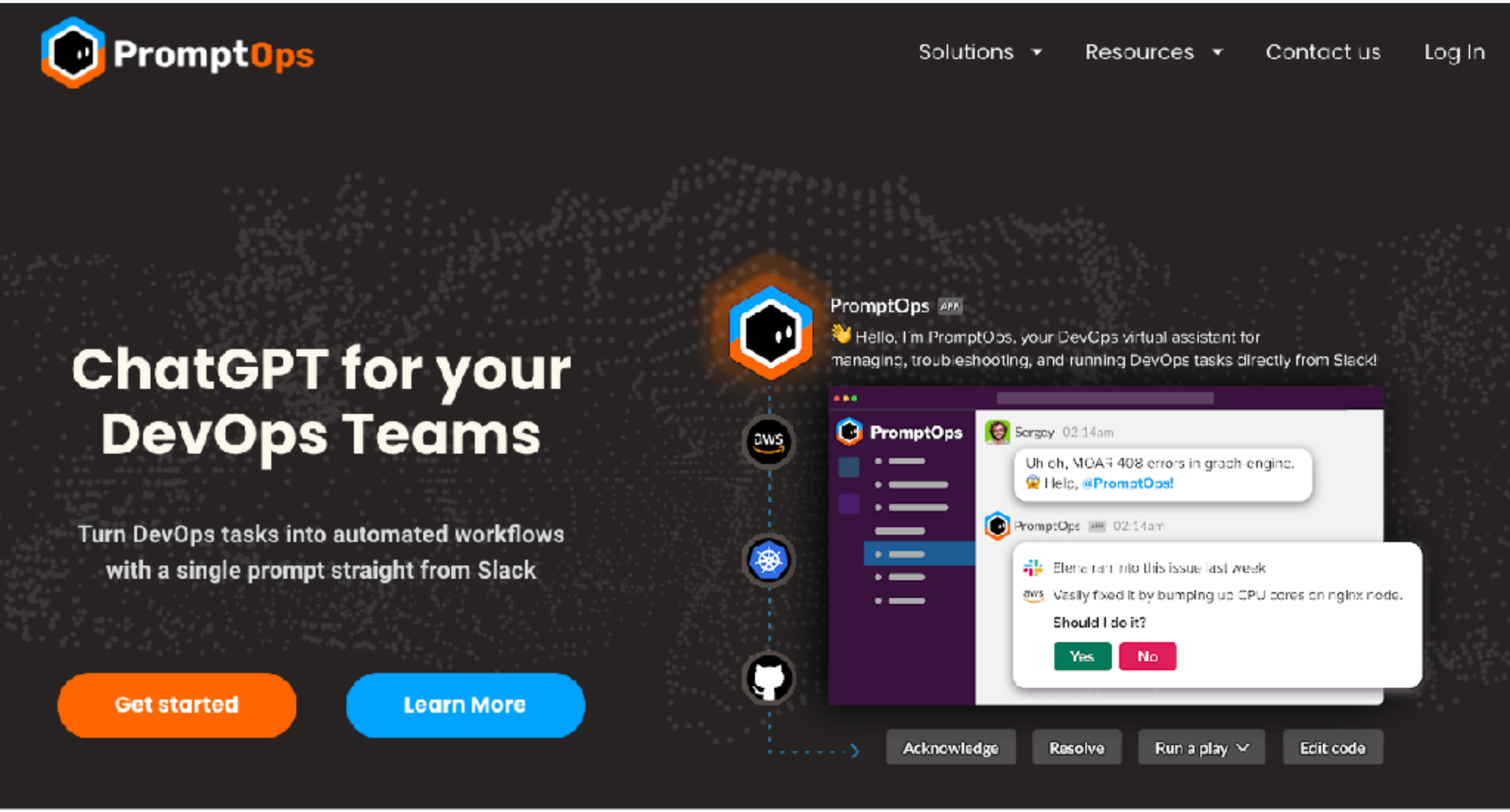
K8sGPT



https://k8sgpt.ai/

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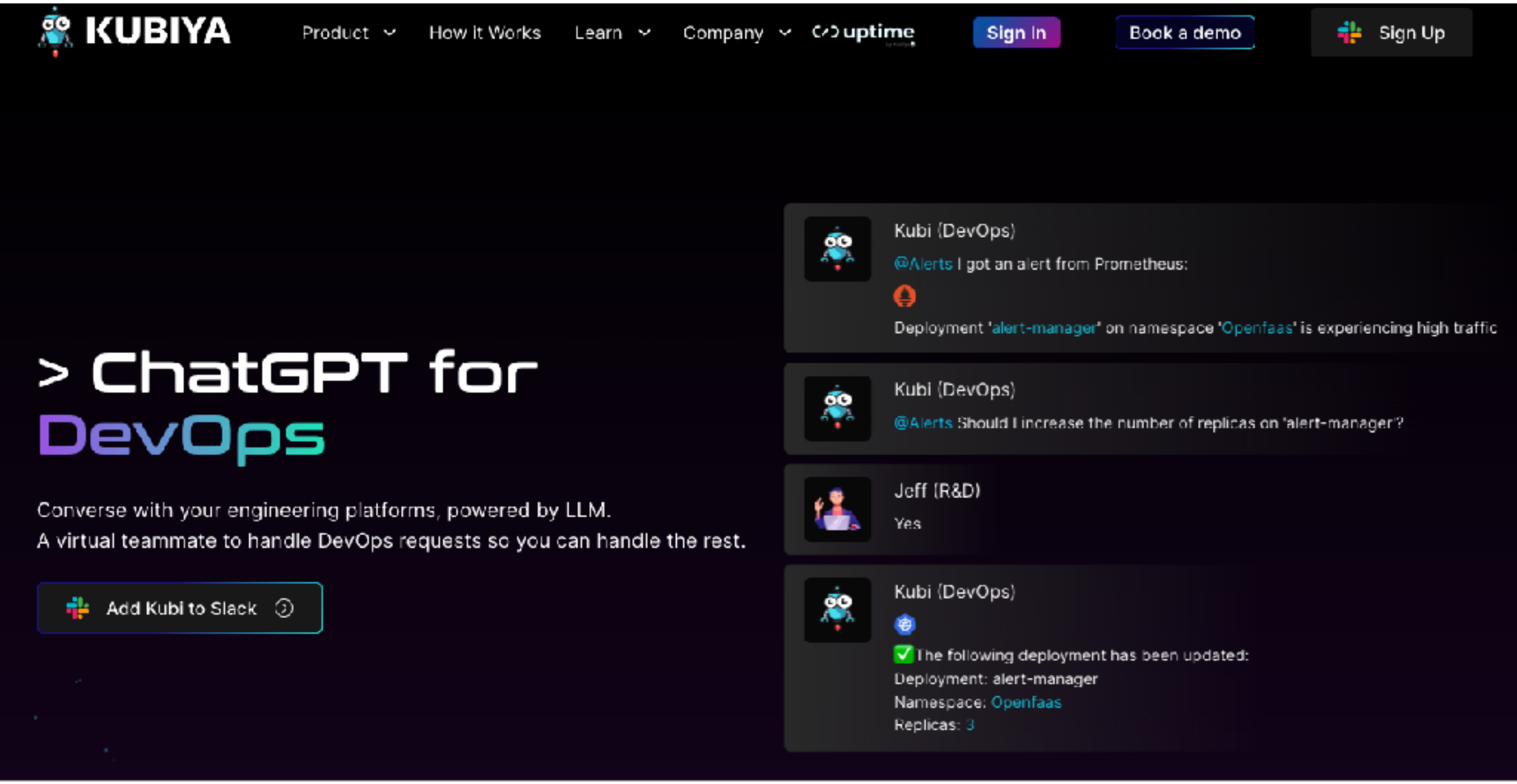
PromptOps



https://www.promptops.com/devops/

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ChatOps for DevOps



https://www.kubiya.ai/

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Risks when using Generative AI



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Risks

Quality of output generated
Explainability of decisions
Security policy !!
Sensitive data !!



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Tips

Understand what you want

Modular approach

Clear and Precise inputs

Make sure you understand the code



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Local LLM



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Local LLM

Run LLM on local machine/device
Try to customize with your requirement

Reduce cost

Data privacy

Responsive

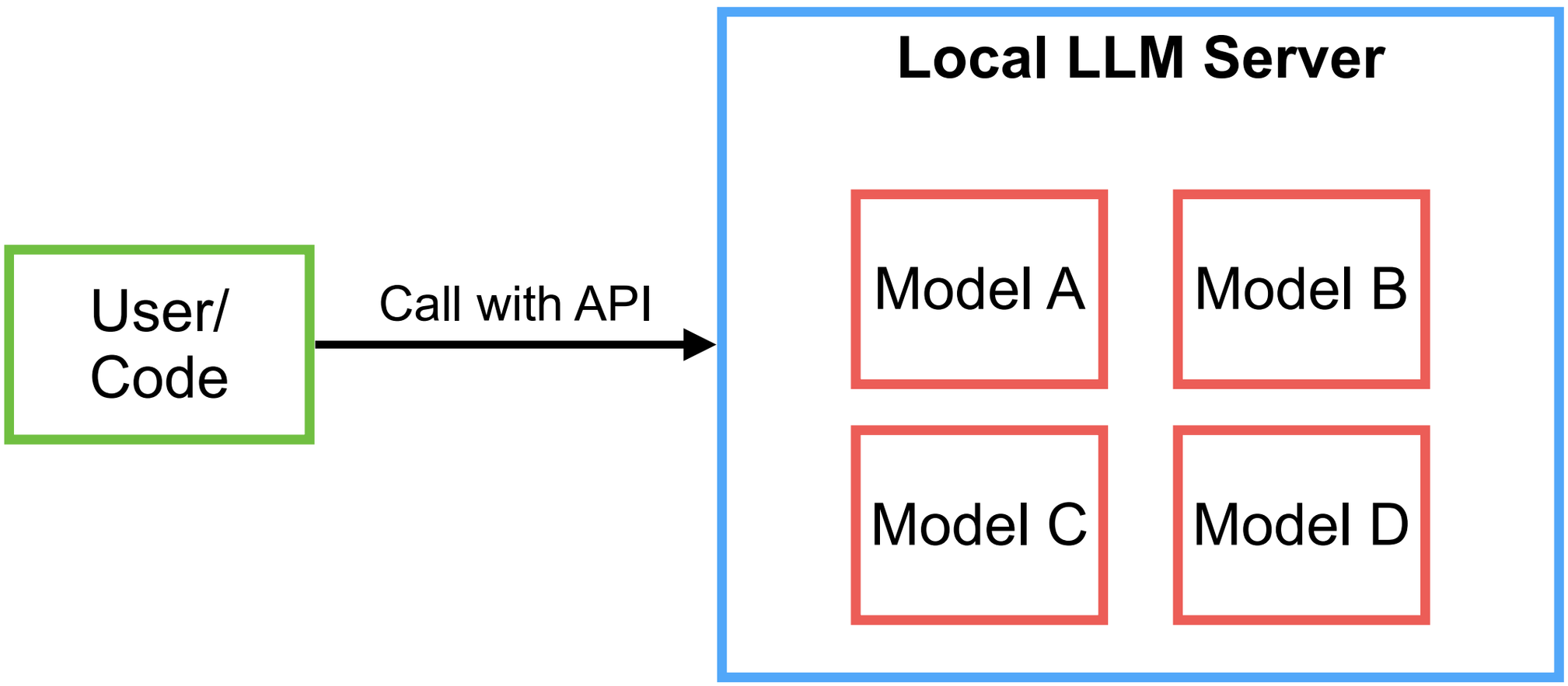
Offline mode



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Local LLM

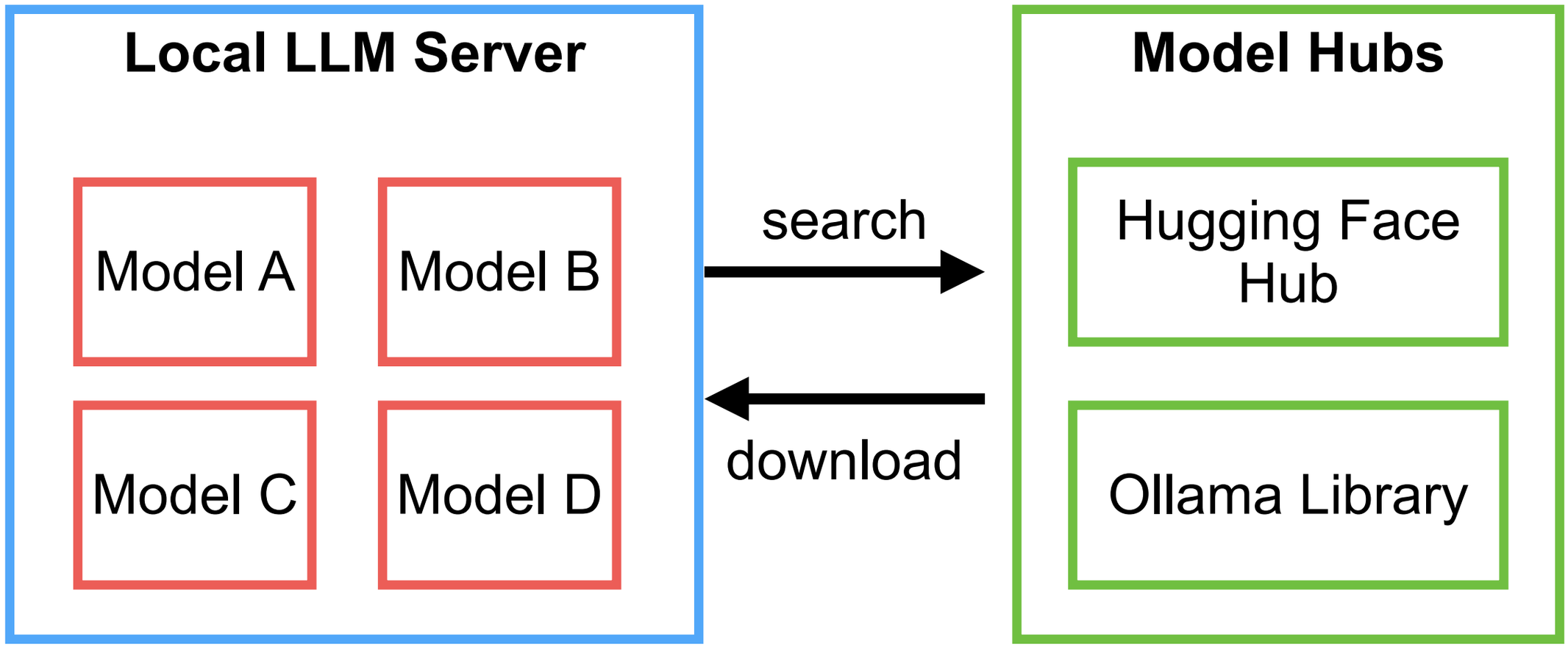
Improve your LLM models, more accurate answer



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Models ?

How to download models ?

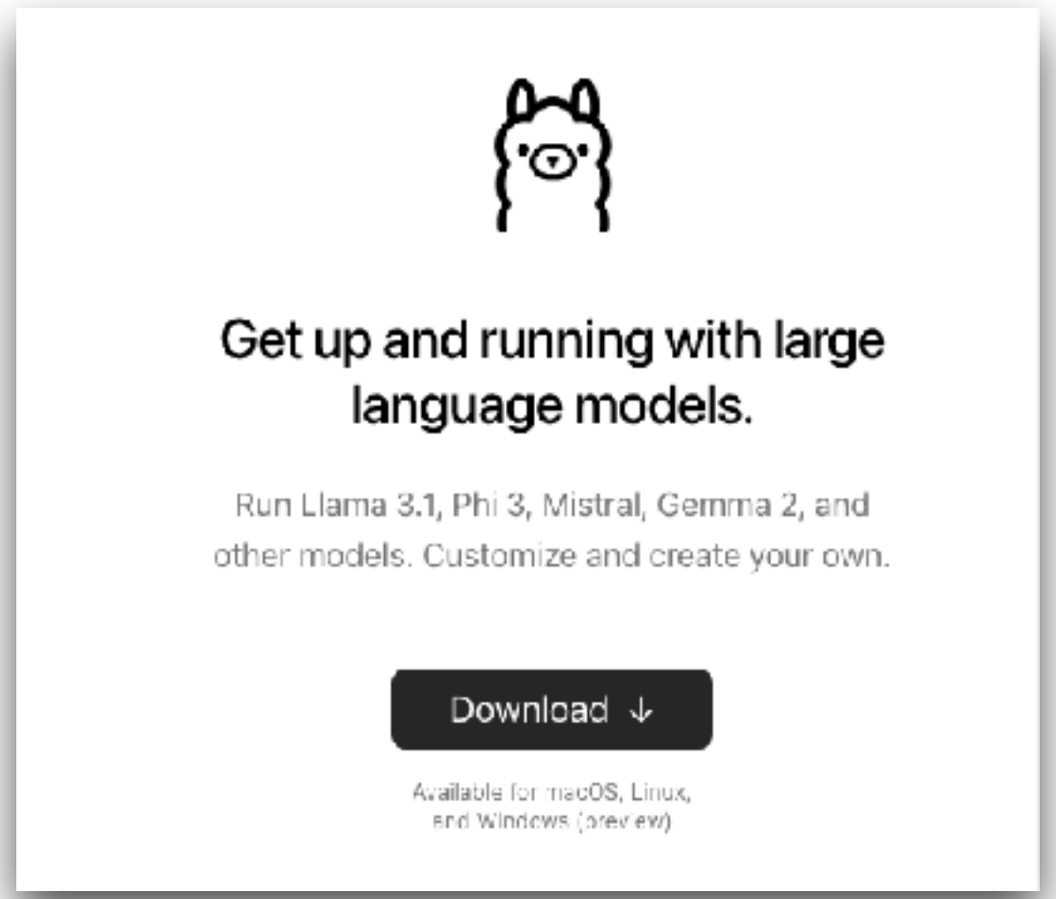


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Local LLM with Ollama

$ollama run **llama3.2**

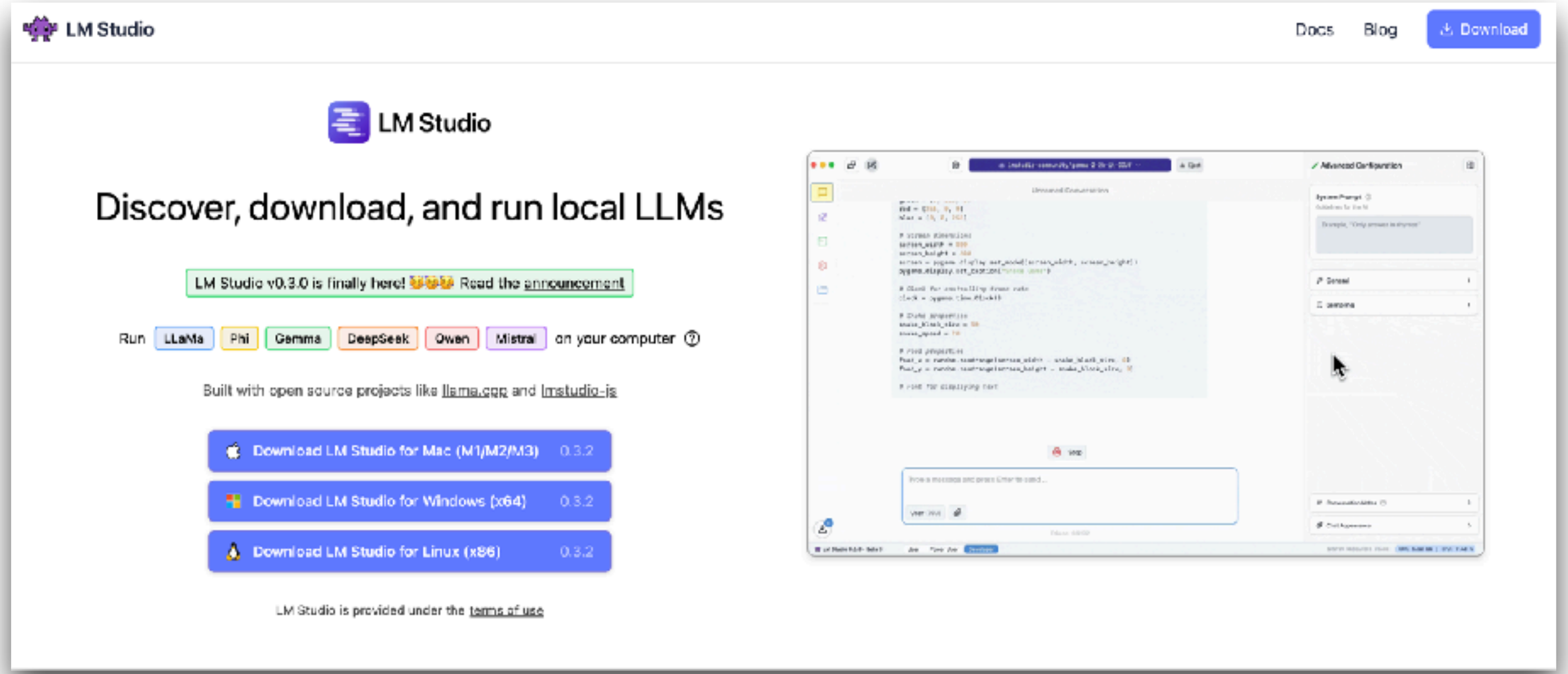
LLM model



https://ollama.com/

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Local LLM with LM Studio

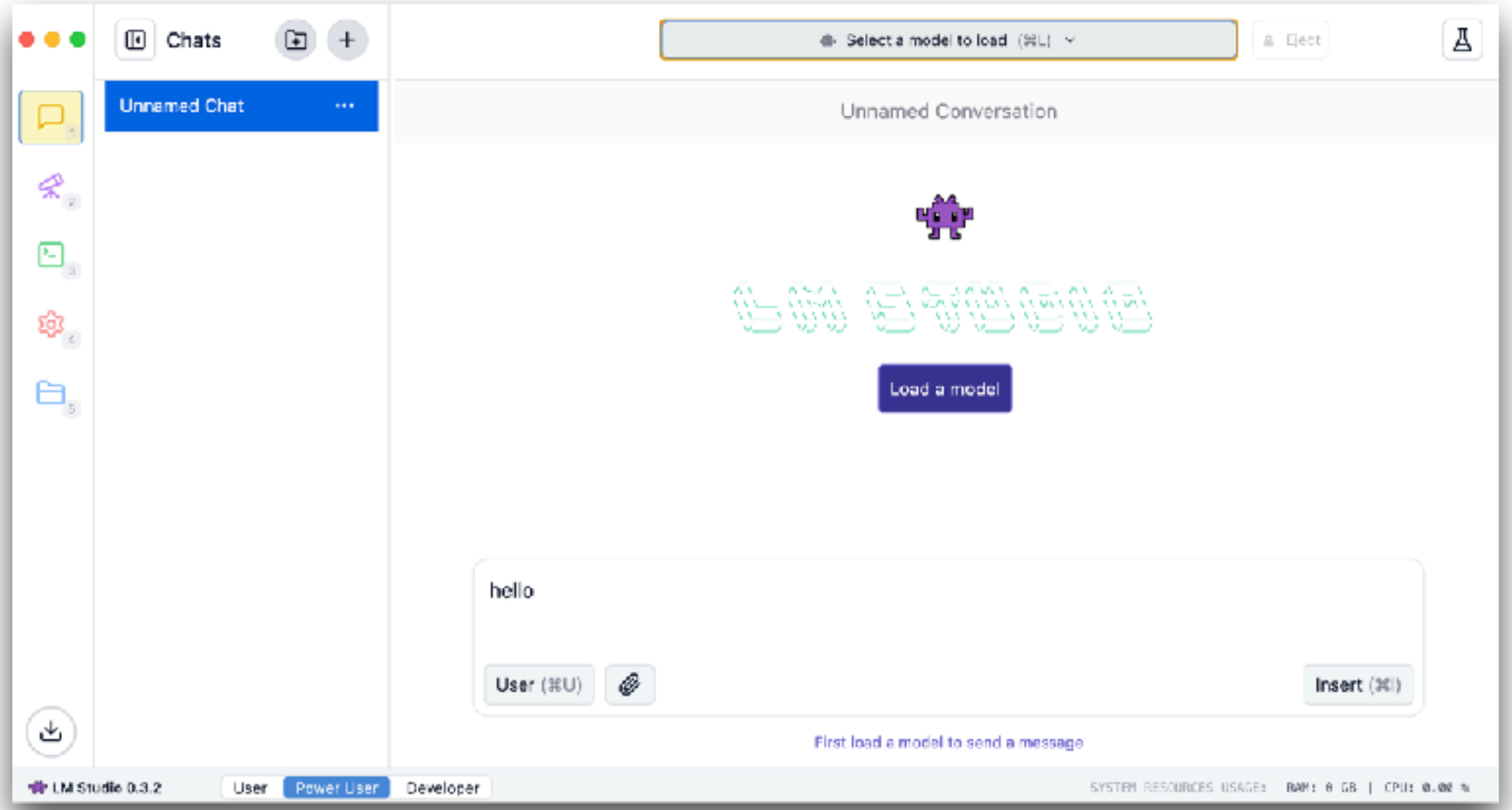


https://lmstudio.ai/

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Local LLM with LM Studio

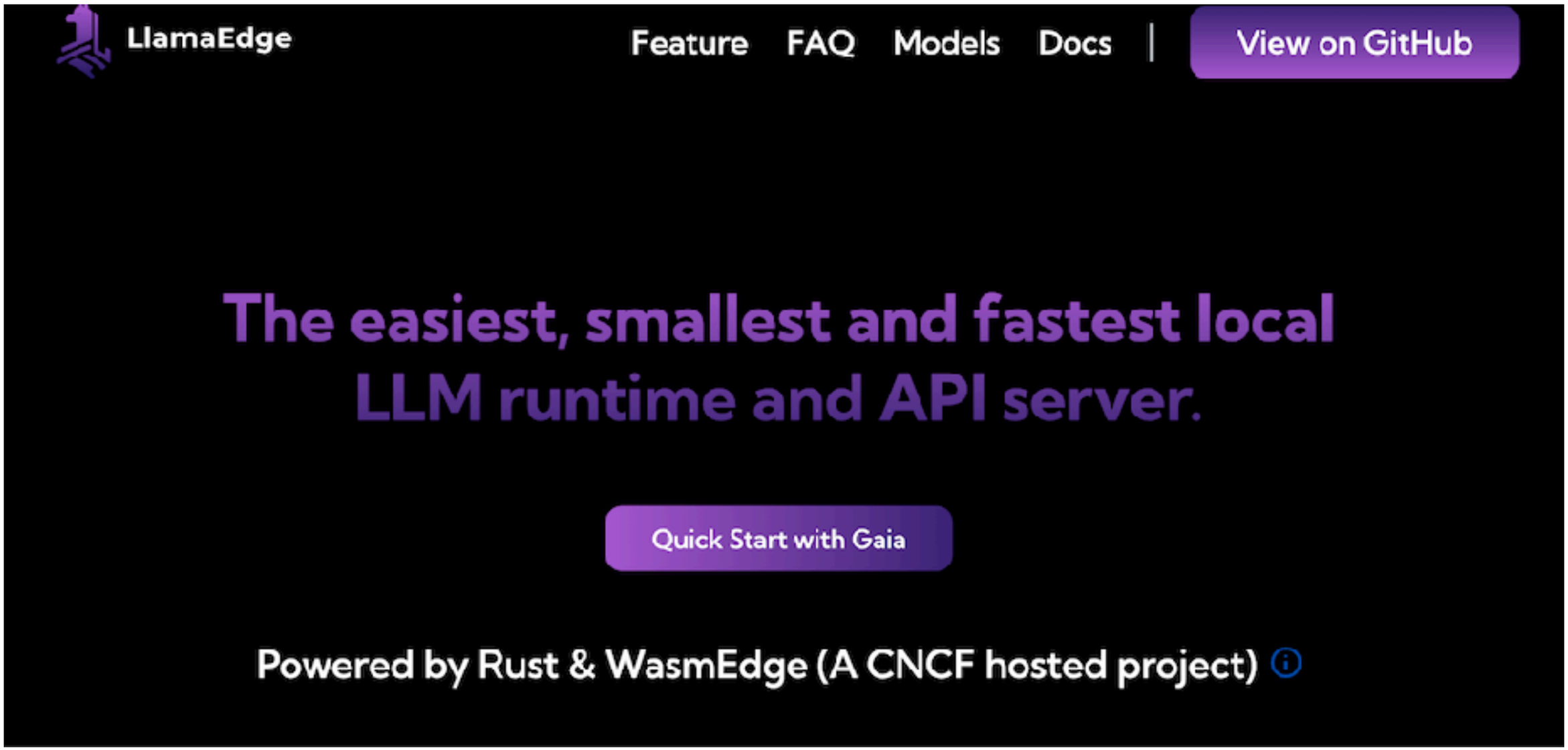
Load model from Hugging Face



https://lmstudio.ai/

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Local LLM with LlamaEdge



https://llamaedge.com/

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Local LLM with LocalAI



https://localai.io/

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More

GPT4All

LlamaFile

Jan.ai

NextChat

Anything LLM

https://github.com/Hannibal046/Awesome-LLM



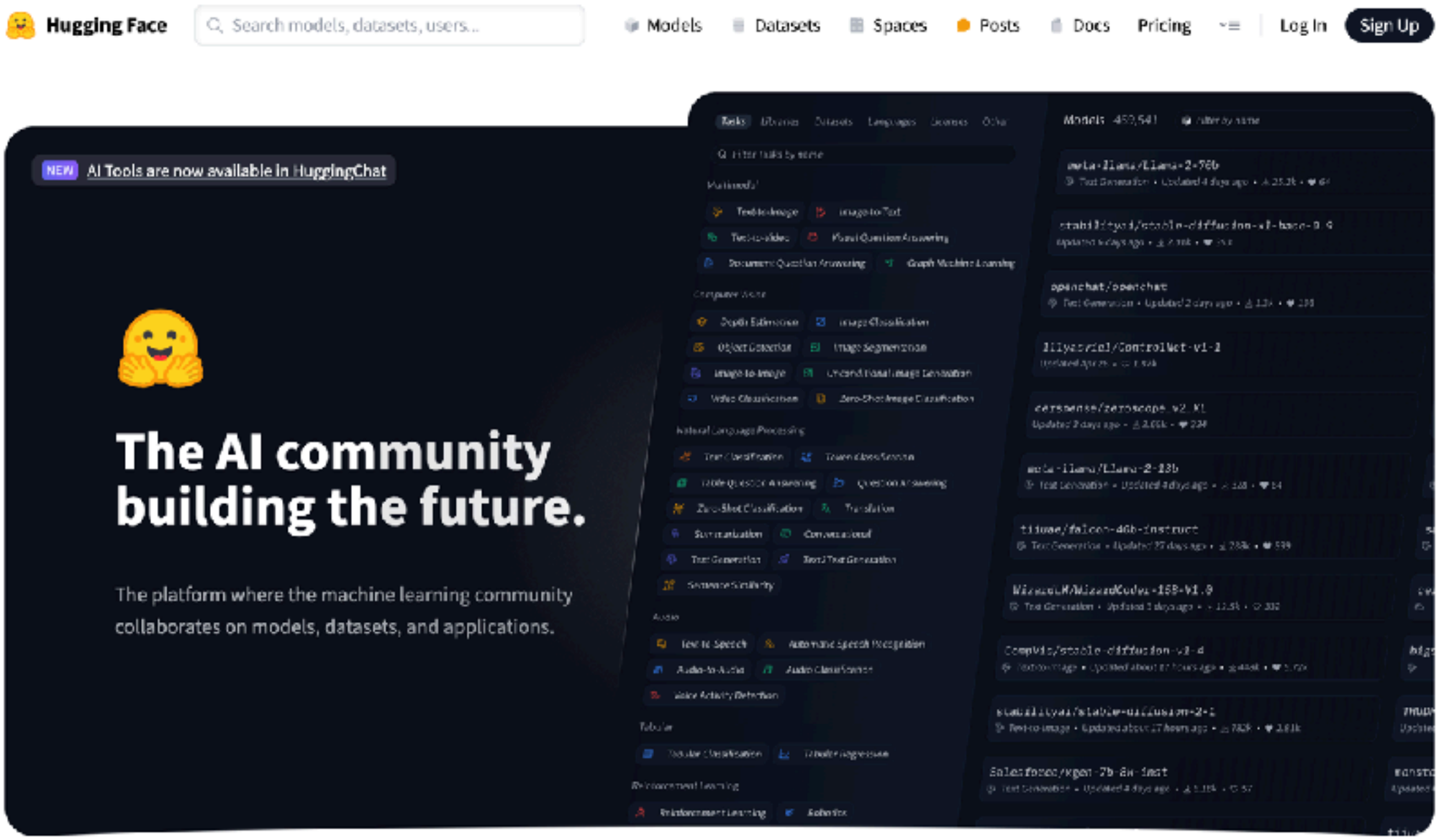
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LLM Models



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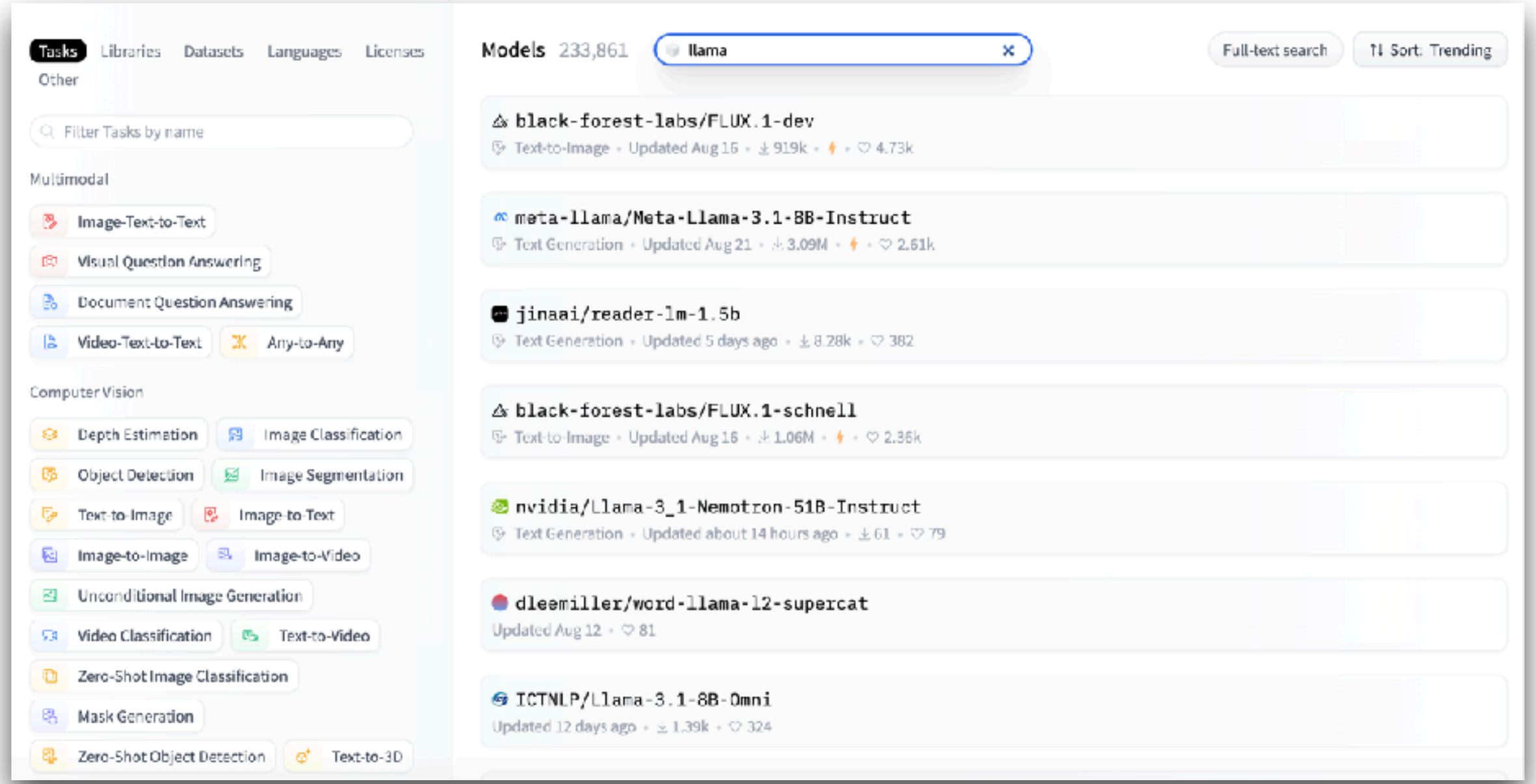
Hugging Face Model Hub



https://huggingface.co/

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Hugging Face :: Model



https://huggingface.co/

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Big Code model leader board

⭐ Big Code Models Leaderboard

Inspired from the 🤗 Open LLM Leaderboard and 🤗 Open LLM-Perf Leaderboard 🏋️, we compare performance of base multilingual code generation models on HumanEval benchmark and MultiPL-E. We also measure throughput and provide information about the models. We only compare open pre-trained multilingual code models, that people can start from as base models for their trainings.

🔍 Evaluation table | 📊 Performance Plot | 📝 About | Submit results 🚀

➡️ See All Columns ◀

🔍 Search for your model and press ENTER...

Filter model types

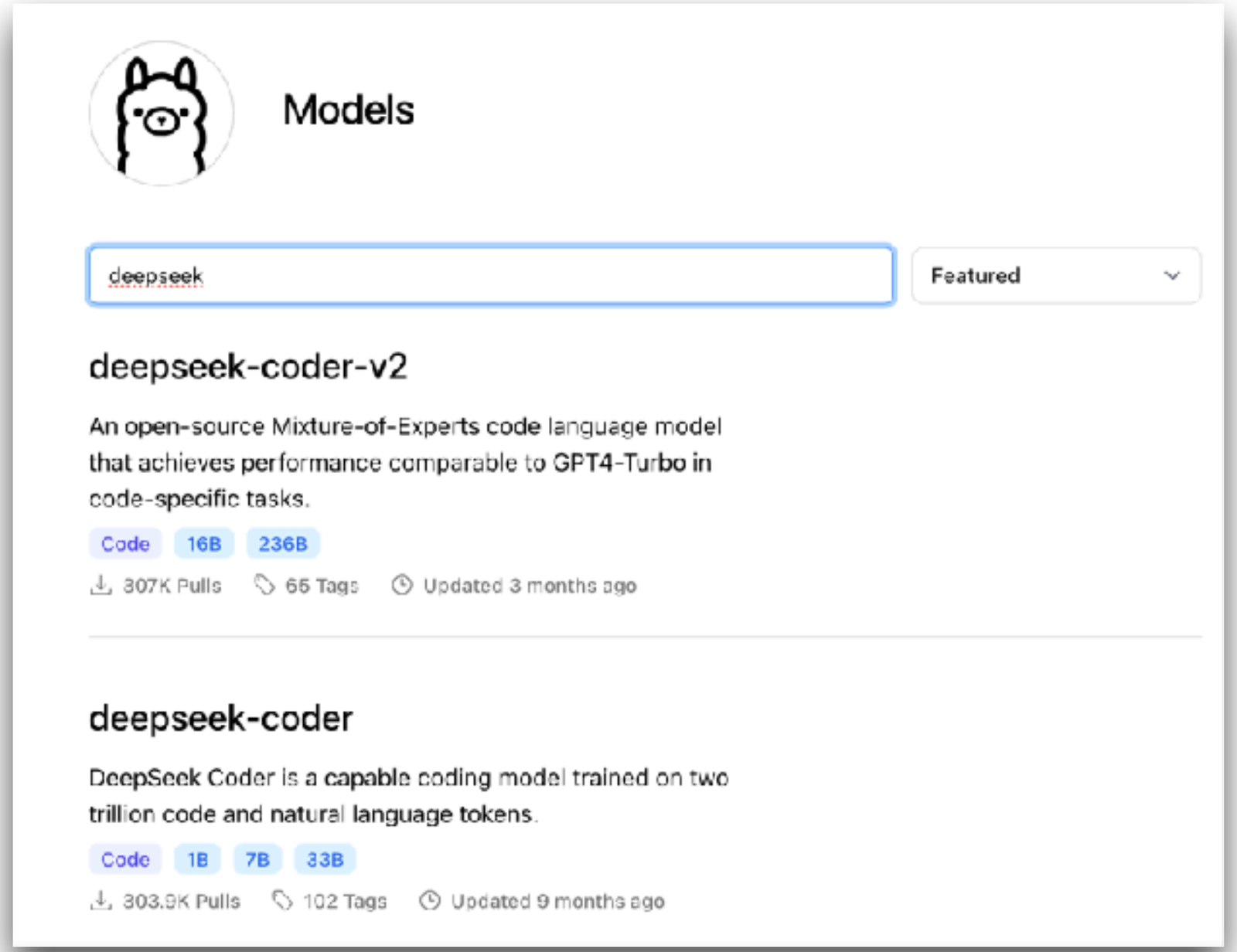
all | base | 🔶 instruction-tuned | EXT external-evaluation

T	Model	Win Rate	humaneval-python	java	javascript	cpp
🔶 EXT	OpenCodeInterpreter-DS-33B	55.83	75.23	54.8	69.06	64.47
🔶 EXT	Nxcode-CQ-7B-orpo	55.42	87.23	60.91	71.69	68.84
🔶	CodeQwen1.5-7B-Chat	55.08	87.2	61.04	70.31	67.85
🔶 EXT	CodeFuse-DeepSeek-33b	54.33	76.83	60.76	66.46	65.22
🔶 EXT	DeepSeek-Coder-33b-instruct	52	80.02	52.03	65.13	62.36
🔶 EXT	Artigenz-Coder-DS-6.7B	51.5	70.89	56.84	66.16	59.75
🔶 EXT	DeepSeek-Coder-7b-instruct	50.33	80.22	53.34	65.8	59.66
🔶 EXT	OpenCodeInterpreter-DS-6.7B	49.67	73.2	51.41	63.85	60.81

https://huggingface.co/spaces/bigcode/bigcode-models-leaderboard

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Model in Ollama

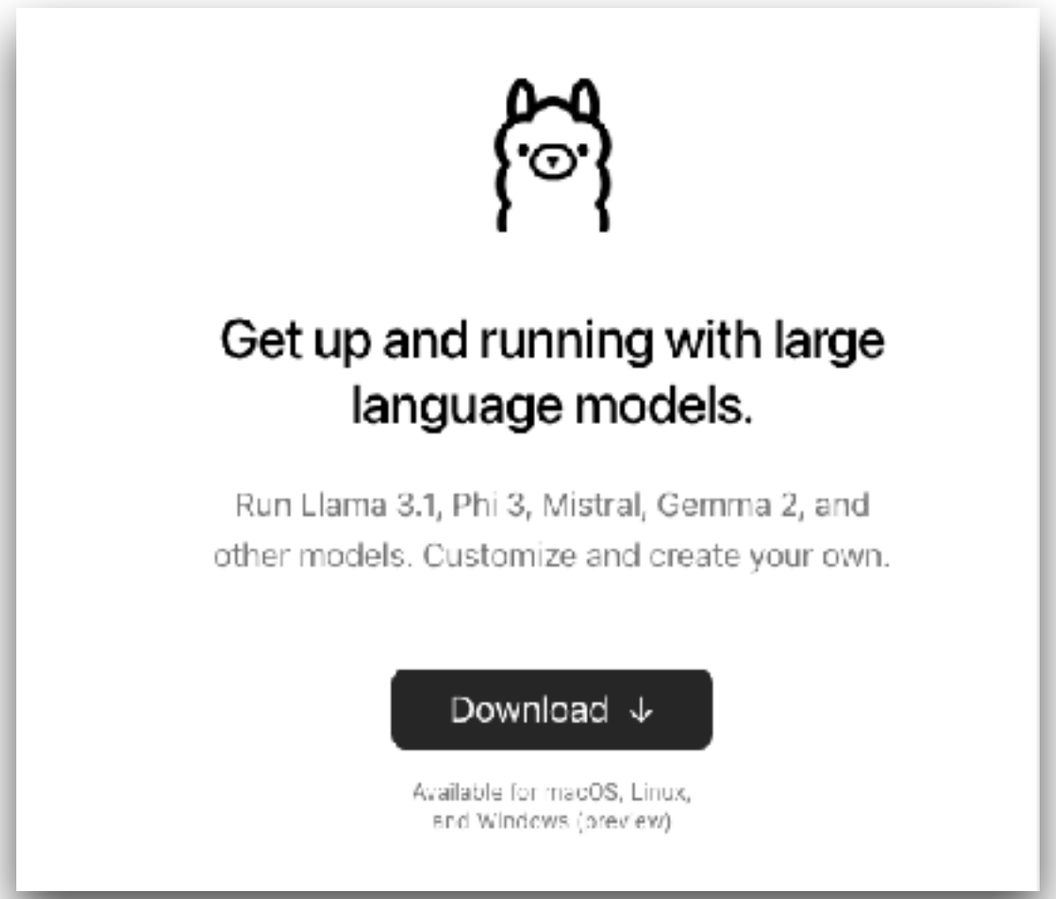


https://ollama.com/library

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Workshop with Ollama

$ollama run **llama3.1**



https://github.com/up1/workshop-ai-with-technical-team/wiki/Local-LLM-with-Ollama

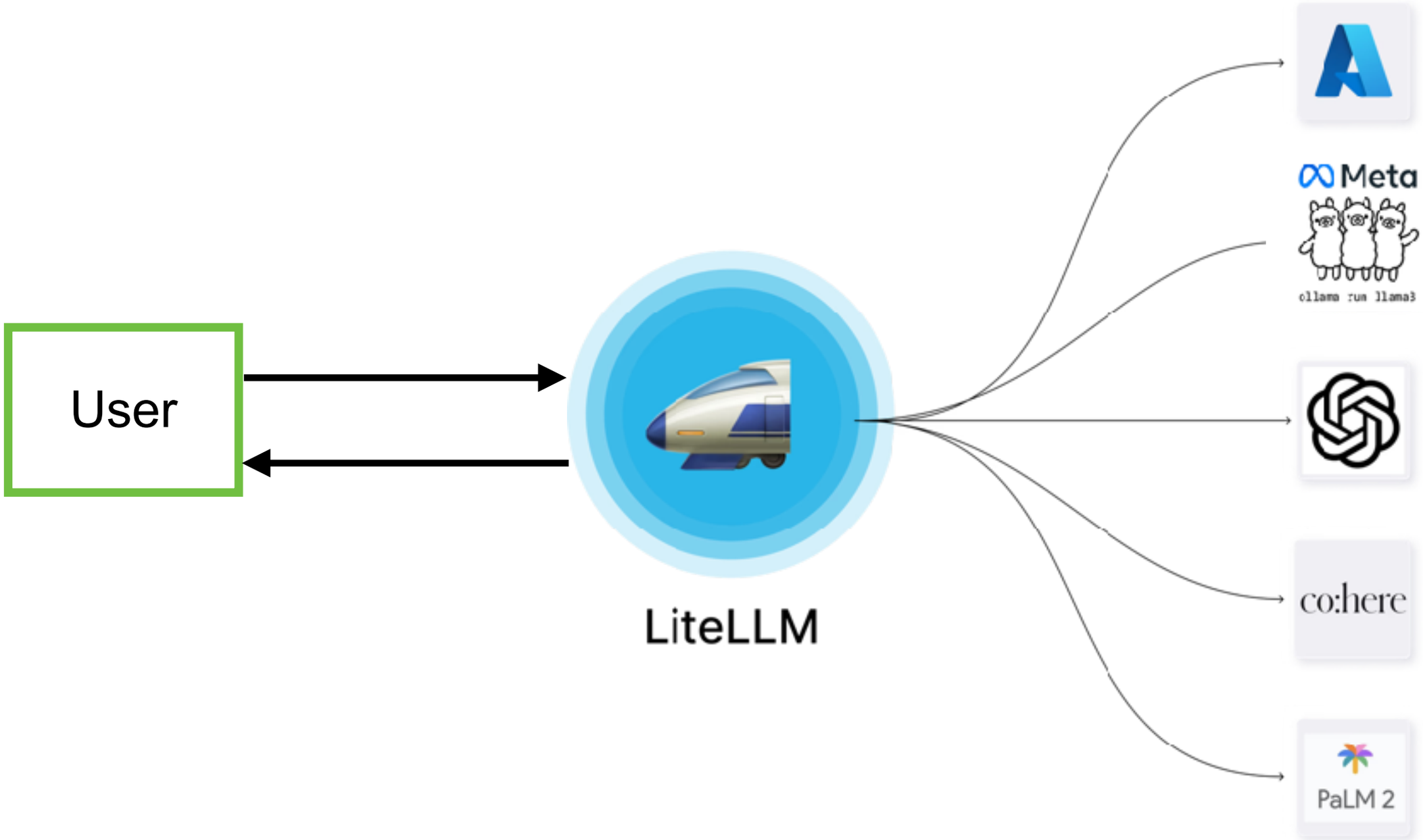
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LiteLLM as a Proxy



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LiteLLM as a Proxy

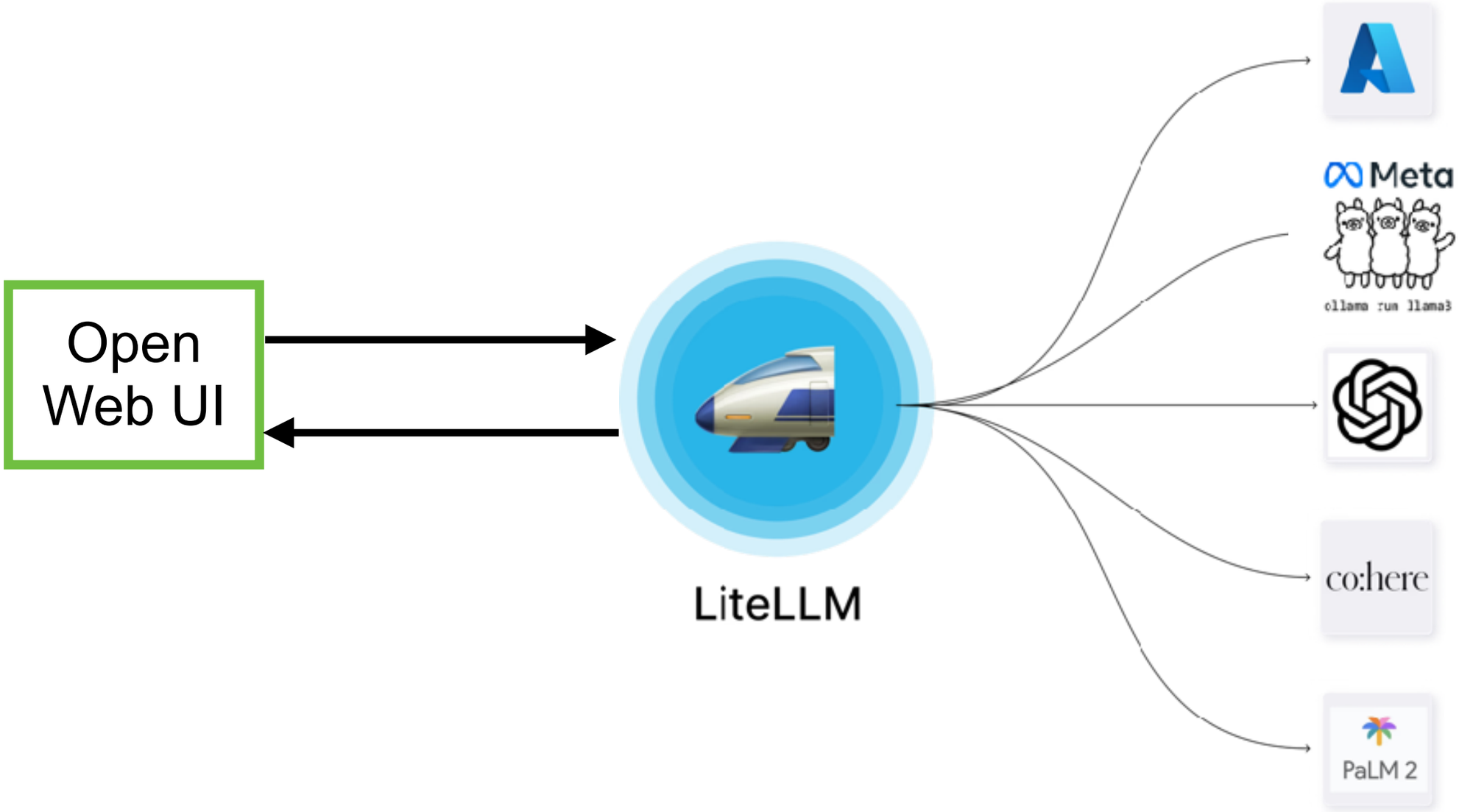


https://www.litellm.ai/

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Workshop

Use docker compose to build and run



https://github.com/up1/workshop-ai-with-technical-team/wiki/LiteLLM-and-WebUI

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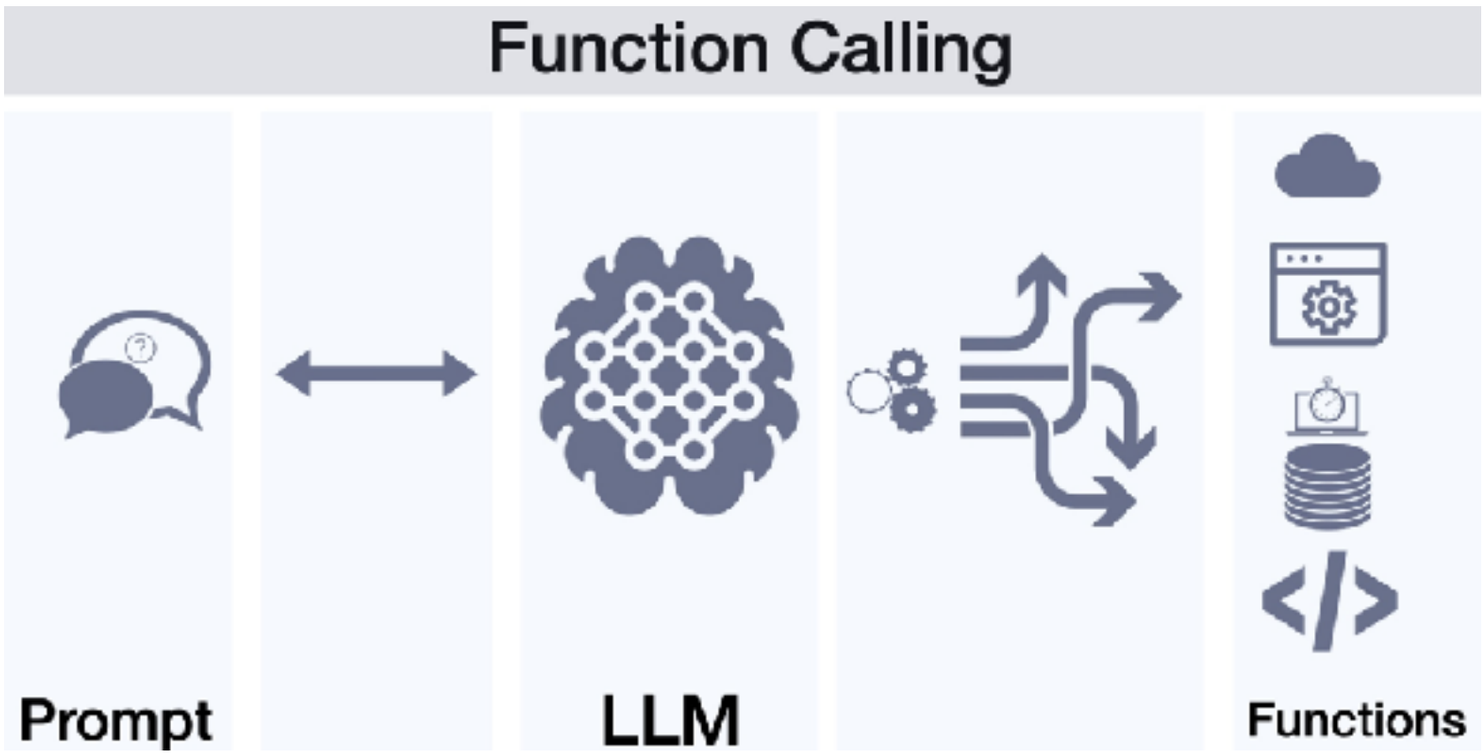
Function Calling ?



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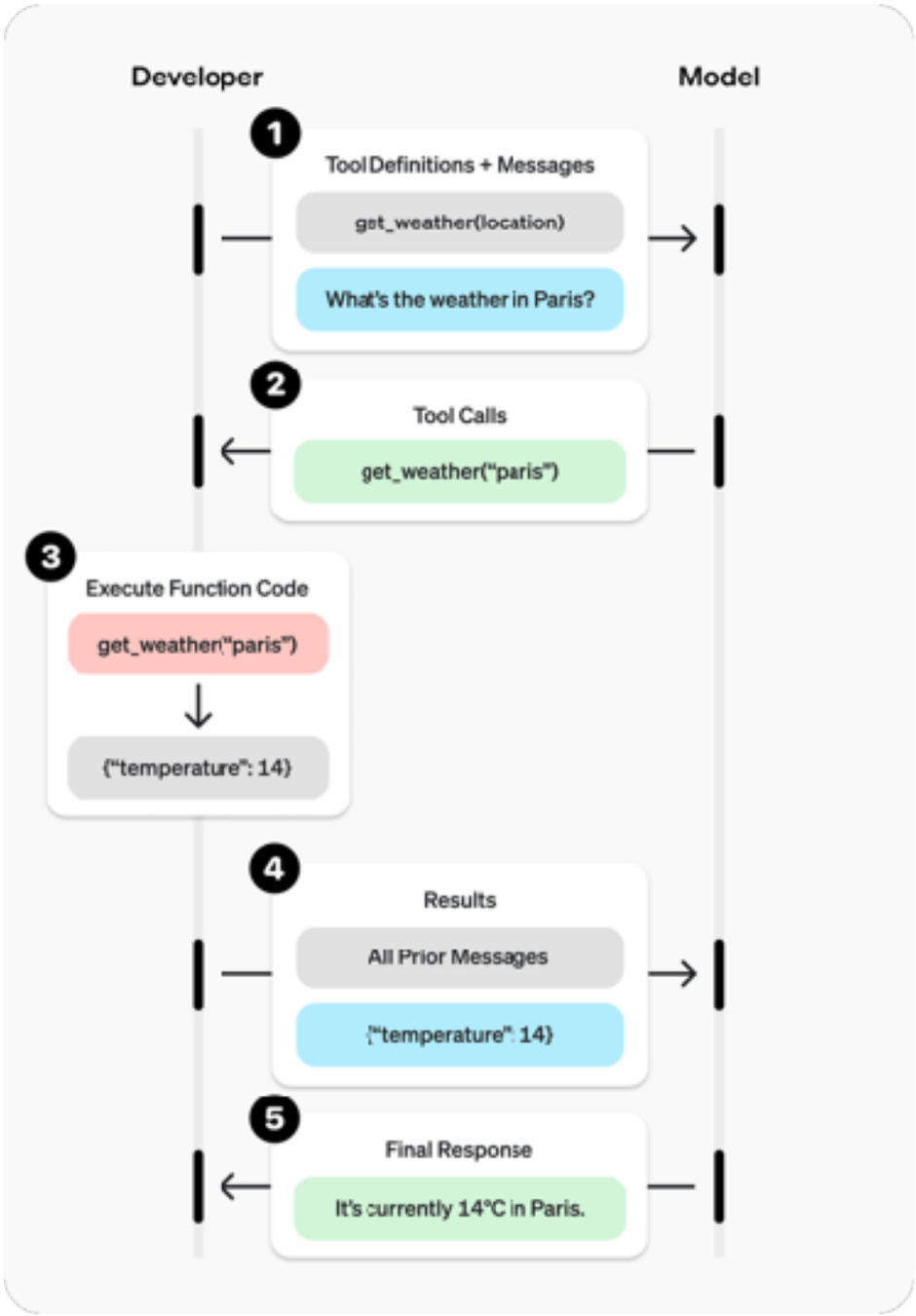
Function Calling ?

Allow LLM to recognize what tool it need based on user's input and when to invoke it.



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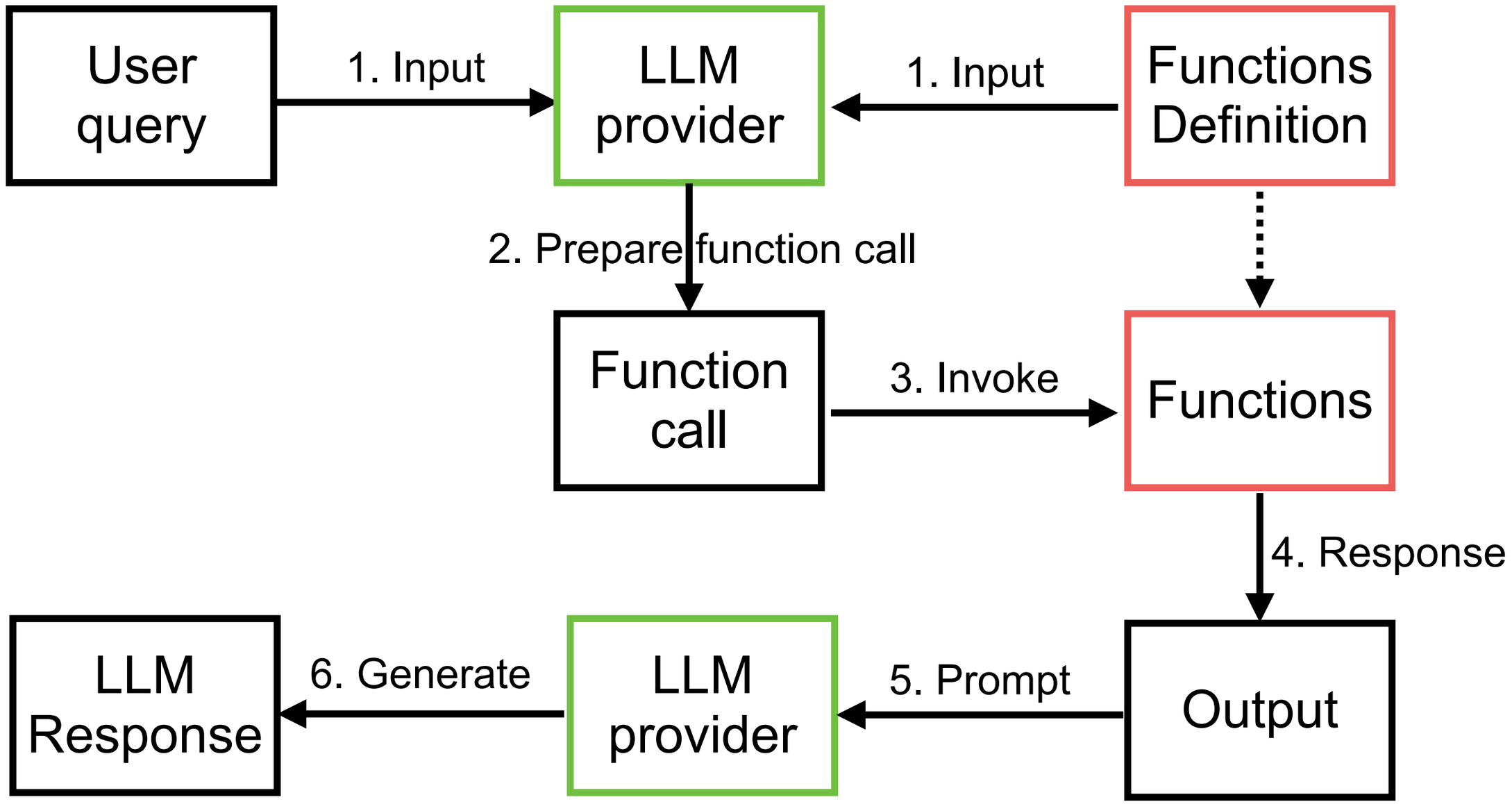
Function calling ?



https://platform.openai.com/docs/guides/function-calling?api-mode=responses

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Function Calling



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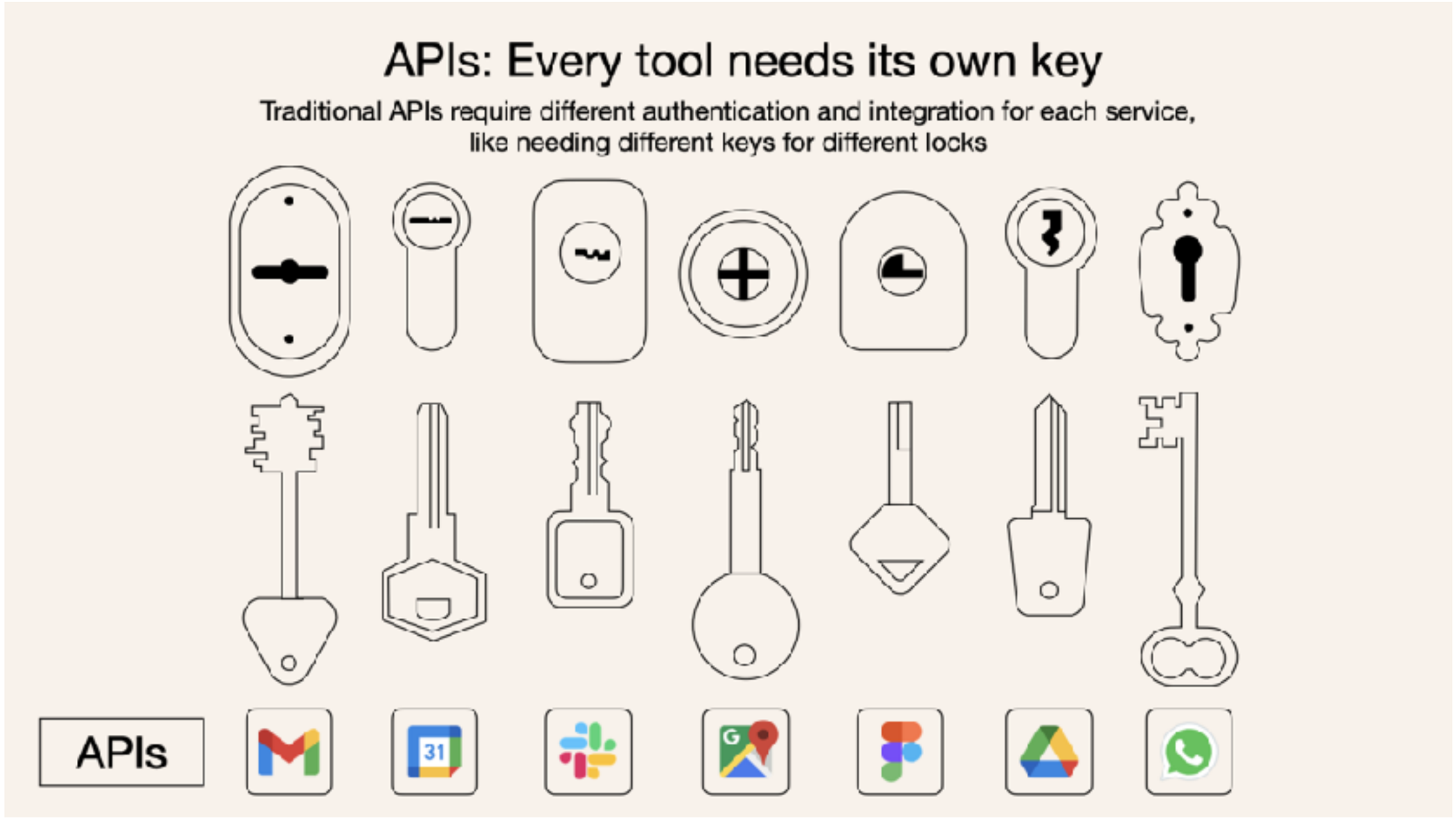
Support function calling

Provider name	Model name
OpenAI	GPT 4
Anthropic	Claude 3 (Sonnet, Haiku, Opus)
Google	Gemini



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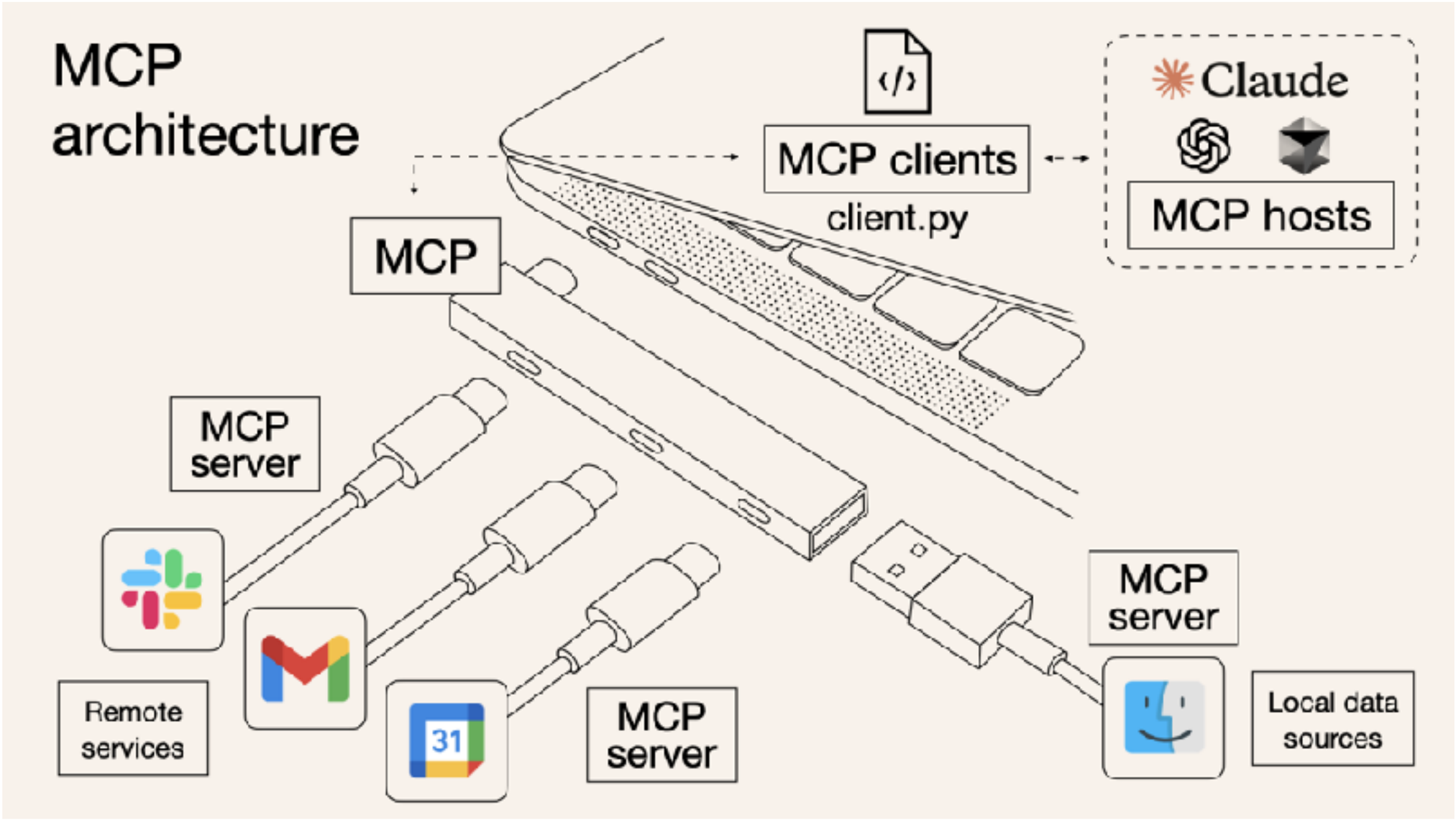
Function calling !!



https://norahsakal.com/blog/mcp-vs-api-model-context-protocol-explained/

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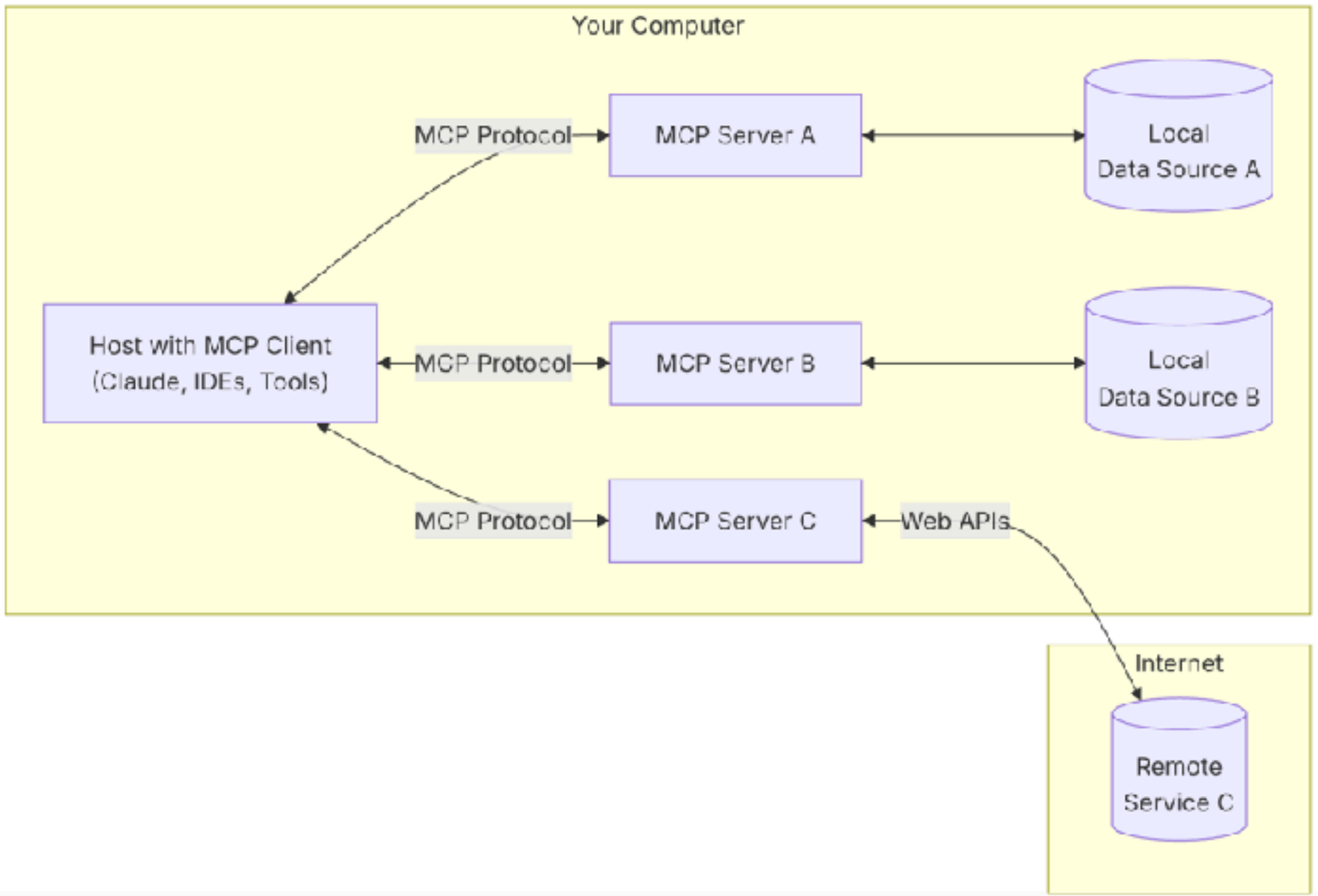
Standardized function calling !!



https://norahsakal.com/blog/mcp-vs-api-model-context-protocol-explained/

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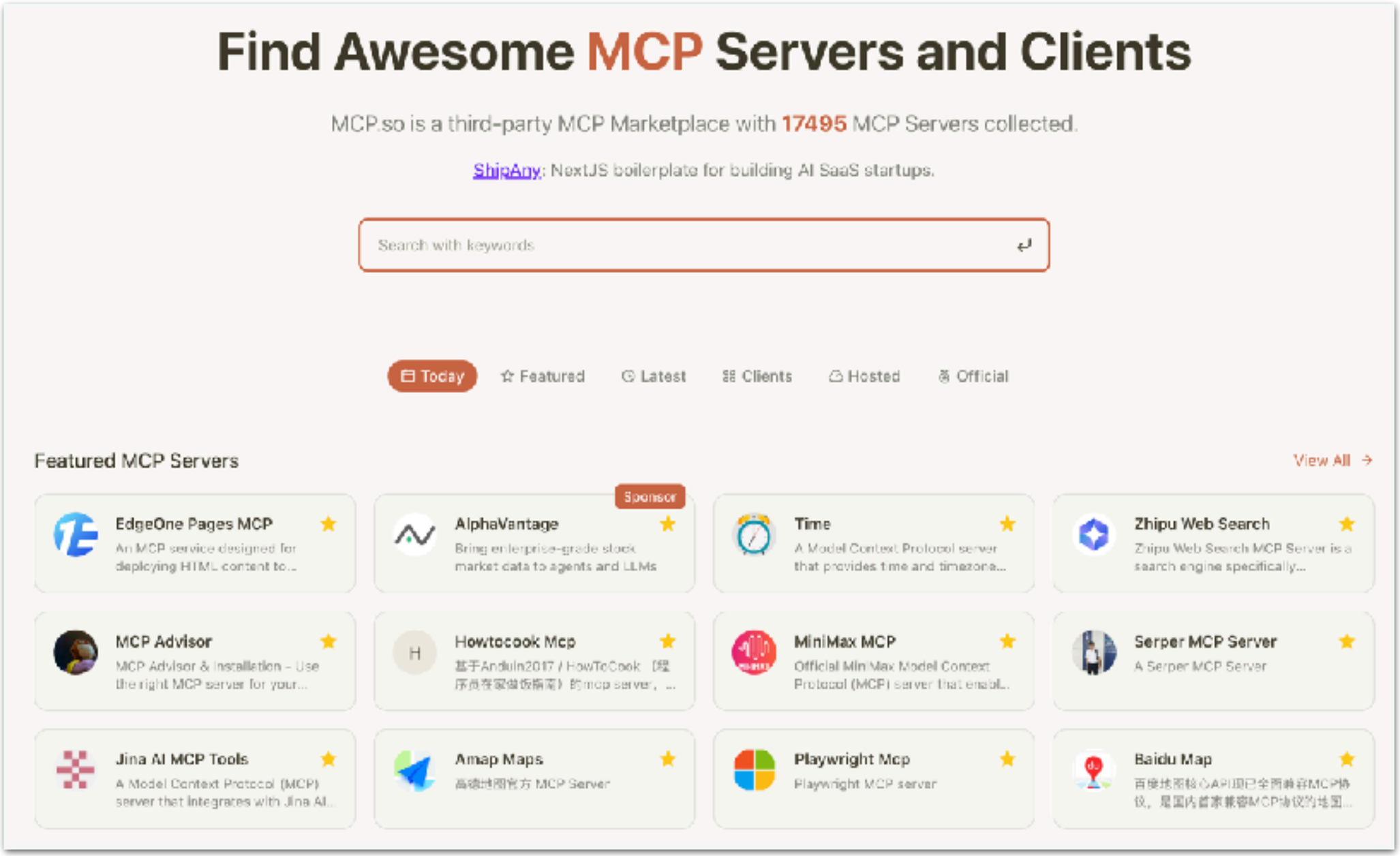
Model Context Protocol



https://modelcontextprotocol.io/

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Awesome MCP



https://mcp.so/

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Retrieval-Augmented Generation (RAG)

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RAG

Enhances LLMs by retrieving external knowledge before generating response

Improve accuracy	Reduce hallucinations	Real-time knowledge updates

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Core Components

Retriever

Fetch relevant documents from a knowledge base

Generator

Uses the retrieved information to generate a response

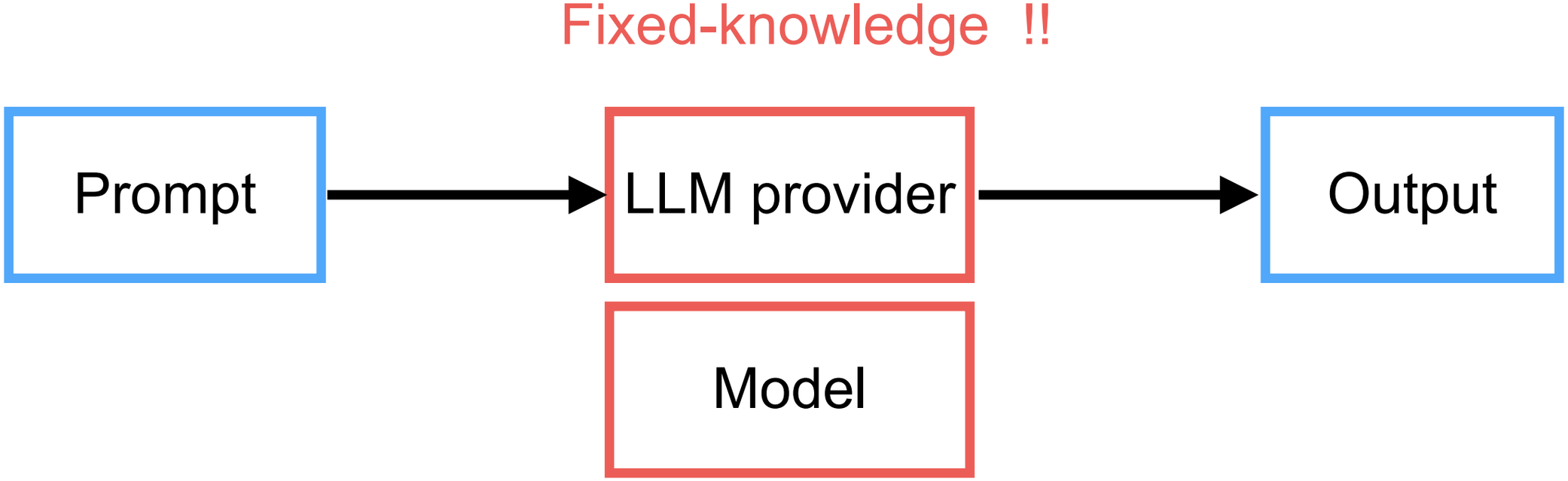
Enhancements

Different RAG architectures modify how retrieval and generation interact to improve performance



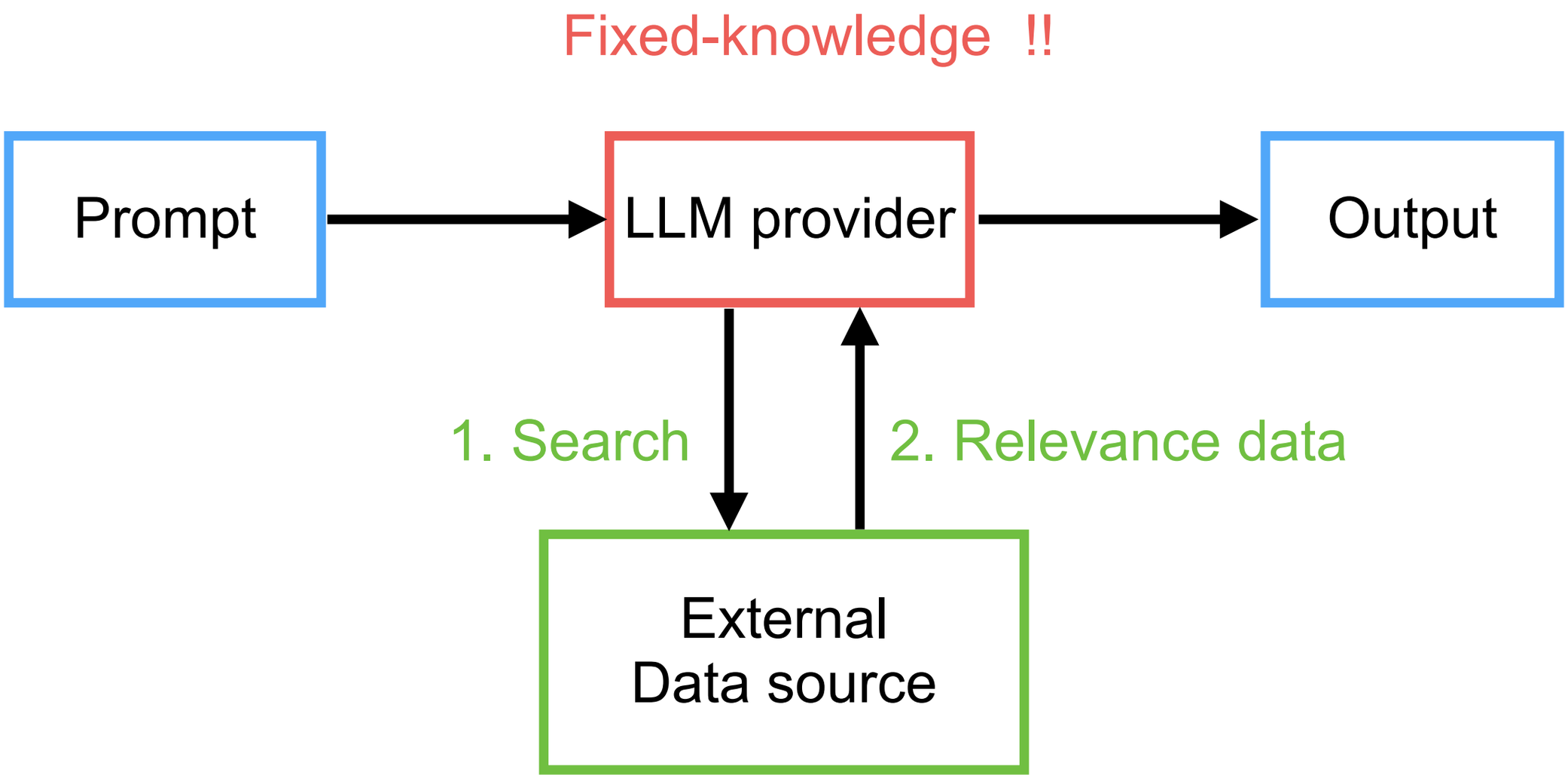
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Limitation of LLM



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RAG ?



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How to search/retrieve data ?

Data source

Data preparation

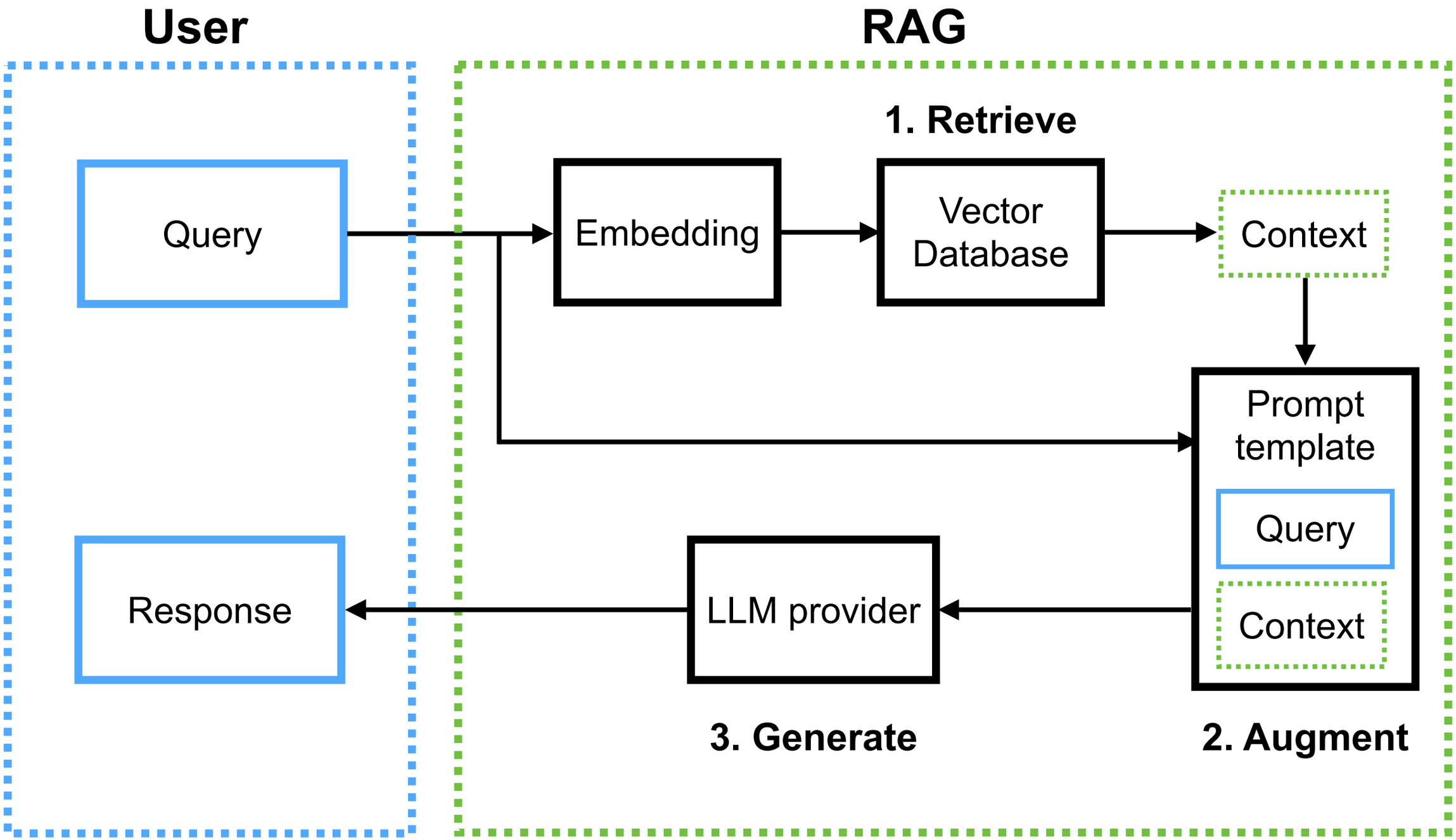
Search

Search result



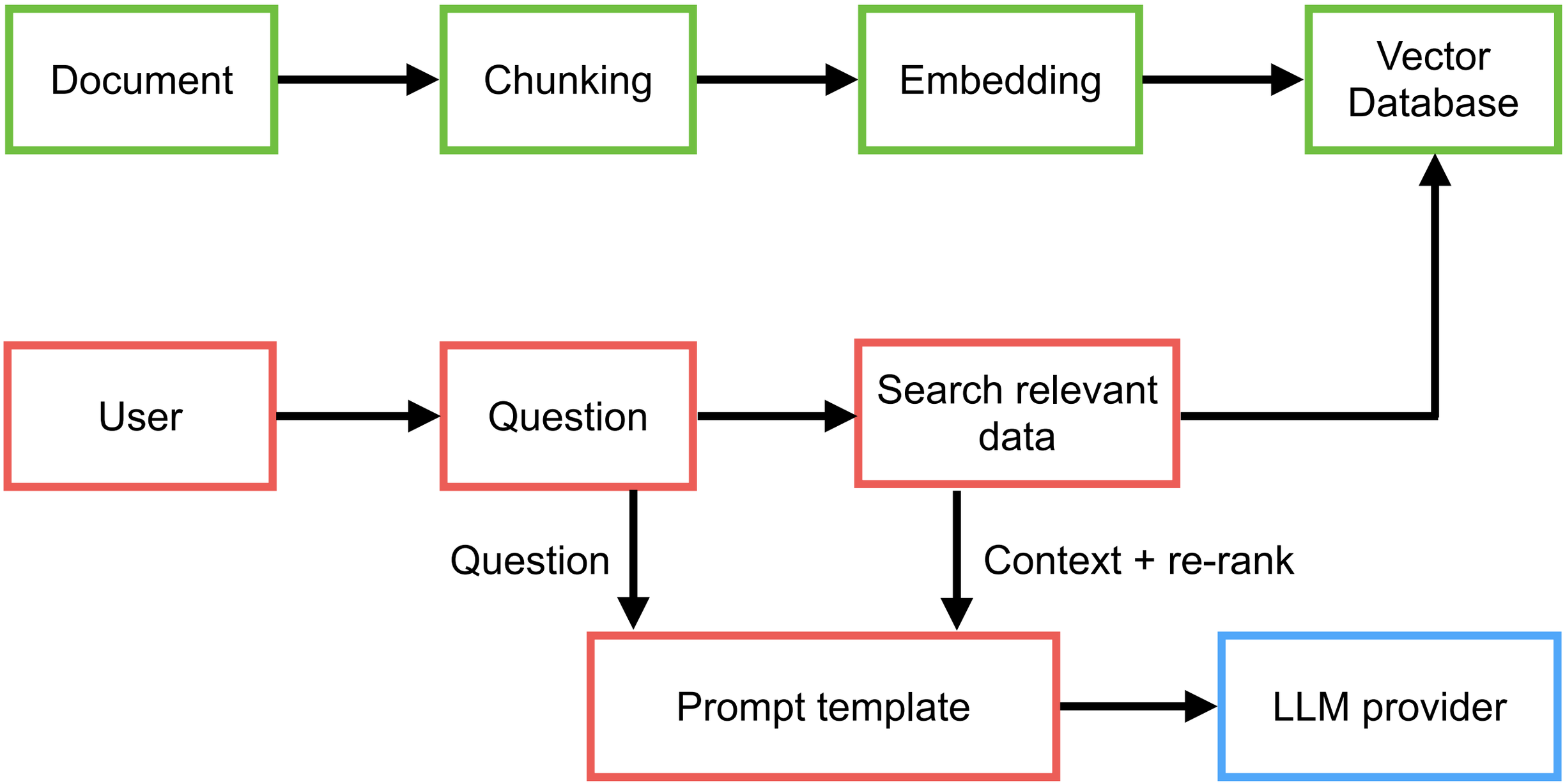
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Basic RAG Architecture



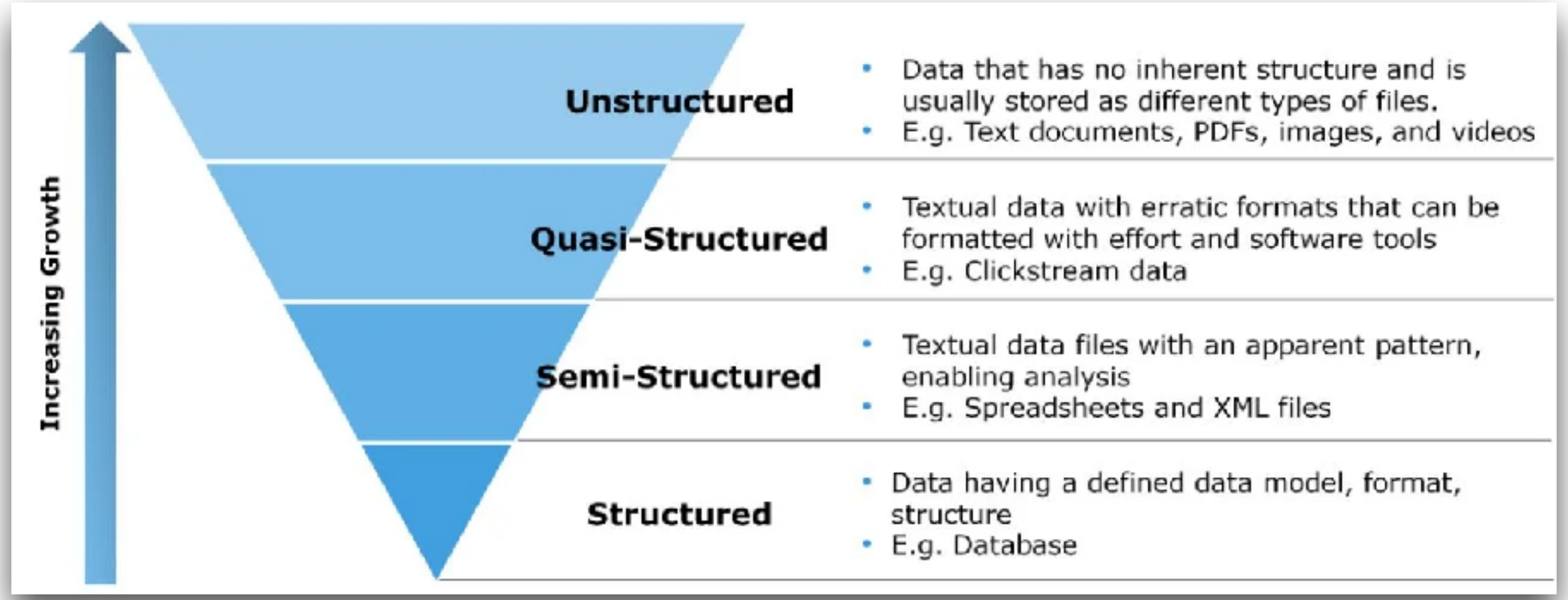
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RAG process



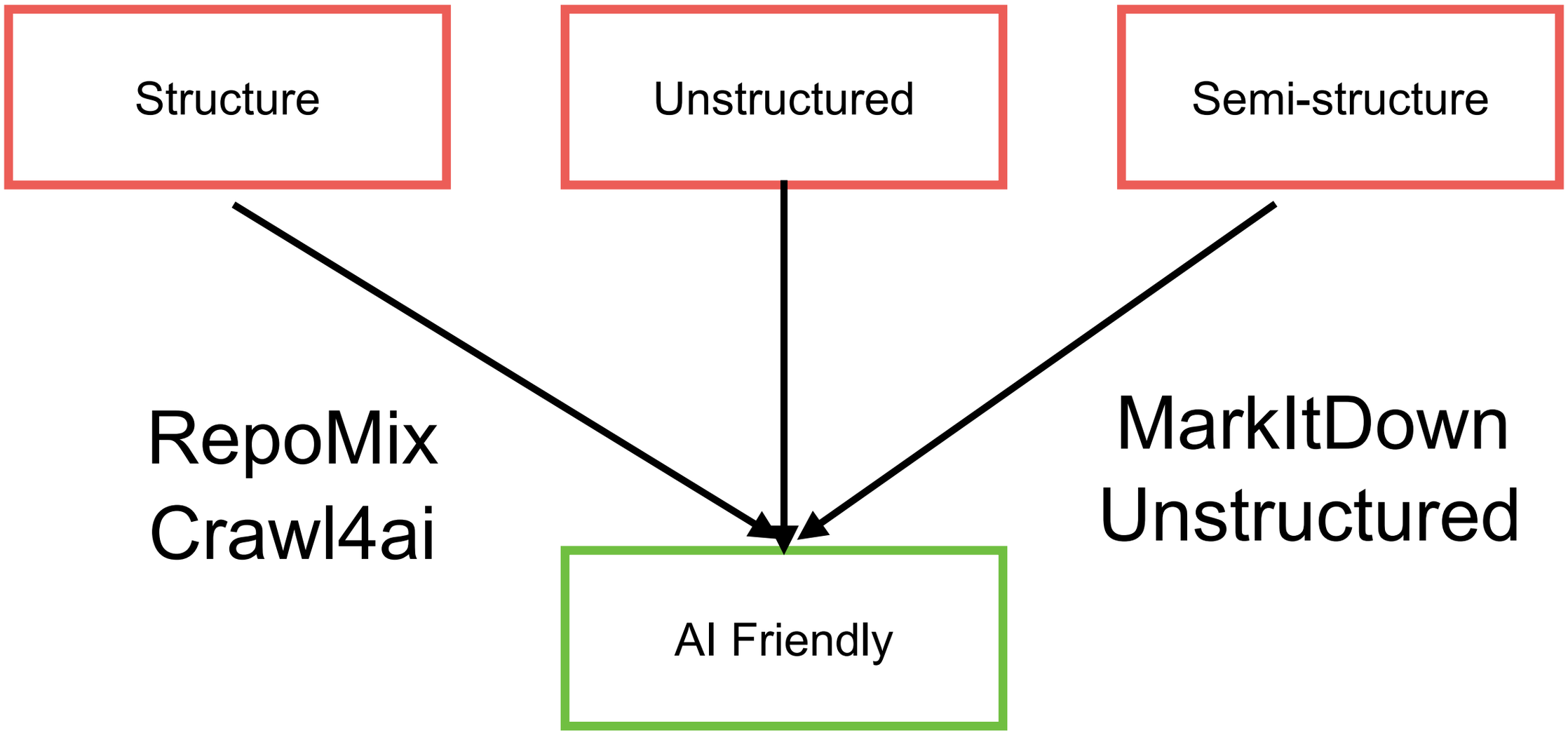
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Structures of Data ?



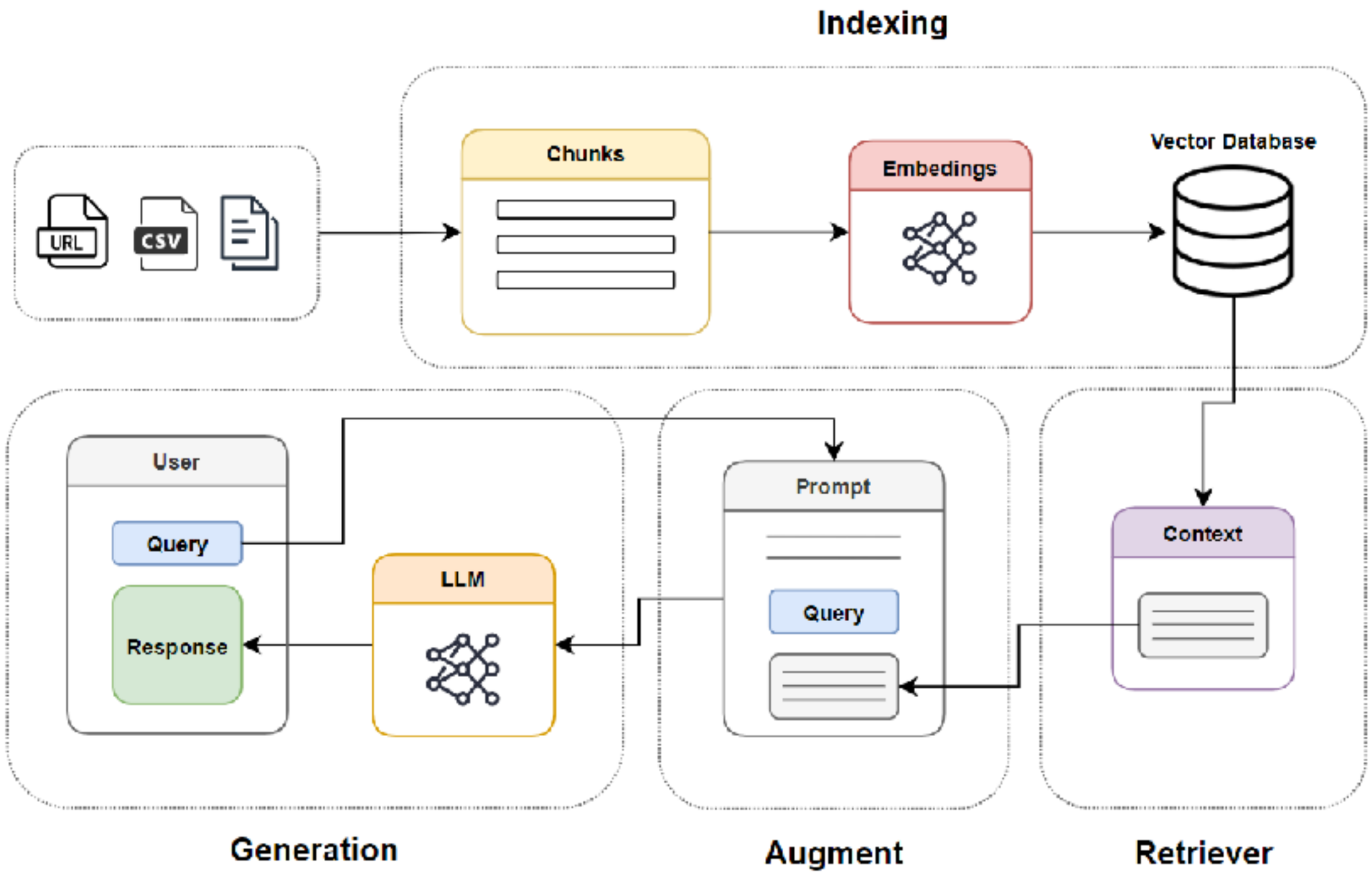
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Friendly Data for LLM/AI



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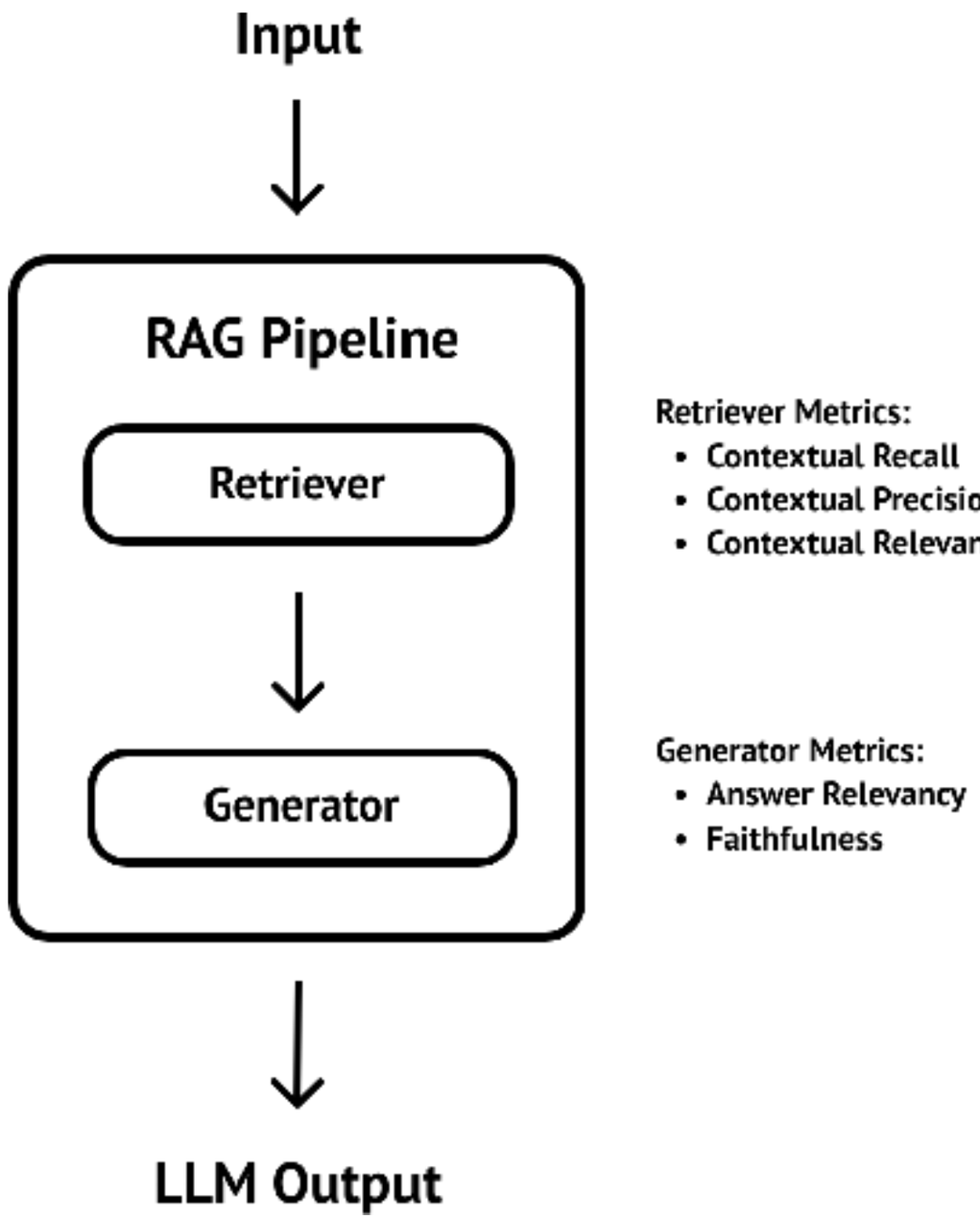
RAG Cookbook (techniques)



https://github.com/athina-ai/rag-cookbooks

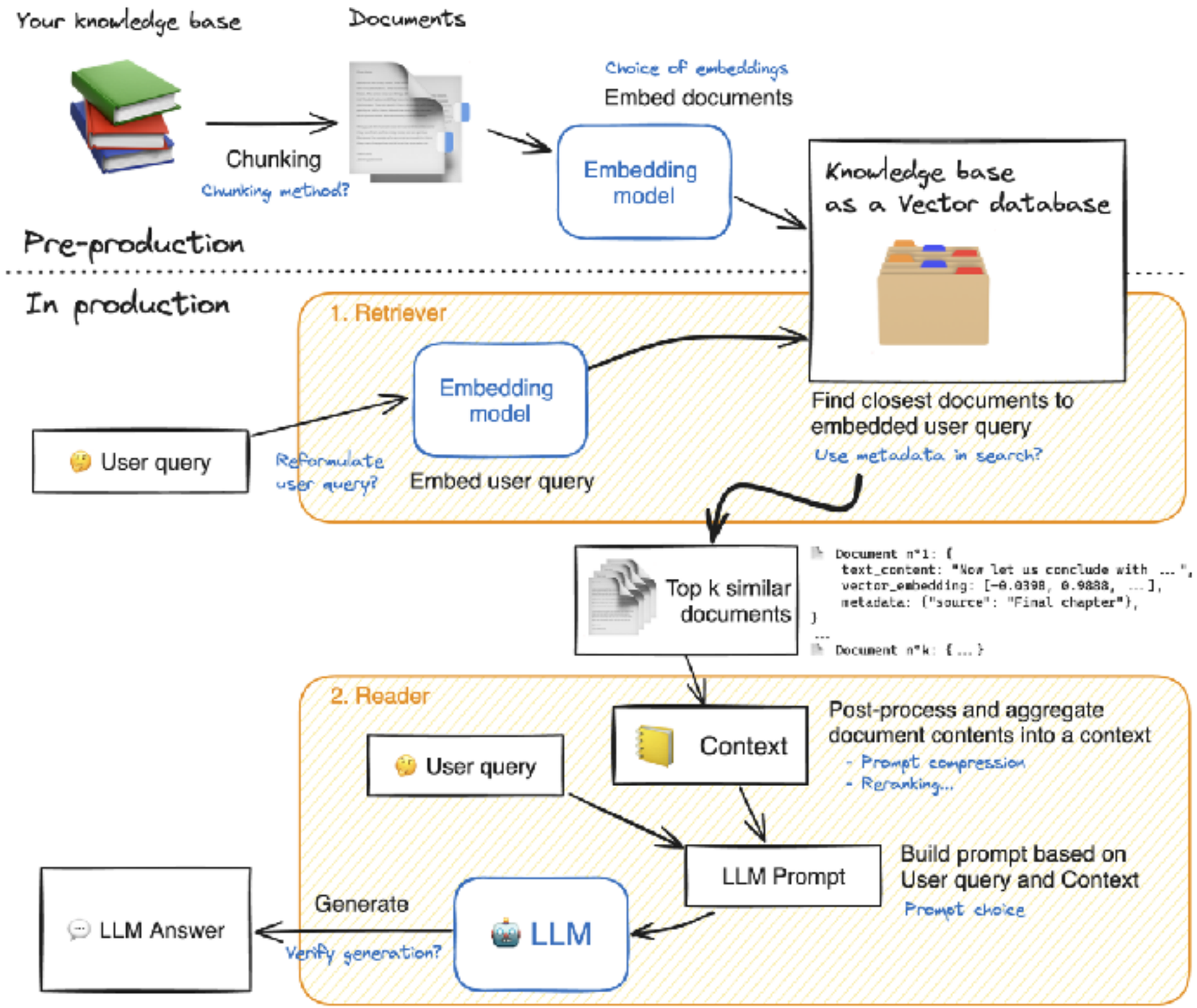
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RAG Evaluation !!



https://www.deepeval.com/guides/guides-rag-evaluation

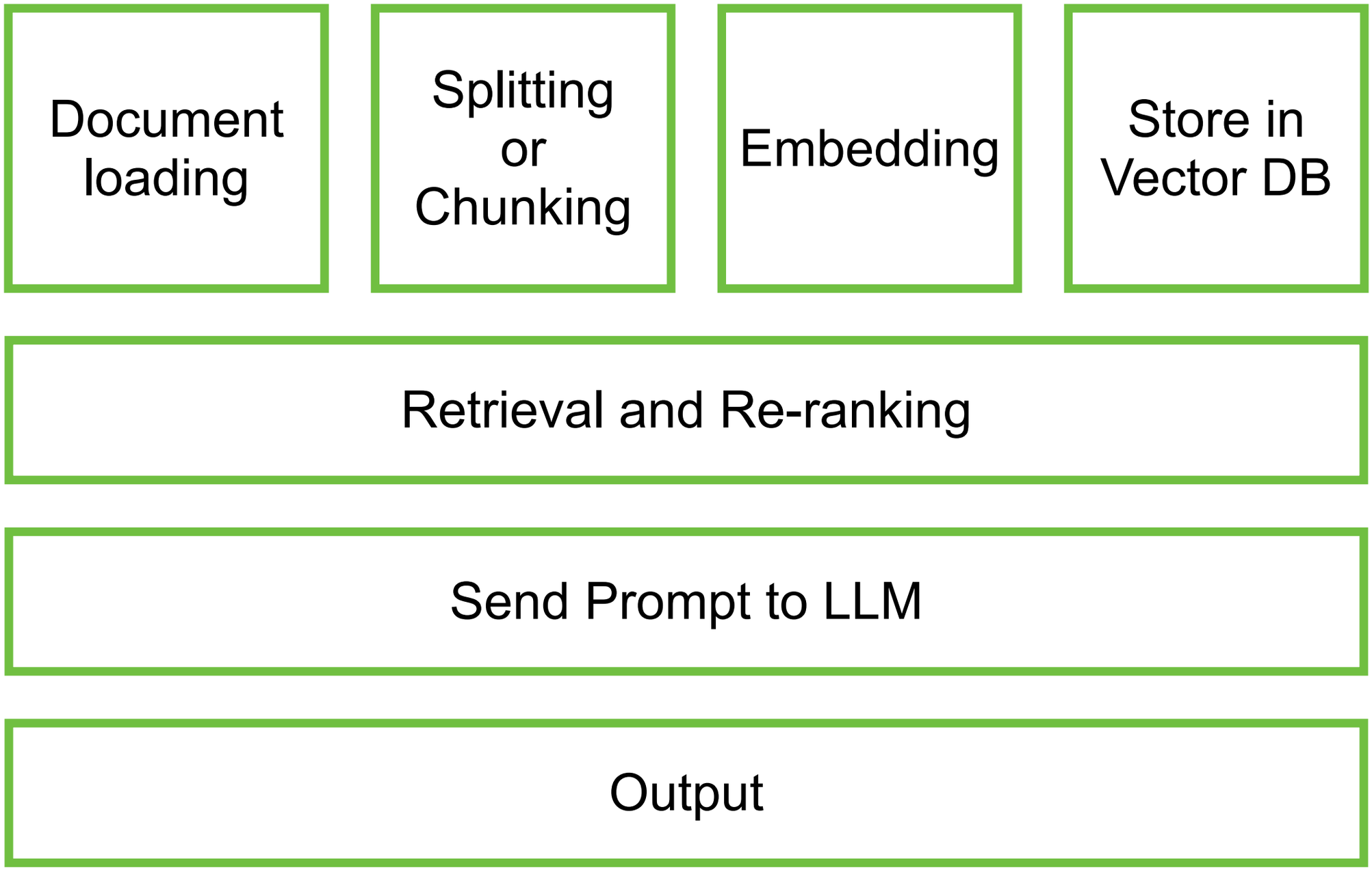
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https://huggingface.co/learn/cookbook/en/rag_evaluation

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RAG Implementation



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Failure Points of RAG

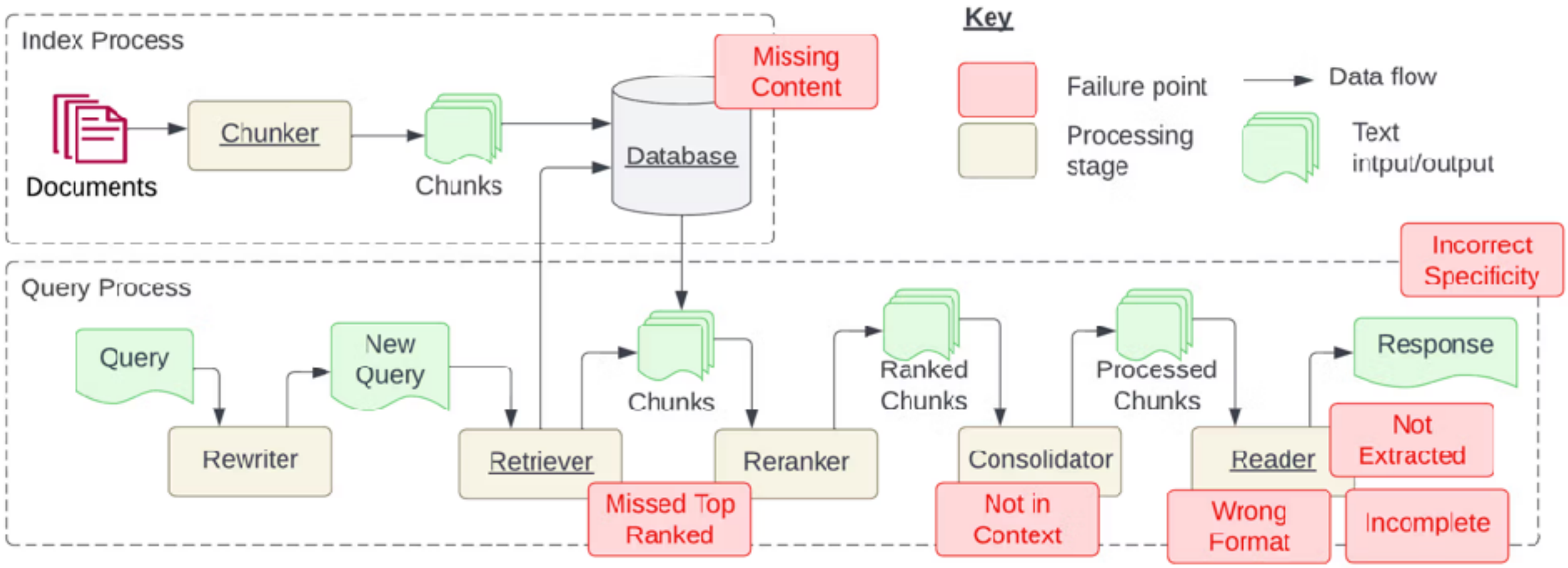


Figure 1: Indexing and Query processes required for creating a Retrieval Augmented Generation (RAG) system. The indexing process is typically done at development time and queries at runtime. Failure points identified in this study are shown in red boxes. All required stages are underlined. Figure expanded from [19].

https://www.galileo.ai/blog/mastering-rag-how-to-architect-an-enterprise-rag-system

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RAG is better

But come with Cost !!

More latency

Retrieval errors

Accuracy !!

More Complexity

Maintenance overhead

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RAG Techniques ?

Semantic chunking

Chunk size selector

Context chunk header

Adaptive RAG

Re-ranking

Graph TAG

https://github.com/FareedKhan-dev/all-rag-techniques



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Pre-Retrieval optimization ?

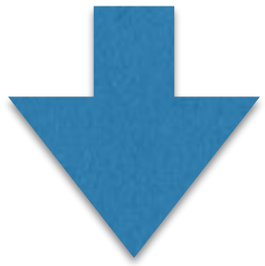
Preprocessing and cleansing data
Chunking strategies
Add metadata
Embedding model selection



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Chunking strategies

Size of Document
> context window size



Chunking

Chunk 1

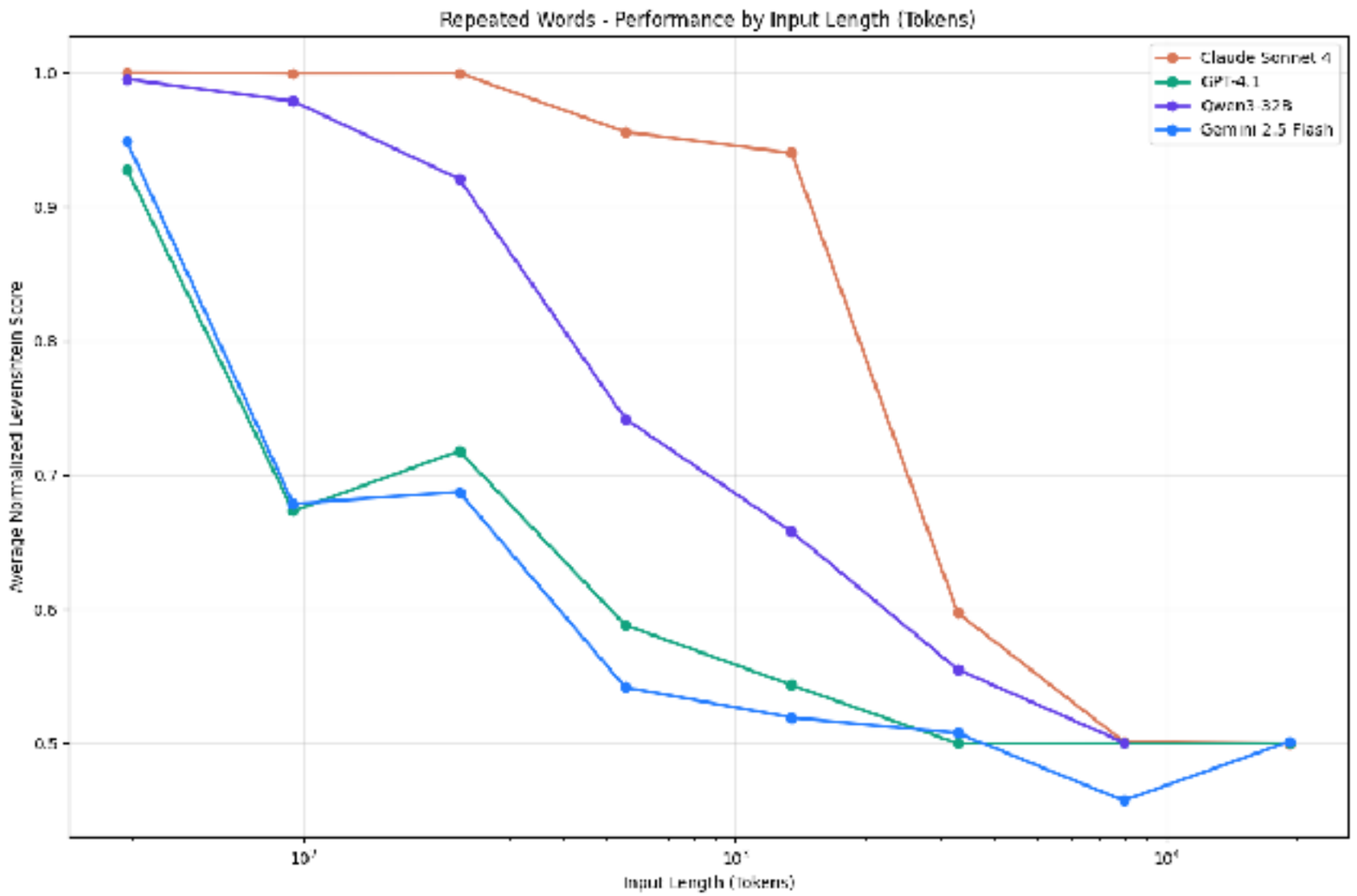
Chunk 2

Chunk 3

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Context Window ?

Number of tokens in the context window increases,
the model's ability to accurately recall information from that context decreases



https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents

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Chunking Strategies !!

Fixed size
Recursive characters
Document structure-based
Semantic chunking
Agentic chunking

...

https://blog.dailydoseofds.com/p/5-chunking-strategies-for-rag



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Good Chunking

Efficiency

Relevance

Context preservation

Improve content generation

Reduce noise

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Bad Chunking

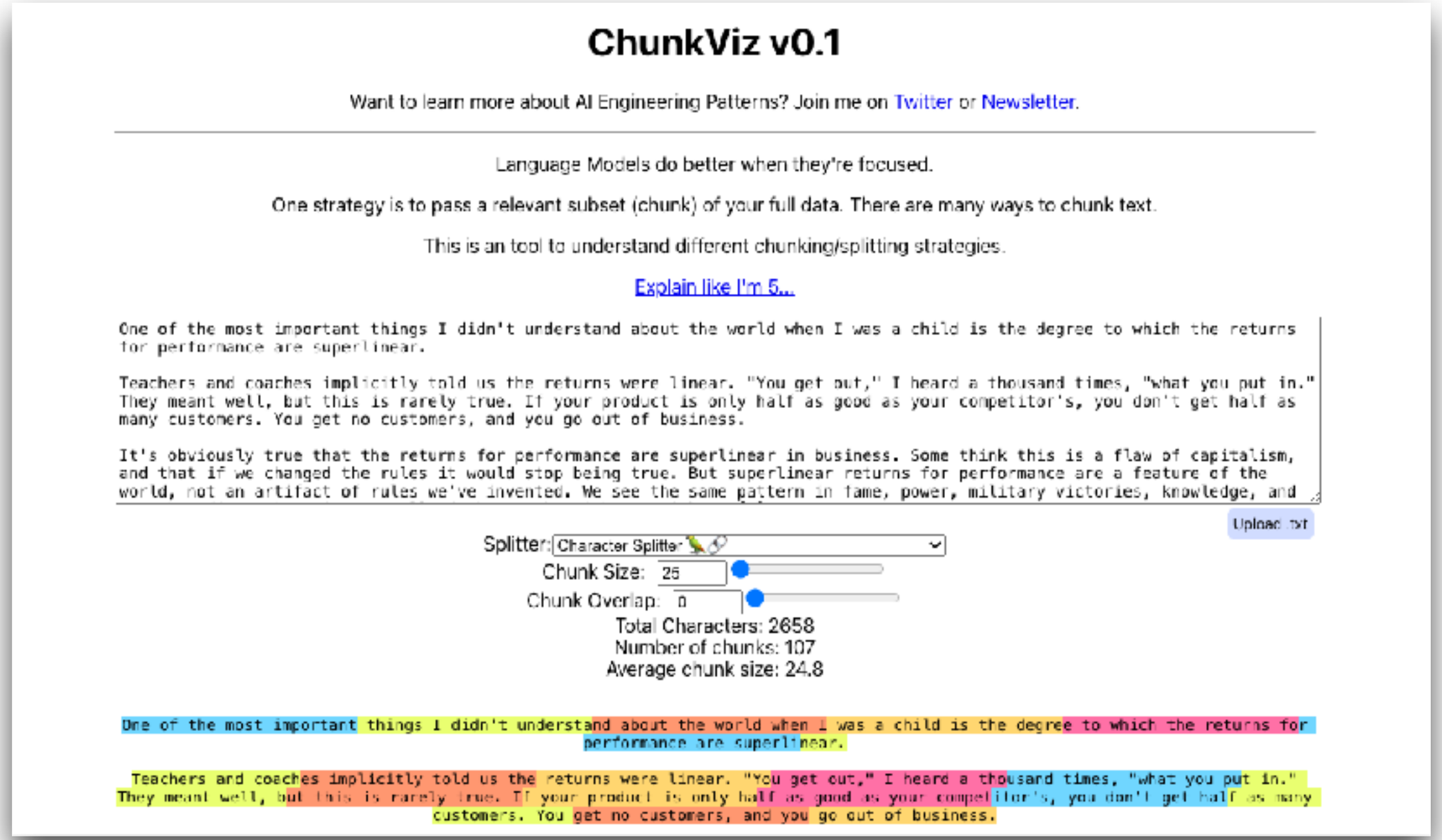
Loss context

Redundancy

Inconsistency

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Chunking Visualization



https://chunkviz.up.railway.app/

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Retrieval optimization ?

Re-ranking

Hybrid search

Query transformation

Multi-vector embedding

Contextual retrieval

Vector database Selection



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RAG Workshop

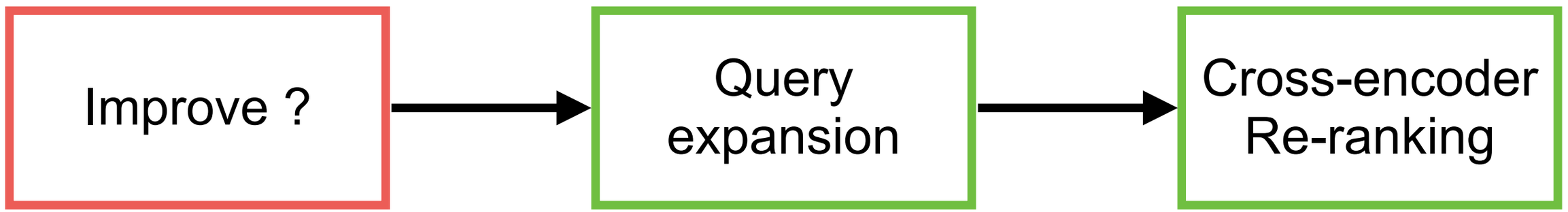
https://github.com/up1/workshop-basic-llm/tree/main/workshop/basic-rag



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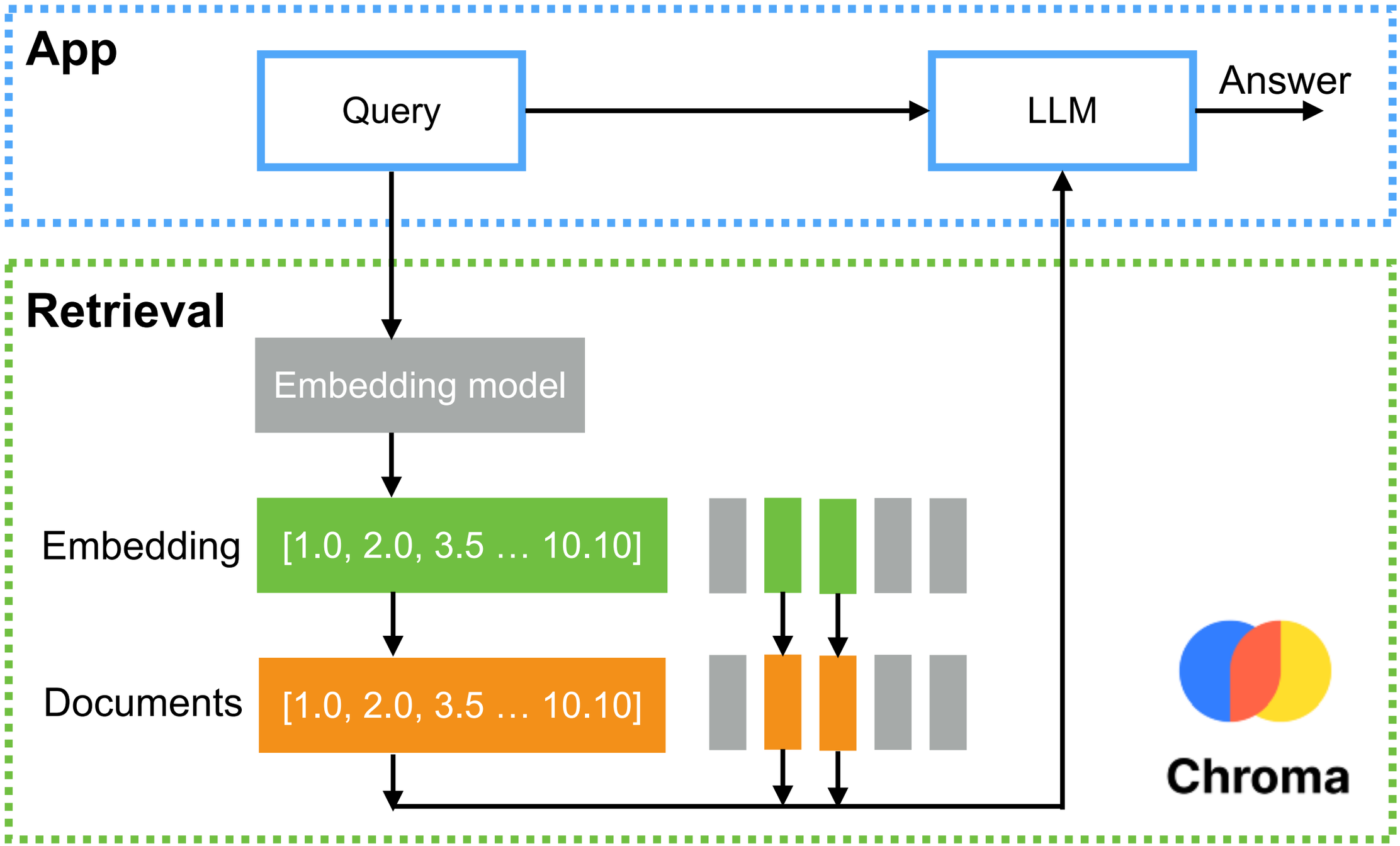
RAG workshop





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Basic RAG



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Query Expansion

Query expansion is a widely used technique to improve the recall of search systems

Ambiguity	Vocabulary mismatch	Lack of context

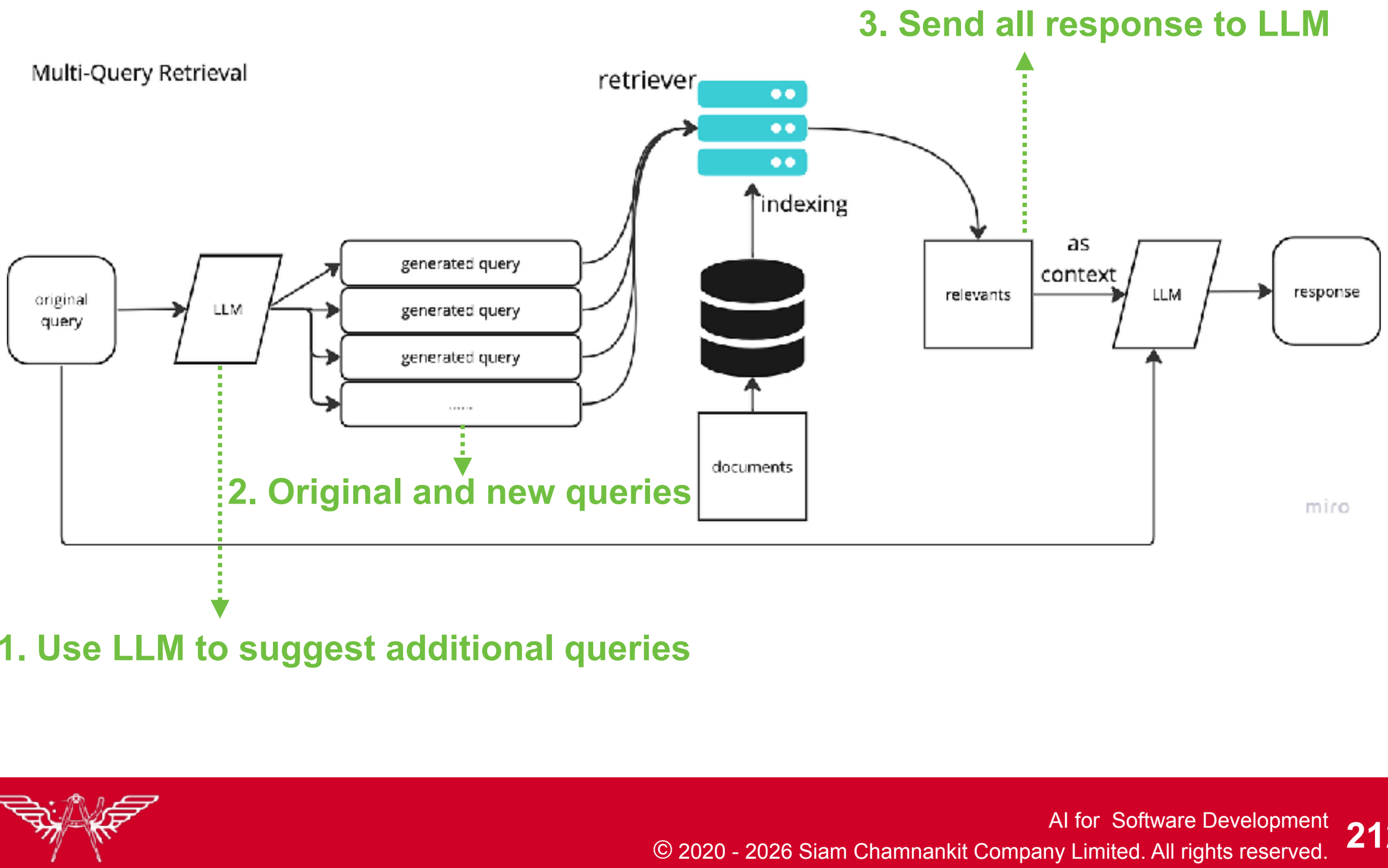
"Add synonyms, related context and contextual terms"

https://arxiv.org/abs/2305.03653



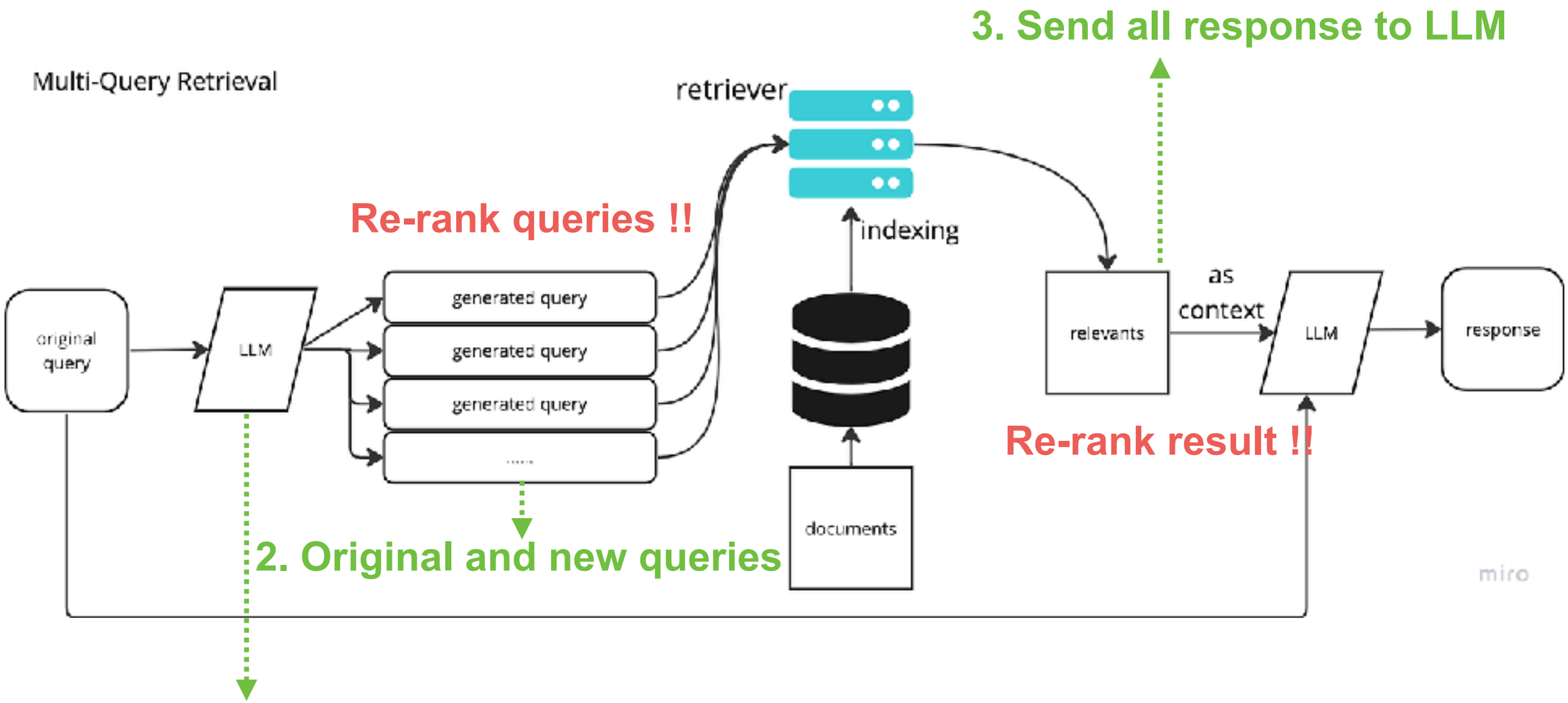
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Query Expansion



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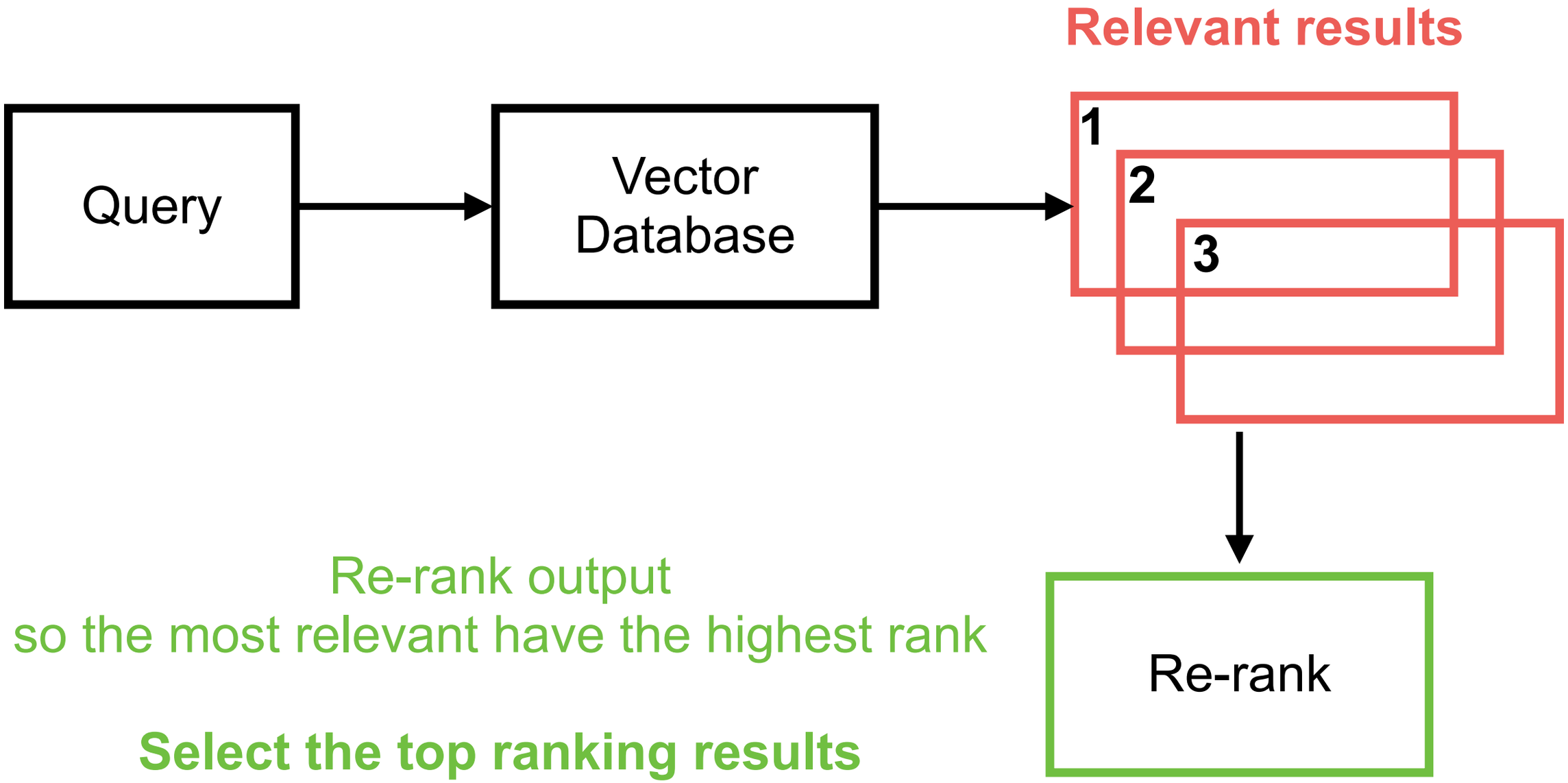
Re-rank !!



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Cross-encoder Re-ranking

How to ordering relevant results !!

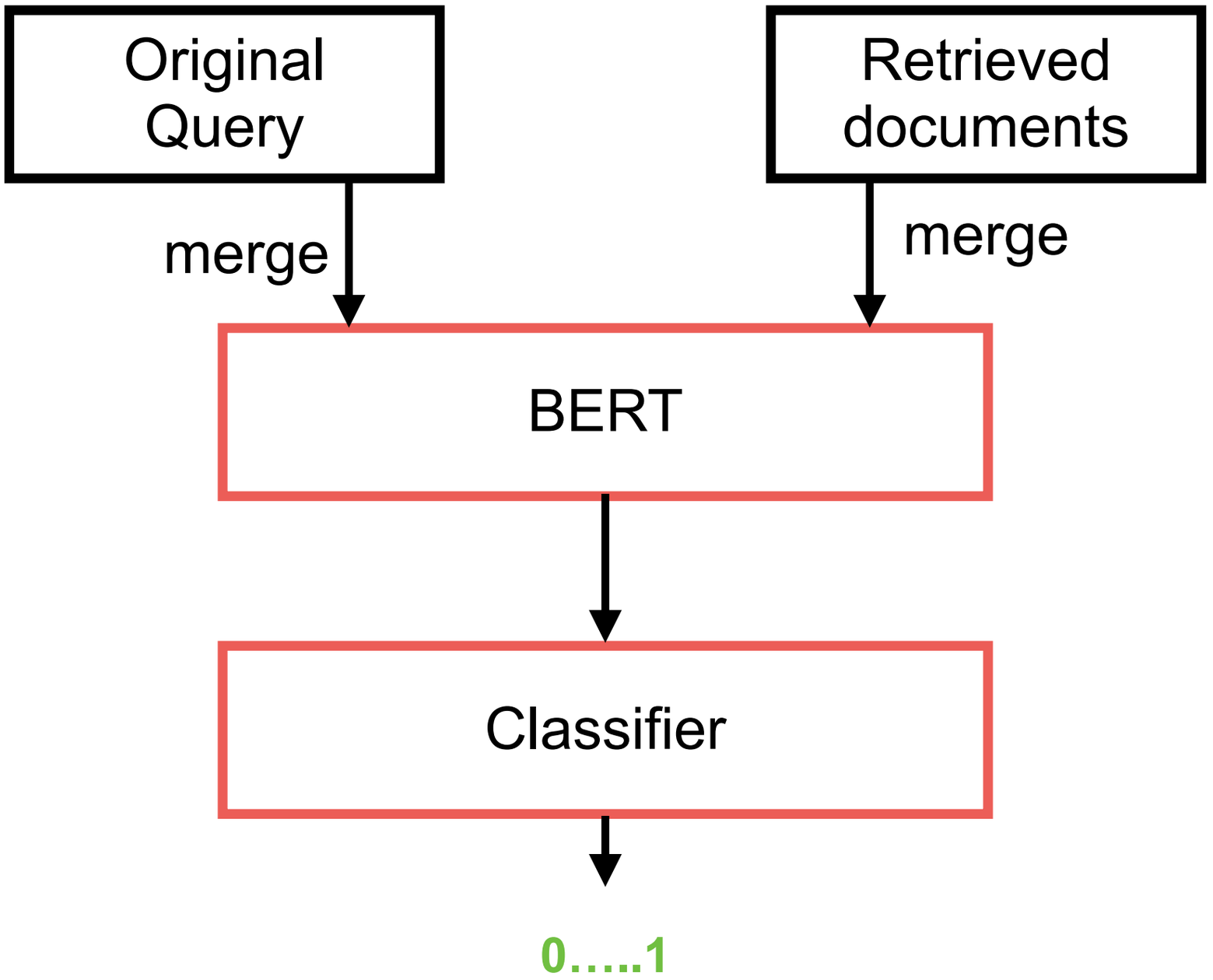


Re-rank output
so the most relevant have the highest rank

Select the top ranking results

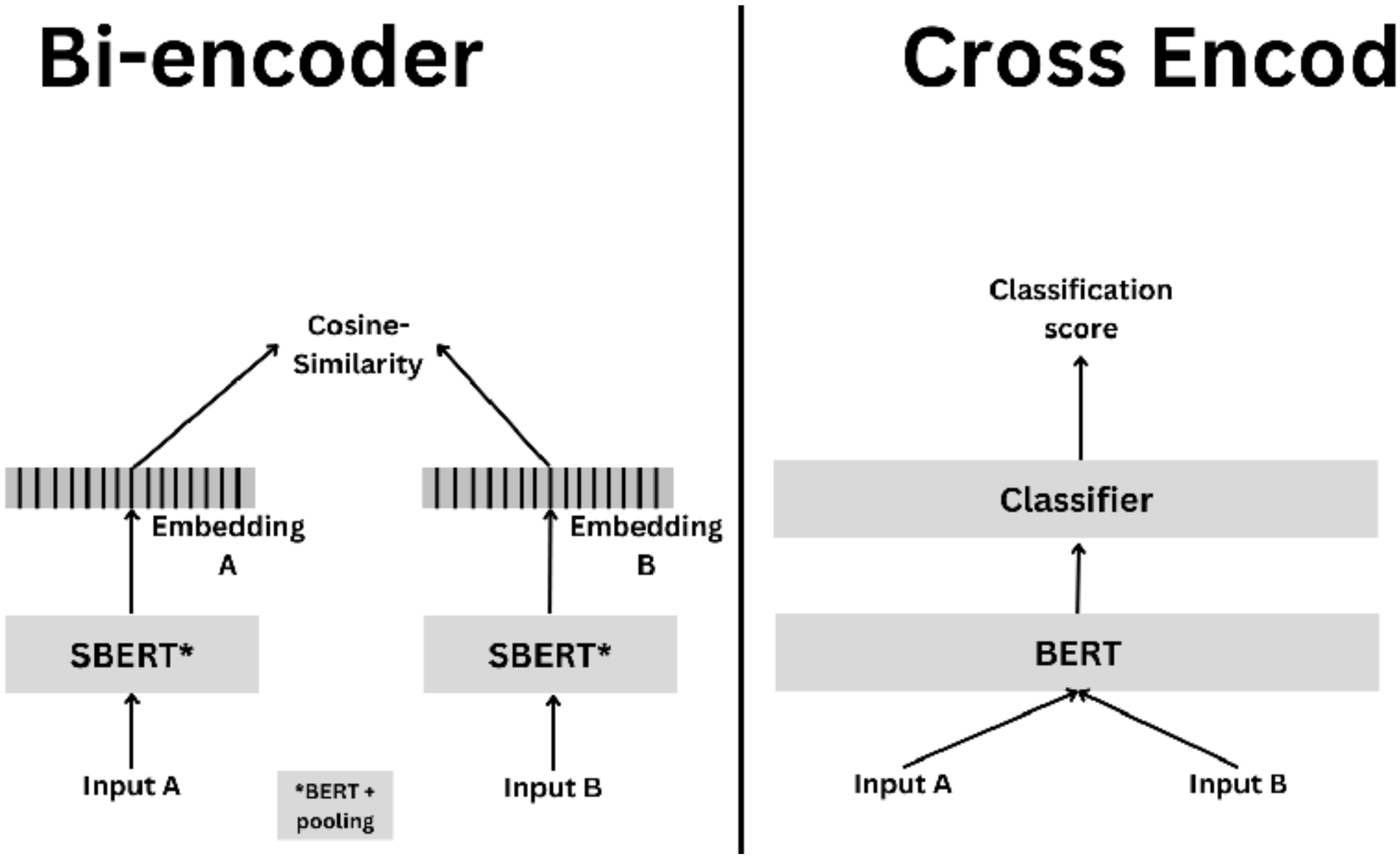
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Cross-Encoder



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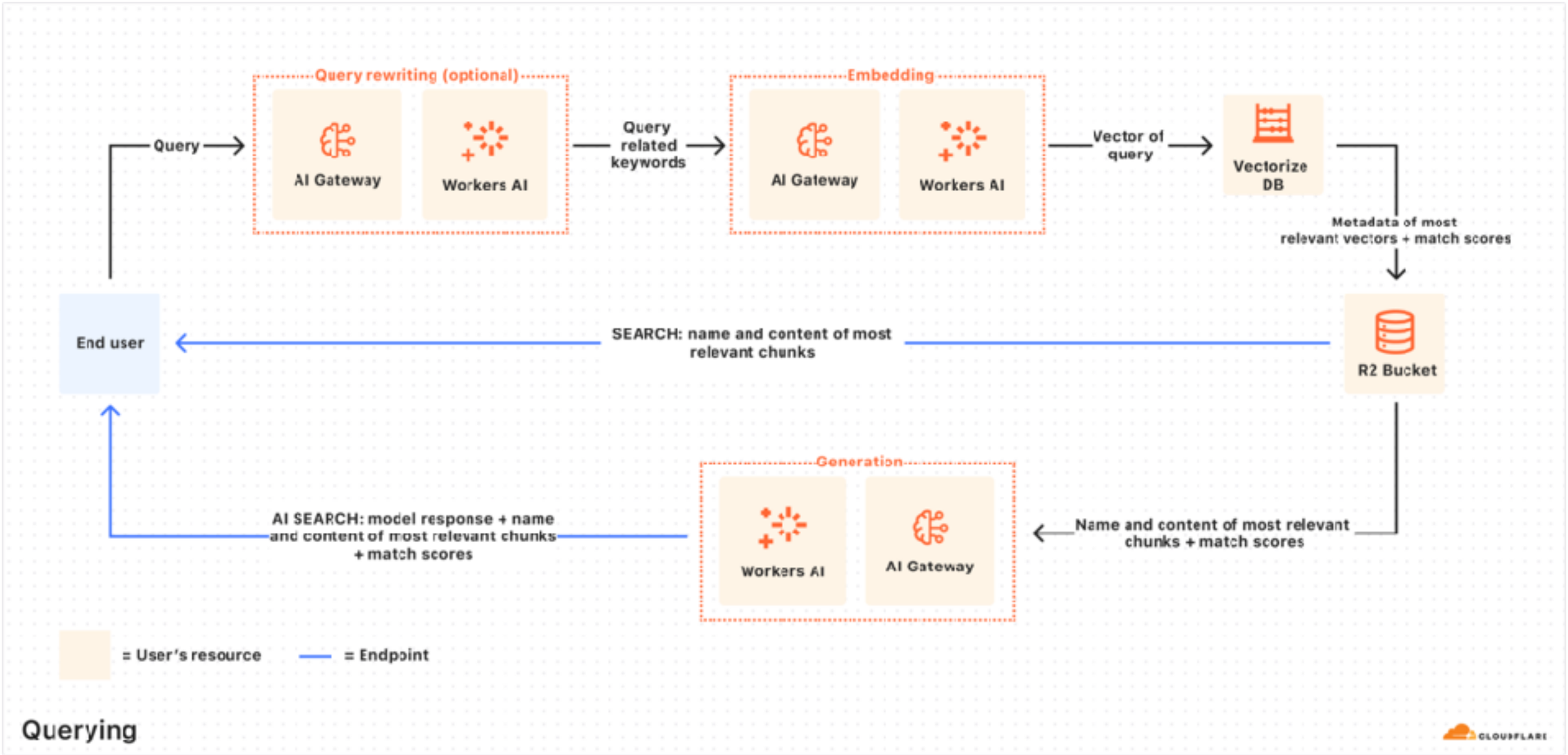
Cross-Encoder !!



https://sbert.net/examples/cross_encoder/applications/README.html

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Cloudflare AutoRAG



https://developers.cloudflare.com/autorag/

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Guardrails



https://github.com/guardrails-ai/guardrails

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Guardrails

Help to build reliable AI applications

Guard for
input and output

Help to generate
Structured output

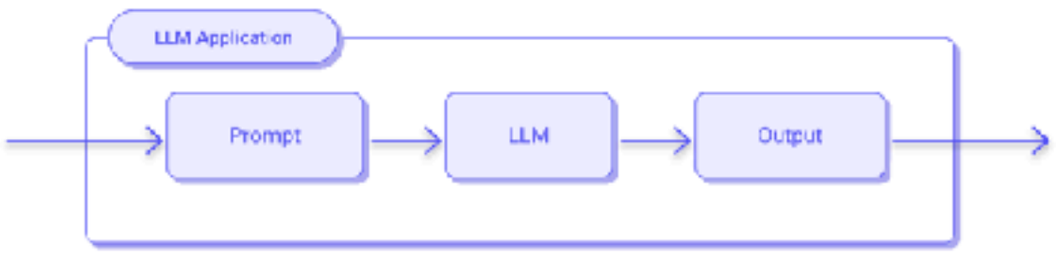
https://github.com/guardrails-ai/guardrails



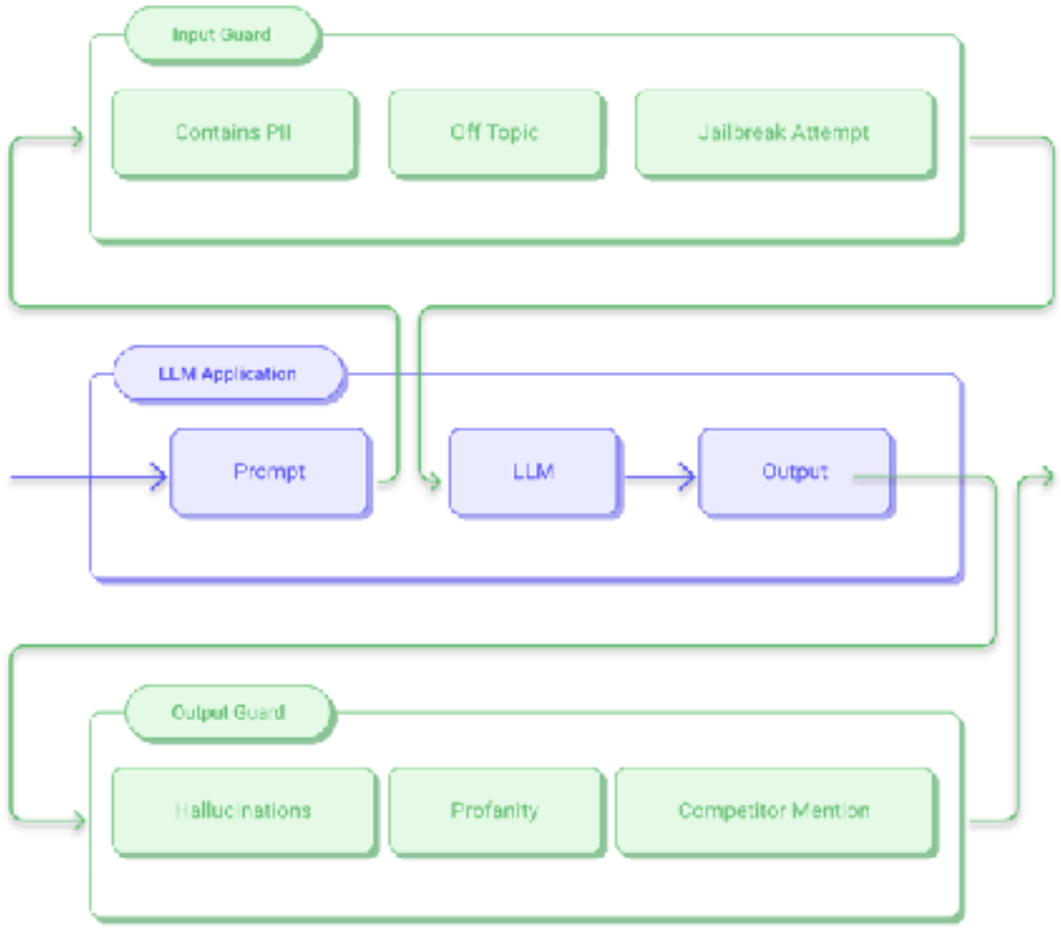
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Guardrails

Without Guardrails

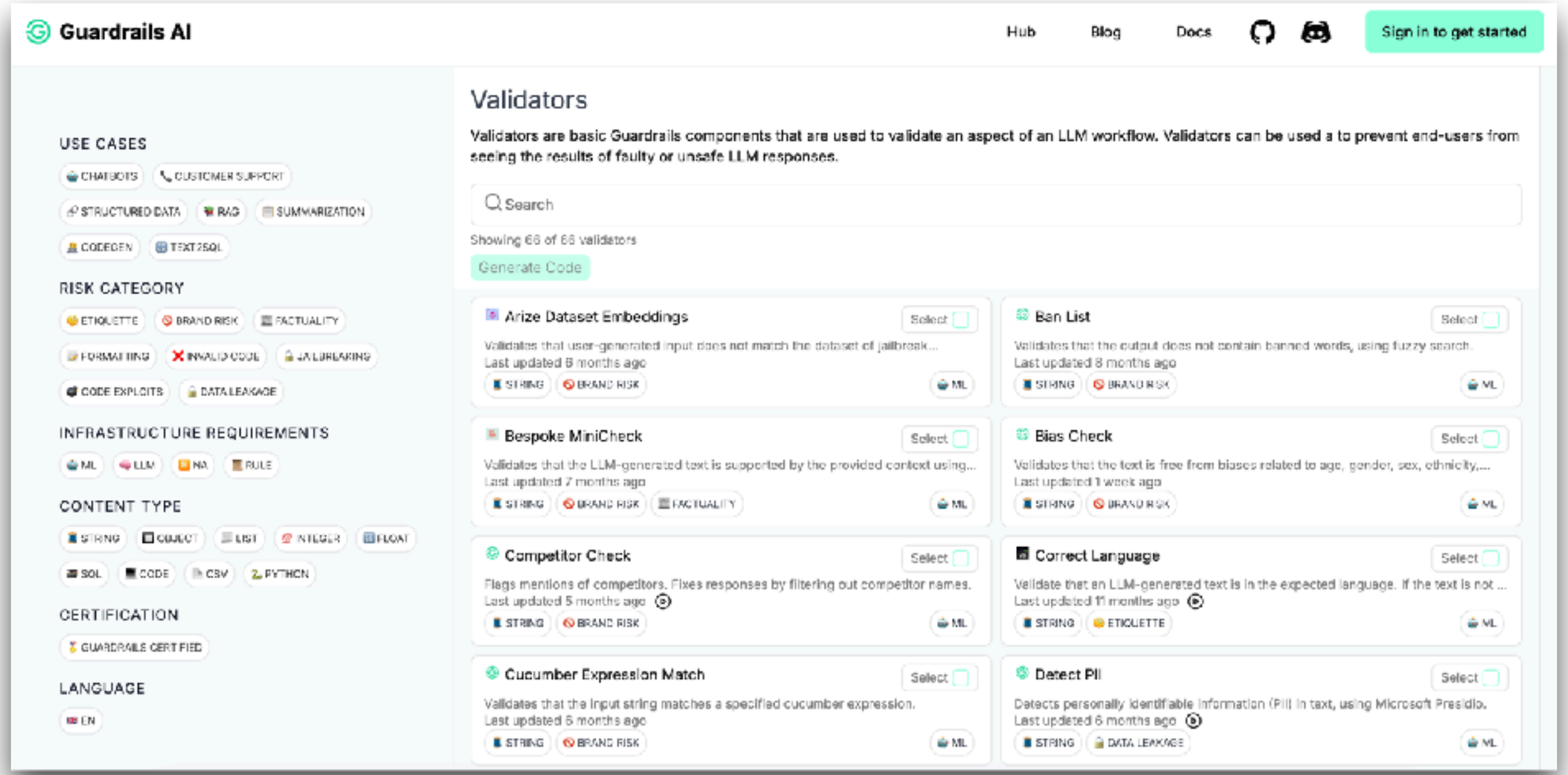


With Guardrails



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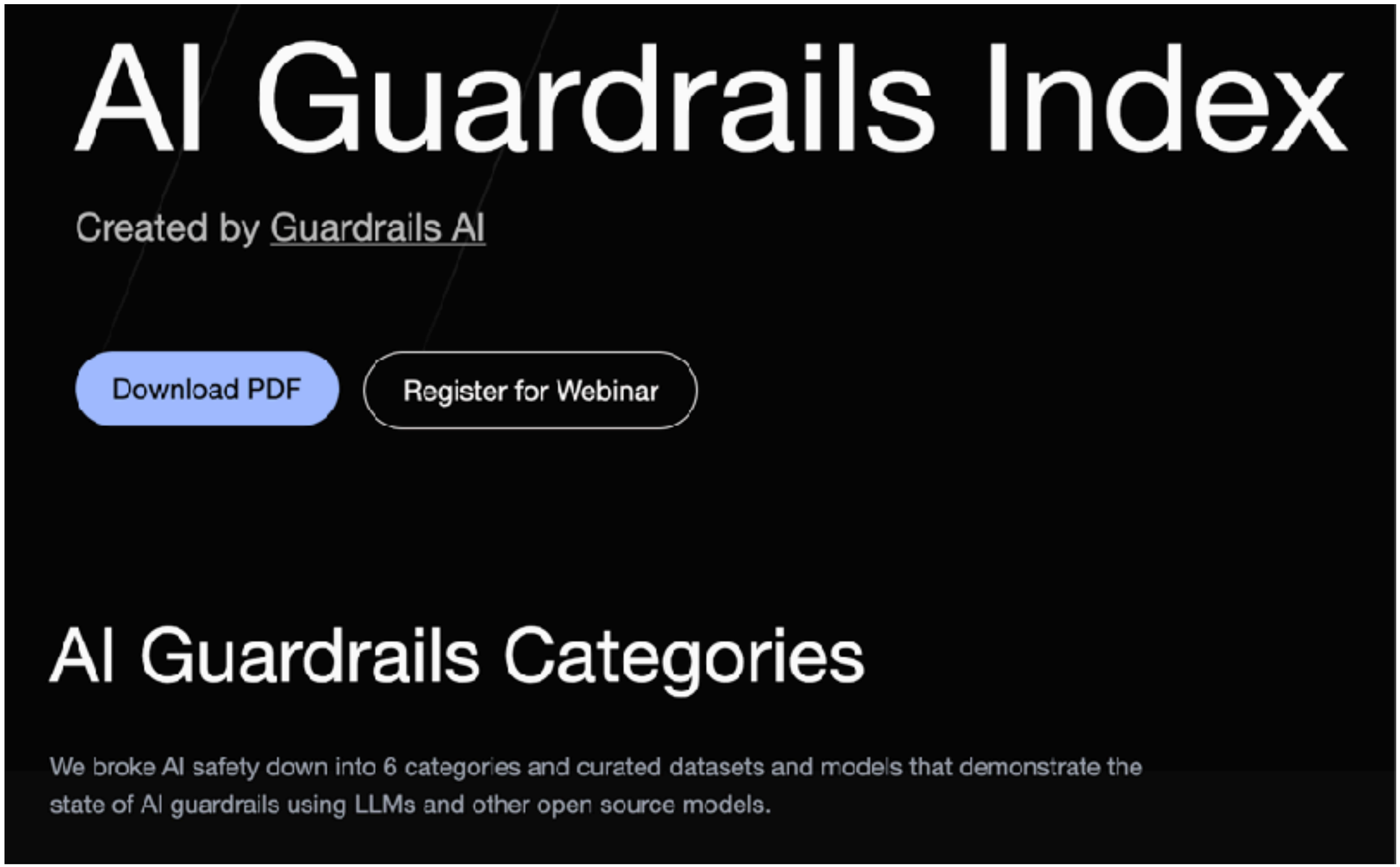
Guardrails Hub



https://hub.guardrailsai.com/

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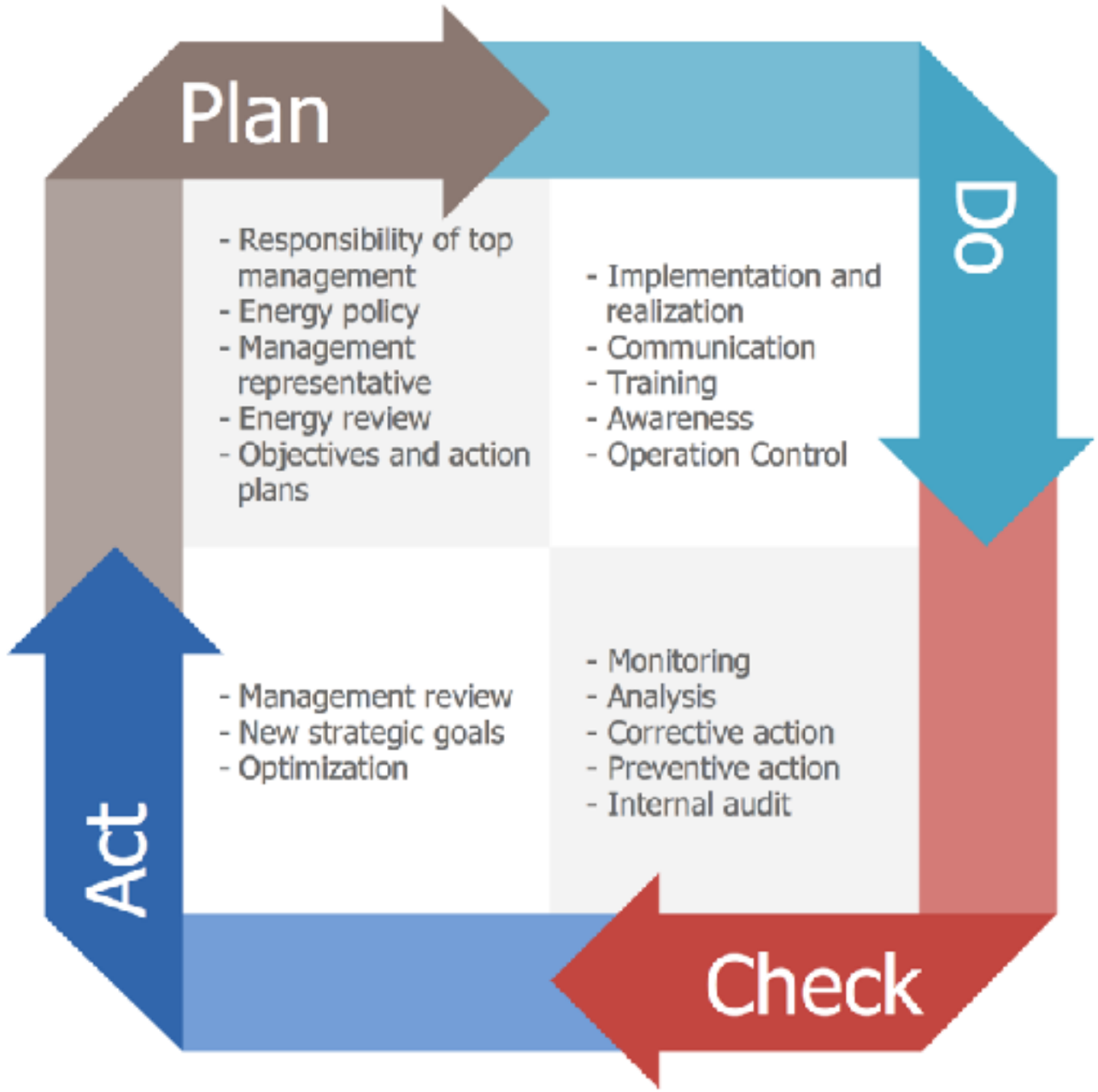
Guardrails Index



https://index.guardrailsai.com/

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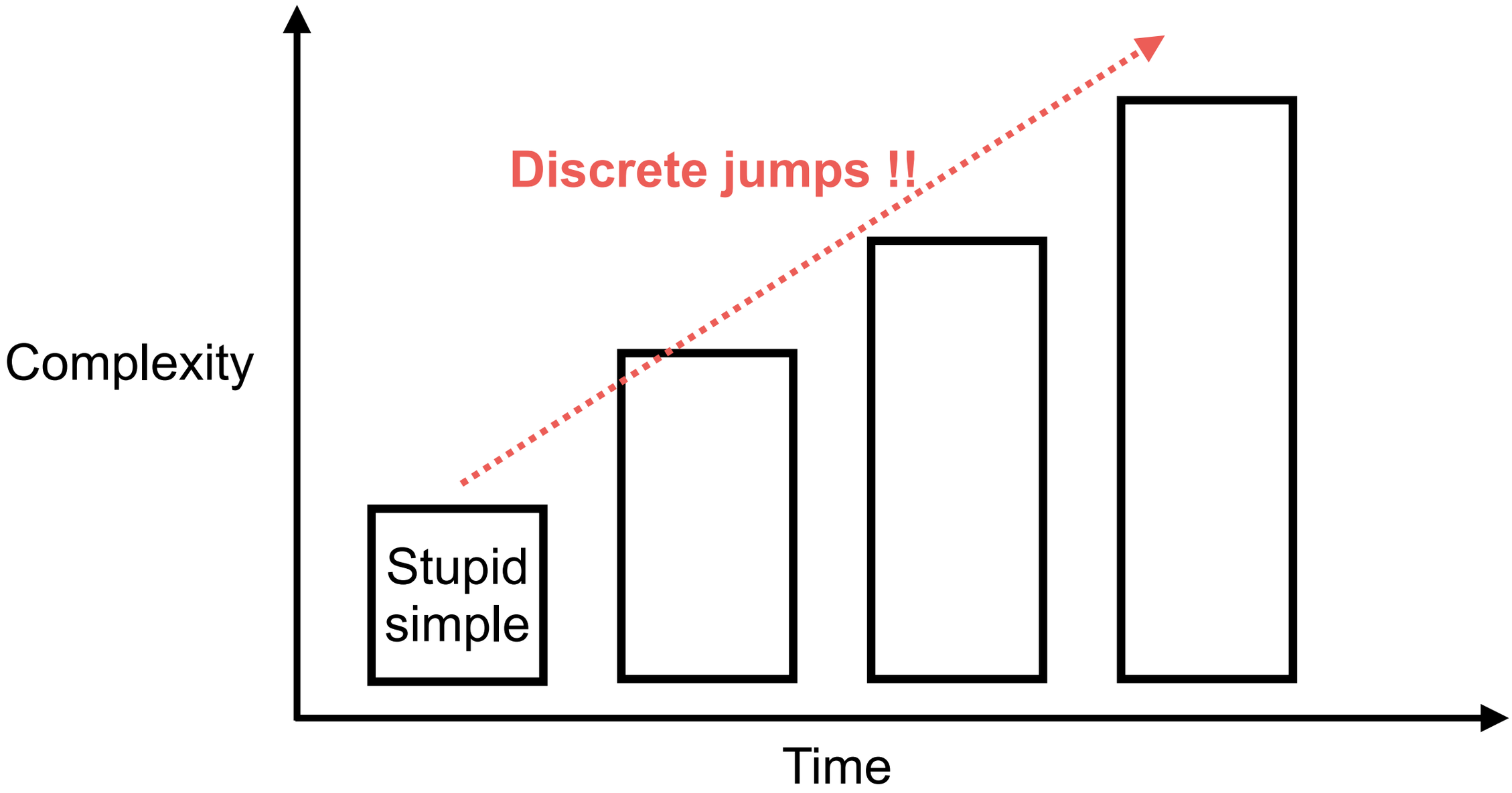
PDCA process



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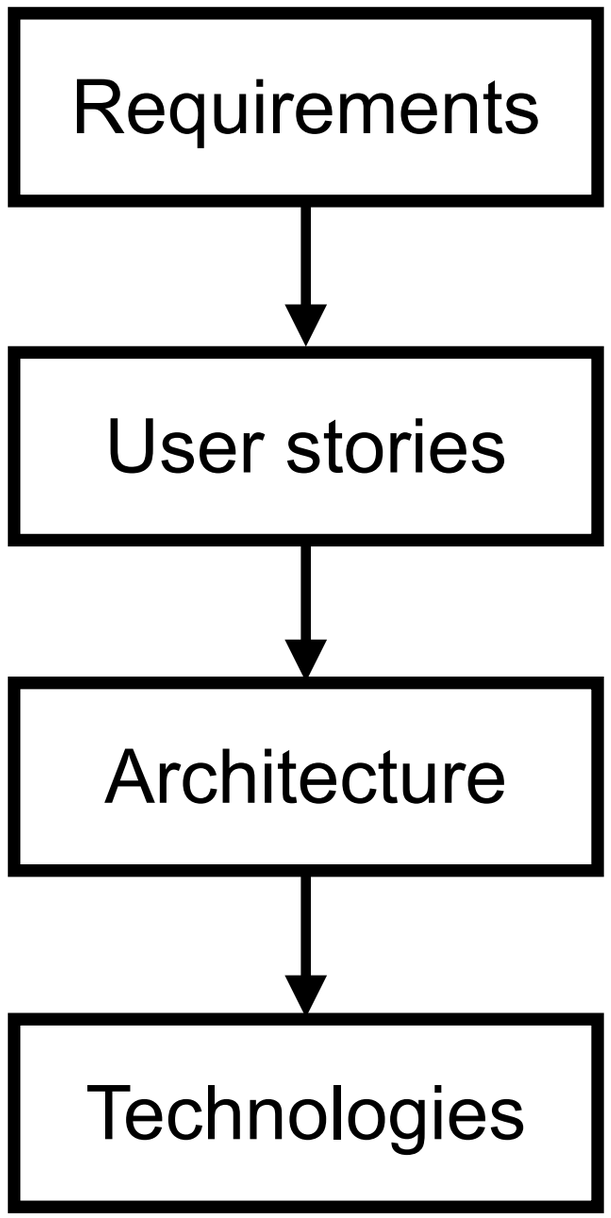
Staged Complexity

Start small and get a feature working first



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Planning workflow



What you’re build ?

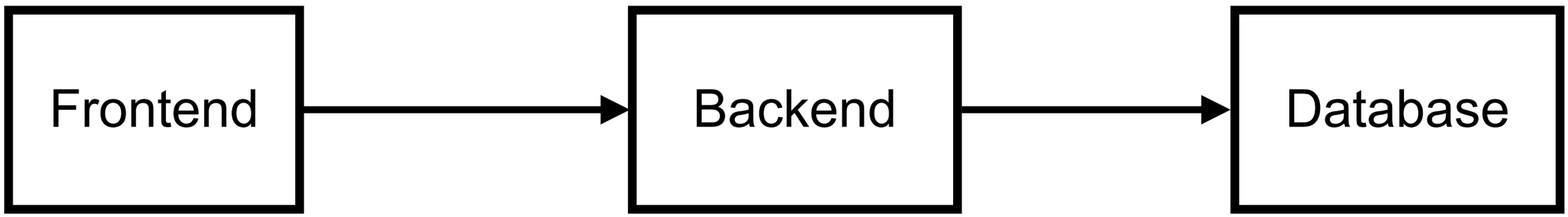
What use need ?

Selection ?

Selection ?

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Architecture of project

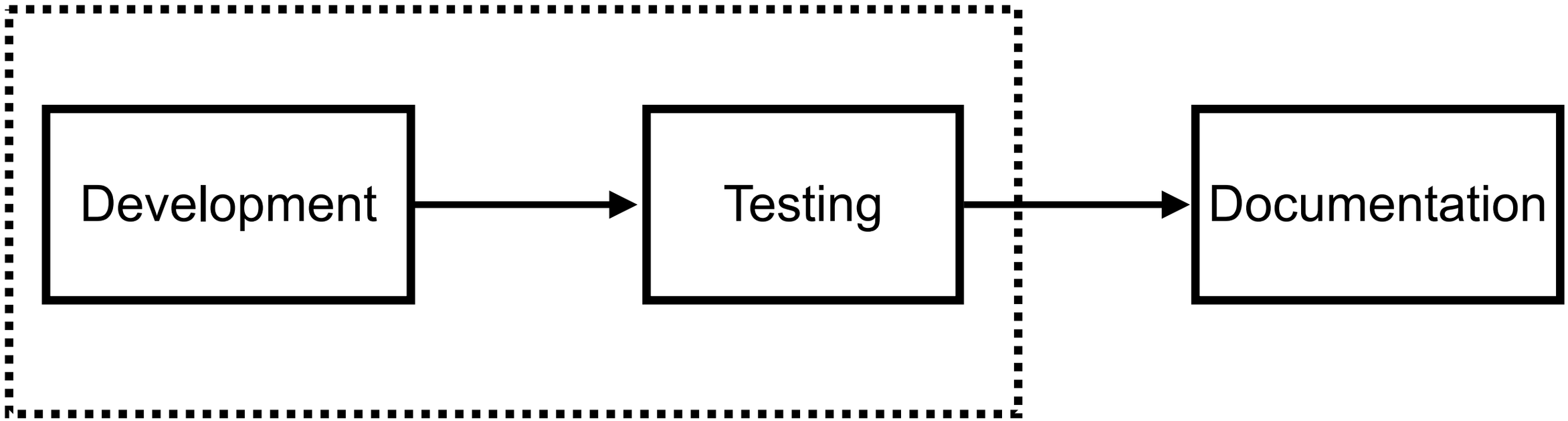


Choose your tech stack !!

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Processes

Iterative and incremental process



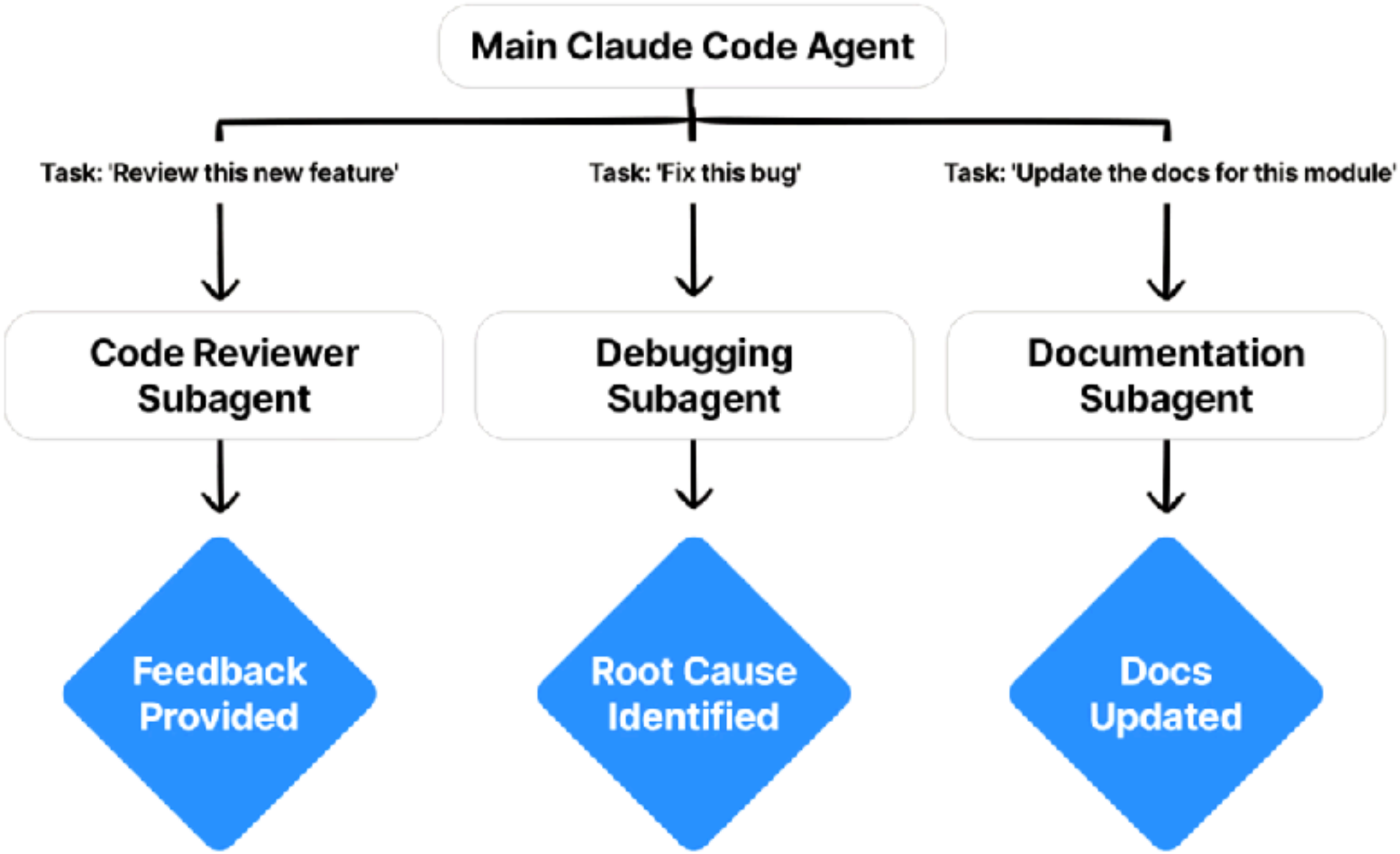
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AI Subagent



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https://code.claude.com/docs/en/sub-agents

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Start your journey

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